DEPLOY A THREE TIER WEB APPLICATION ON AWS

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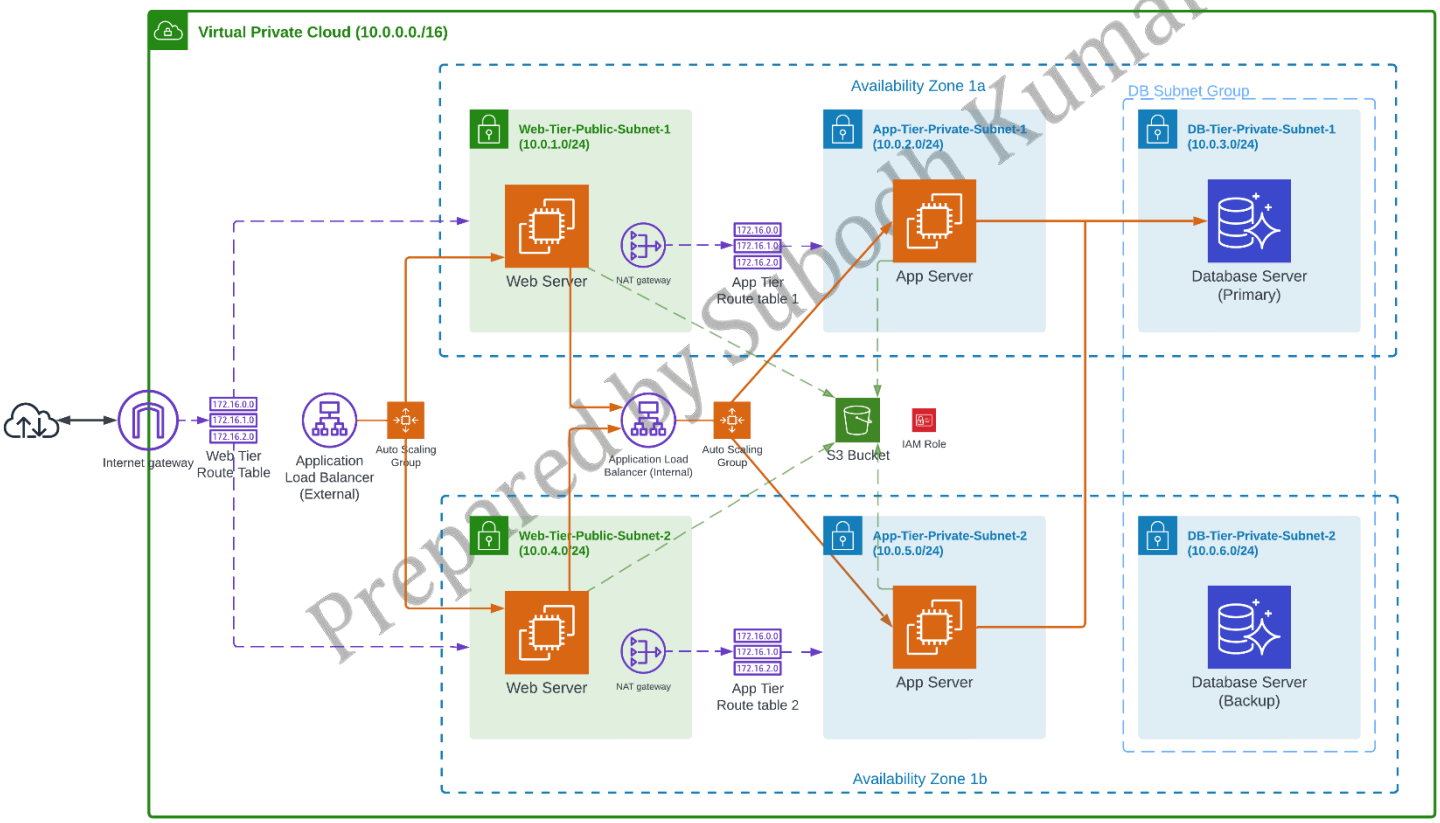

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The three-tier web application architecture is a robust and scalable design pattern that separates an application into three distinct layers: the presentation/web tier, the application tier, and the database tier. This architecture enhances security, manageability, and flexibility by isolating each layer's responsibilities.

In this documentation, let's see how to deploy a three-tier web application on AWS

Introduction

A three-tier architecture is a common architectural pattern used in software development and deployment, particularly in cloud environments like AWS. As I mentioned before, it separates an application into three main layers or tiers: the presentation/Web tier, the application tier, and the database tier.

1. **Presentation/Web Tier (Frontend)** - This is the user interface layer where users interact with the application. It's responsible for presenting information to the user and collecting user inputs.

2. **Application Tier (Backend)** - This is the logic layer where the actual application logic, processing, and control flow take place. It processes user inputs, performs calculations, and makes decisions.

3. **Database Tier (Database)** - This layer is responsible for data storage and management. It stores, retrieves, and manages data that the application tier needs.

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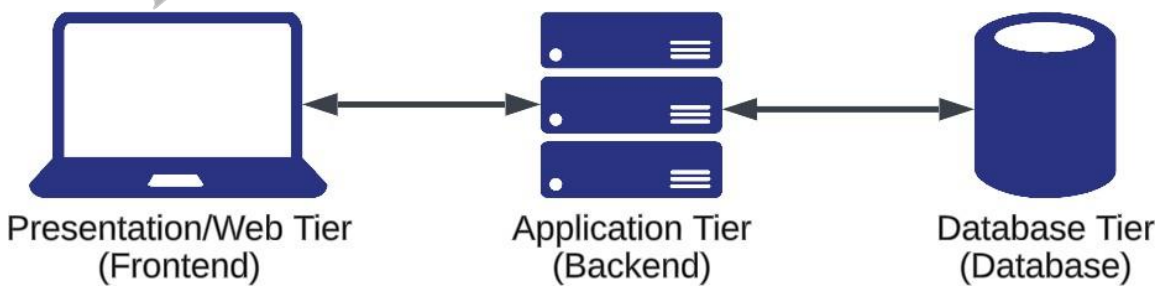
Users interact with the application via the presentation tier. These requests are sent to the application tier, where the logic processes the request. The application tier interacts with the database tier to retrieve or store data. The processed data or output is sent back to the presentation tier, which displays the results to the user.



Project Overview

In this project we are implementing **Highly Available, Highly Secured, Highly Scalable** and **Fault tolerant** three tier application that AWS

Throughout this project, we have to interact with the following components on AWS.

1. Amazon Elastic Compute Cloud (EC2)
2. Amazon Simple Storage Service (S3)
3. Virtual Private Cloud (VPC)
4. Auto Scaling Group (ASG)
5. Application Load Balancer (ALB)
6. Identity and Access Management (IAM)
7. Amazon Relational Database Service (RDS)
8. Amazon Machine Image (AMI)
9. Launch Templates
10. Target Groups
11. Internet Gateway (IGW)
12. NAT Gateway
13. Route Table
14. Security Groups

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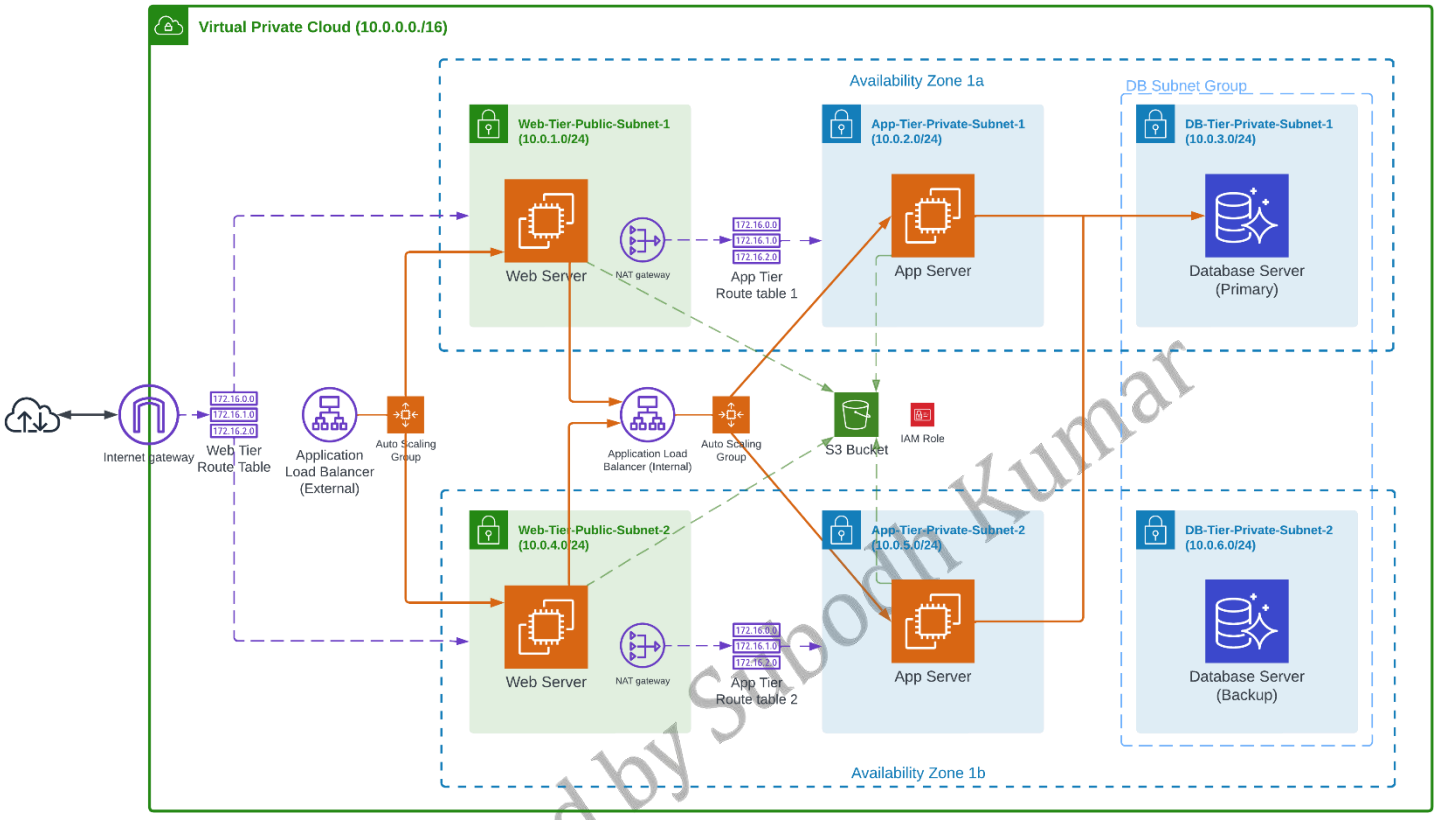
The following diagram shows the overview of the project. It shows connection between each component and how they are distributed in two availability zones Let's have a look at architecture diagram



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As you can see, we use public subnets for web tier because it should be open to the internet to let users interact with the application. App tier and Database tier should be secured. Therefore, private subnets are assigned for them. Resources are distributed in two availability zones, and all are included in our custom VPC. Observe how CIDR blocks are assigned for each subnet.

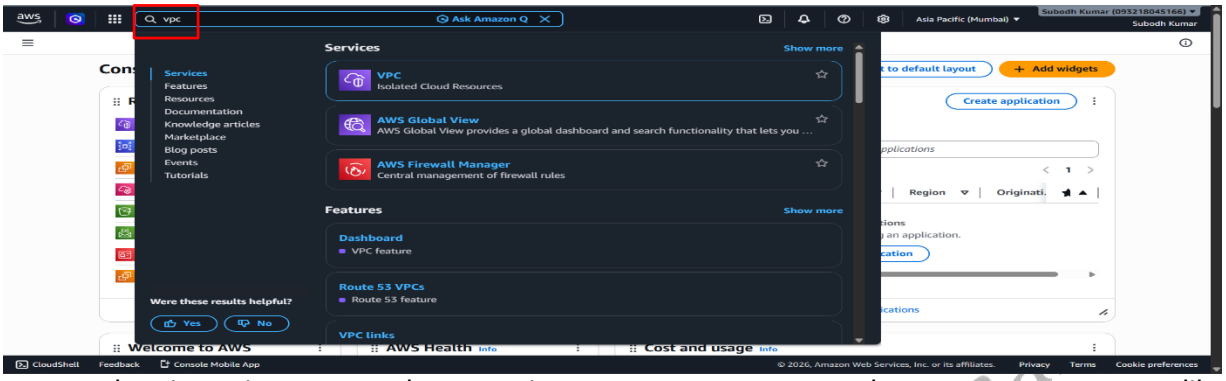
A S3 Bucket has been used to store the source codes of the application. I will make easier the process of server configuration.

Source Codes:https://github.com/SubodhK143/three-tier-app-on-aws.git

Let's get start the implementation

Project Implementation

Go to **AWS Console** and **choose a region** you like (I choose Mumbai region) and create the following services as shown in figures. For go to each services use search bar and search for them.



Note: When inserting names when creating components, you can always use names as you like

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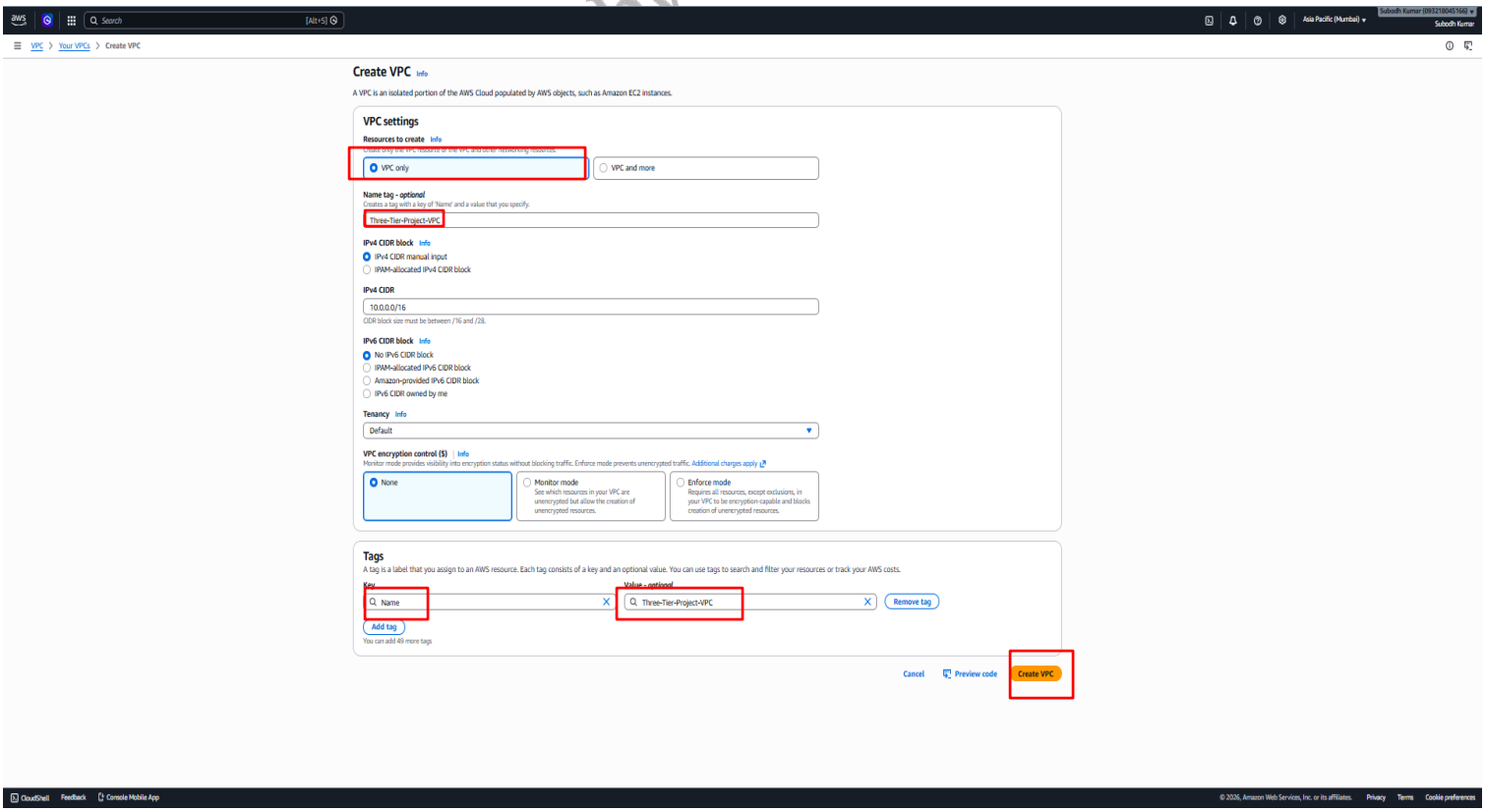
1. Create VPC and Subnets

Create VPC

We have to create a VPC. Search for VPC and go to **Your VPCs > Create VPC**. Setup it as follows. Use **10.0.0.0/16** as CIDR block.



Create Subnets

Next, we Have to create six subnets in this VPC. (Refer the diagram)

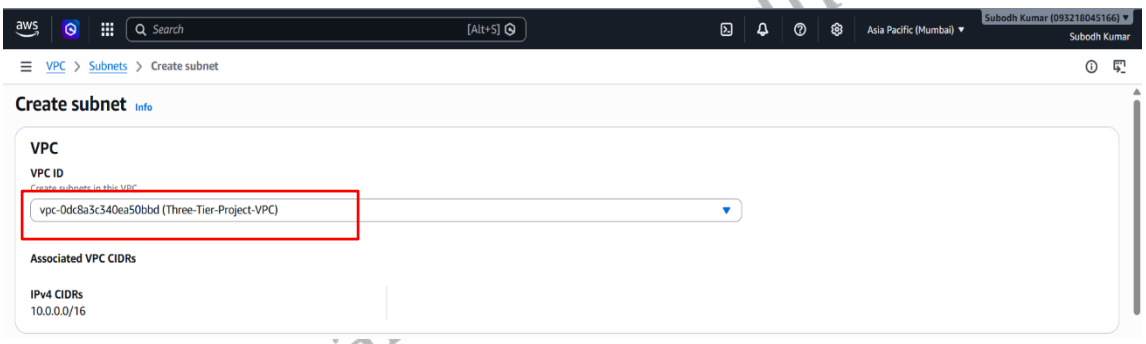
In Availability Zone 1a

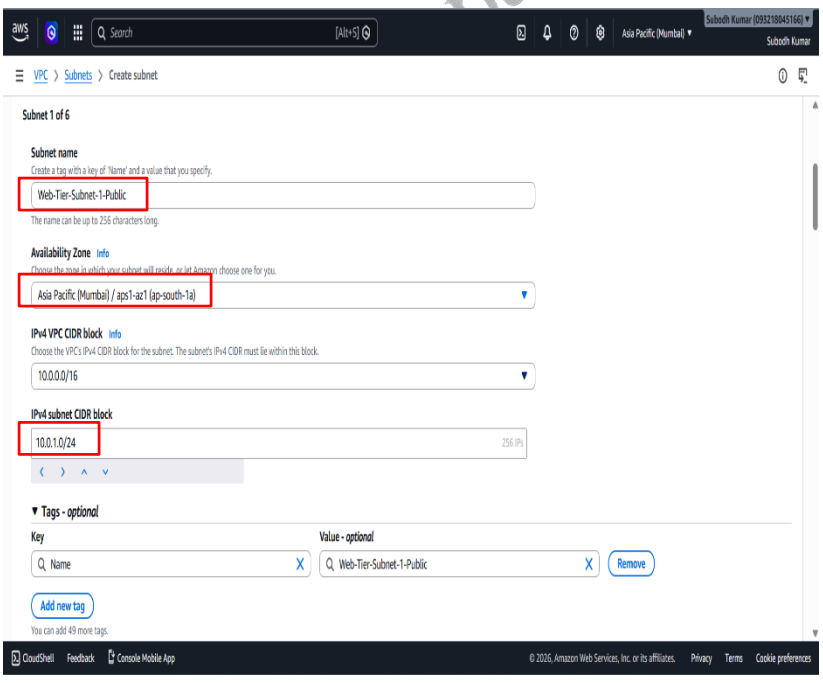
1. Web-Tier-Subnet-1-Public
2. App-Tier-Subnet-1-Private
3. DB-Tier-Subnet-1-Private

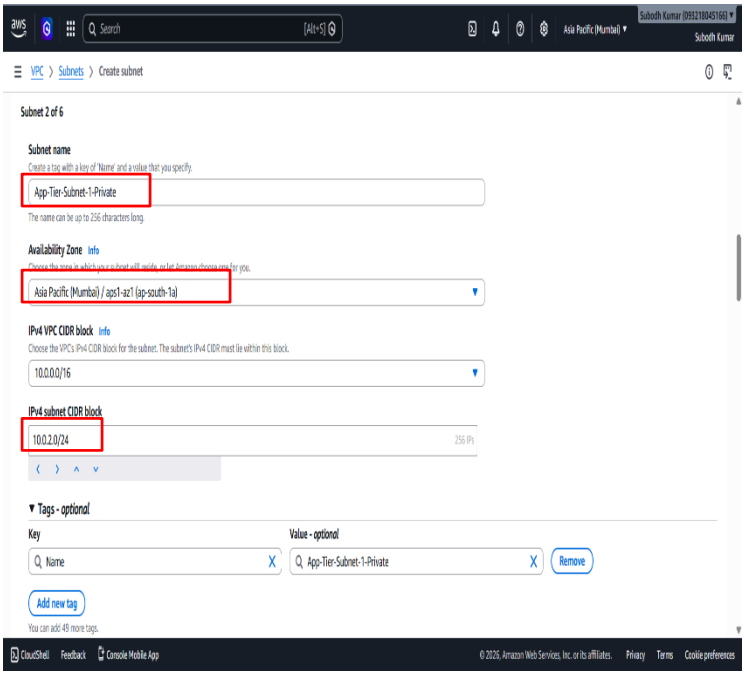
In Availability Zone 1b

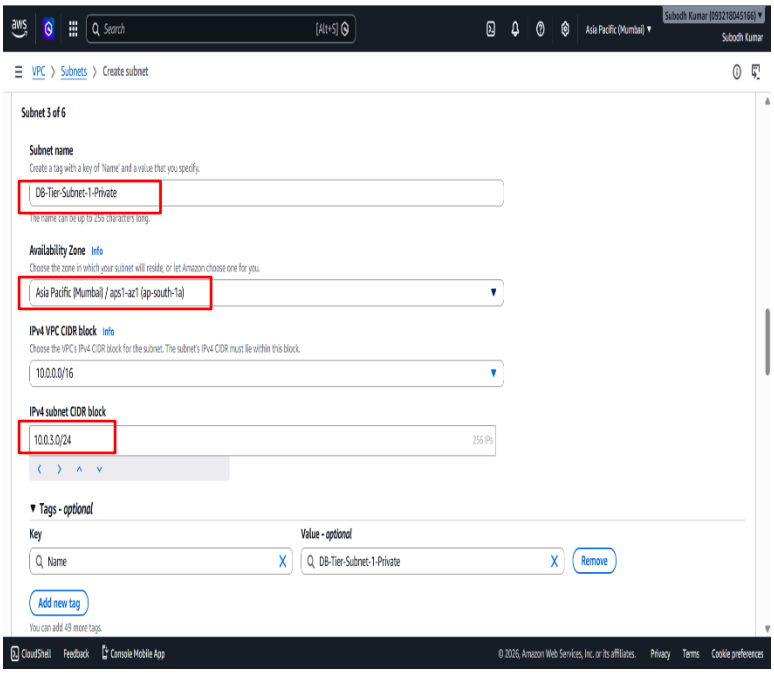
4. Web-Tier-Subnet-2-Public
5. App-Tier-Subnet-2-Private
6. DB-Tier-Subnet-2-Private

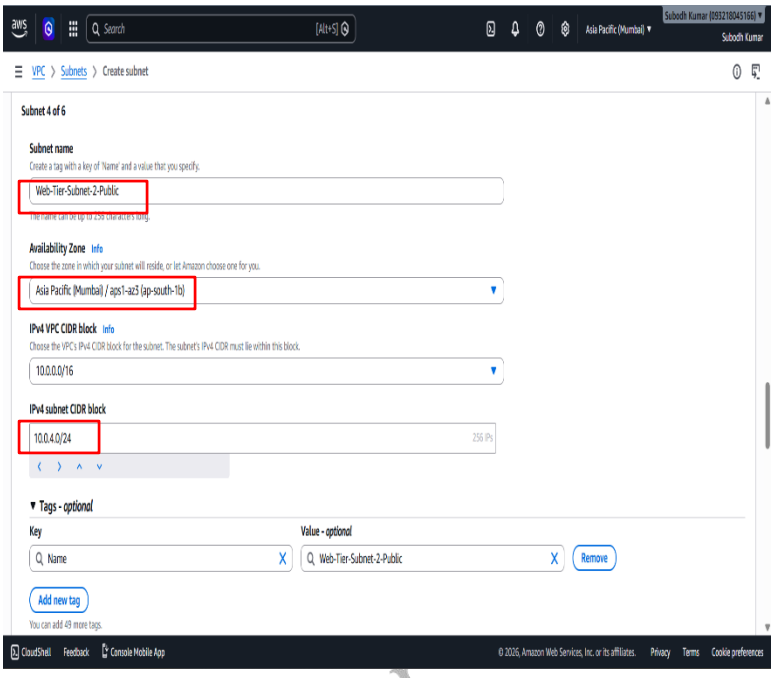
Go to **VPC dashboard > Subnets > Create subnets.** And setup them as follows. And Click create subnets

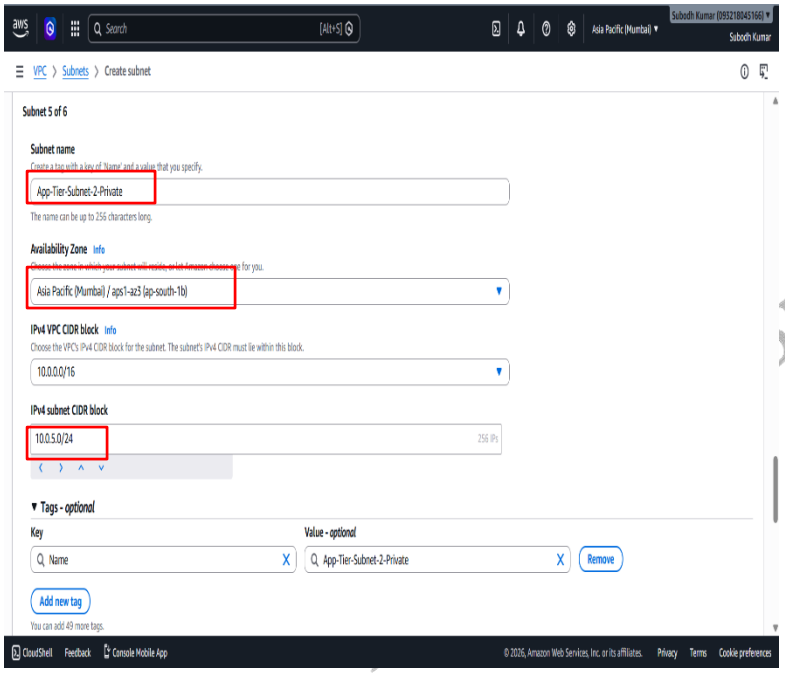


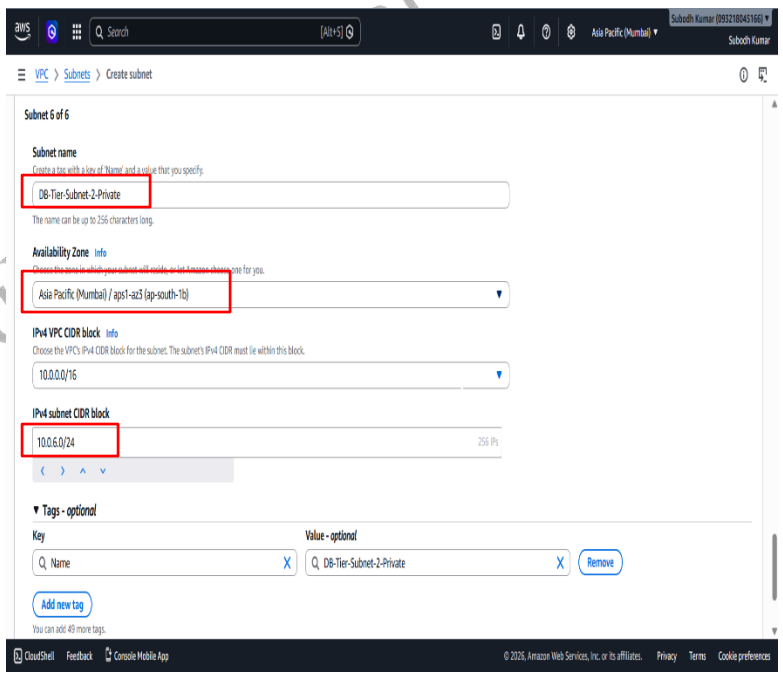




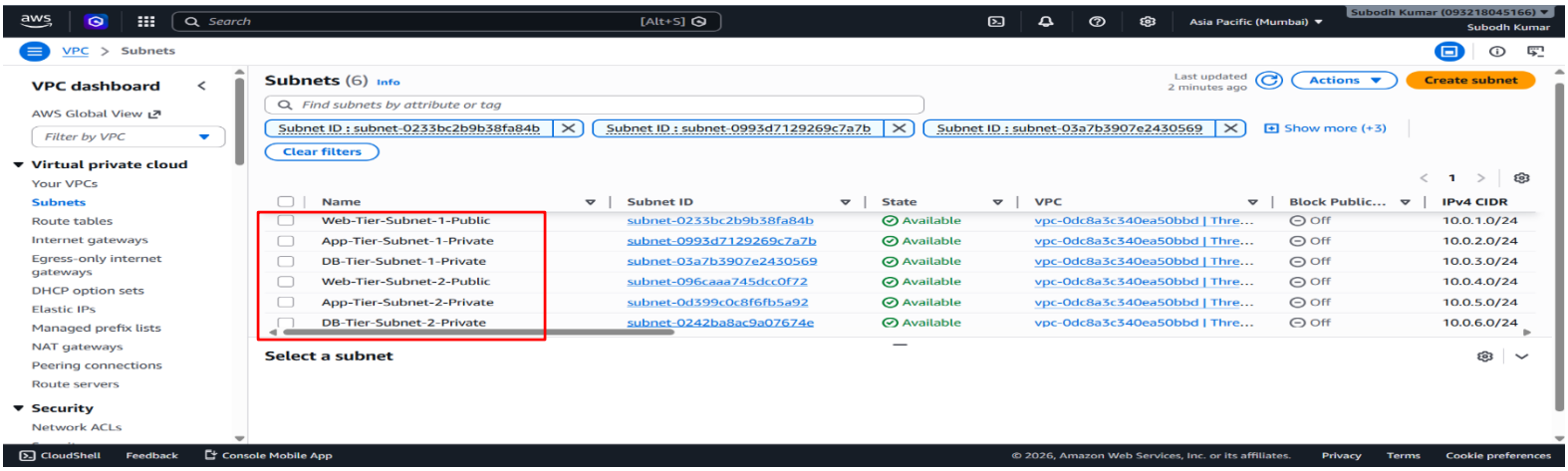




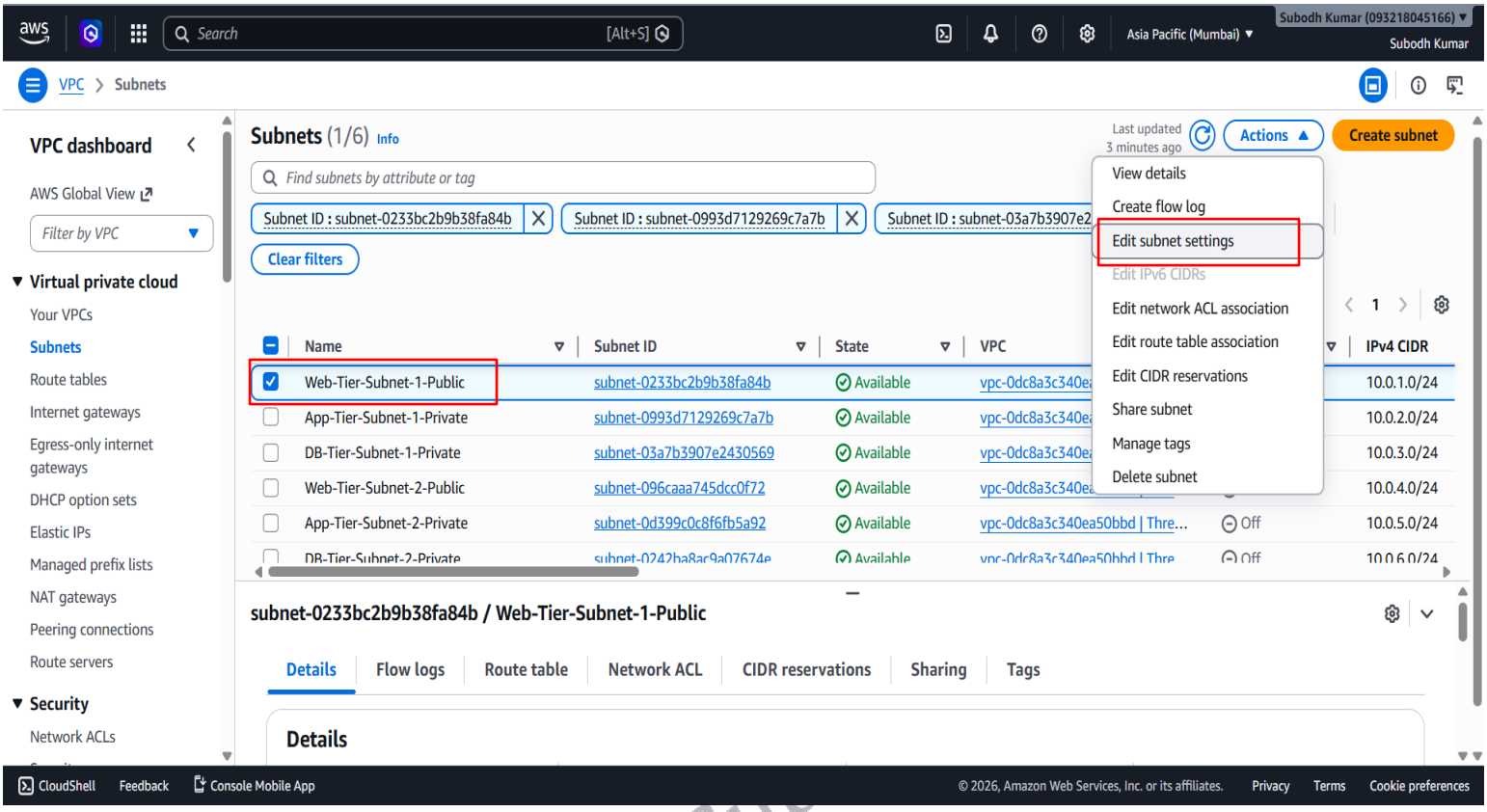


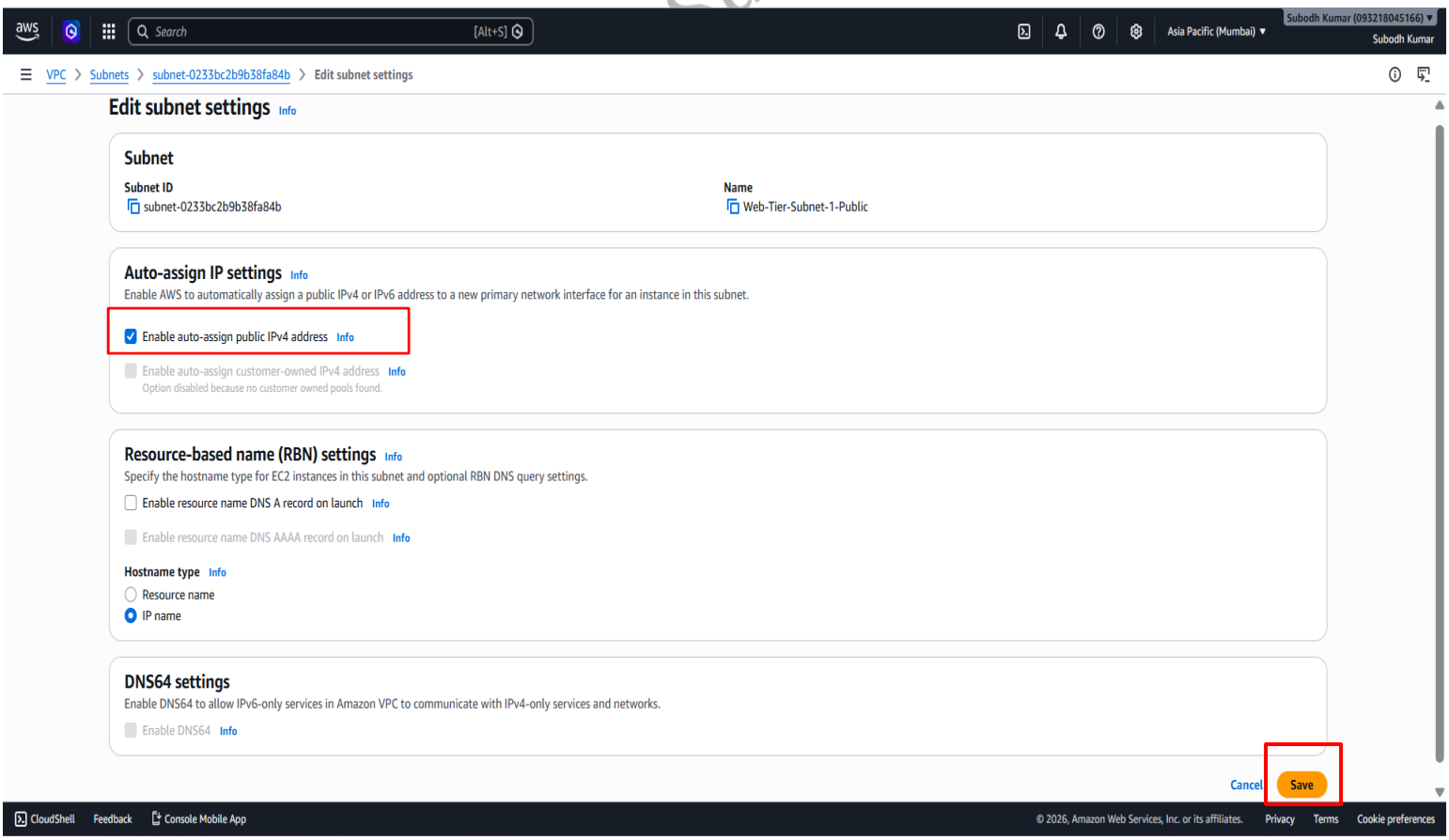


They will create as follows

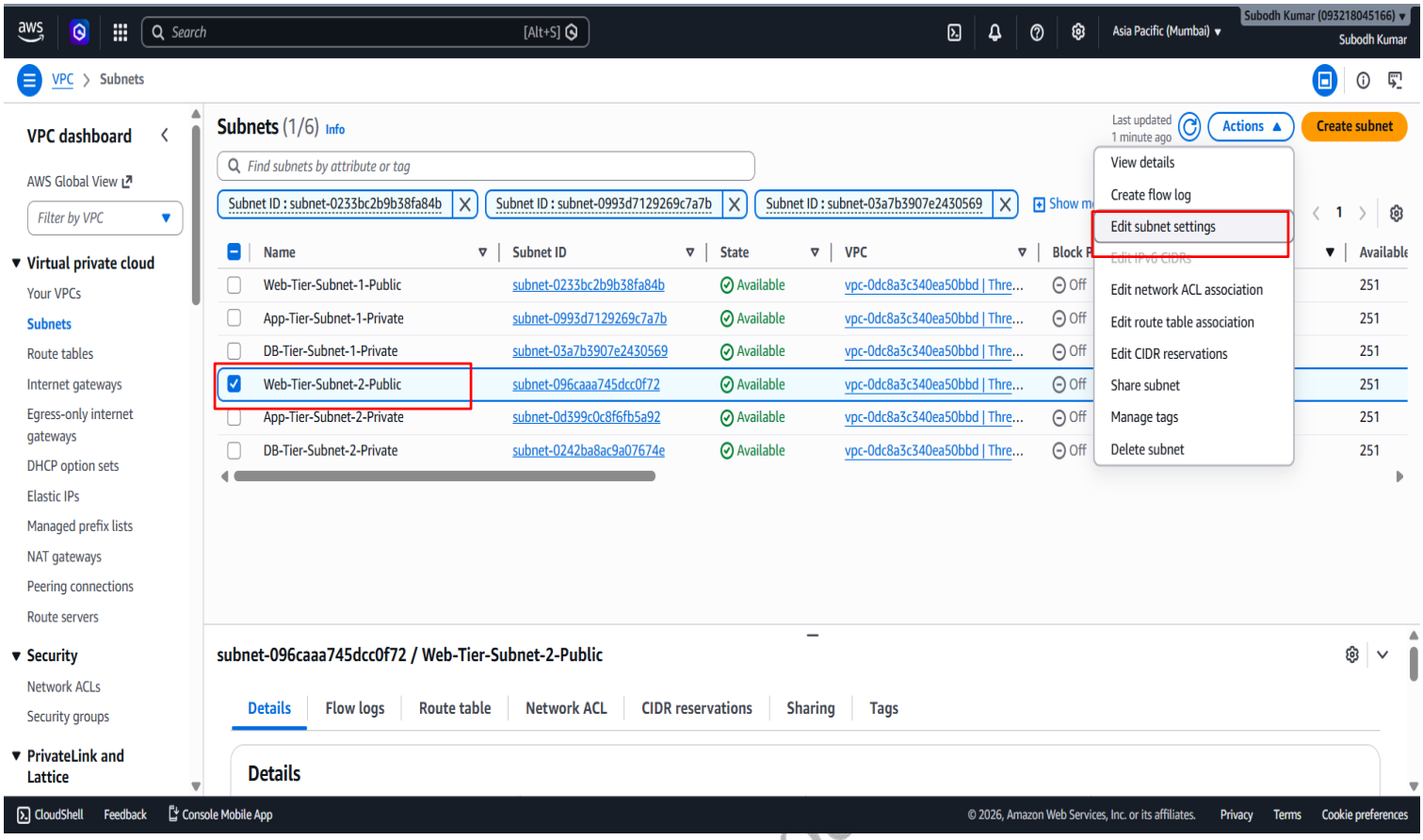


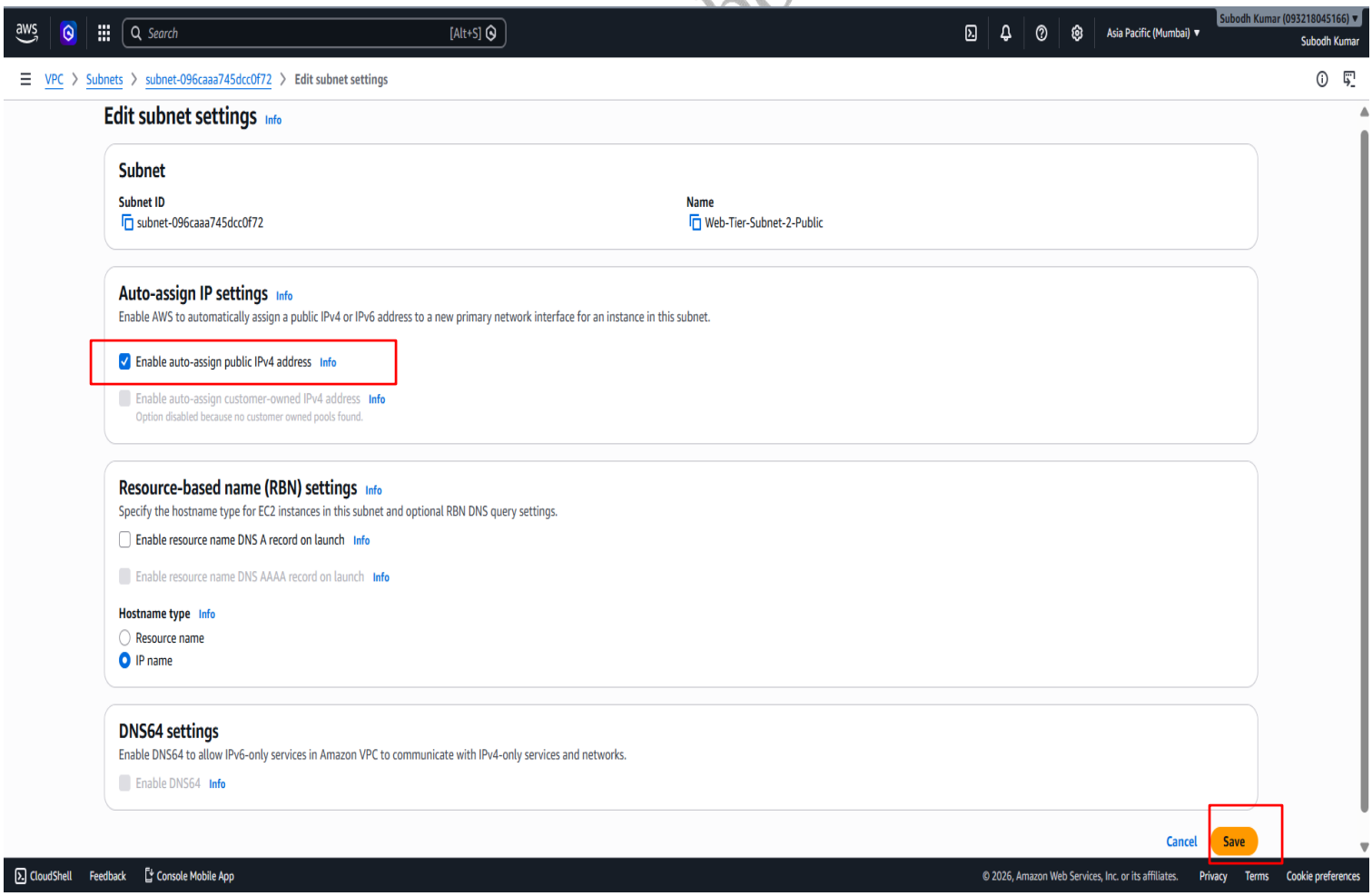
Now select **Web-Tier-Subnet-1-Public.** Go to **Action > Edit subnet settings.** Under Auto-assign IP settings, select **Enable auto-assign public IPv4 address** and save**.**





Do the same for **Web-Tier-Subnet-2-Public** as well.

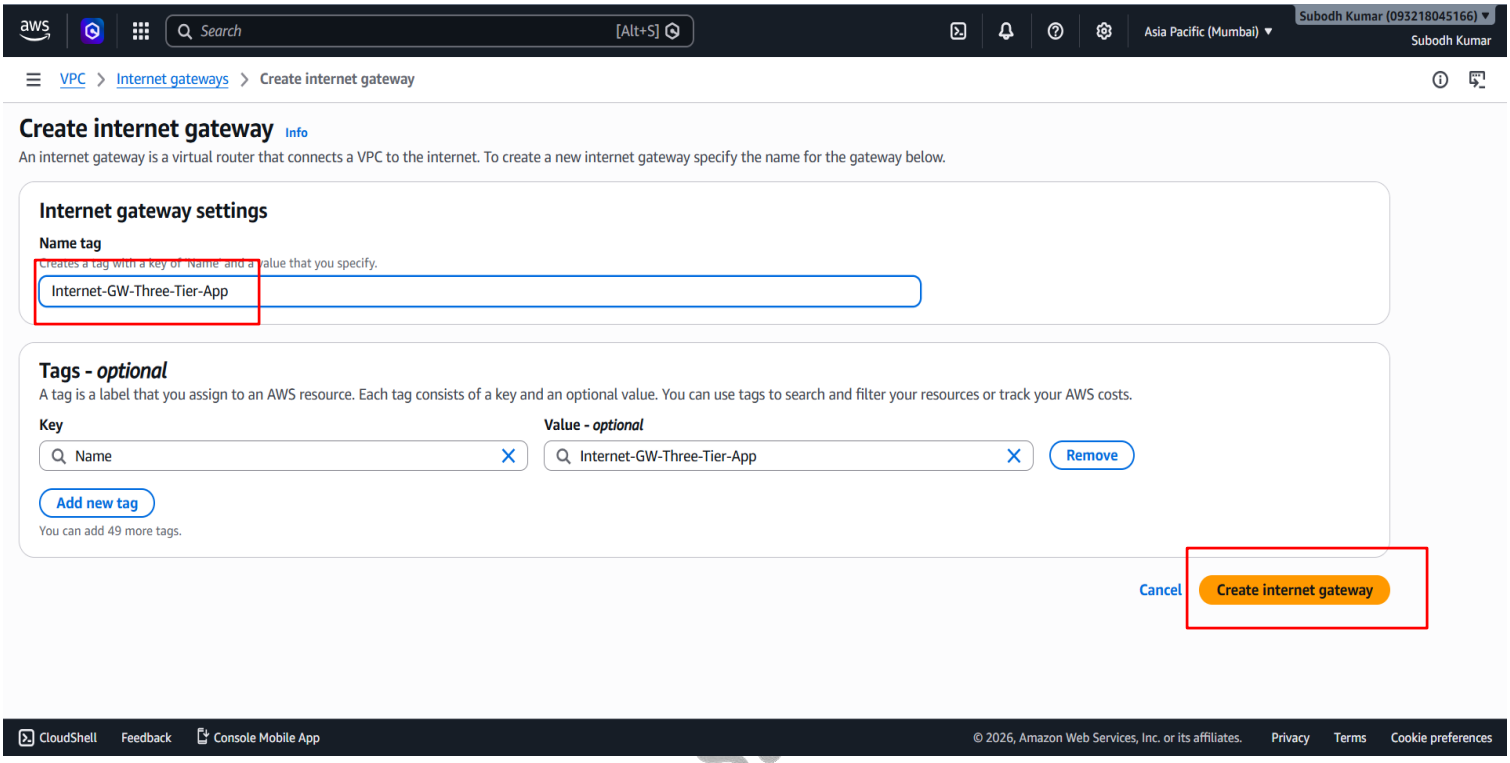




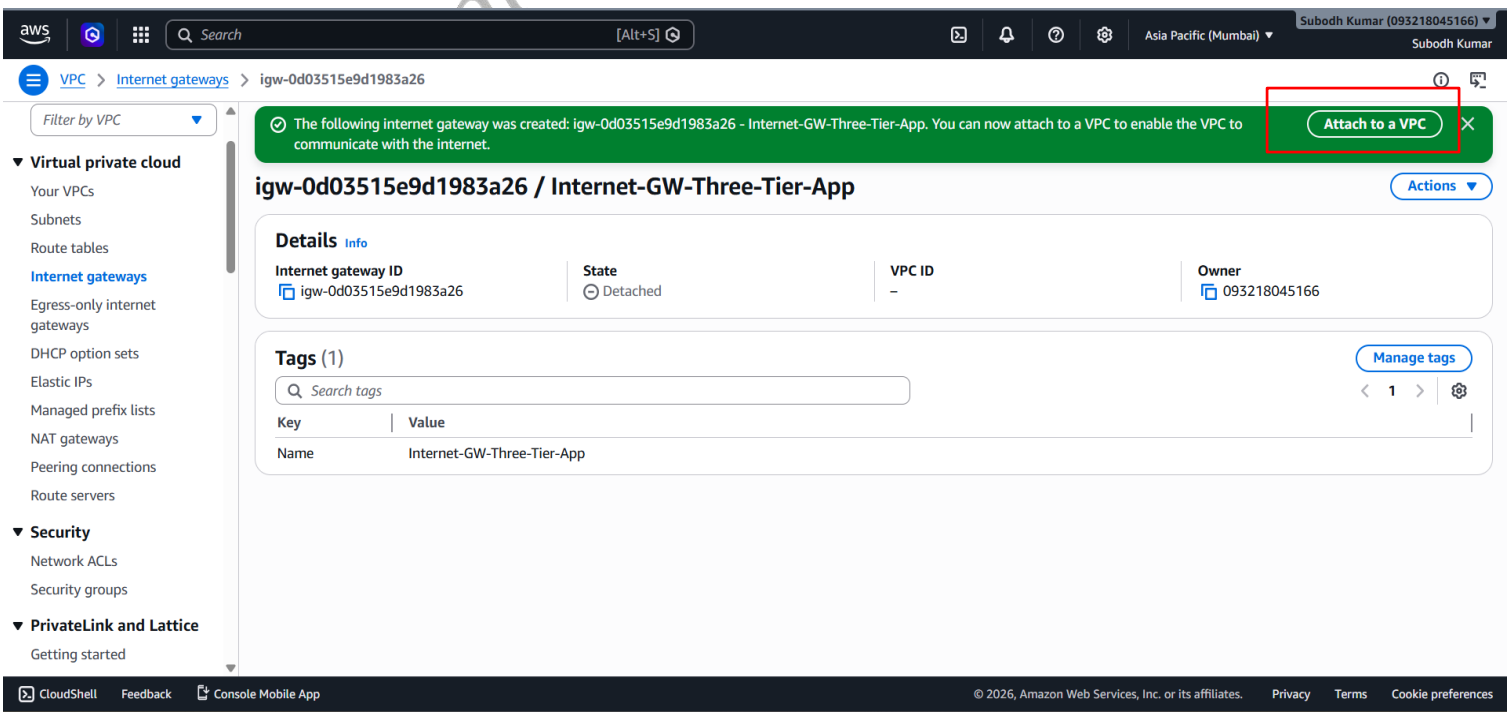
2. Create Internet Gateway, NAT Gateway and Rout tables.

Create Internet Gateway

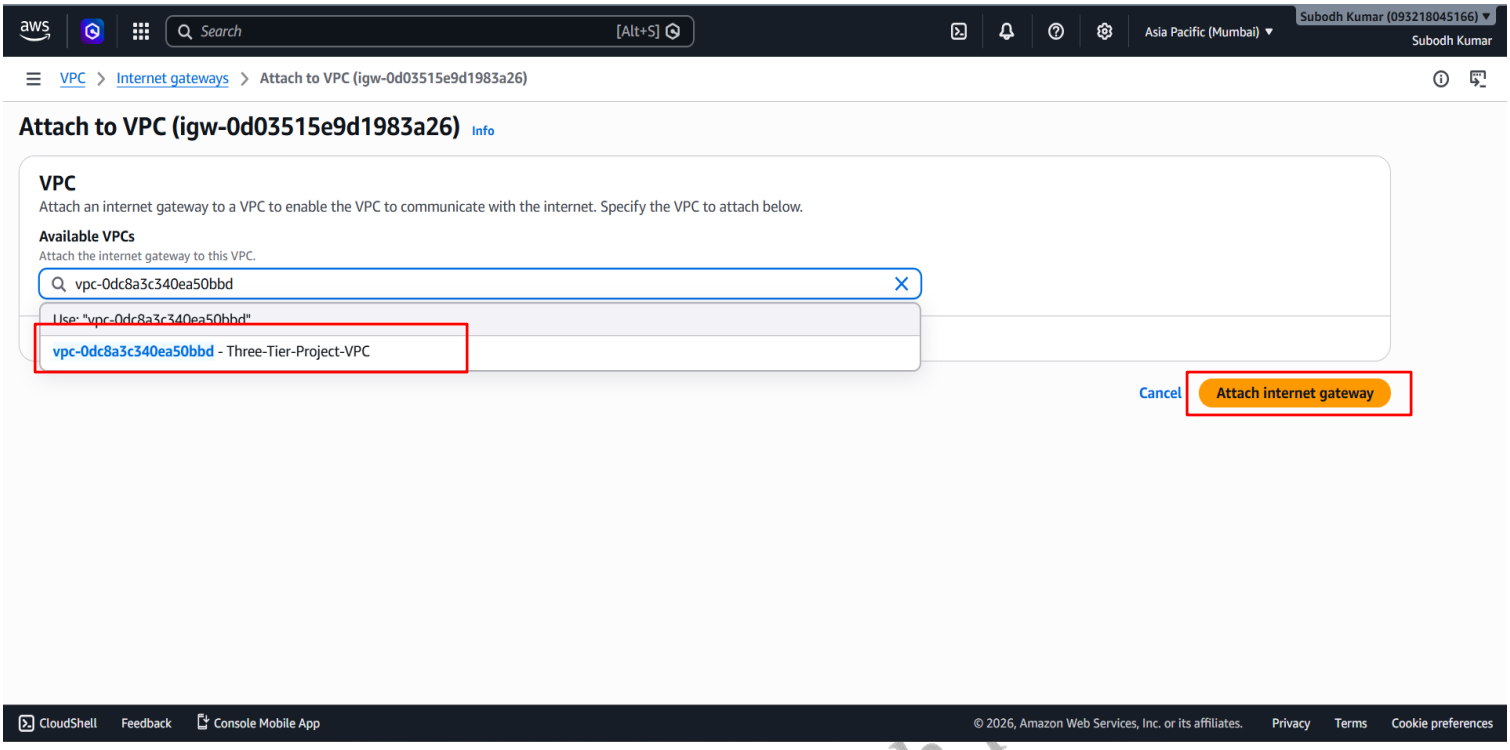
Go to VPC Dashboard **Internet gateways > Create internet Gateway.** Enter a name then click on Create internet gateway icon.



Then Go to internet gateways list and select our internet gateway and go to **Actions > Attach to VPC or Click directly on pop-up > Attach to VPC or Directly click on Pop-up menu.**

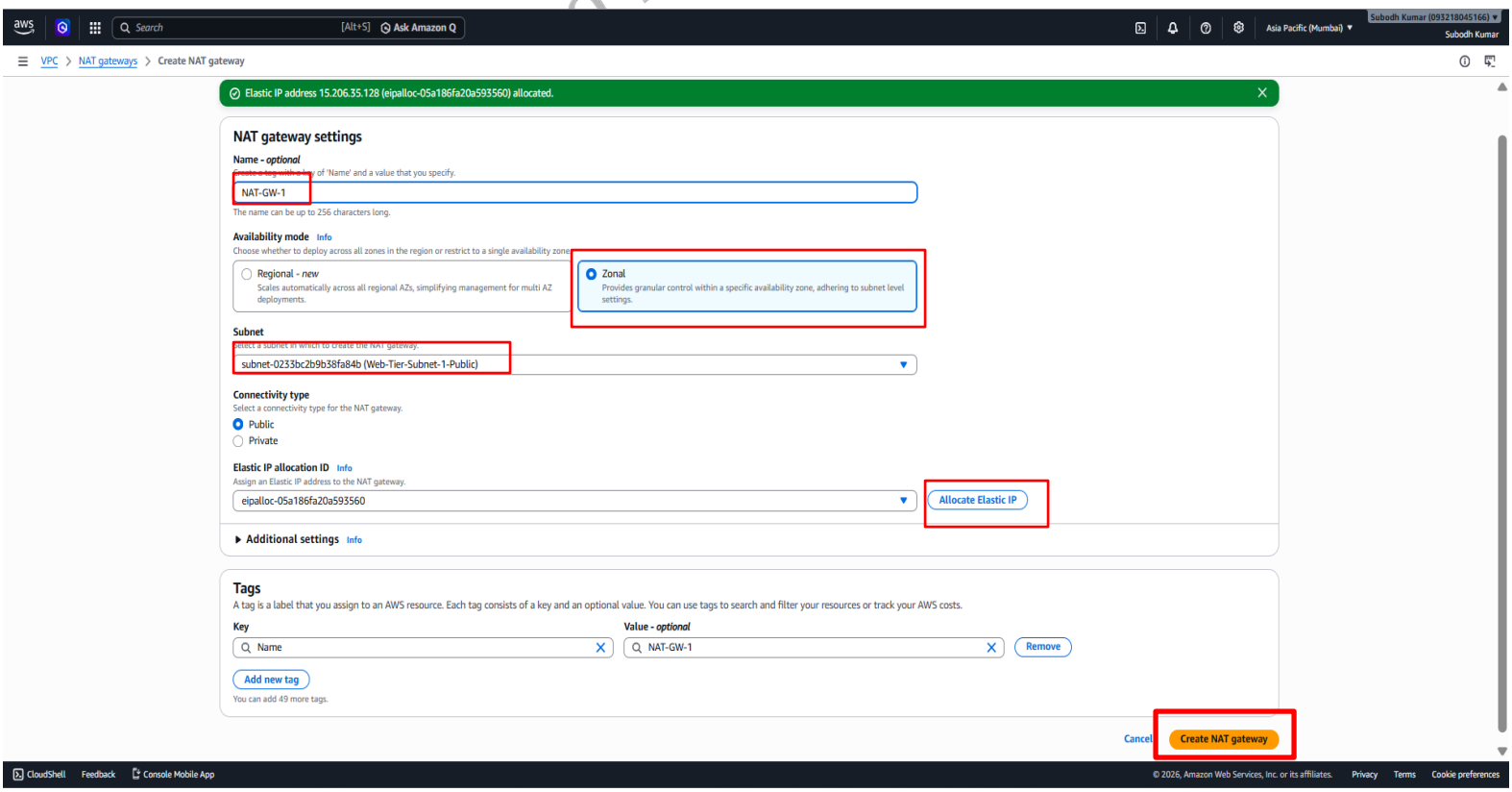


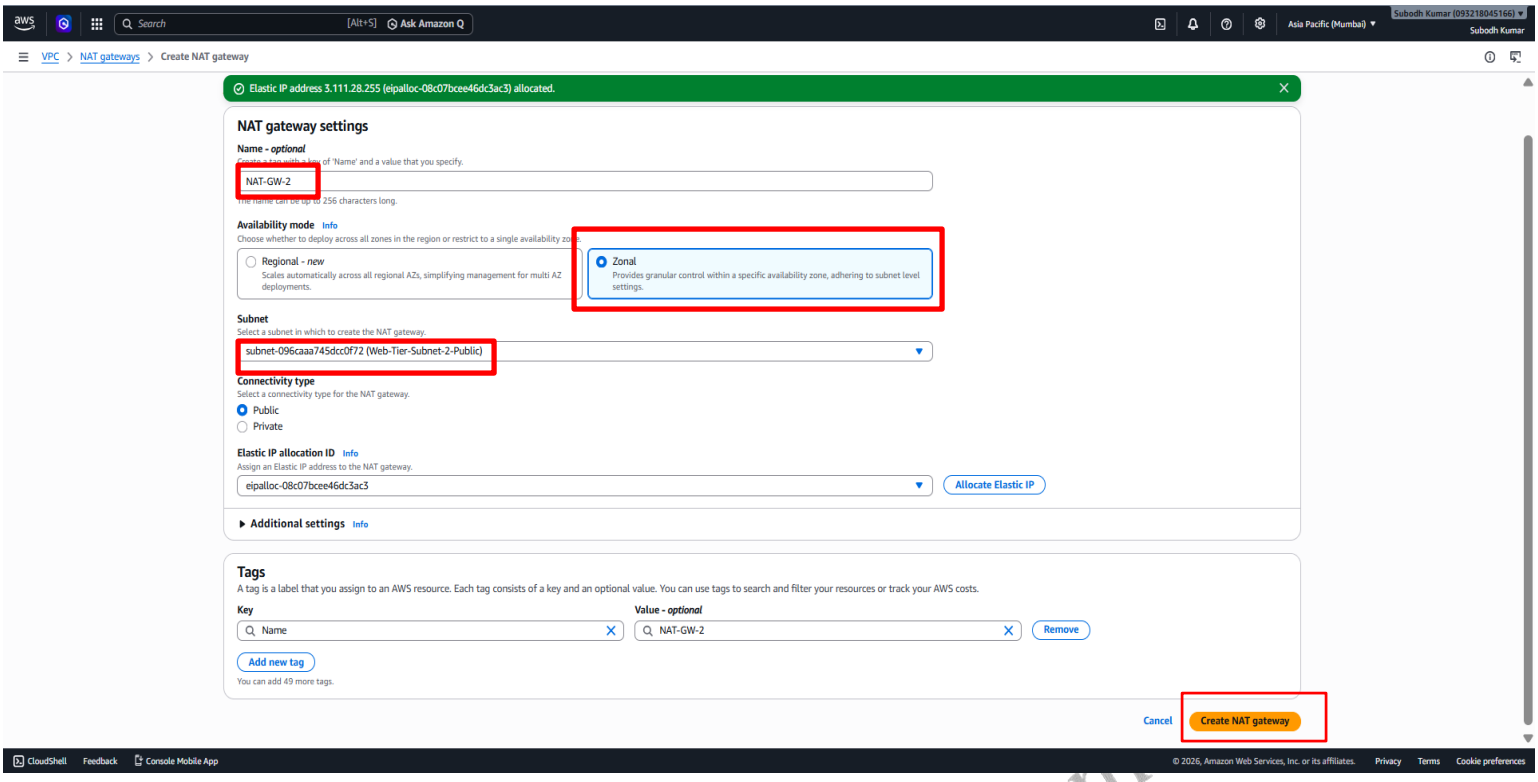
Then select our **VPC**. Finally, click on **Attach internet gateway** icon.



Create NAT Gateways

Go to VPC Dashboard **NAT gateways > Create NAT Gateway.** Create two NAT Gateways. Create it as follows.



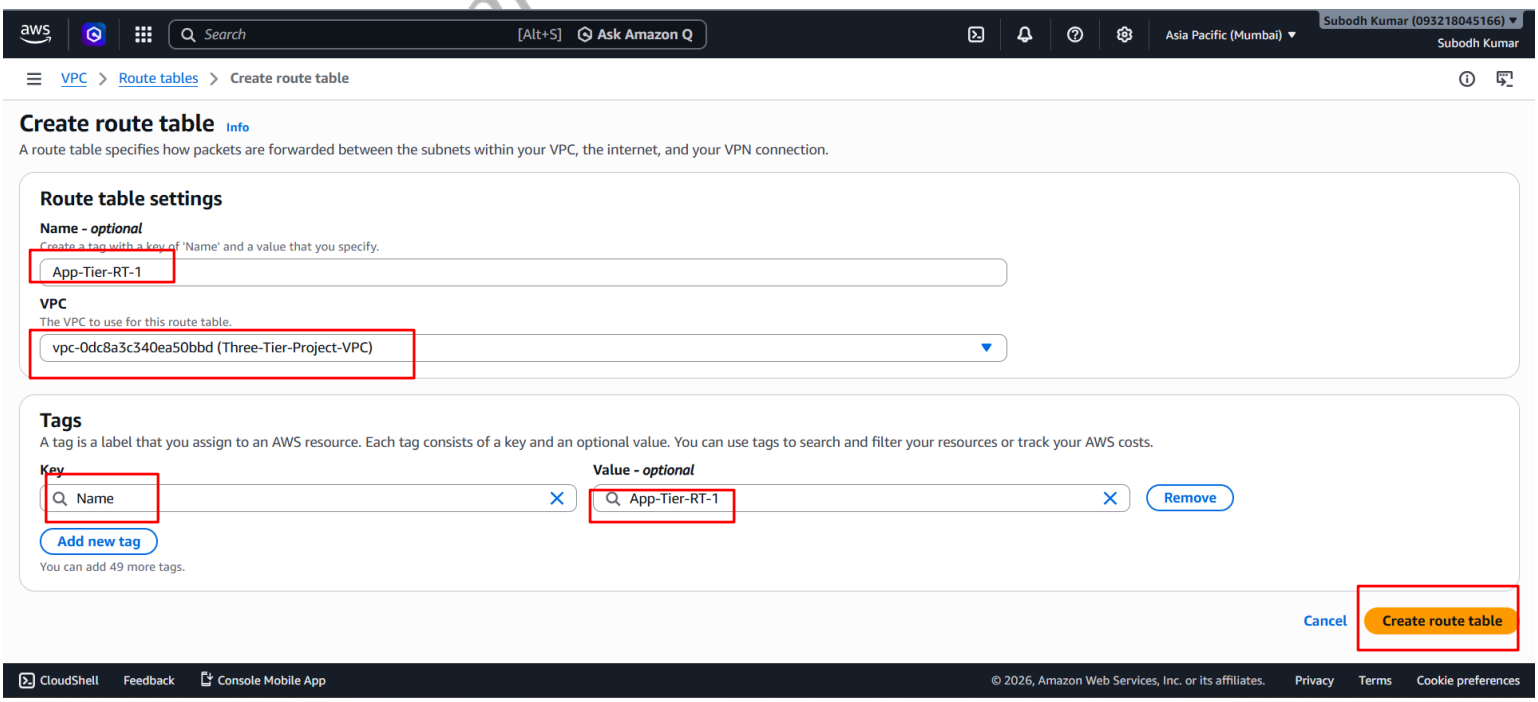


Create Route Tables

Now Let's create **Route tables**. According to our diagram we have to create **three route tables**.

1. **App-Tier-RT-1**
2. **App-Tier-RT-2**
3. **Web-Tier-RT**

Now got to VPC dashboard **Route tables > Create route table**. Create Route tables as follows.



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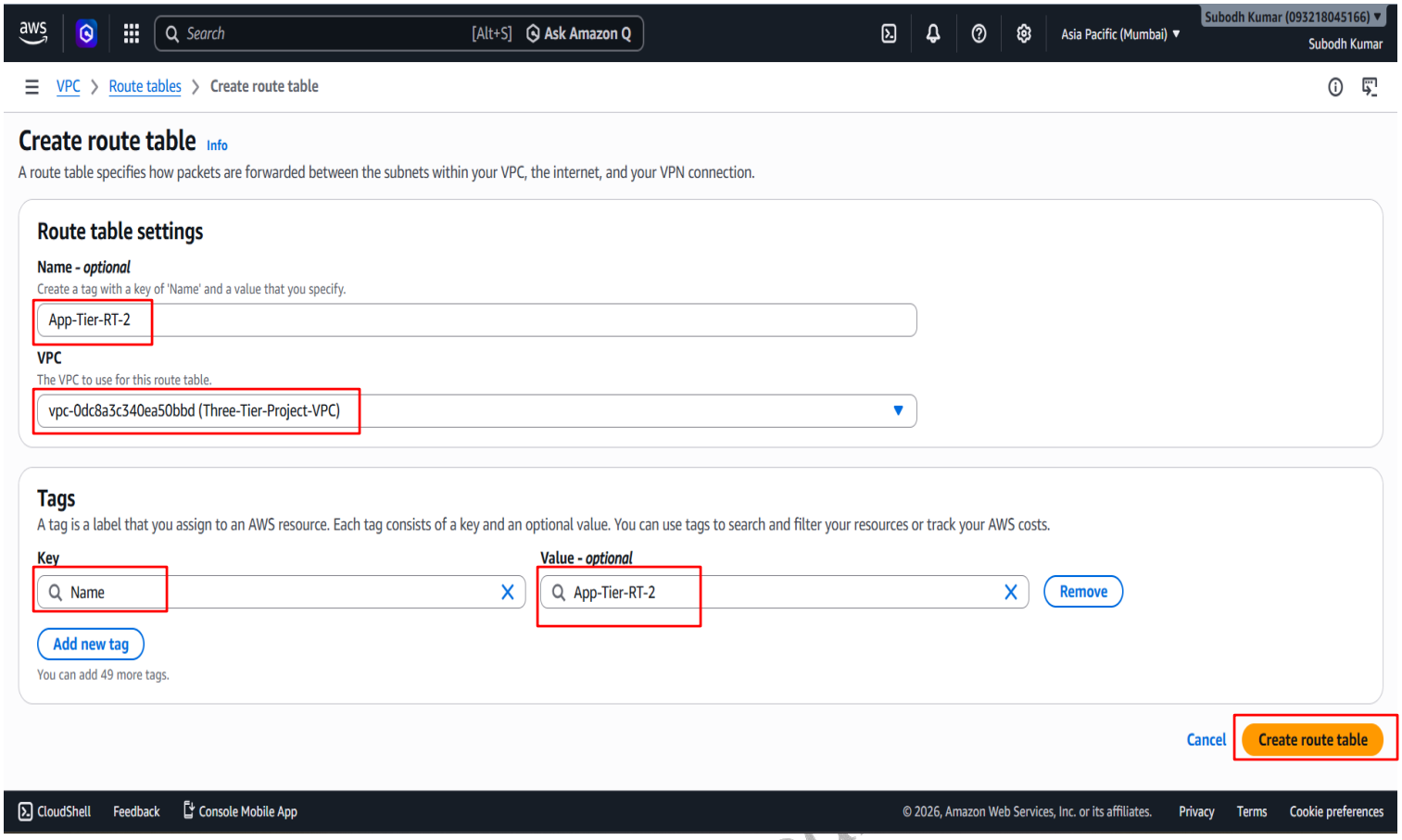
Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
VPC > Route tables > Create route table
Create route table Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
App-Tier-RT-2
VPC
The VPC to use for this route table.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
App-Tier-RT-2
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table
CloudShell
Feedback
Console Mobile App
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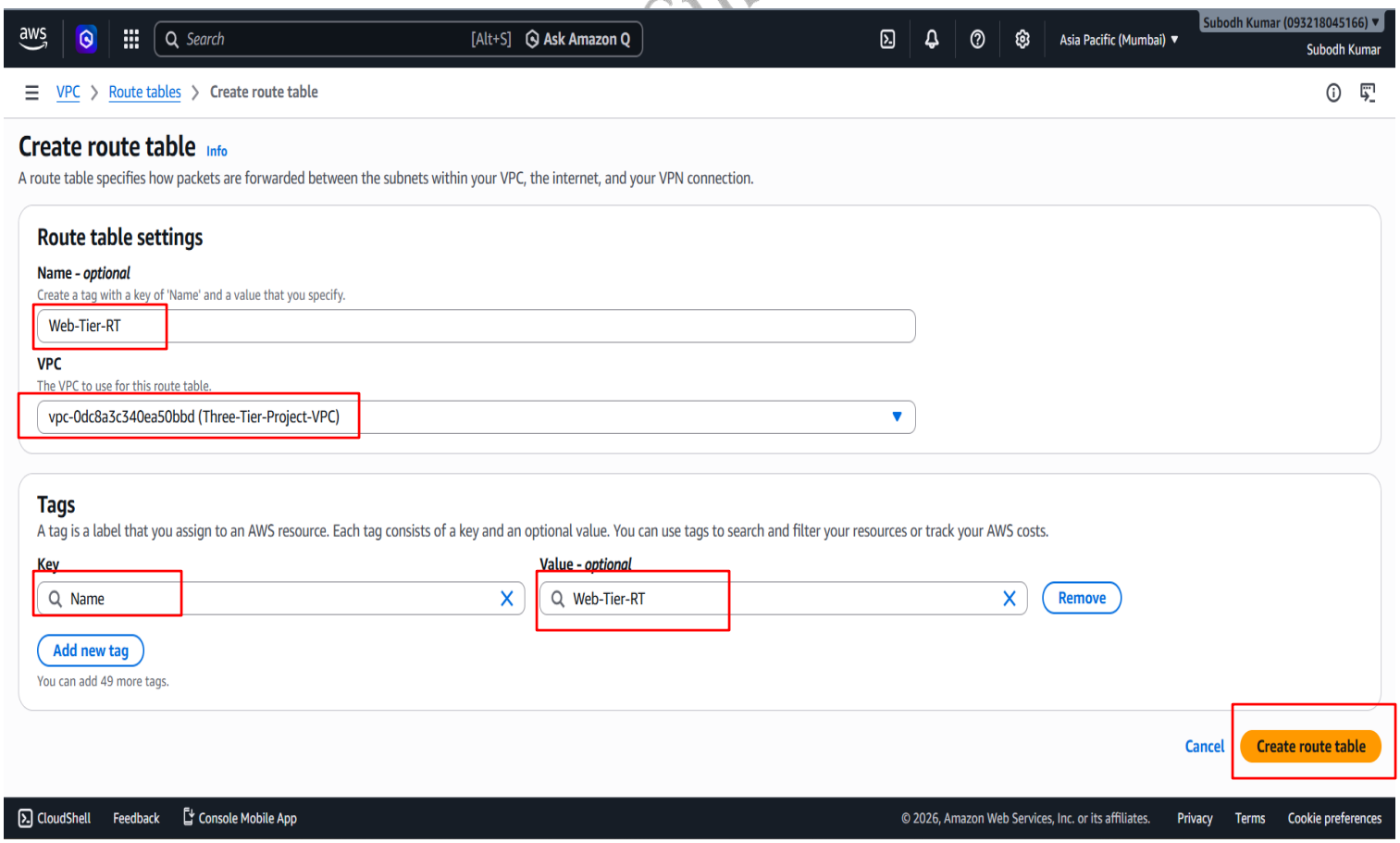
Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
VPC > Route tables > Create route table
Create route table Info
A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.
Route table settings
Name - optional
Create a tag with a key of 'Name' and a value that you specify.
Web-Tier-RT
VPC
The VPC to use for this route table.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Name
Value - optional
Web-Tier-RT
Remove
Add new tag
You can add 49 more tags.
Cancel
Create route table
CloudShell
Feedback
Console Mobile App
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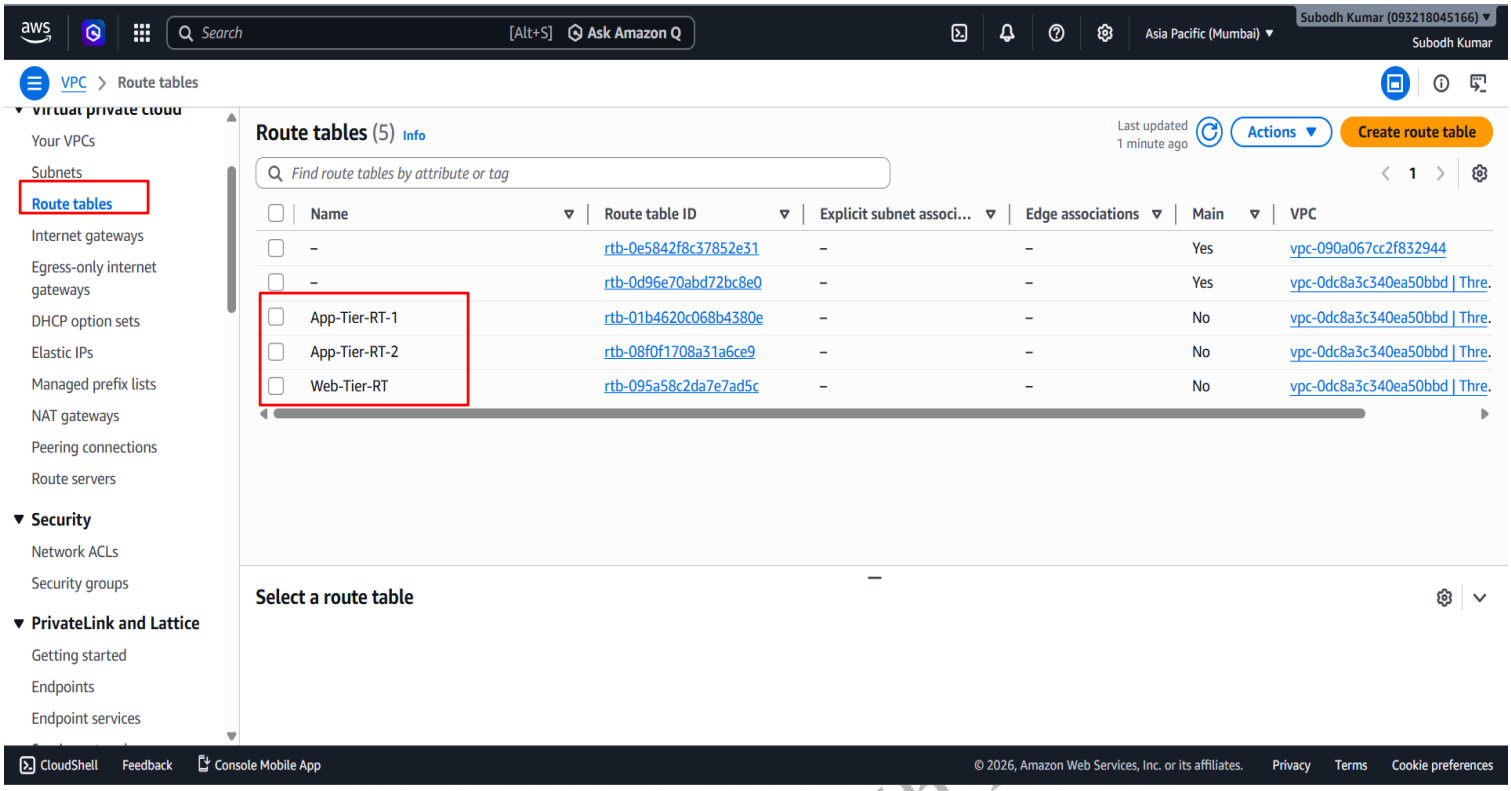
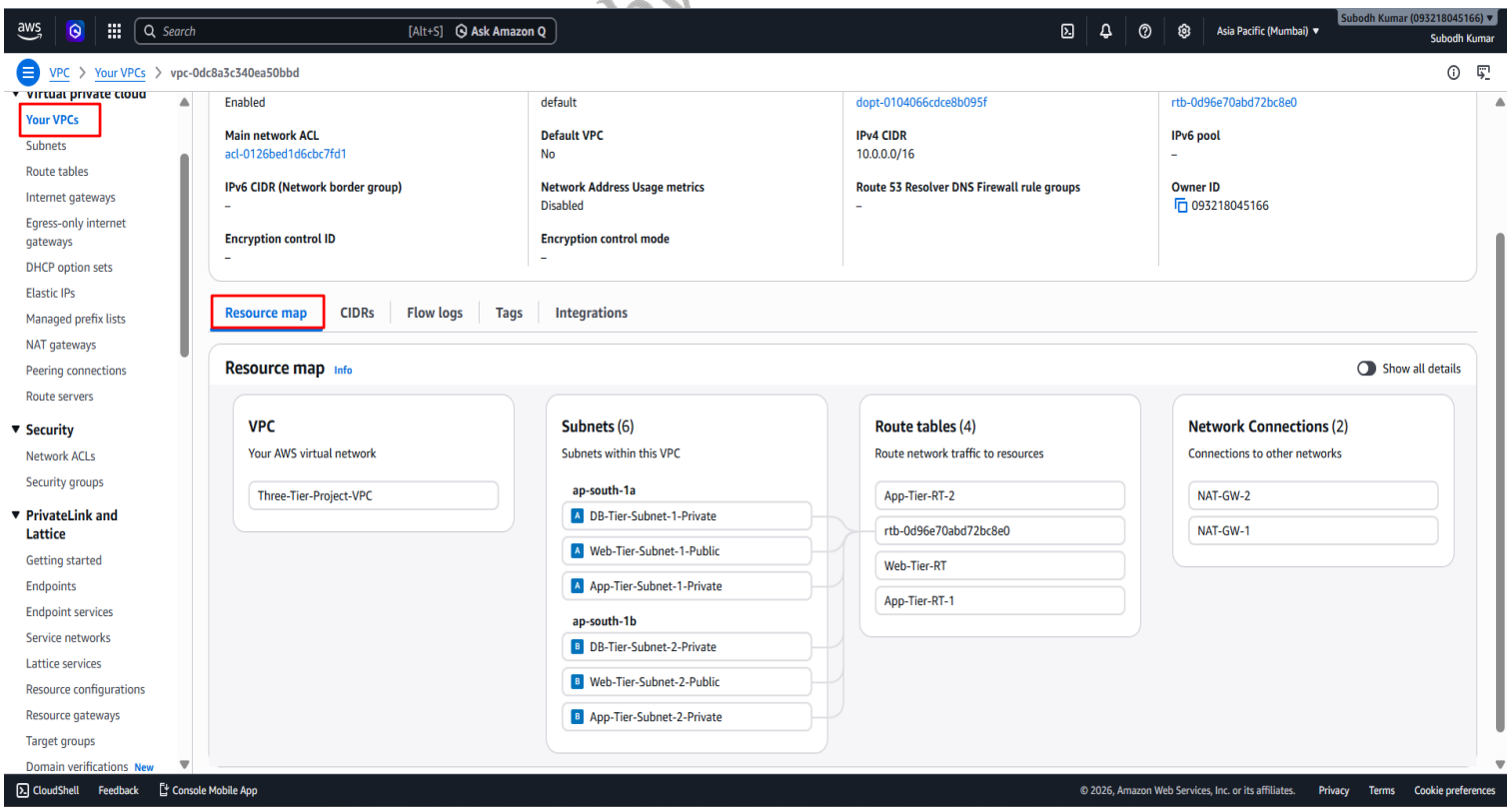
Now you will see all the route tables in the menu.

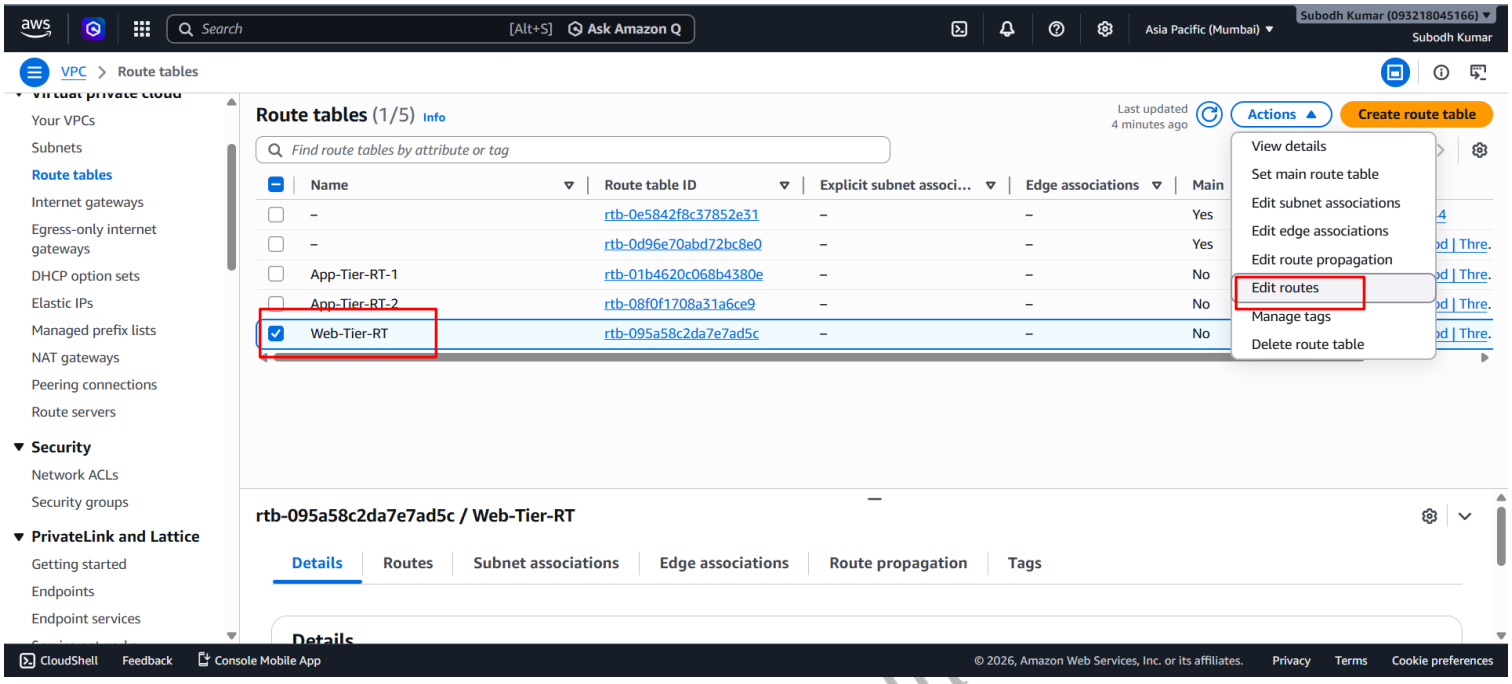


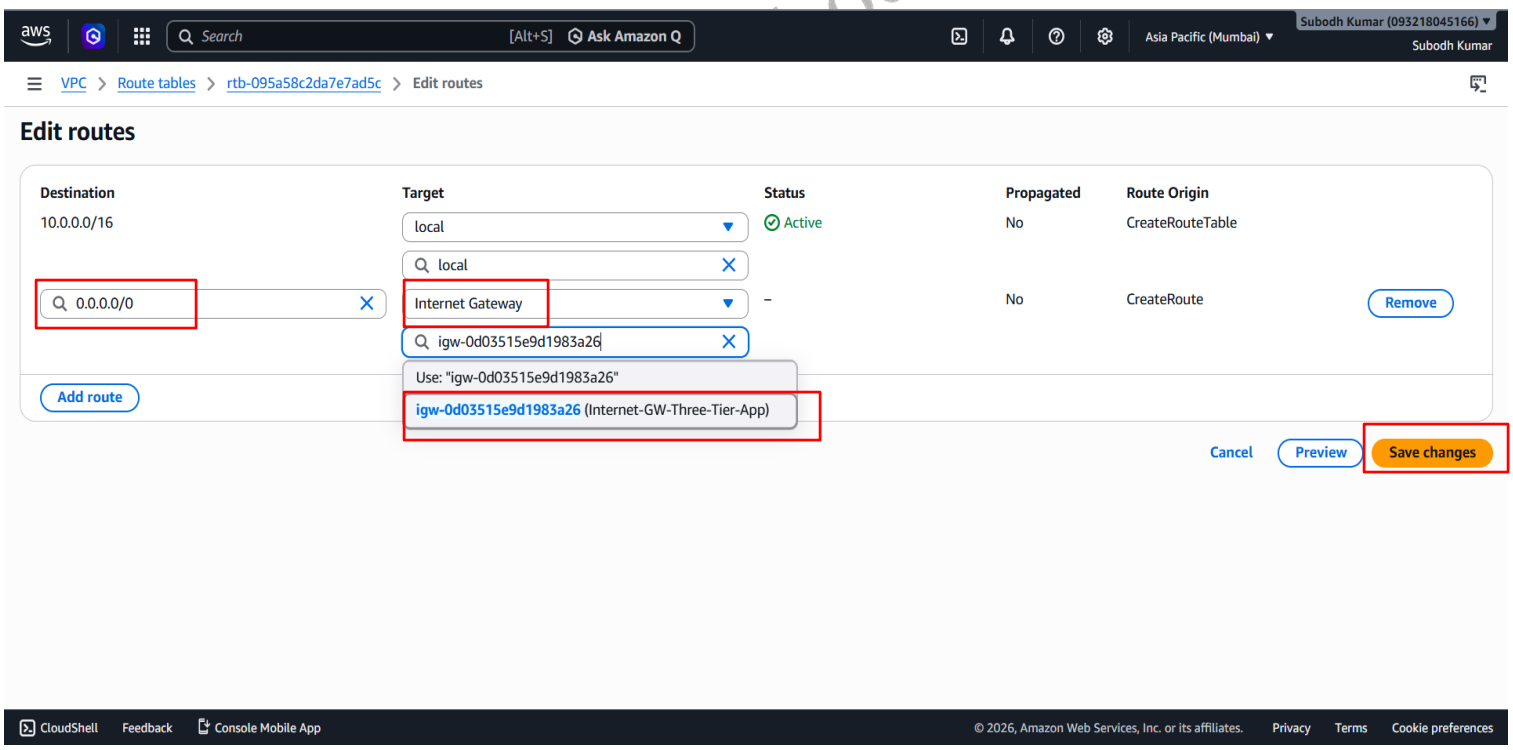
If you go to **VPC** dashboard **> Your VPCs**. And select **Three-Tier-Project-VPC**. Then if you go to **Resource map** you can see **Route tables** are **created** but there are **no connections.**



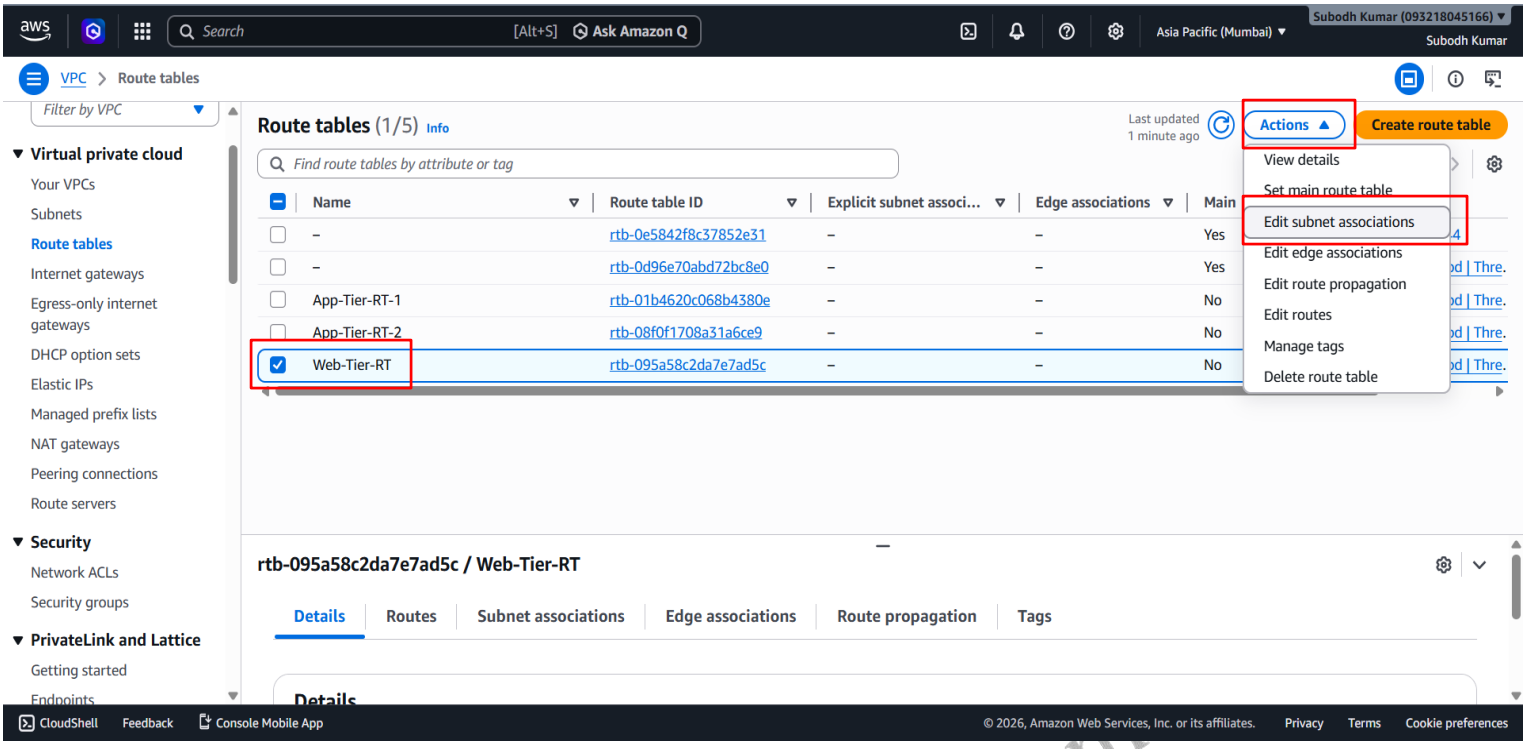
Edit Routes

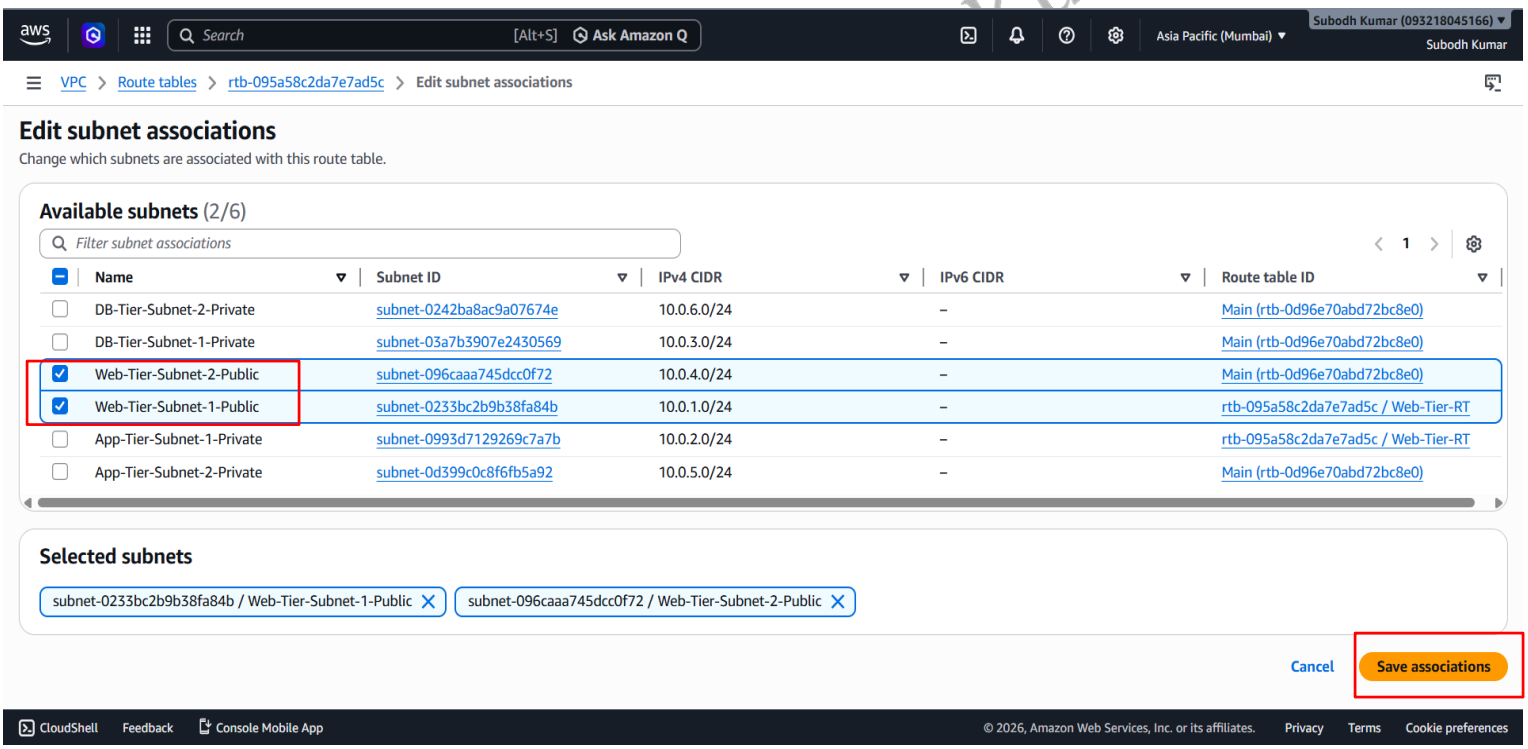
1. Now Select Web-Tier-RT Go to **Actions > Edit routes**. Add route as follows. Destination 0.0.0.0/0, target internet gateway > select our igw. Then Save changes.



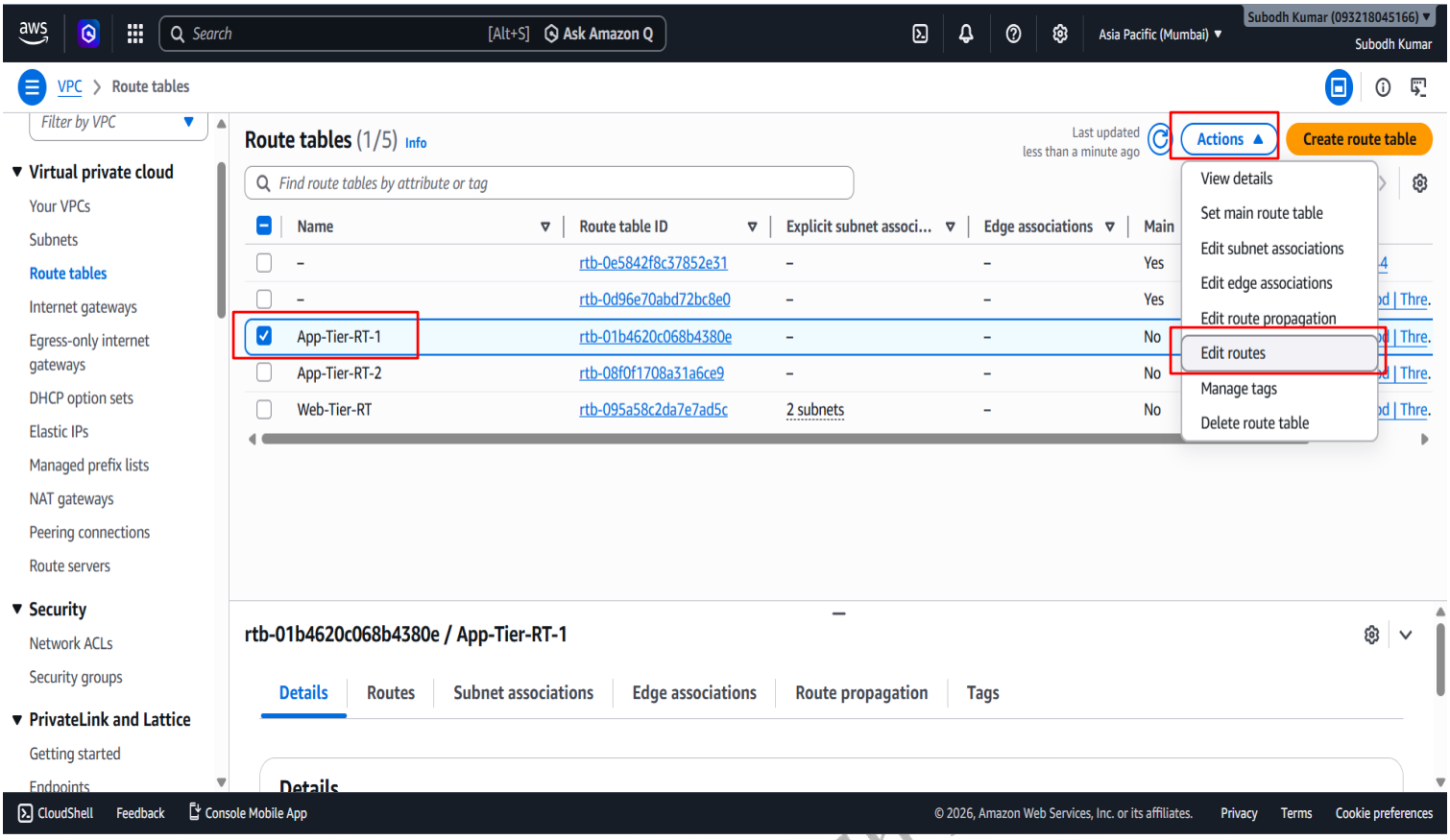


Again, select the Web-Tier-RT and go to **Actions > Edit subnet associations.** Select Web-Tier-Subnet-1-Public and Web-Tier-Subnet-2-Public and Save associations.





2. Now Select **App-Tier-RT-1** Go to **Actions > Edit routes**. Add route as follows. Destination 0.0.0.0/0, target NAT gateway > select our NAT-GW-1. Then Save changes.


Search [Alt+S] Ask Amazon Q
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VPC > Route tables
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Route tables (1/5) Info
Last updated less than a minute ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
rtb-0e5842f8c37852e31
rtb-0d96e70abd72bc8e0
App-Tier-RT-1
rtb-01b4620c068b4380e
App-Tier-RT-2
rtb-08f0f1708a31a6ce9
Web-Tier-RT
rtb-095a58c2da7e7ad5c
2 subnets
View details
Set main route table
Edit subnet associations
Edit edge associations
Edit route propagation
Edit routes
Manage tags
Delete route table
rtb-01b4620c068b4380e / App-Tier-RT-1
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
CloudShell
Feedback
Console Mobile App
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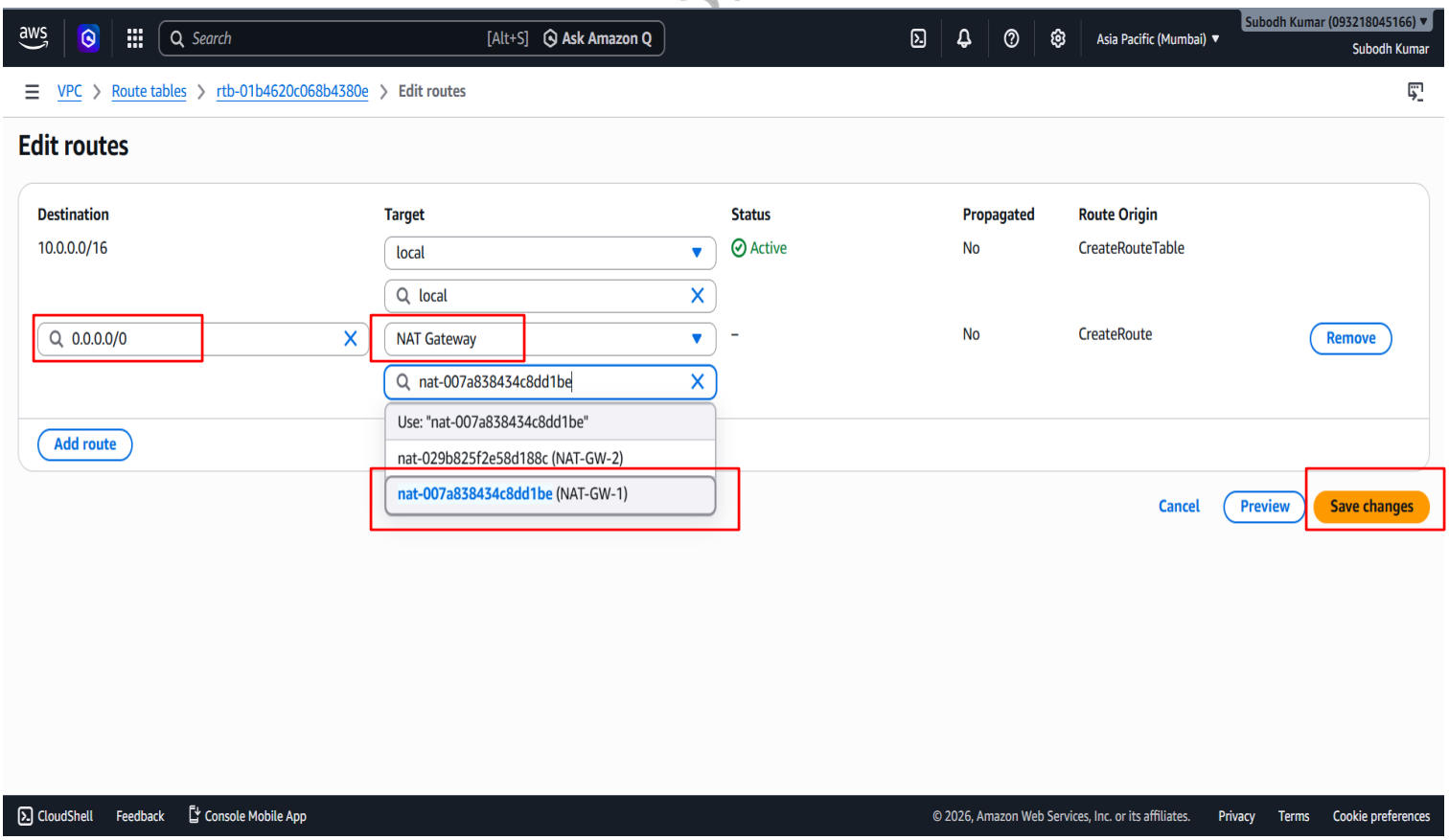
Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
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Subodh Kumar
VPC > Route tables > rtb-01b4620c068b4380e > Edit routes
Edit routes
Destination
Target
Status
Propagated
Route Origin
10.0.0.0/16
local
Active
No
CreateRouteTable
local
0.0.0.0/0
NAT Gateway
No
CreateRoute
Remove
nat-007a838434c8dd1be
Use: "nat-007a838434c8dd1be"
nat-029b825f2e58d188c (NAT-GW-2)
nat-007a838434c8dd1be (NAT-GW-1)
Add route
Cancel
Preview
Save changes
CloudShell
Feedback
Console Mobile App
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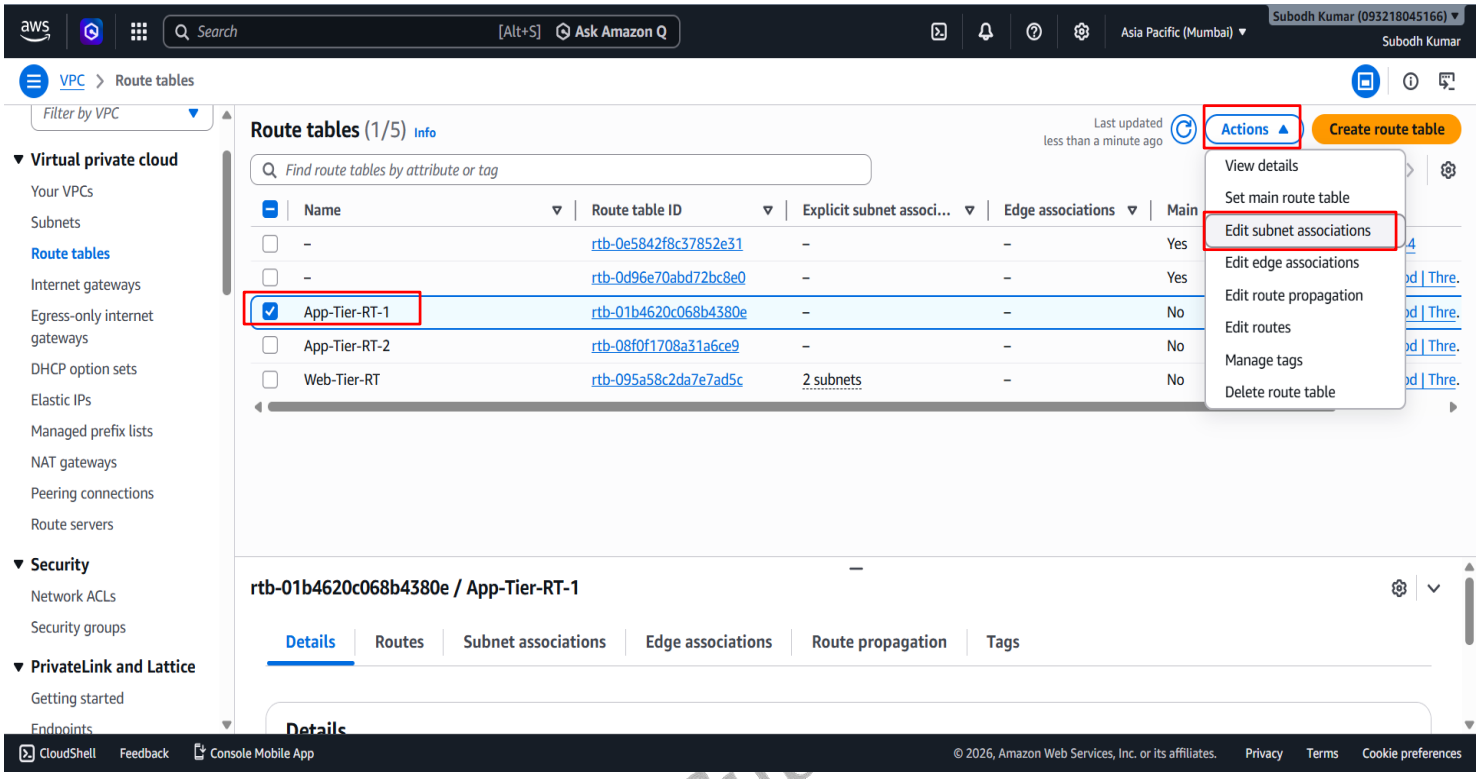
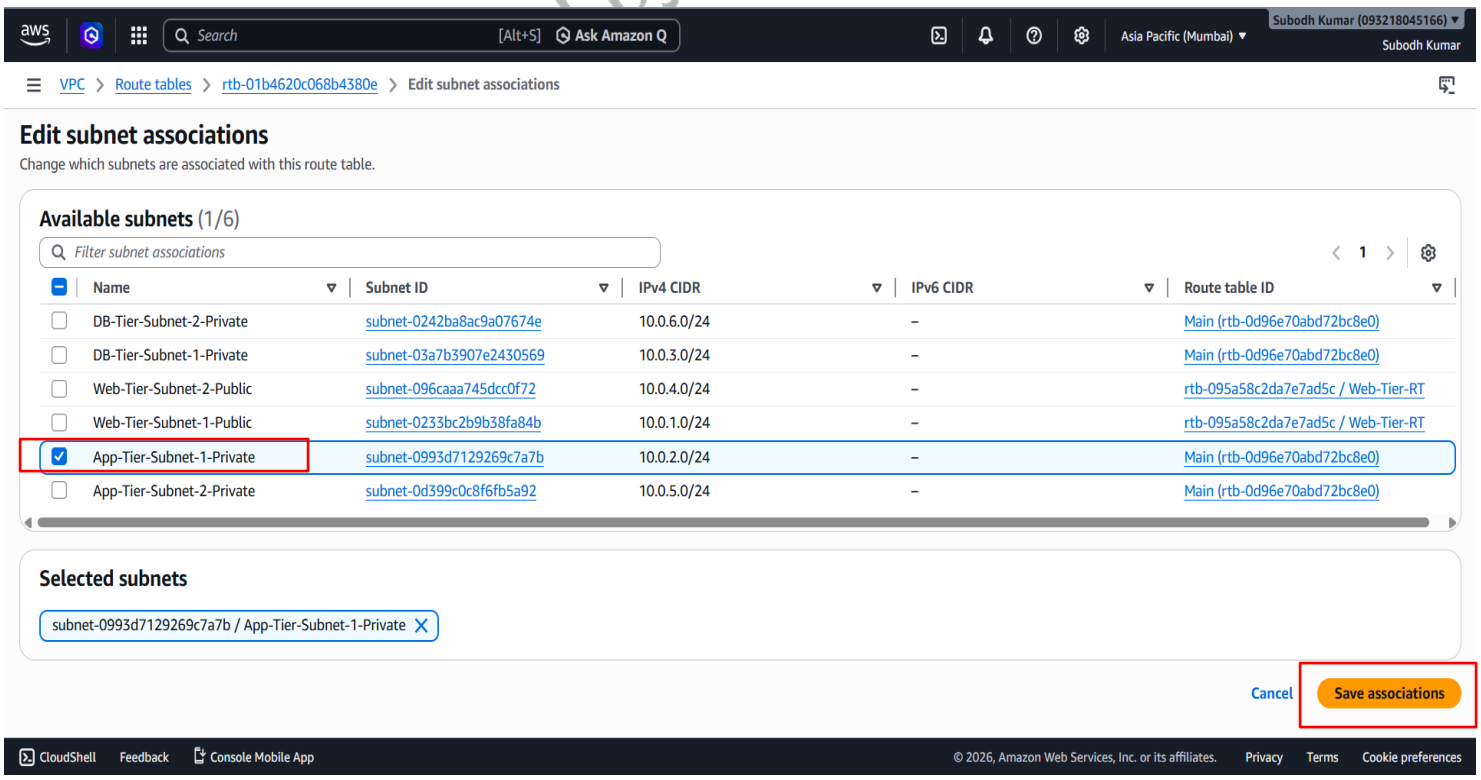
Subodh

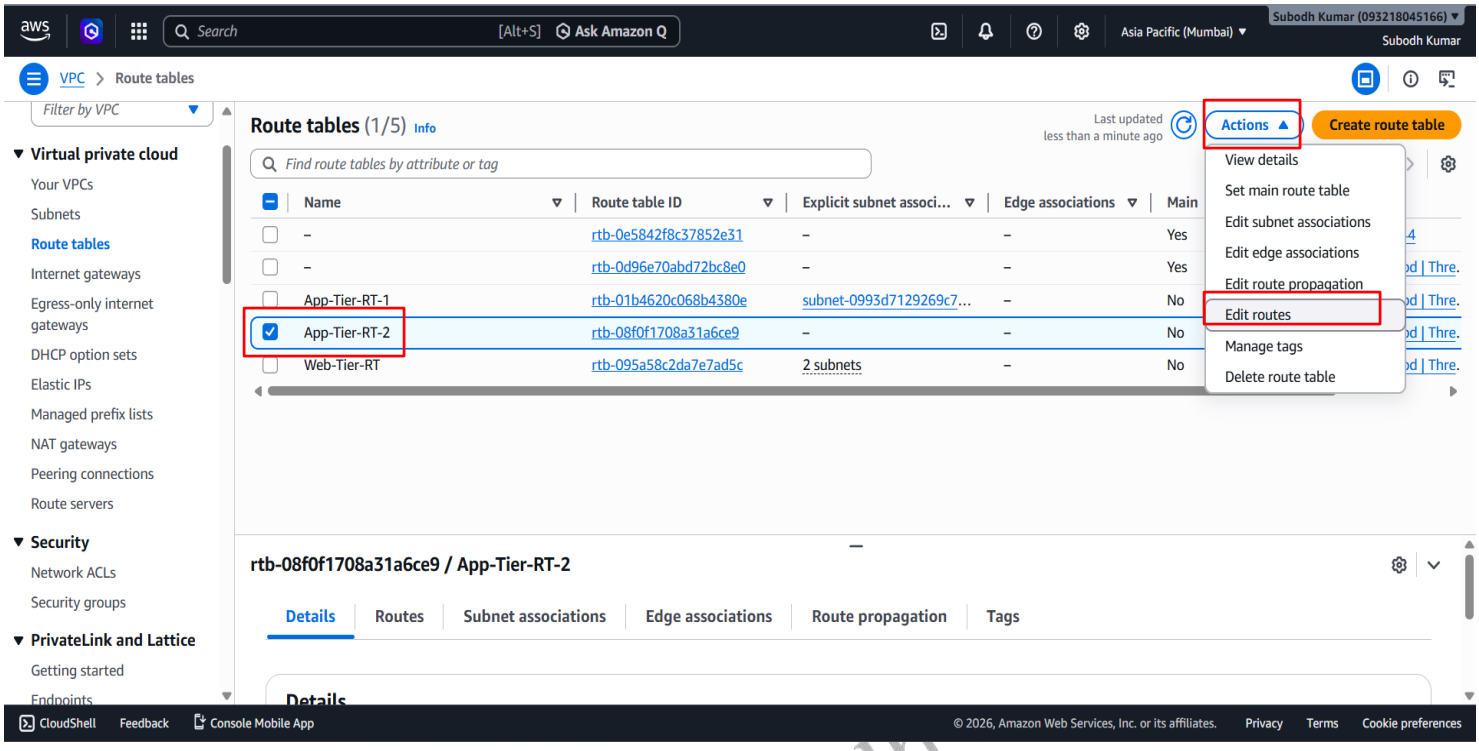
Again, select the App-Tier-RT-1 and go to **Actions > Edit subnet associations.**

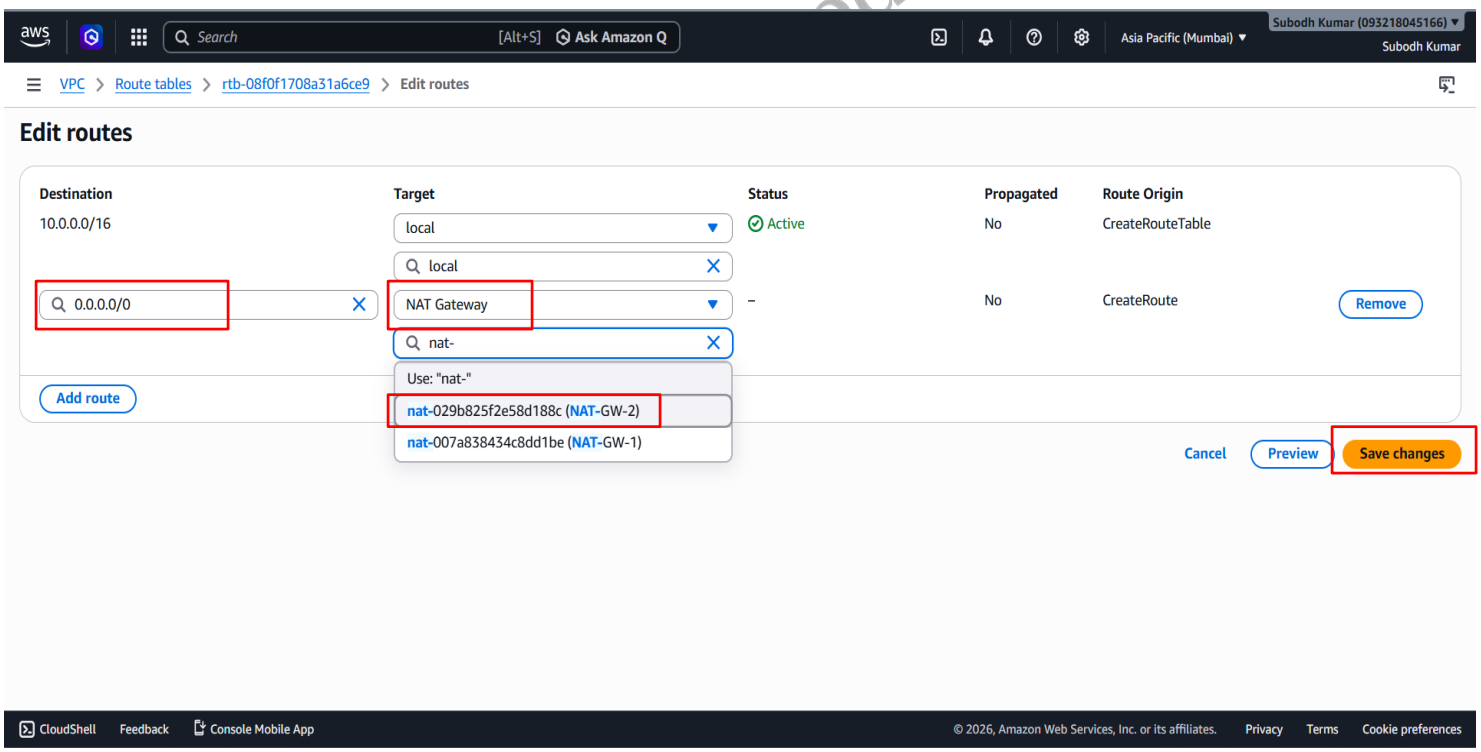
Select App-Tier-Subnet-1-Private and Save associations.



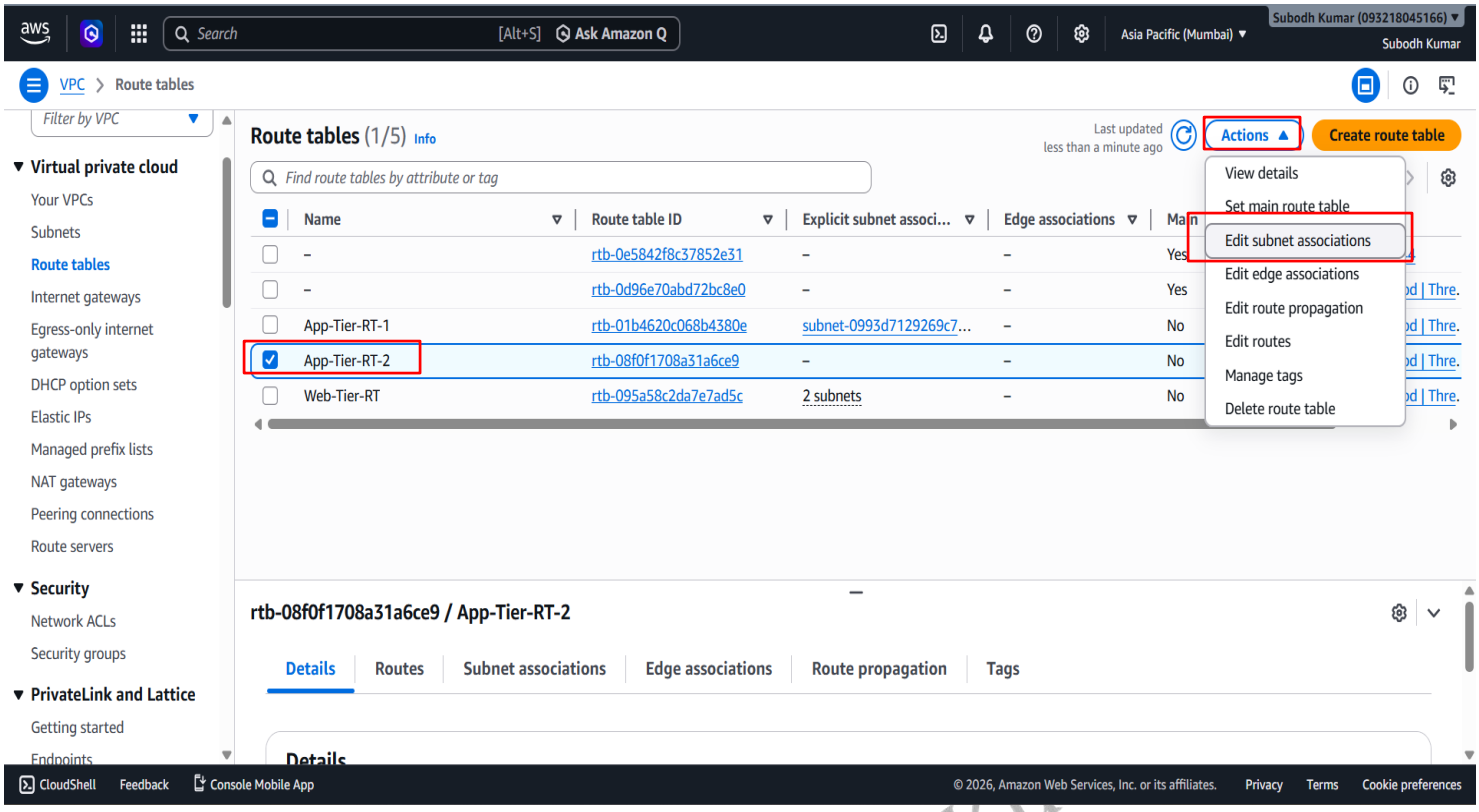


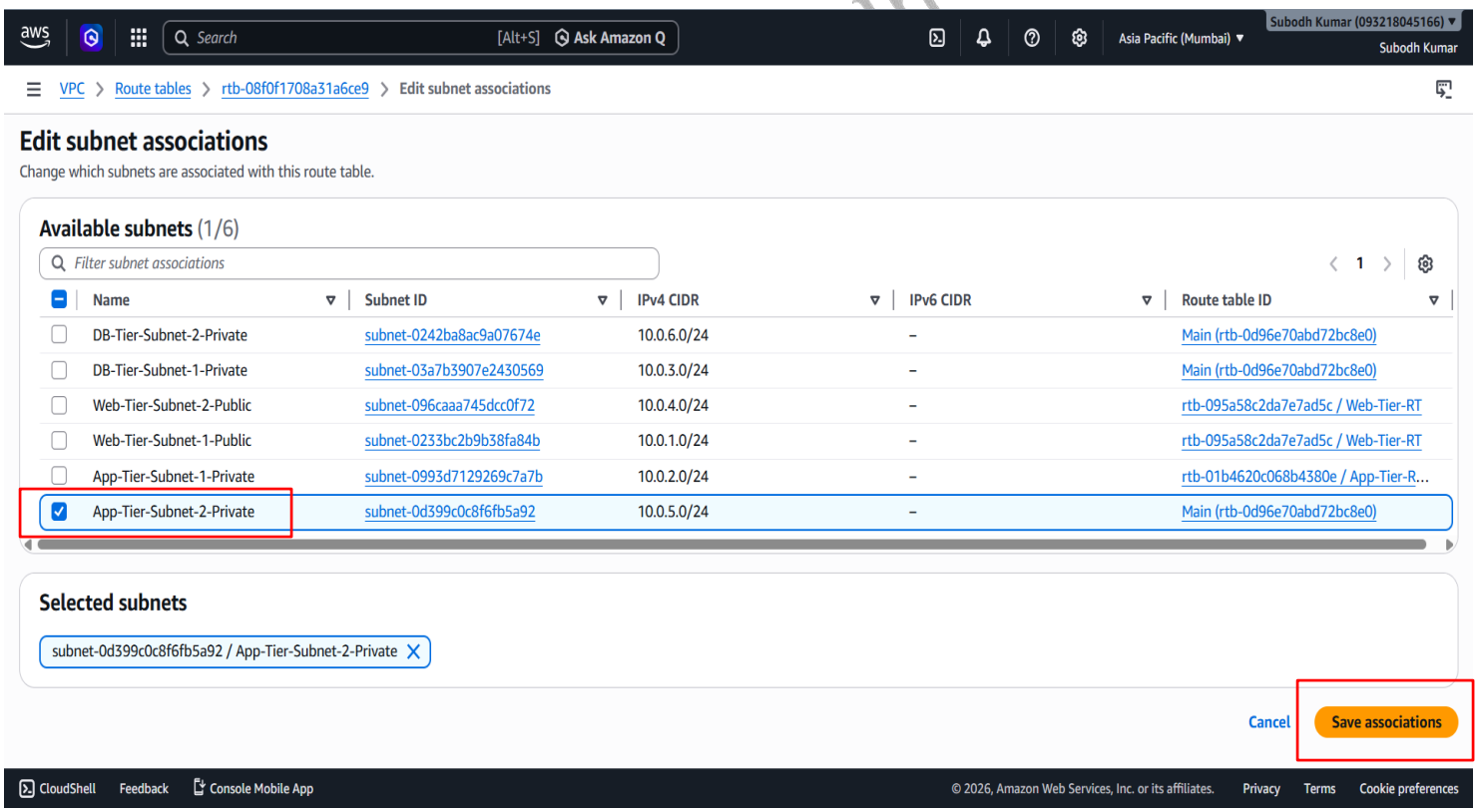
3. Now Select **App-Tier-RT-2** Go to **Actions > Edit routes**. Add route as follows. Destination 0.0.0.0/0, target NAT gateway > select our NAT-GW-2. Then Save changes.



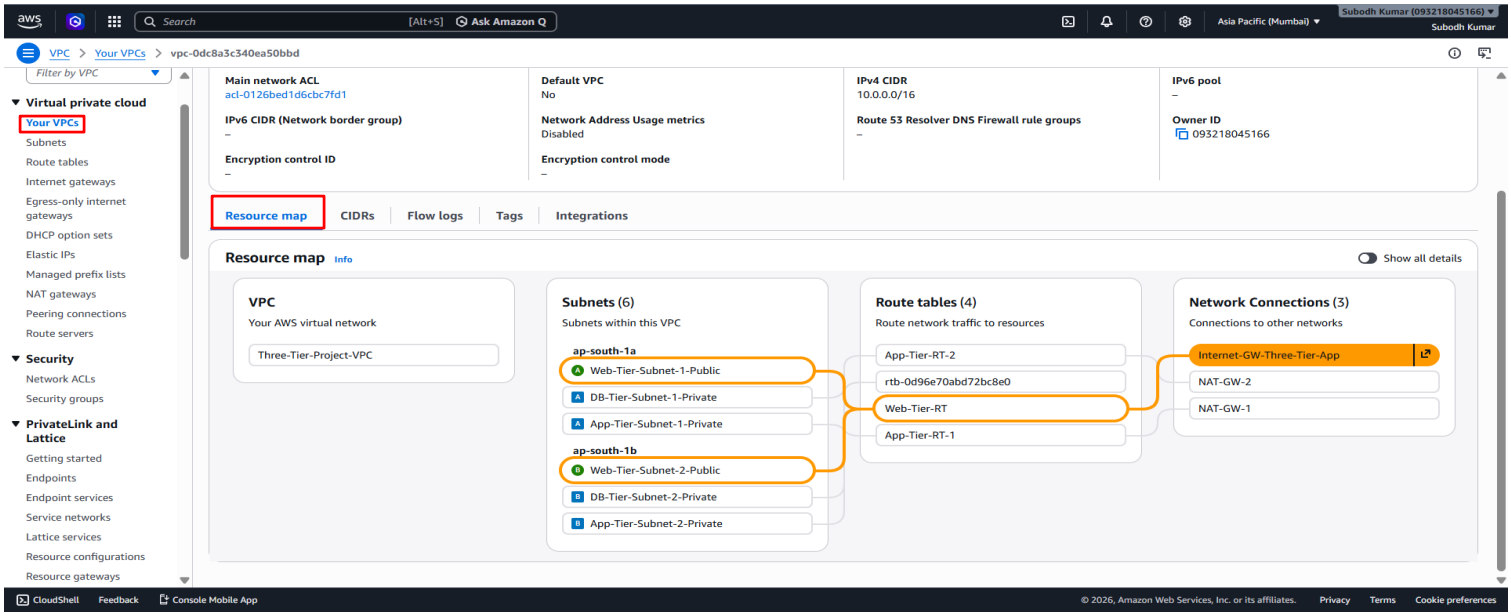


Again, select the App-Tier-RT-2 and go to **Actions > Edit subnet associations.** Select App-Tier-Subnet-2-Private and Save associations.





Now if you **check** the **Resource map** you can see that connections have been made. **Check** if there are **any mistakes** according to our architecture diagram by hovering over each element.


Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
VPC > Your VPCs > vpc-0dc8a3c340ea50bbd
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services
Resource configurations
Resource gateways
Main network ACL
acl-0126bed1d6cbc7fd1
Default VPC
No
IPv4 CIDR
10.0.0.0/16
IPv6 pool
IPv6 CIDR (Network border group)
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
Owner ID
093218045166
Encryption control ID
Encryption control mode
Resource map
CIDRs
Flow logs
Tags
Integrations
Resource map Info
Show all details
VPC
Your AWS virtual network
Three-Tier-Project-VPC
Subnets (6)
Subnets within this VPC
ap-south-1a
Web-Tier-Subnet-1-Public
DB-Tier-Subnet-1-Private
App-Tier-Subnet-1-Private
ap-south-1b
Web-Tier-Subnet-2-Public
DB-Tier-Subnet-2-Private
App-Tier-Subnet-2-Private
Route tables (4)
Route network traffic to resources
App-Tier-RT-2
rtb-0d96e70abd72bc8e0
Web-Tier-RT
App-Tier-RT-1
Network Connections (3)
Connections to other networks
Internet-GW-Three-Tier-App
NAT-GW-2
NAT-GW-1
CloudShell
Feedback
Console Mobile App
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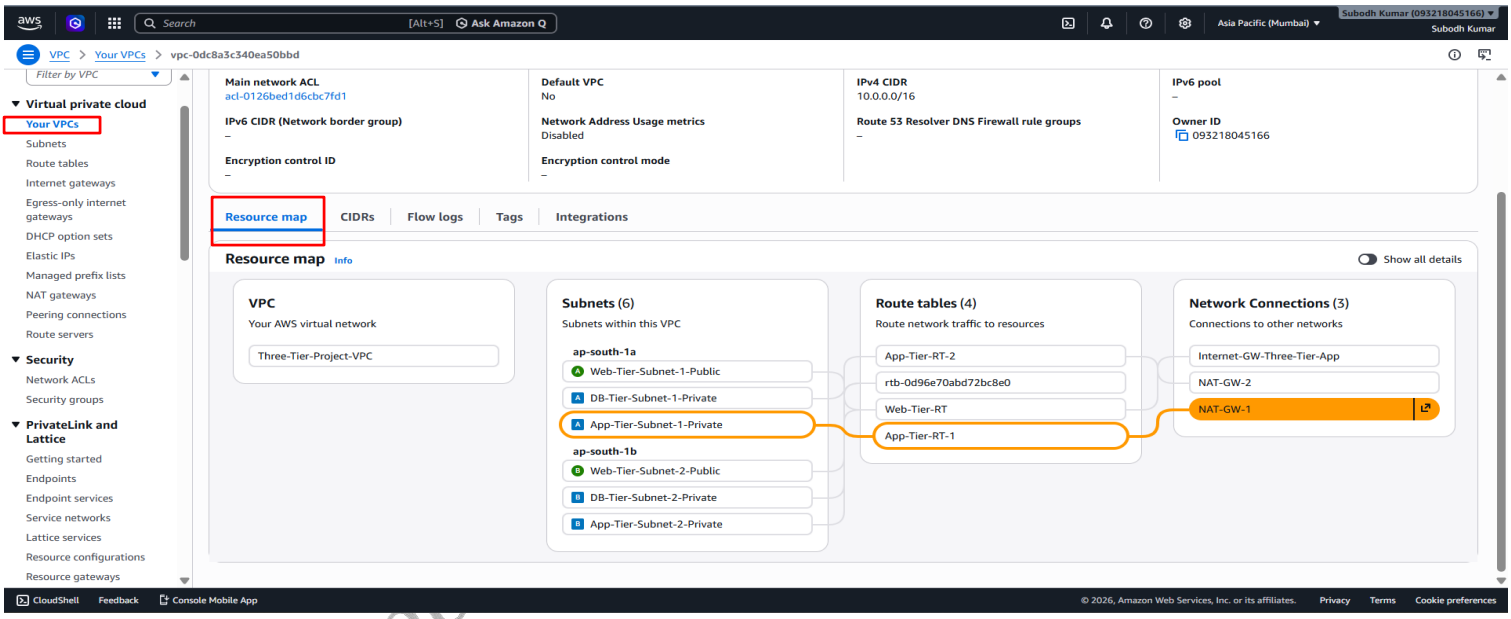
Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
VPC > Your VPCs > vpc-0dc8a3c340ea50bbd
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
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IPv6 CIDR (Network border group)
Network Address Usage metrics
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Encryption control ID
Encryption control mode
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NAT-GW-2
NAT-GW-1
CloudShell
Feedback
Console Mobile App
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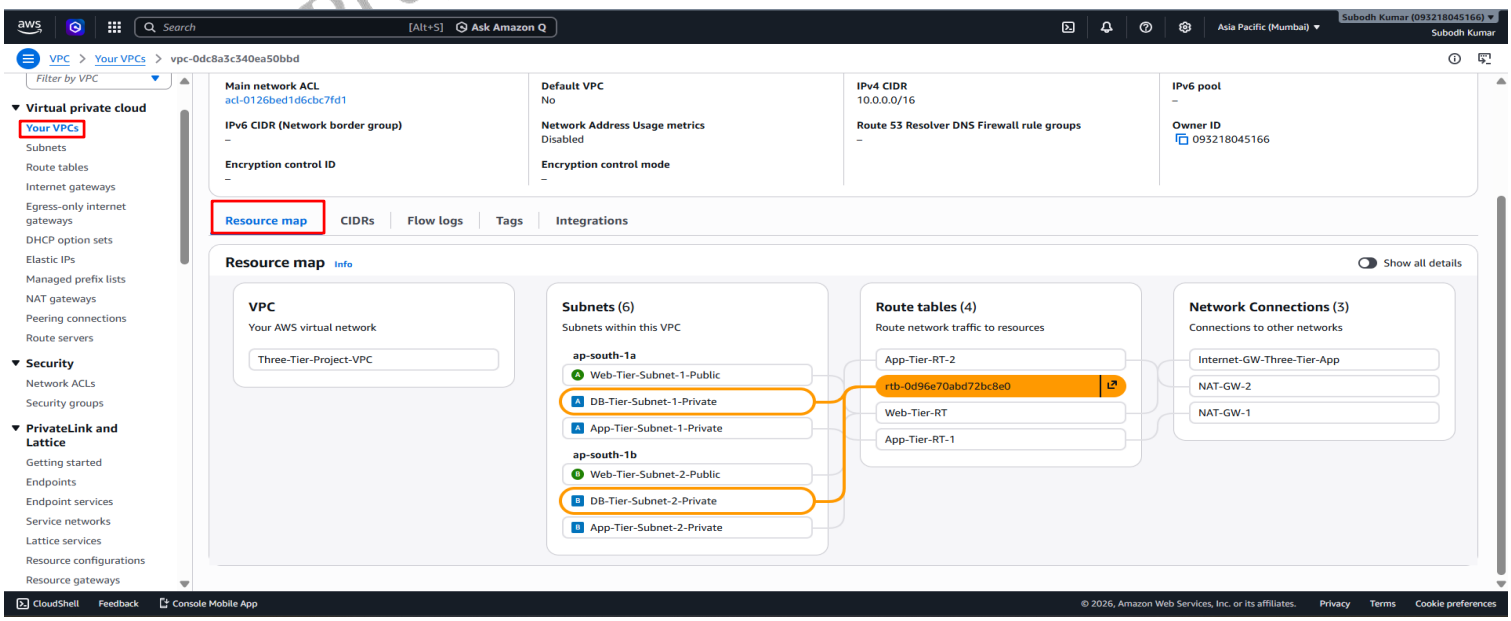
Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
VPC > Your VPCs > vpc-0dc8a3c340ea50bbd
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
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ap-south-1b
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rtb-0d96e70abd72bc8e0
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Internet-GW-Three-Tier-App
NAT-GW-2
NAT-GW-1
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Good Job . Now Let's Create Security Groups.

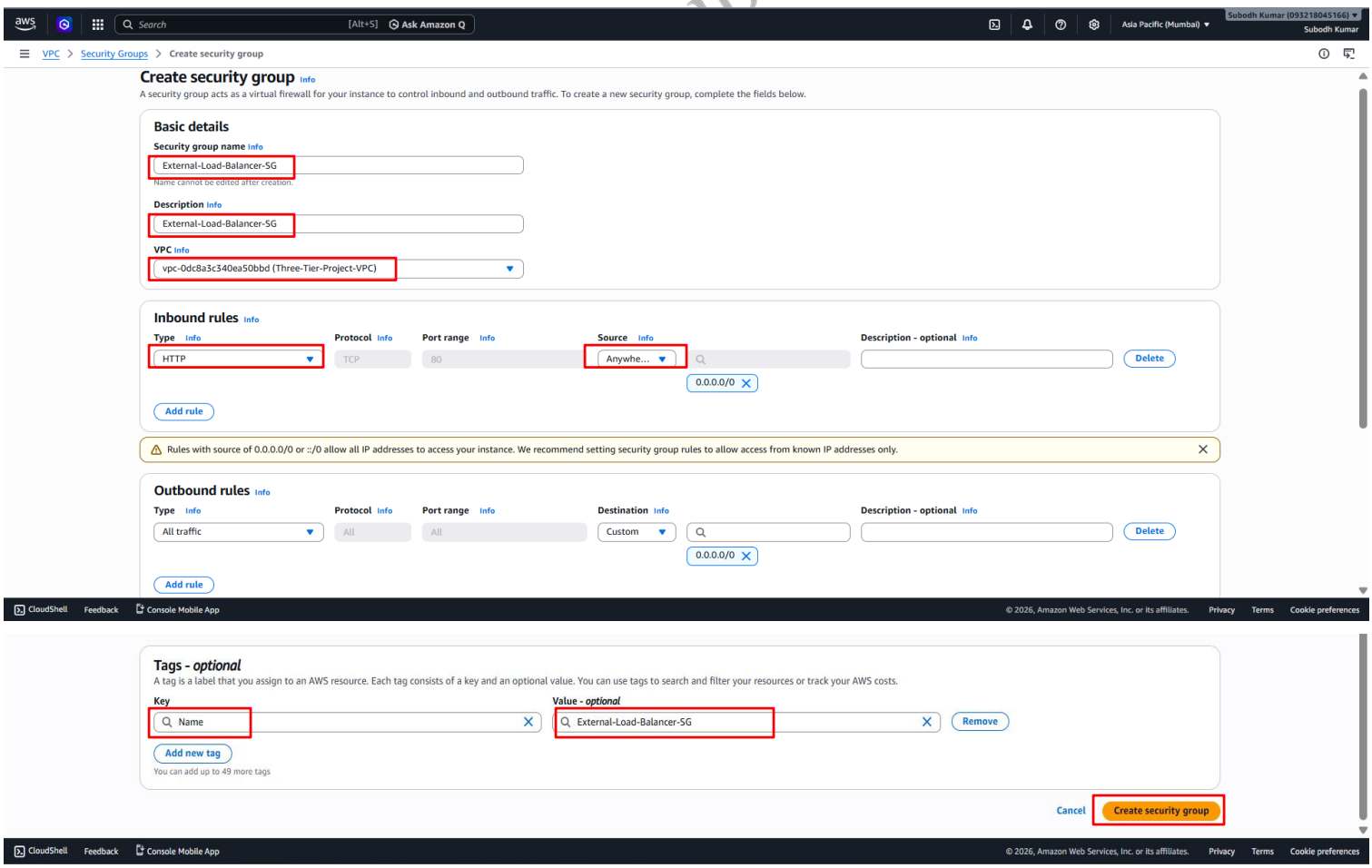
3. Create Security Groups

We have to create the following Security Groups.

Security Group	Type (Port Range)	Source
External-Load-Balancer-SG	HTTP (80)	0.0.0.0/0
Web-Tier-SG	HTTP (80)	External-Load-Balancer-SG
Internal-Load-Balancer-SG	HTTP (80)	Web-Tier-SG
App-Tier-SG	Custom TCP (4000)	Internal-Load-Balancer-SG
DB-Tier-SG	MYSQL (3306)	App-Tier-SG

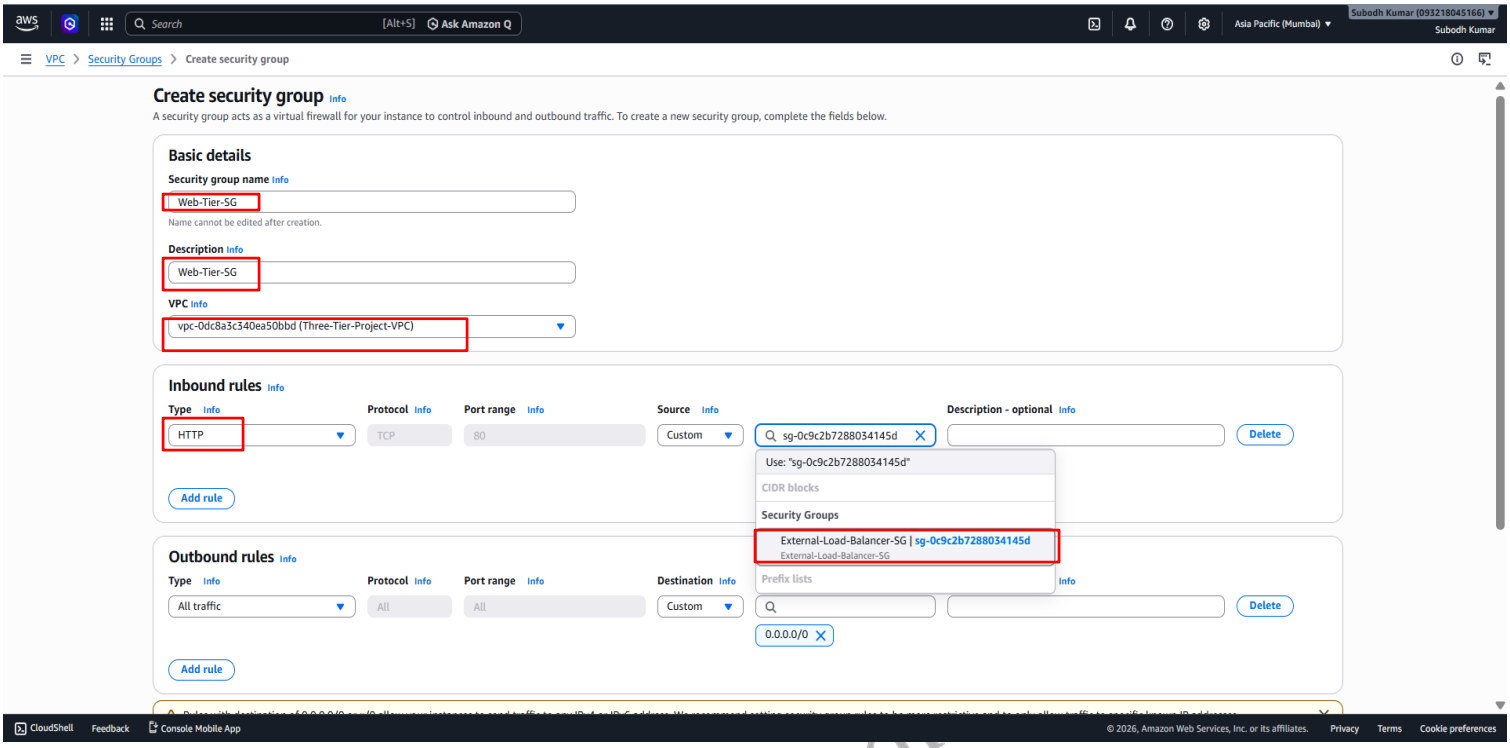
Subodh Kumar

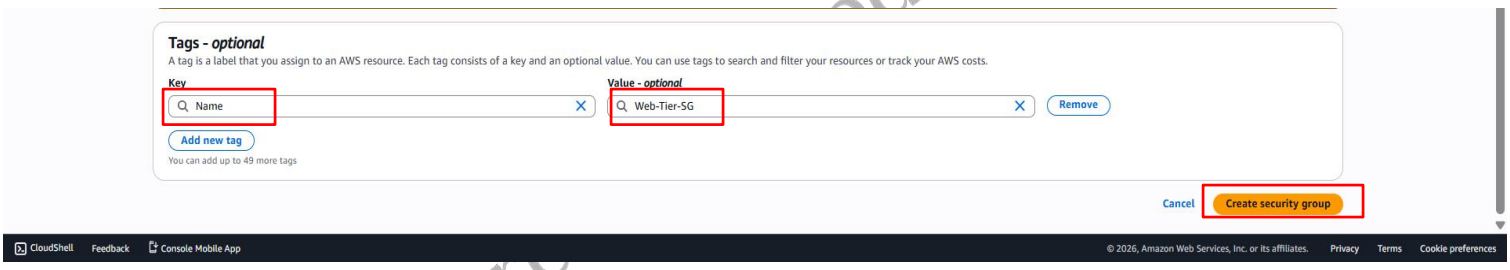
Go to VPC dashboard **Security Groups > Create security group**. Create **External-Load-Balancer-SG** and add inbound rule as follows by referring the above table. Then, Click **Create security group** icon.



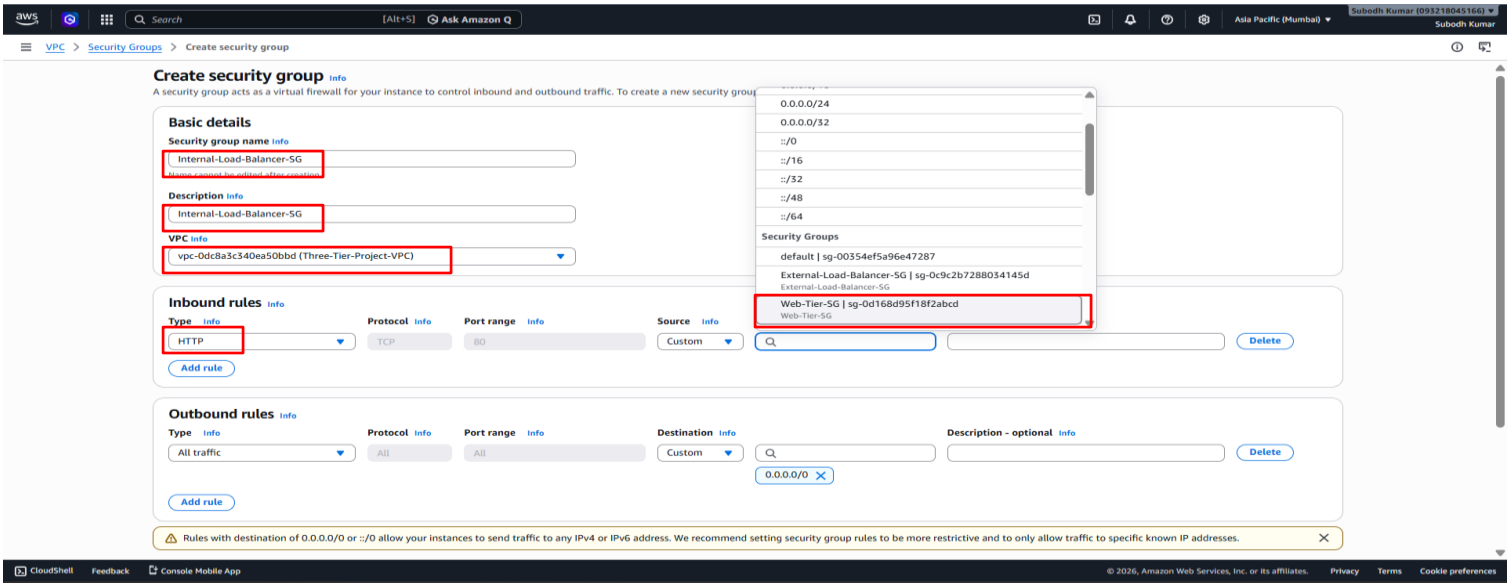
Creating other Security groups as well as shown in the table. The process is same.

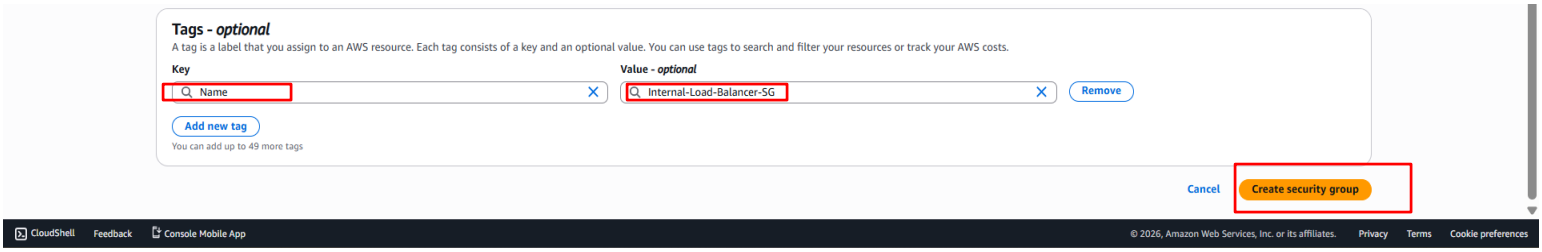
Go to VPC dashboard **Security Groups > Create security group**. Create **Web-Tier-SG** and add inbound rule as follows by referring the above table. Then, Click **Create security group** icon.



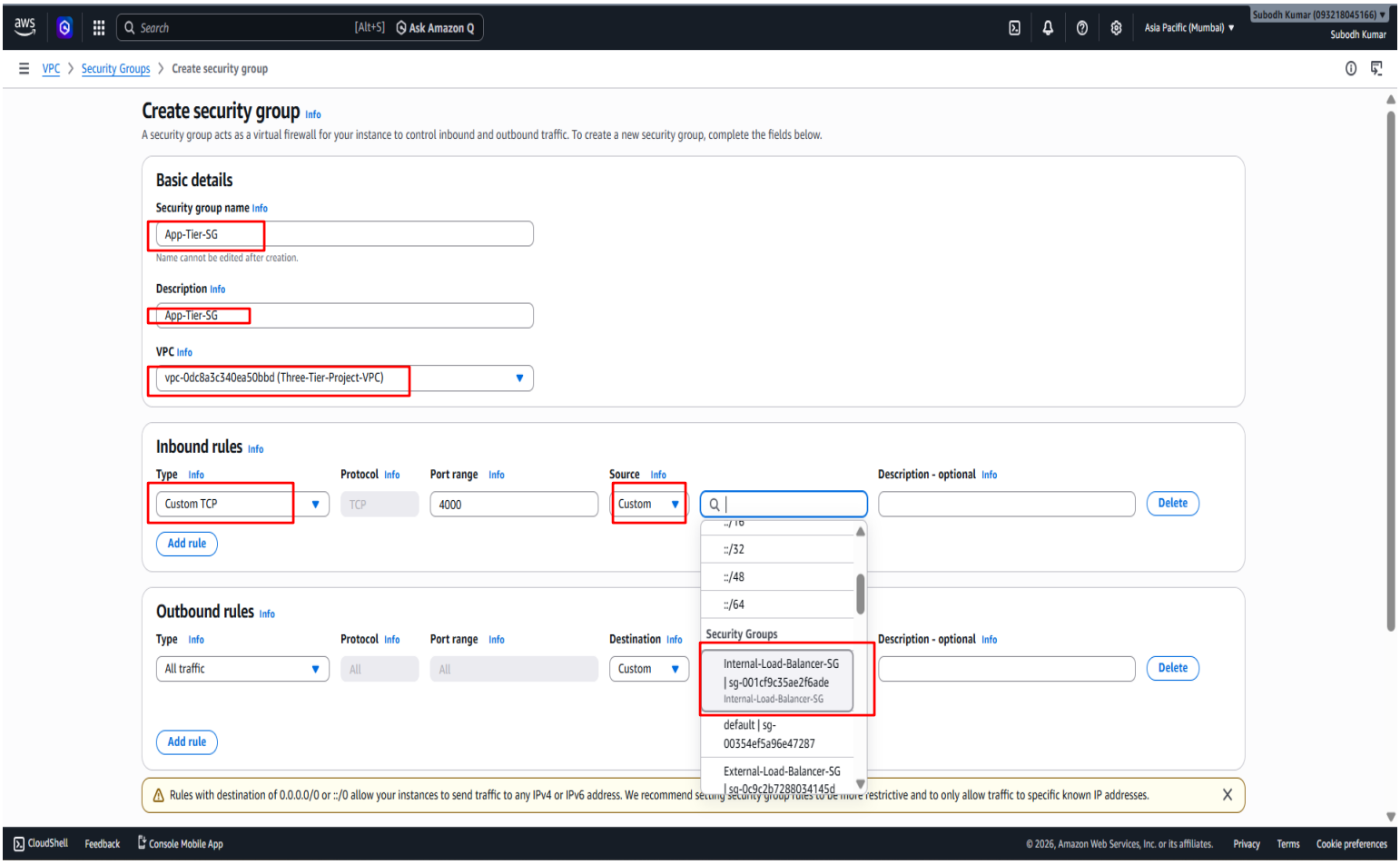


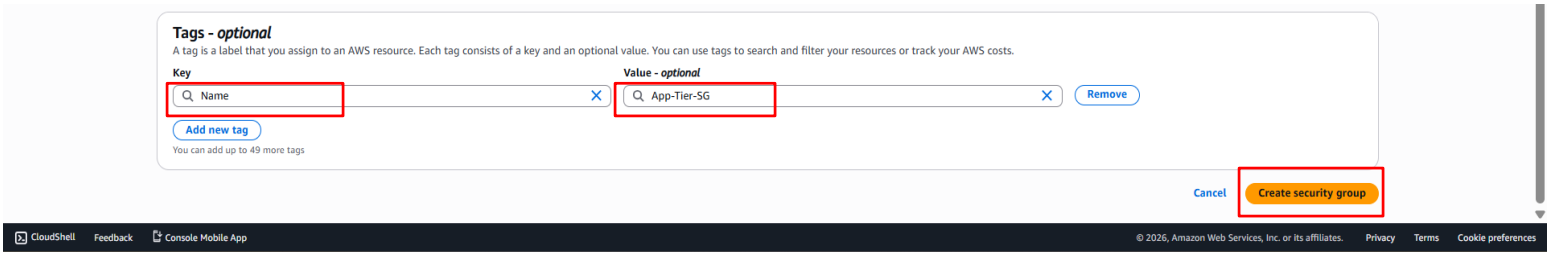
Go to VPC dashboard **Security Groups > Create security group**. Create **Internal- Load-Balancer-SG** and add inbound rule as follows by referring the above table. Then, Click **Create security group** icon.



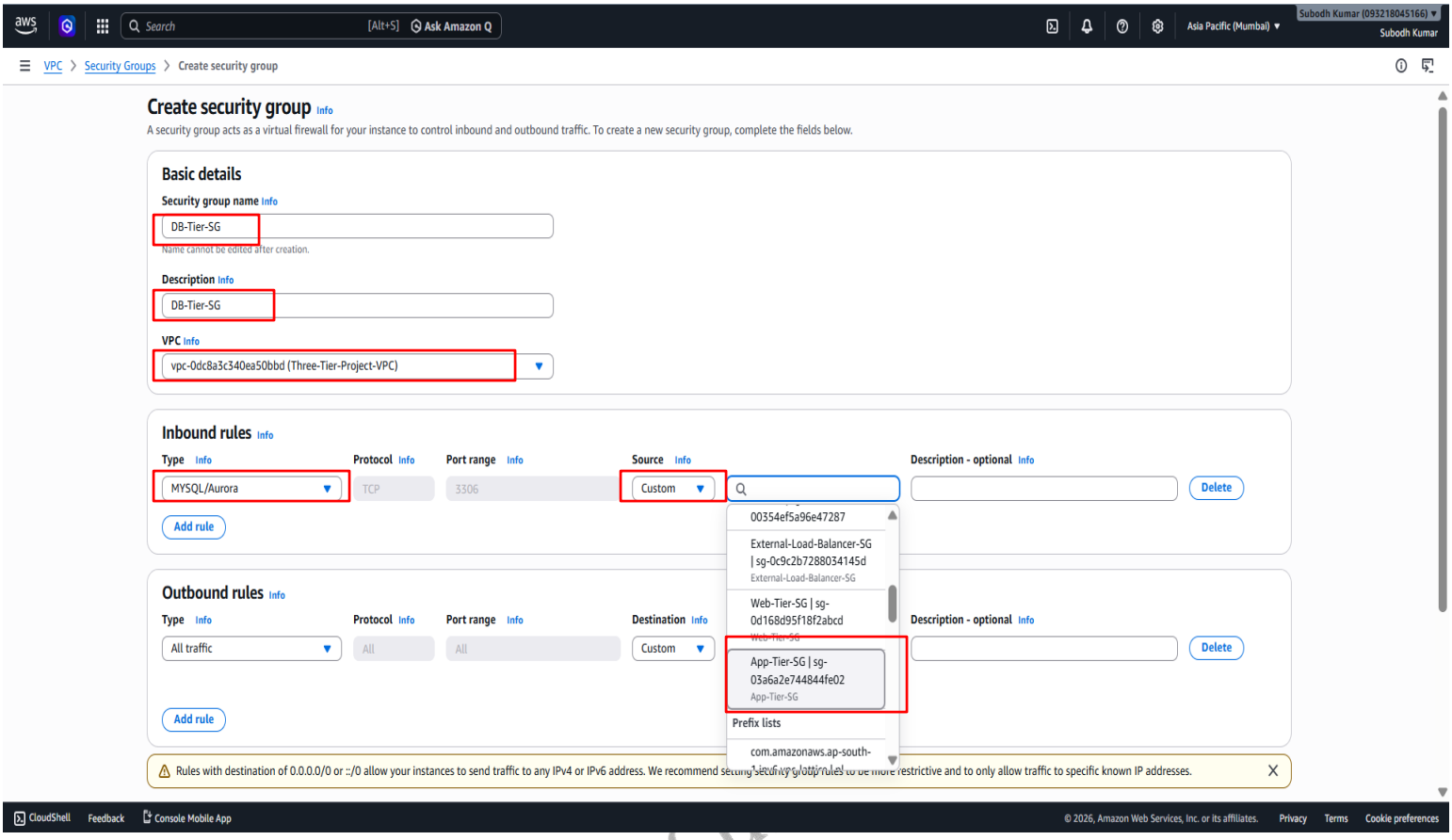


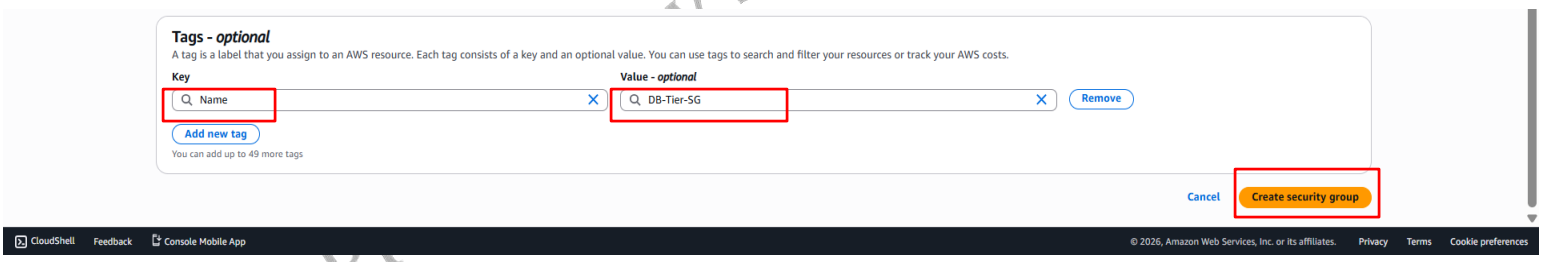
Go to VPC dashboard **Security Groups > Create security group**. Create **App-Tier-SG** and add inbound rule as follows by referring the above table. Then, Click on **Create security group** icon.



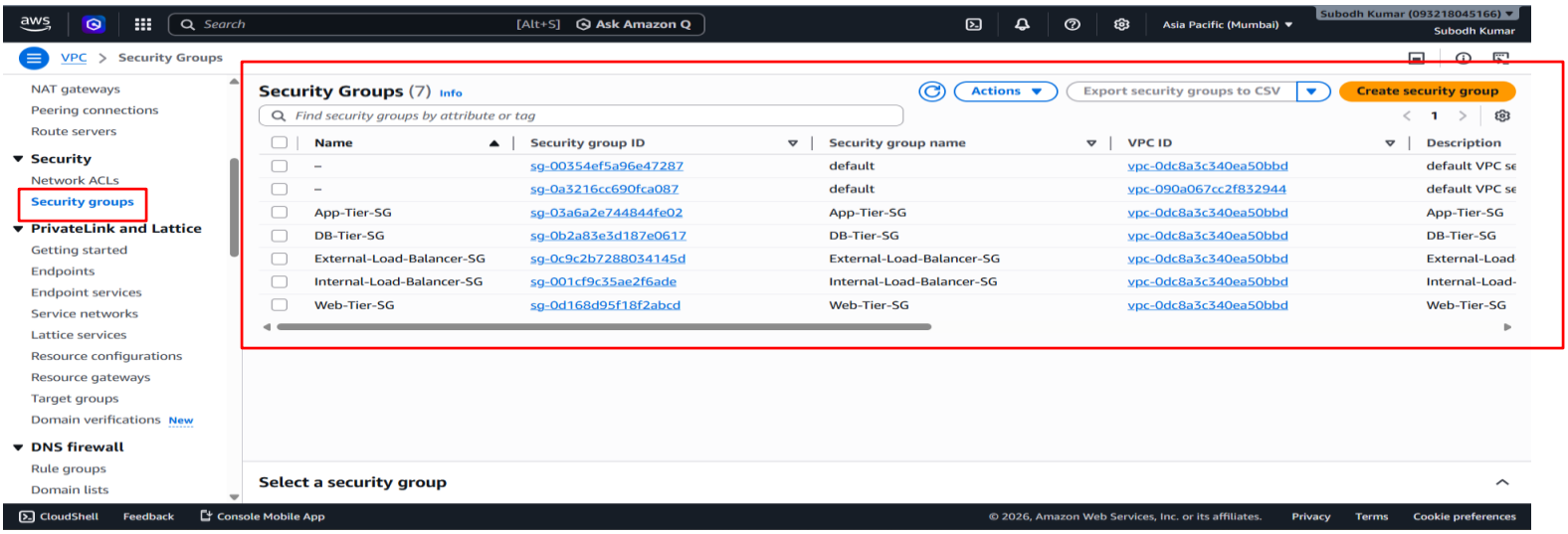


Go to VPC dashboard **Security Groups > Create security group**. Create **DB-Tier-SG** and add inbound rule as follows by referring the above table. Then, Click on **Create security group** icon.





After creating all Security groups you can see them like this.



Well Done . Now Let's Move on to setup the database servers.

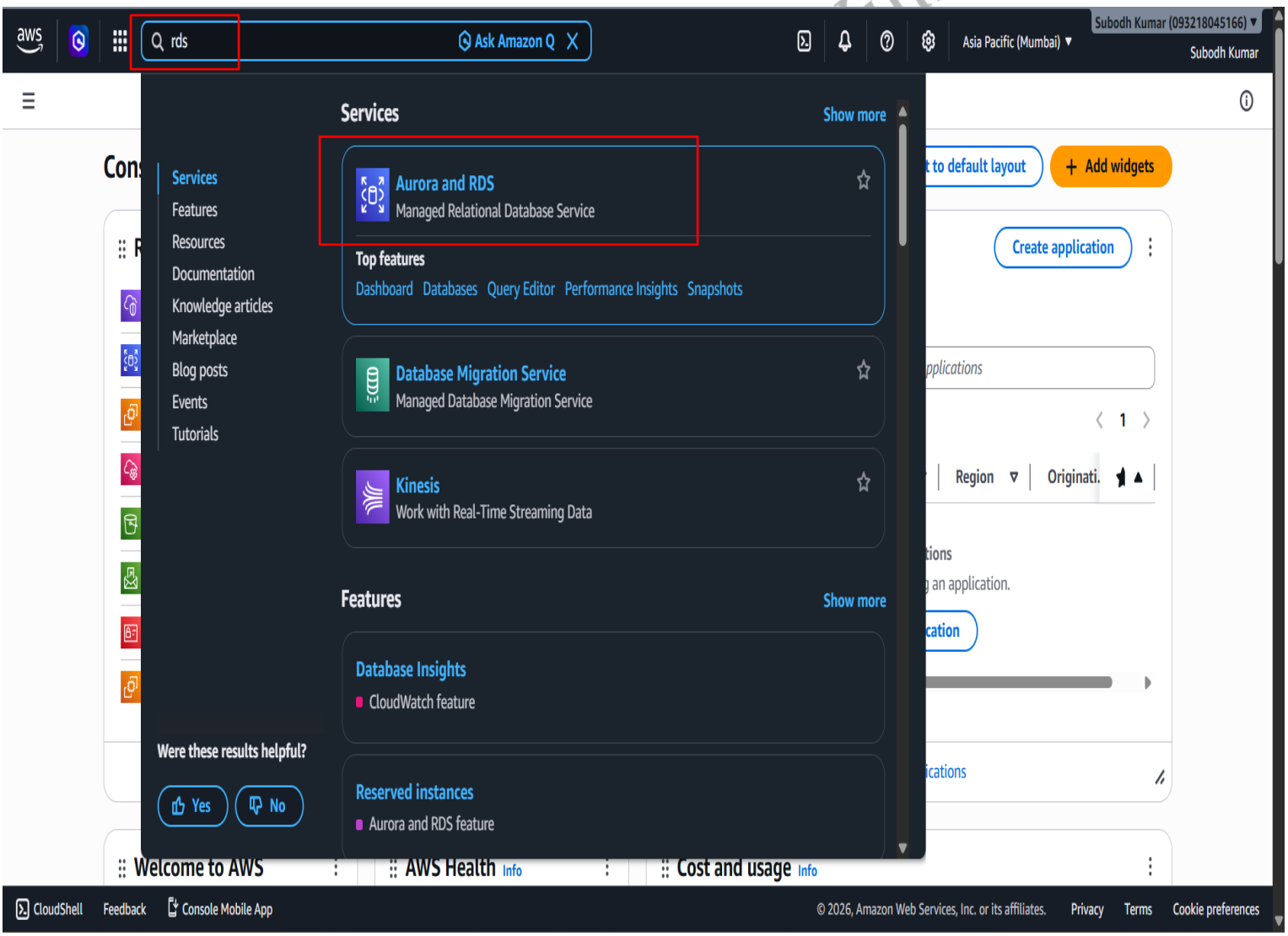
4. Create Database Subnet Group and RDS

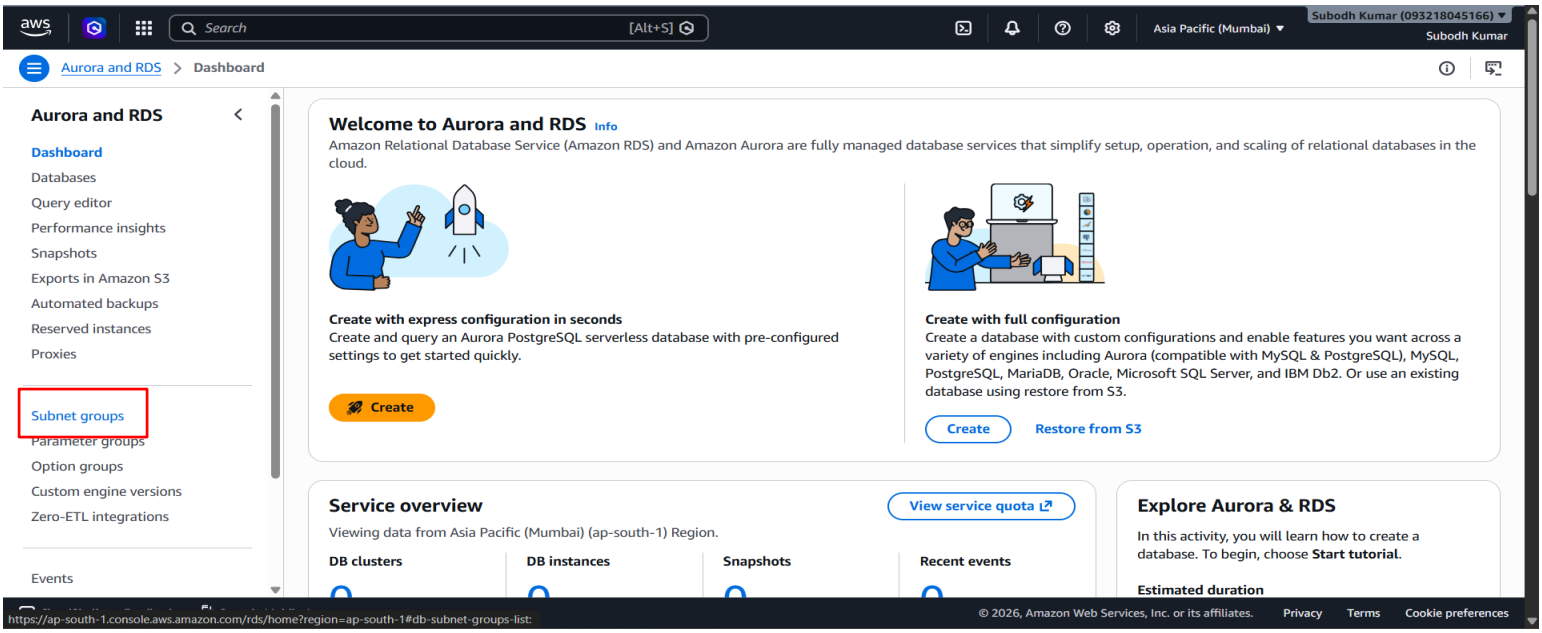
First, we need to **create** a **subnet group**. But **why?**

> Because, when we **deploy** an **RDS instance** in a **Multi-AZ configuration**, AWS **automatically creates** a **primary database instance** in **one Availability Zone (AZ)** and **a standby replica** in **another AZ** within the **same region.** This setup **requires subnet groups** because **each subnet** in the **group** must belong to a **different AZ**, **ensuring** that your **database instances** are **spread across multiple AZs** for **redundancy** and **failover protection**.

Cool

Now search for **RDS** in the **search bar**. In the Amazon RDS go to **Subnet groups > Create DB subnet group.** Setup it as follows.




Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Aurora and RDS > Dashboard
Aurora and RDS
Dashboard
Databases
Query editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
Events
Welcome to Aurora and RDS Info
Amazon Relational Database Service (Amazon RDS) and Amazon Aurora are fully managed database services that simplify setup, operation, and scaling of relational databases in the cloud.
Create with express configuration in seconds
Create and query an Aurora PostgreSQL serverless database with pre-configured settings to get started quickly.
Create
Create with full configuration
Create a database with custom configurations and enable features you want across a variety of engines including Aurora (compatible with MySQL & PostgreSQL), MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server, and IBM Db2. Or use an existing database using restore from S3.
Create
Restore from S3
Service overview
View service quota
Viewing data from Asia Pacific (Mumbai) (ap-south-1) Region.
DB clusters
DB instances
Snapshots
Recent events
Explore Aurora & RDS
In this activity, you will learn how to create a database. To begin, choose Start tutorial.
Estimated duration
https://ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#db-subnet-groups-list:
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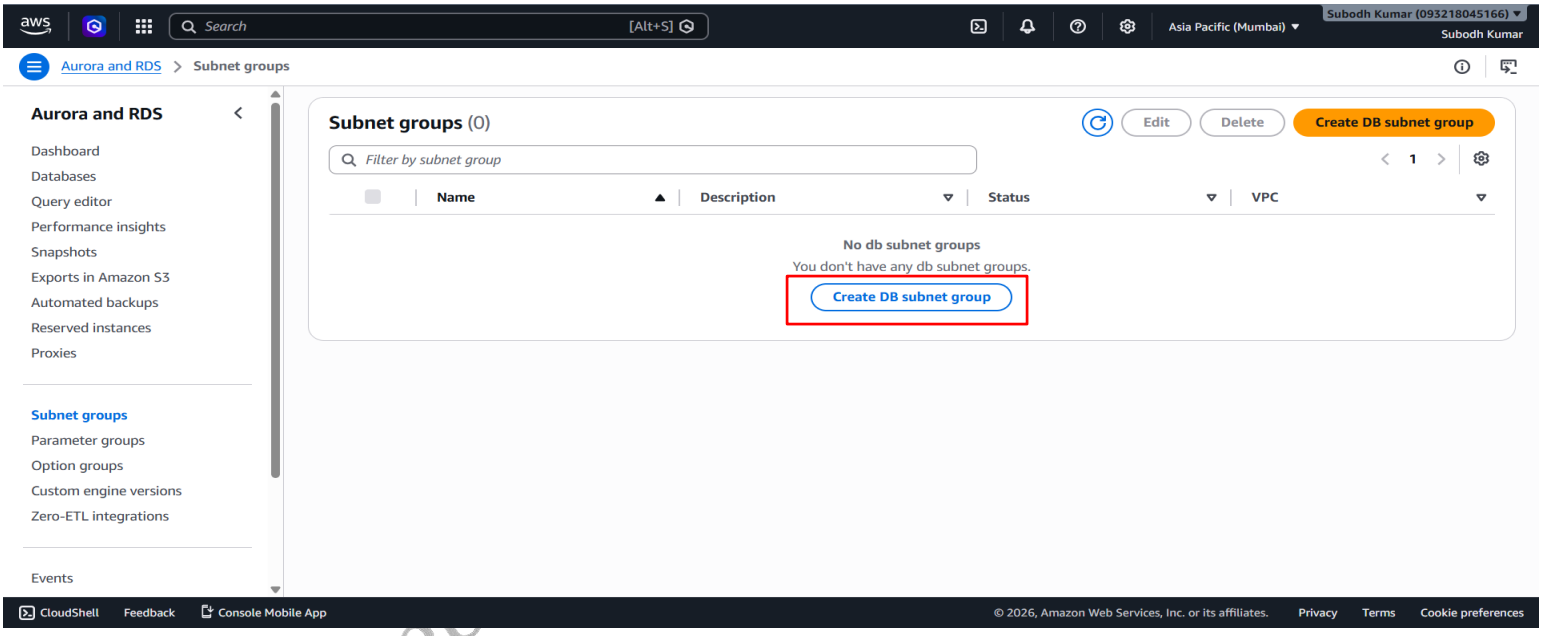
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Aurora and RDS > Subnet groups
Aurora and RDS
Dashboard
Databases
Query editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
Events
Subnet groups (0)
Edit
Delete
Create DB subnet group
Filter by subnet group
Name
Description
Status
VPC
No db subnet groups
You don't have any db subnet groups.
Create DB subnet group
CloudShell
Feedback
Console Mobile App
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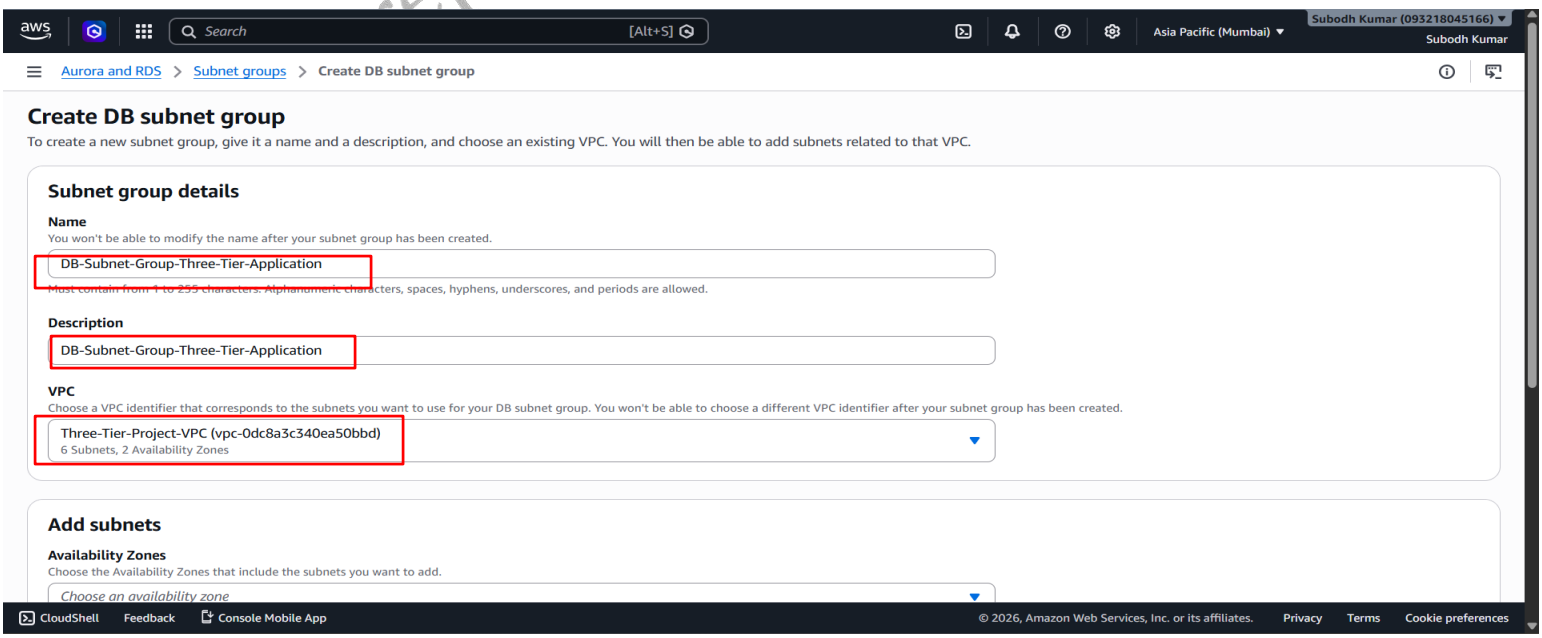
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Aurora and RDS > Subnet groups > Create DB subnet group
Create DB subnet group
To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.
Subnet group details
Name
You won't be able to modify the name after your subnet group has been created.
DB-Subnet-Group-Three-Tier-Application
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.
Description
DB-Subnet-Group-Three-Tier-Application
VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
Three-Tier-Project-VPC (vpc-0dc8a3c340ea50bbd)
6 Subnets, 2 Availability Zones
Add subnets
Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone
CloudShell
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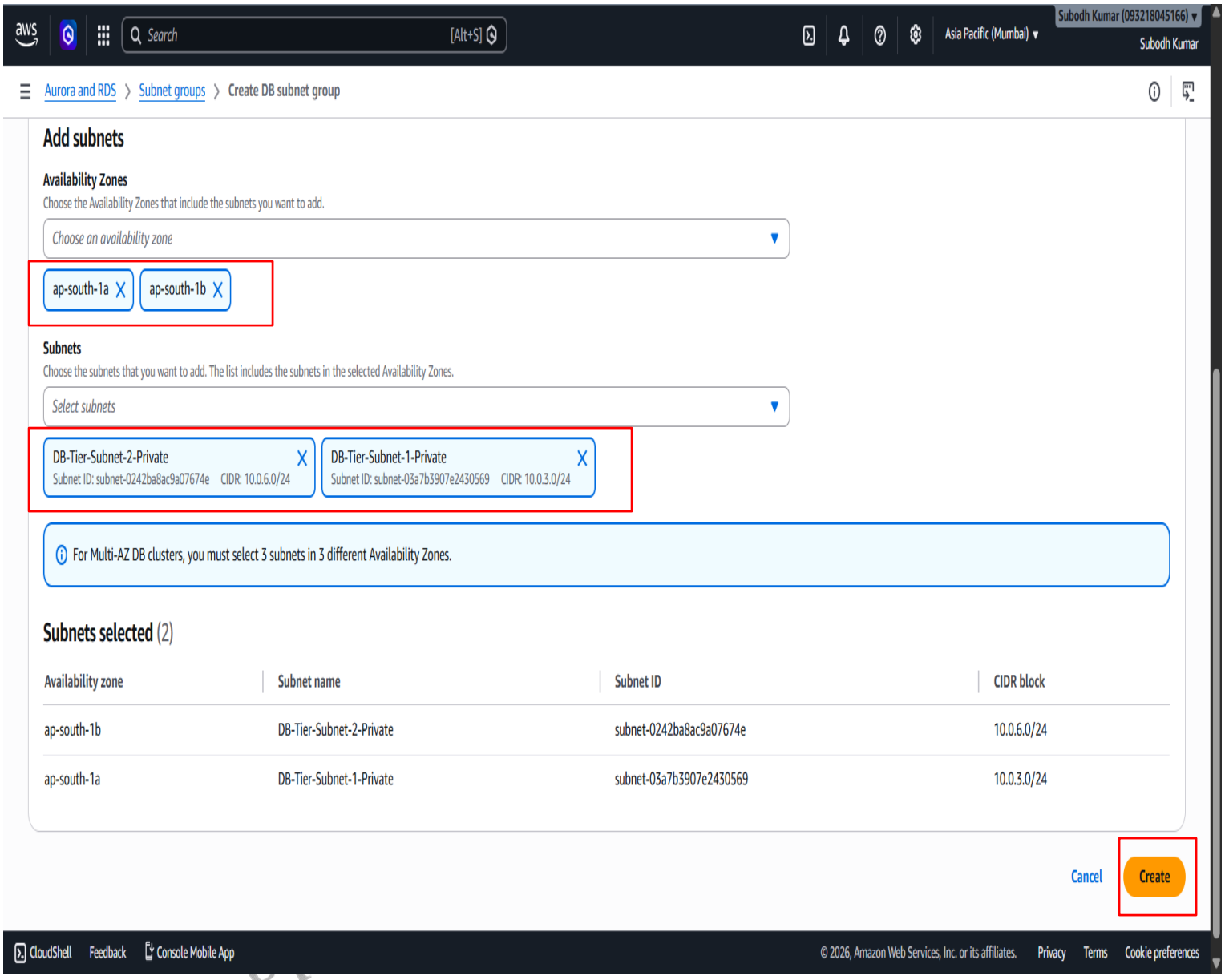
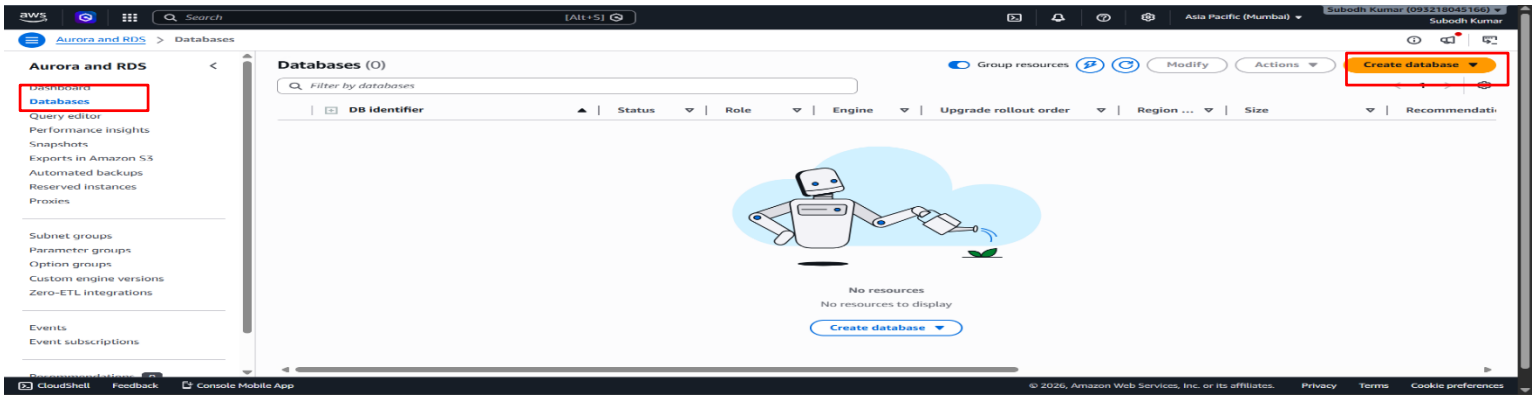
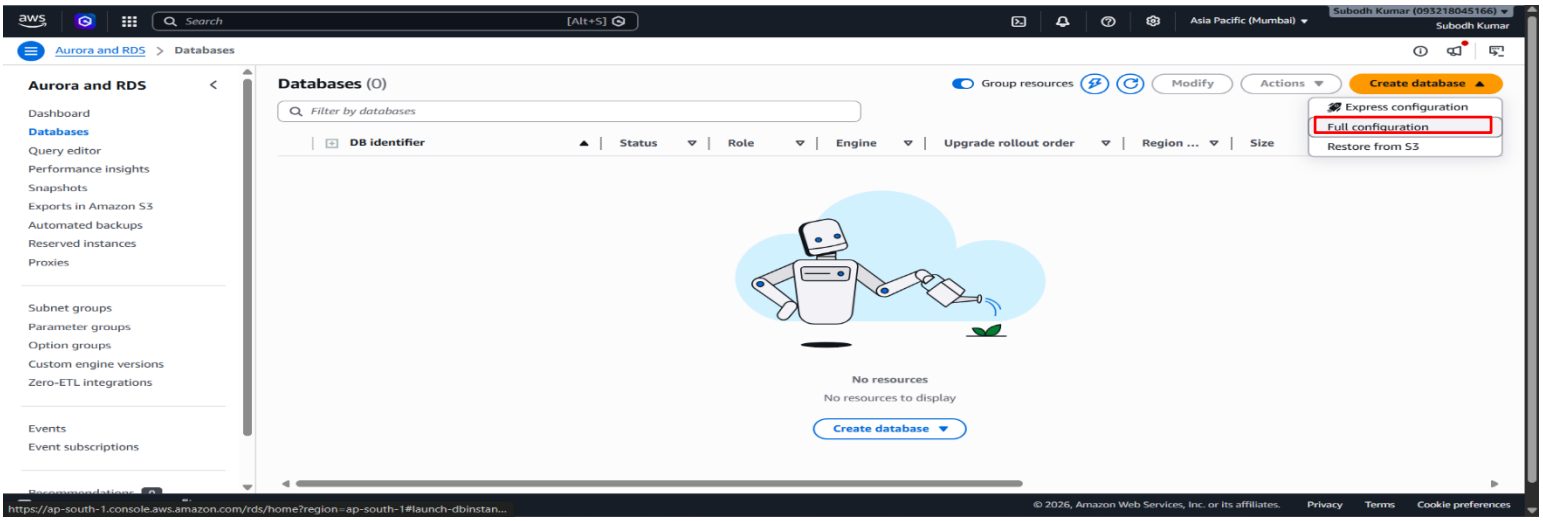
Choose **Availability Zones** that show in the diagram (**1a, 1b**).
Choose **Subnet** which are **assigned** to the **DB** (**10.0.3.0/24, 10.0.6.0/24**).



Now go to Amazon RDS **Databases > Create database**. Setup it as follows (In our application we have use MySQL therefore we use **Aurora (MySQL Compatible)** option. We can use MySQL as well).




Aurora and RDS > Databases
Aurora and RDS
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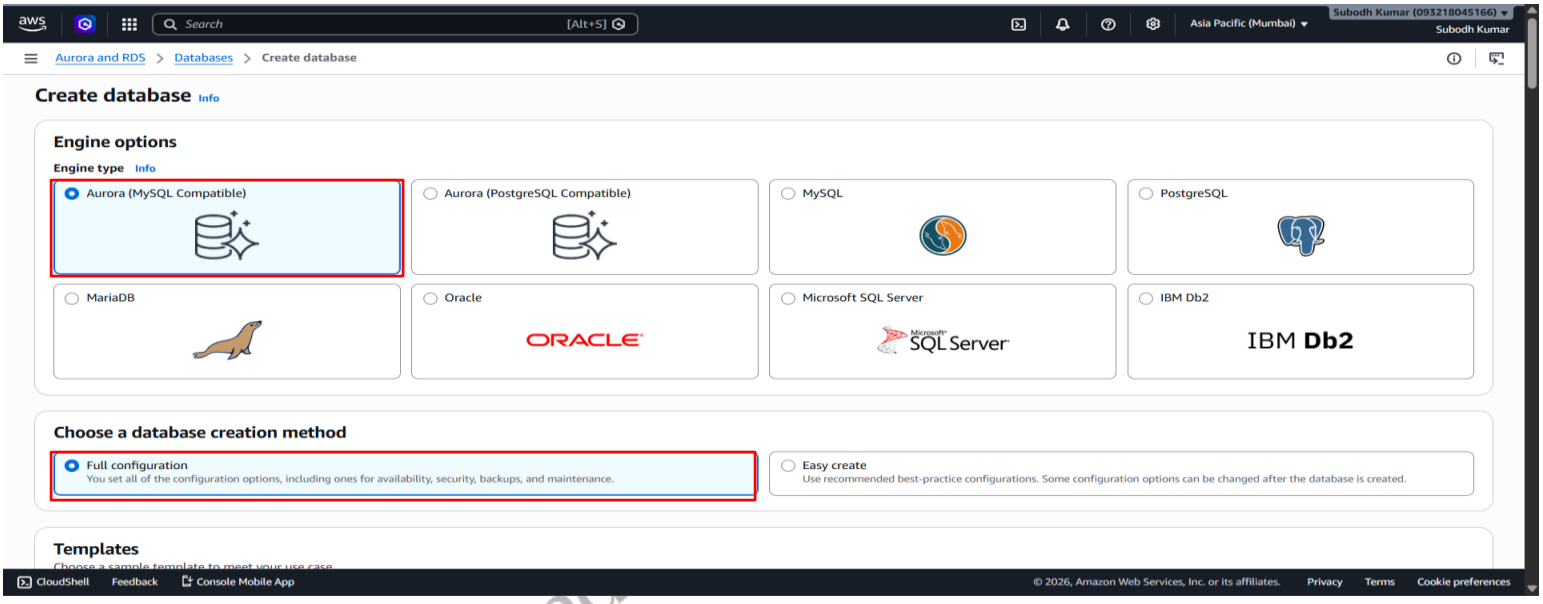
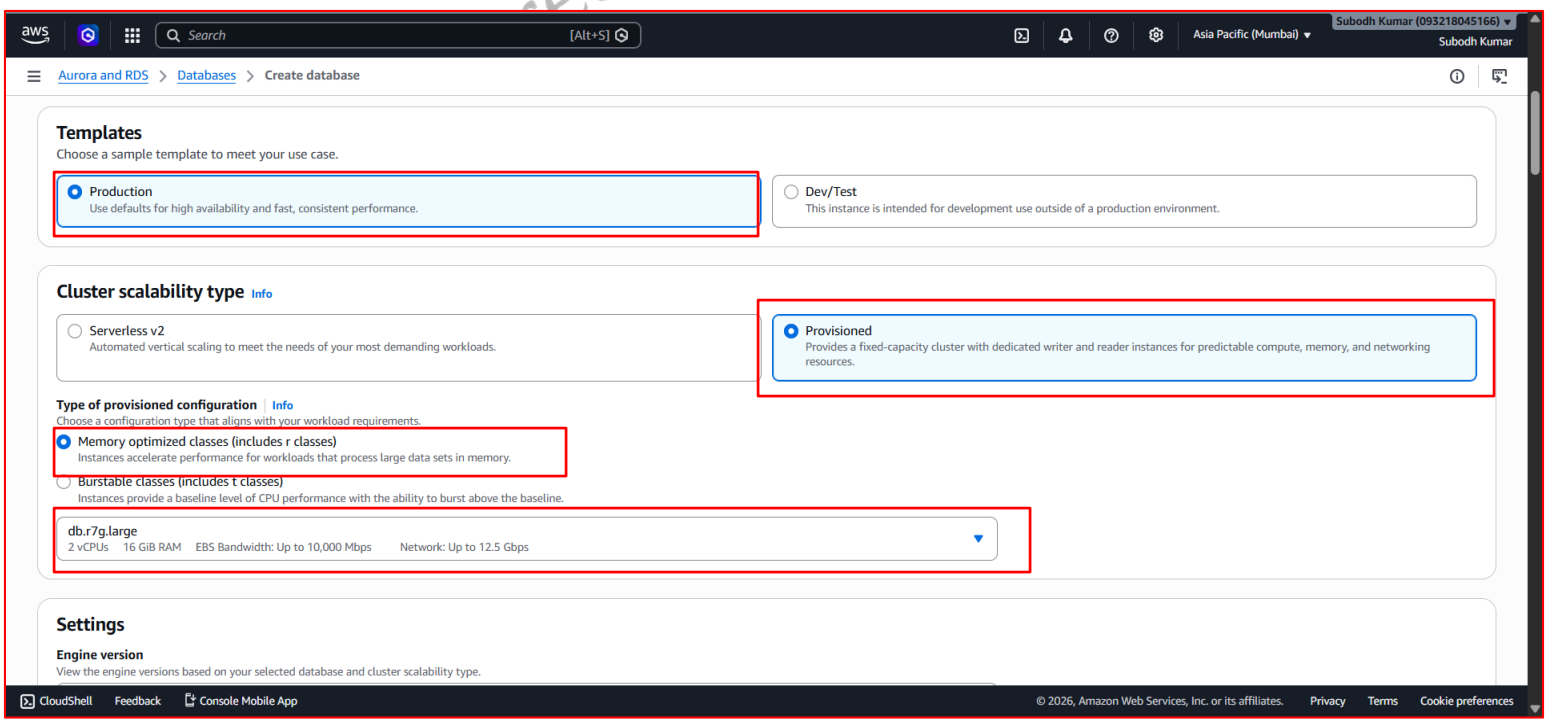
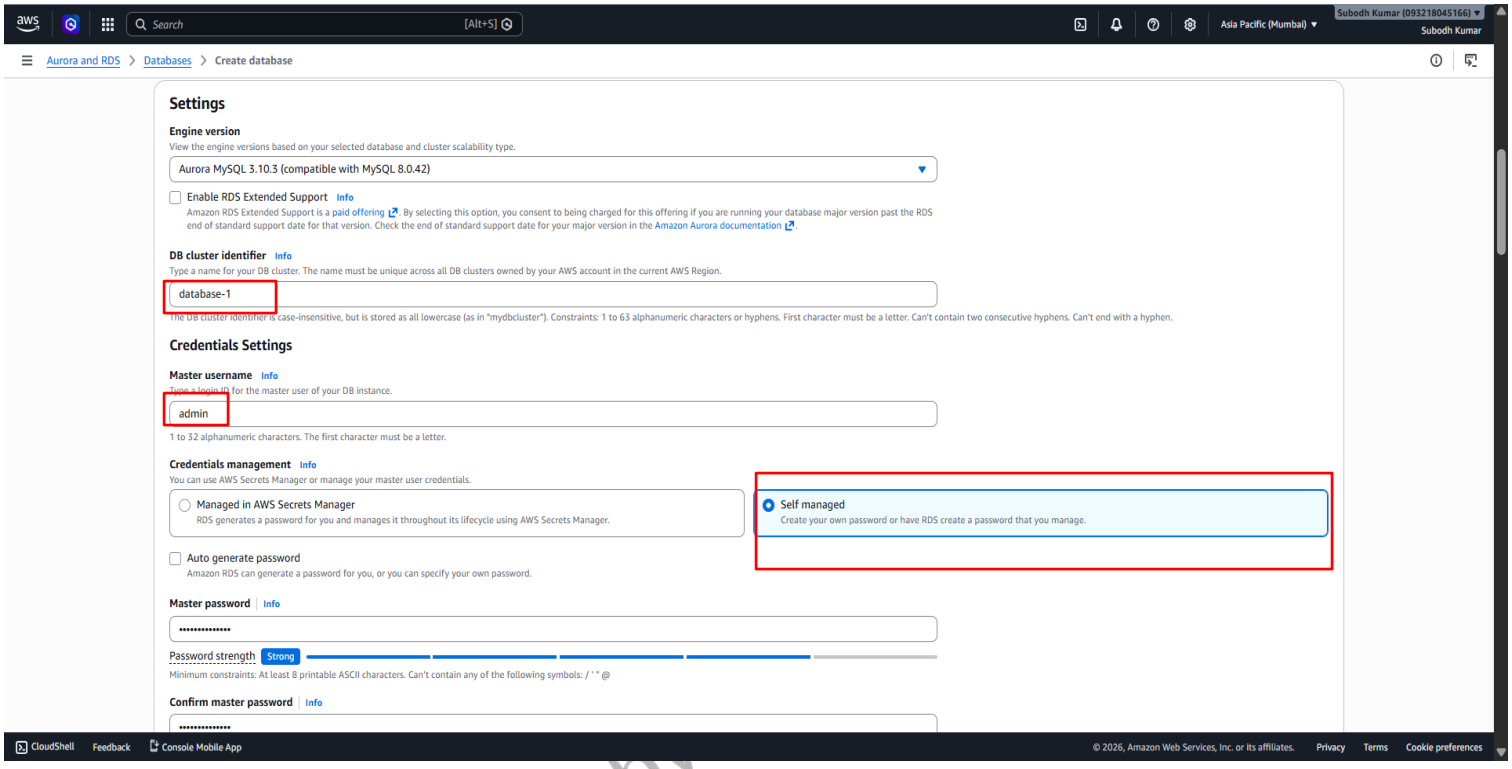
Databases (0)
Filter by databases
DB identifier
Status
Role
Engine
Upgrade rollout order
Region ...
Size
Group resources
Modify
Actions
Create database
Express configuration
Full configuration
Restore from S3
No resources
No resources to display
Create database

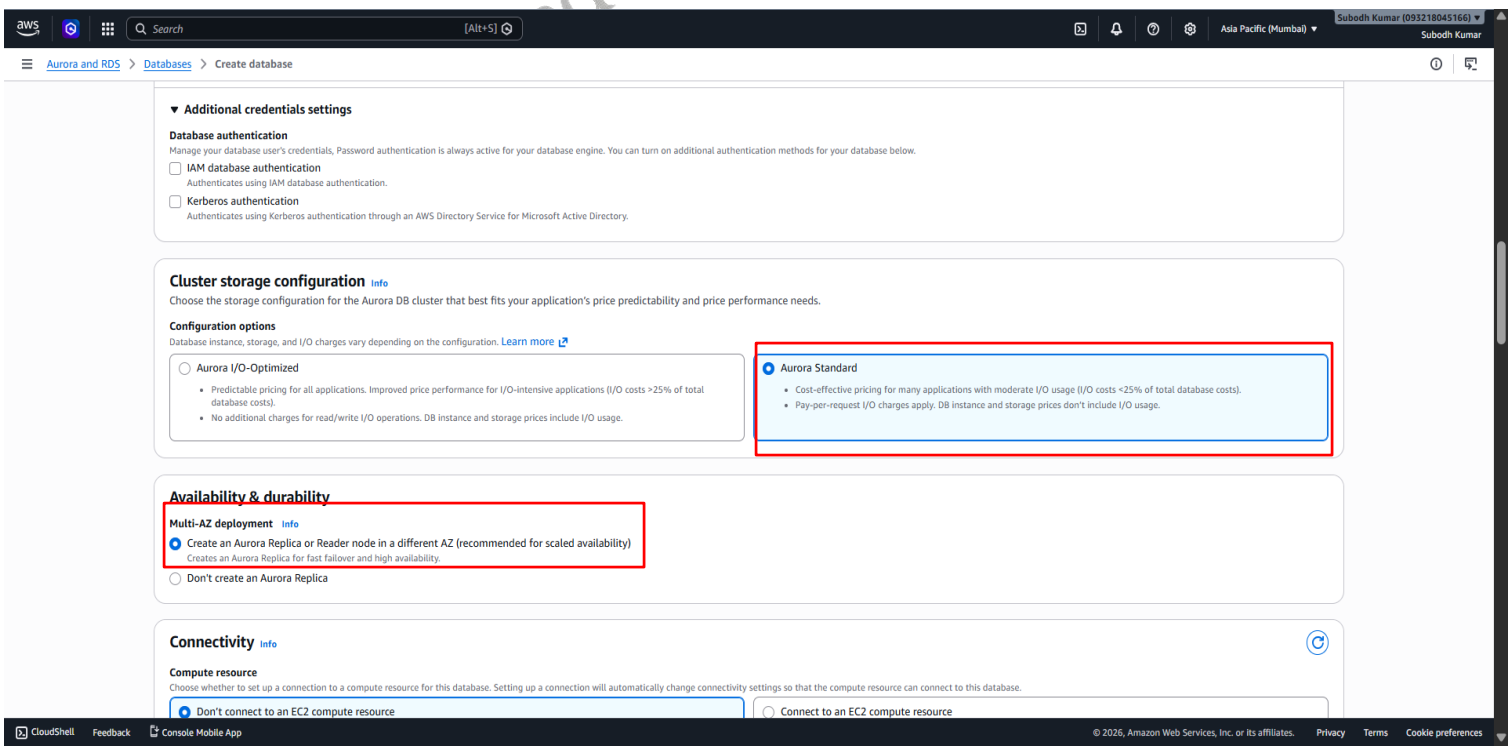

Aurora and RDS > Databases > Create database
Create database Info
Engine options
Engine type Info
Aurora (MySQL Compatible)
Aurora (PostgreSQL Compatible)
MySQL
PostgreSQL
MariaDB
Oracle
Microsoft SQL Server
IBM Db2
Choose a database creation method
Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.
Templates

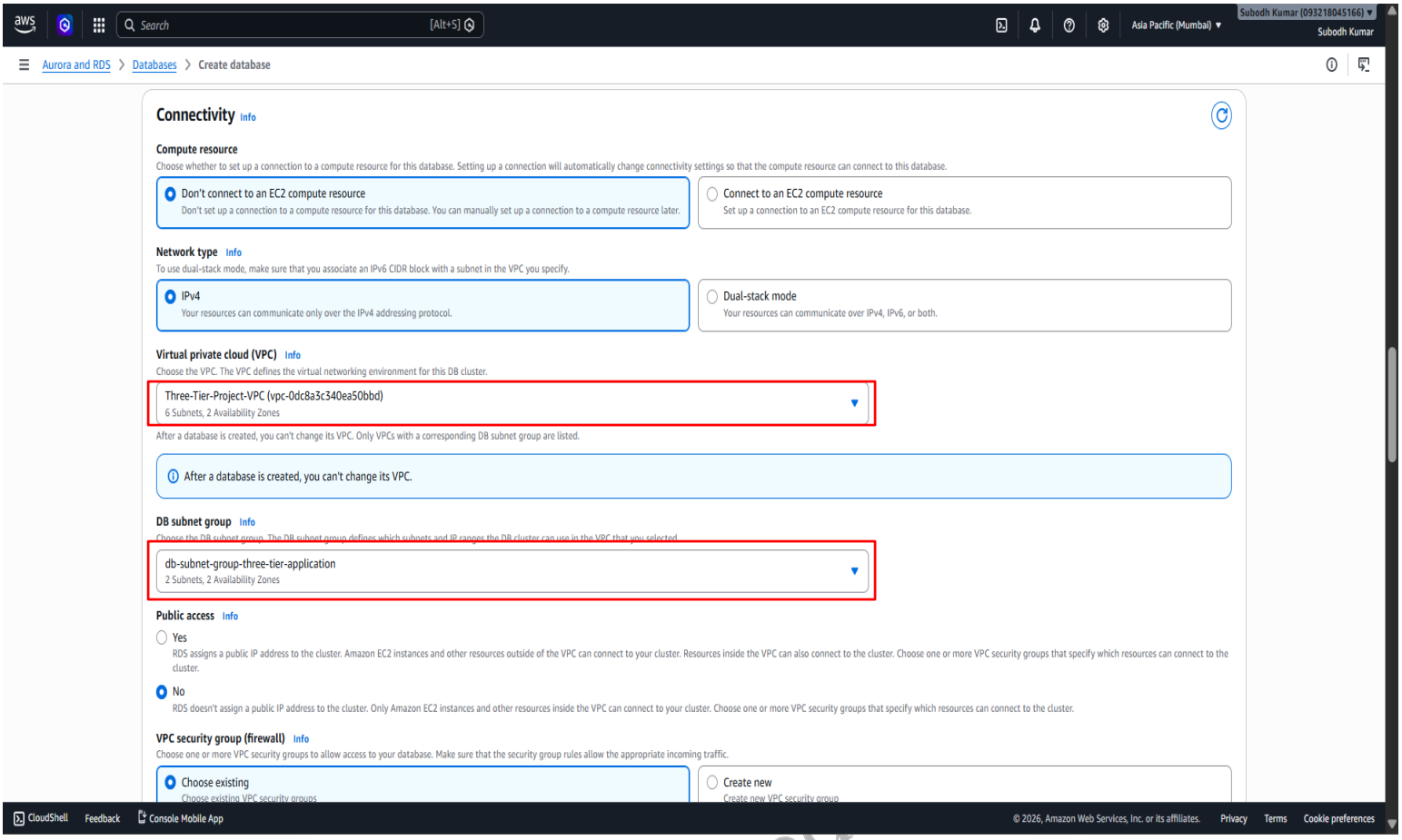

Aurora and RDS > Databases > Create database
Templates
Choose a sample template to meet your use case.
Production
Use defaults for high availability and fast, consistent performance.
Dev/Test
This instance is intended for development use outside of a production environment.
Cluster scalability type Info
Serverless v2
Automated vertical scaling to meet the needs of your most demanding workloads.
Provisioned
Provides a fixed-capacity cluster with dedicated writer and reader instances for predictable compute, memory, and networking resources.
Type of provisioned configuration Info
Choose a configuration type that aligns with your workload requirements.
Memory optimized classes (includes r classes)
Instances accelerate performance for workloads that process large data sets in memory.
Burstable classes (includes t classes)
Instances provide a baseline level of CPU performance with the ability to burst above the baseline.
db.r7g.large
2 vCPUs 16 GiB RAM EBS Bandwidth: Up to 10,000 Mbps Network: Up to 12.5 Gbps
Settings
Engine version
View the engine versions based on your selected database and cluster scalability type.

In the following step remember the username and password. We require it when **we access** the **DB Server** in **upcoming steps.**

(In my case: username: admin, Password: subodhdatabase)






Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Aurora and RDS > Databases > Create database
Connectivity Info
Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.
Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.
Network type Info
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.
IPv4
Your resources can communicate only over the IPv4 addressing protocol.
Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.
Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB cluster.
Three-Tier-Project-VPC (vpc-0dc8a3c340ea50bbd)
6 Subnets, 2 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.
DB subnet group Info
db-subnet-group-three-tier-application
2 Subnets, 2 Availability Zones
Public access Info
Yes
RDS assigns a public IP address to the cluster. Amazon EC2 instances and other resources outside of the VPC can connect to your cluster. Resources inside the VPC can also connect to the cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.
No
RDS doesn't assign a public IP address to the cluster. Only Amazon EC2 instances and other resources inside the VPC can connect to your cluster. Choose one or more VPC security groups that specify which resources can connect to the cluster.
VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group
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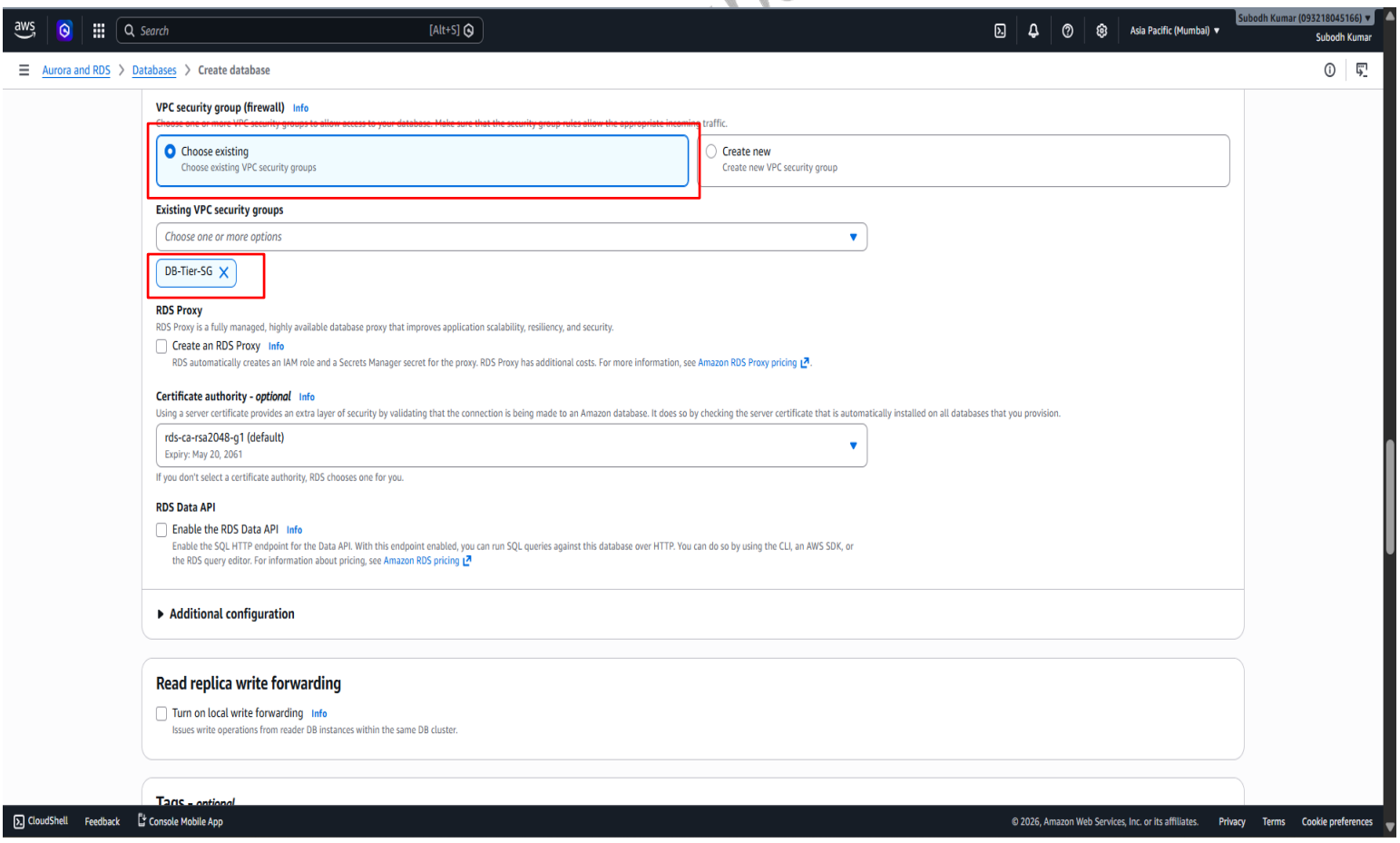
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Aurora and RDS > Databases > Create database
VPC security group (firewall) Info
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group
Existing VPC security groups
Choose one or more options
DB-Tier-SG
RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
Create an RDS Proxy Info
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.
Certificate authority - optional Info
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061
If you don't select a certificate authority, RDS chooses one for you.
RDS Data API
Enable the RDS Data API Info
Enable the SQL HTTP endpoint for the Data API. With this endpoint enabled, you can run SQL queries against this database over HTTP. You can do so by using the CLI, an AWS SDK, or the RDS query editor. For information about pricing, see Amazon RDS pricing
Additional configuration
Read replica write forwarding
Turn on local write forwarding Info
Issues write operations from reader DB instances within the same DB cluster.
Tags - optional
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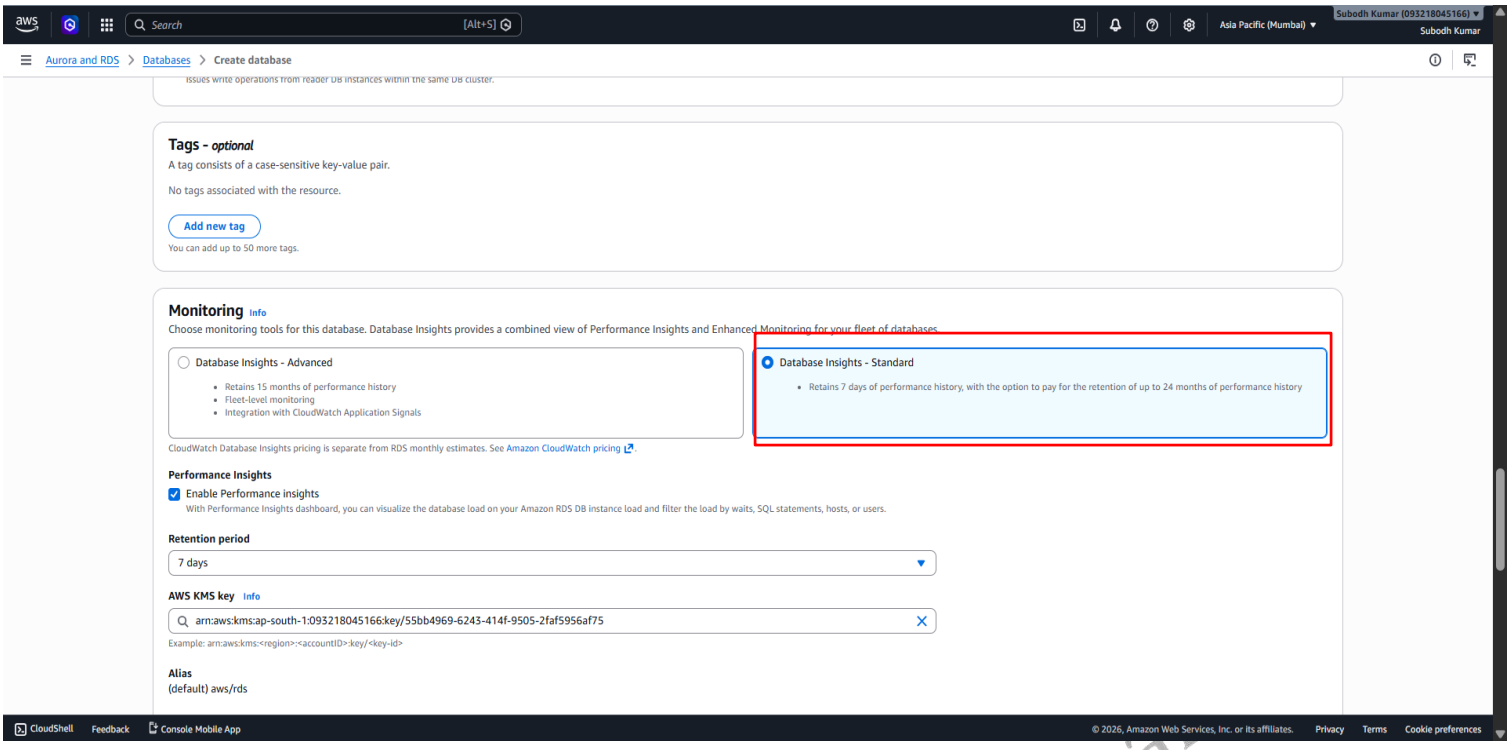
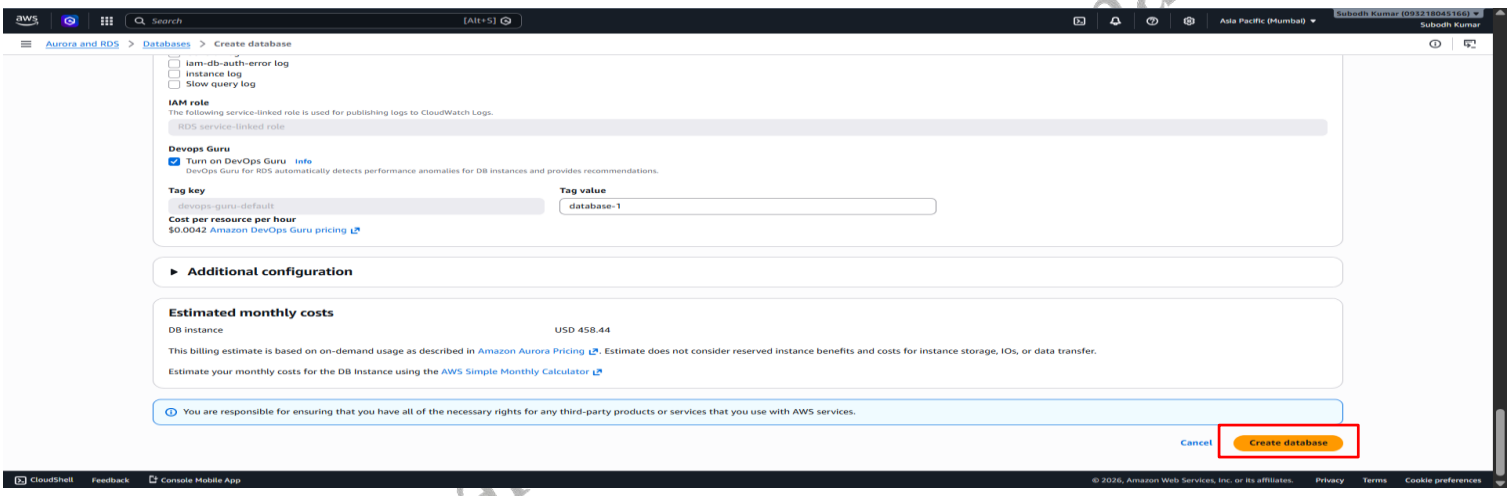
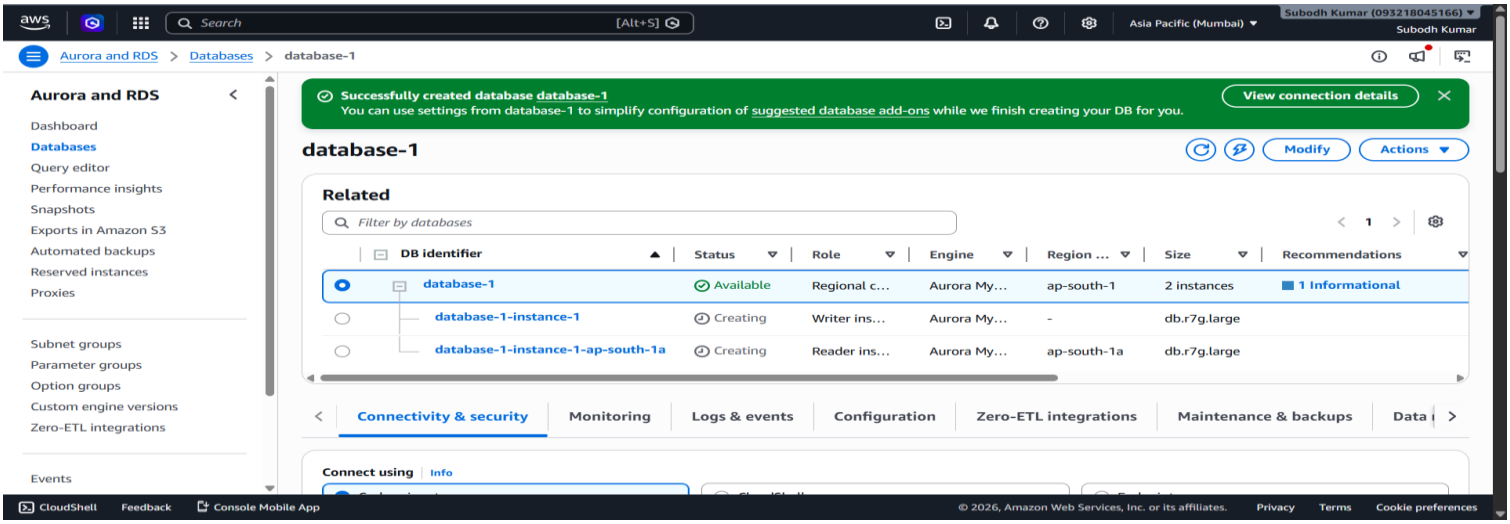




Keep other settings as default. This DB Server creation will cost you some money. But since we delete as soon as project is implemented, it will be few dollars.

Now Click **Create database**. It will take some time to create the server.



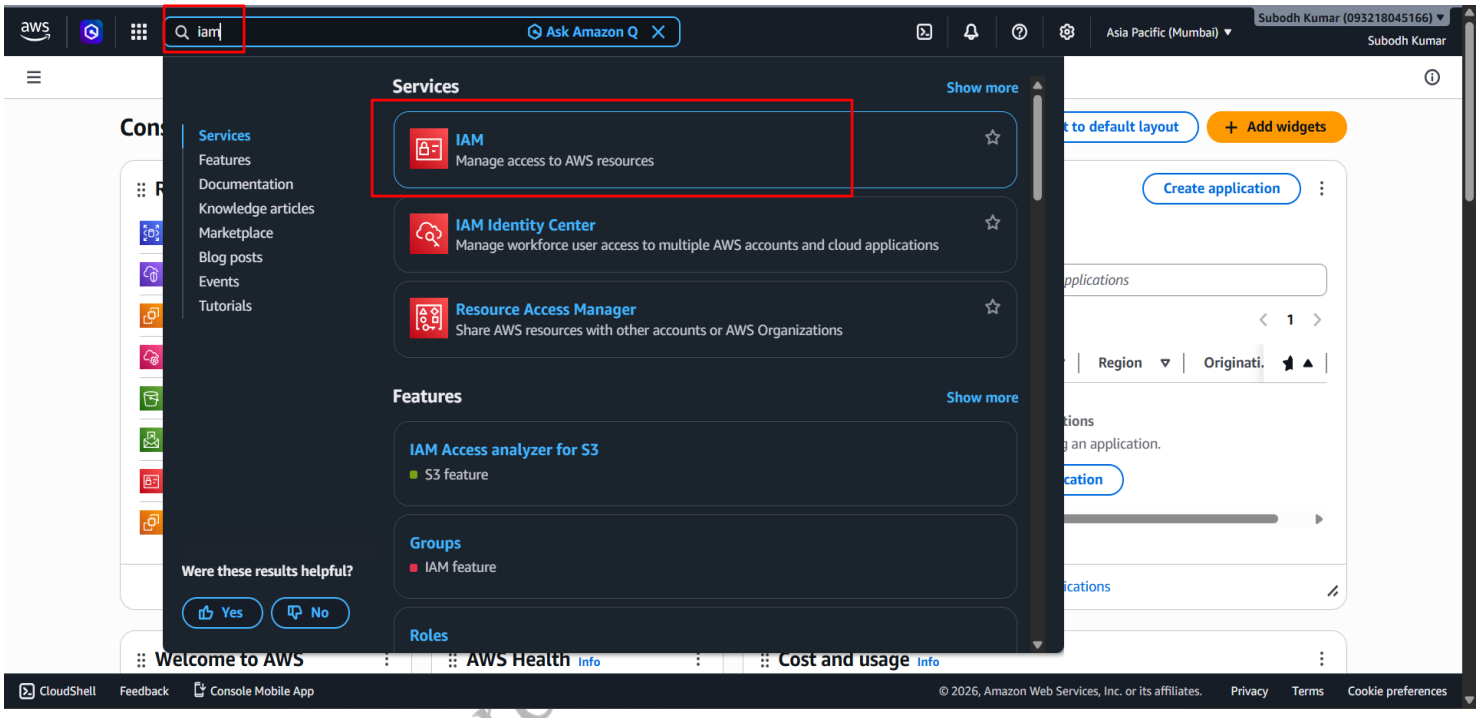
Great we created the database server

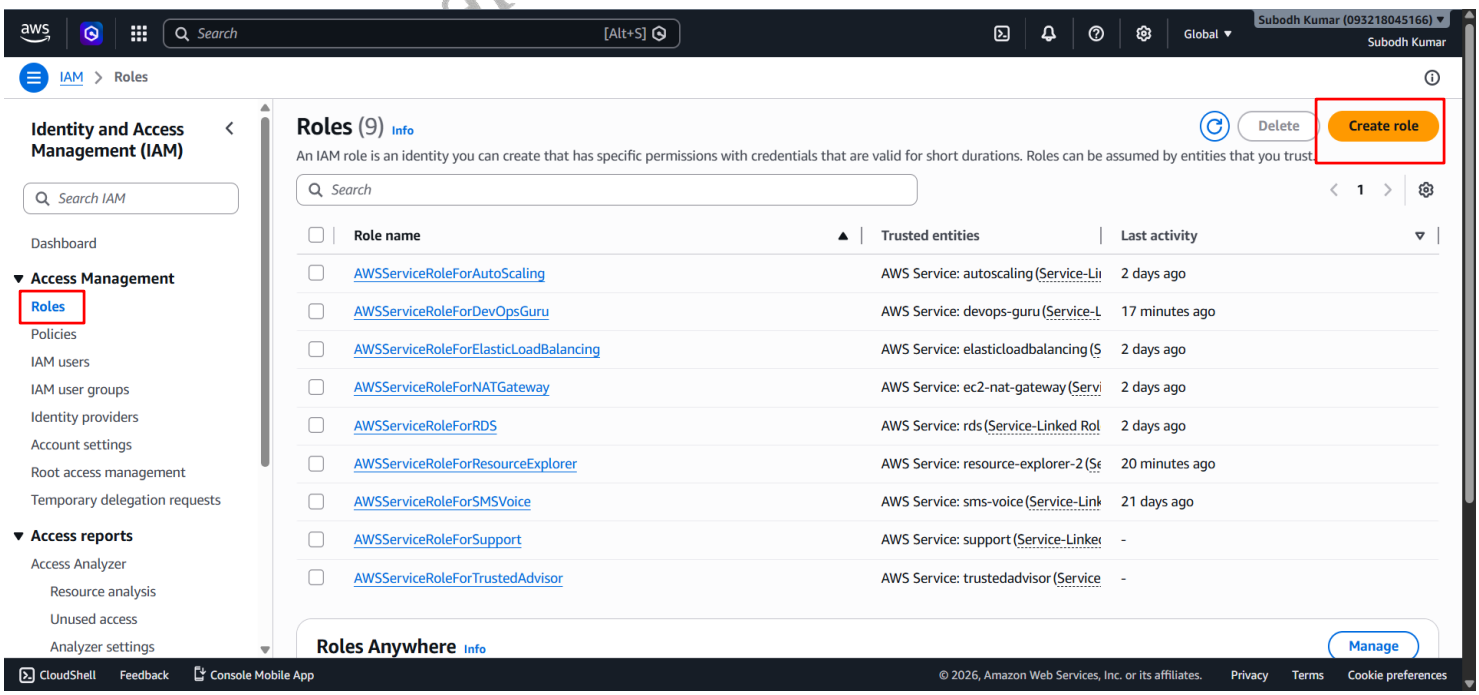
5. Create IAM Role

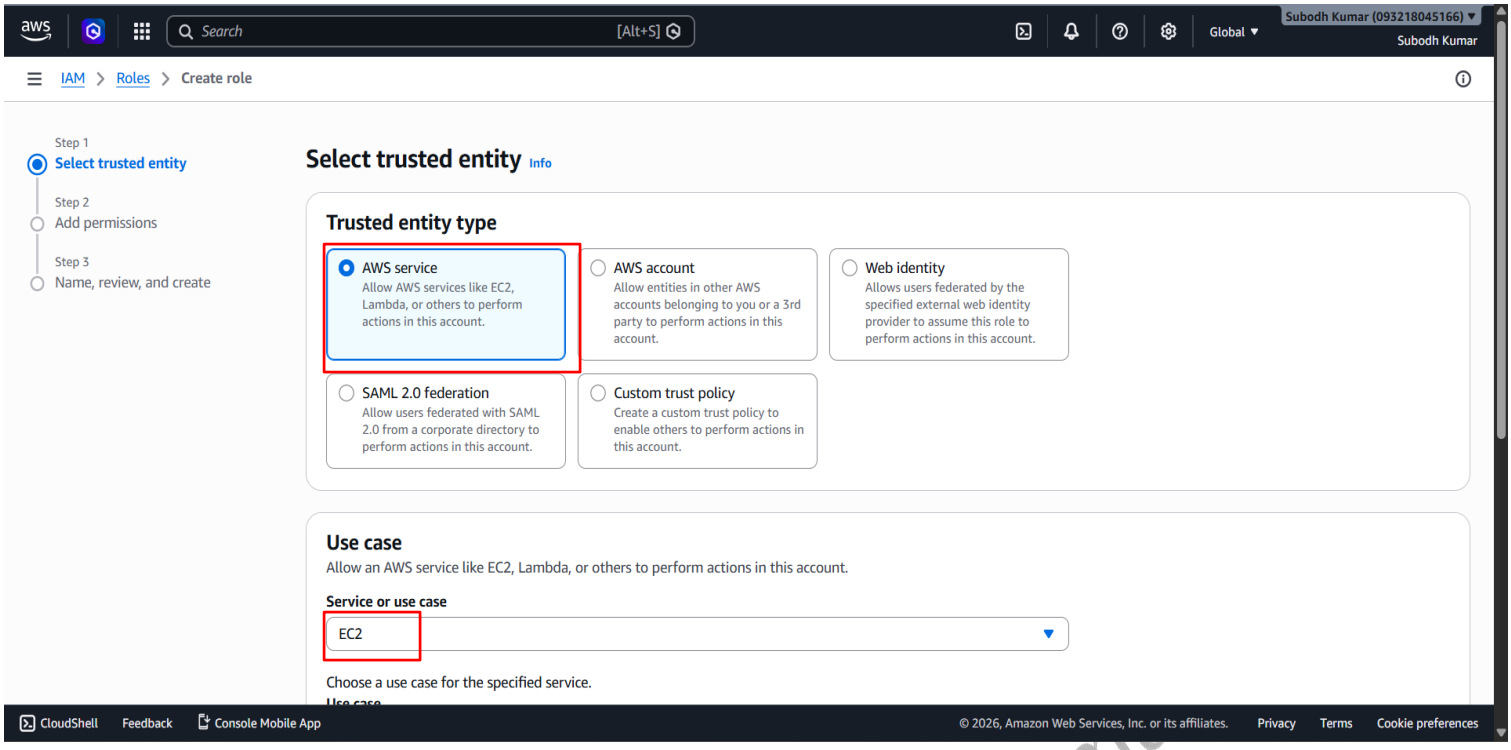
Our application should be Highly secured. In IAM Role we have to use two permission policies in upcoming steps. **S3 read only Access** for access the S3 bucket and **SSM managed instance core** for grants the necessary permissions for an EC2 instance to communicate with the Systems Manager service.

Let's create them

Search for IAM in the search bar. Go to **Roles > Create role**






Search [Alt+S]
Global
Subodh Kumar (093218045166)
Subodh Kumar
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Select trusted entity Info
Trusted entity type
AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.
Service or use case
EC2
Choose a use case for the specified service.
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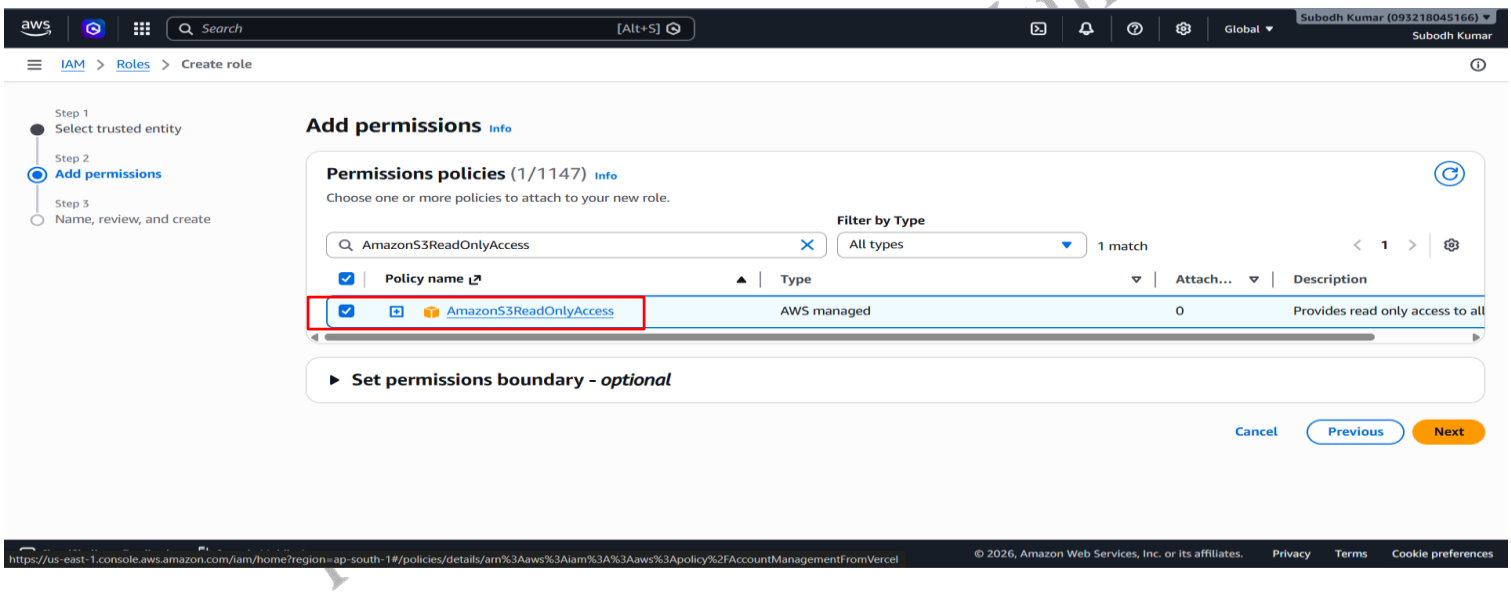
Search [Alt+S]
Global
Subodh Kumar (093218045166)
Subodh Kumar
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Add permissions Info
Permissions policies (1/1147) Info
Choose one or more policies to attach to your new role.
Filter by Type
AmazonS3ReadOnlyAccess
All types
1 match
Policy name
Type
Attach...
Description
AmazonS3ReadOnlyAccess
AWS managed
0
Provides read only access to all
Set permissions boundary - optional
Cancel
Previous
Next
https://us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1#/policies/details/arn%3Aaws%3Aiam%3A%3Aaws%3Apolicy%2FAccountManagementFromVercel
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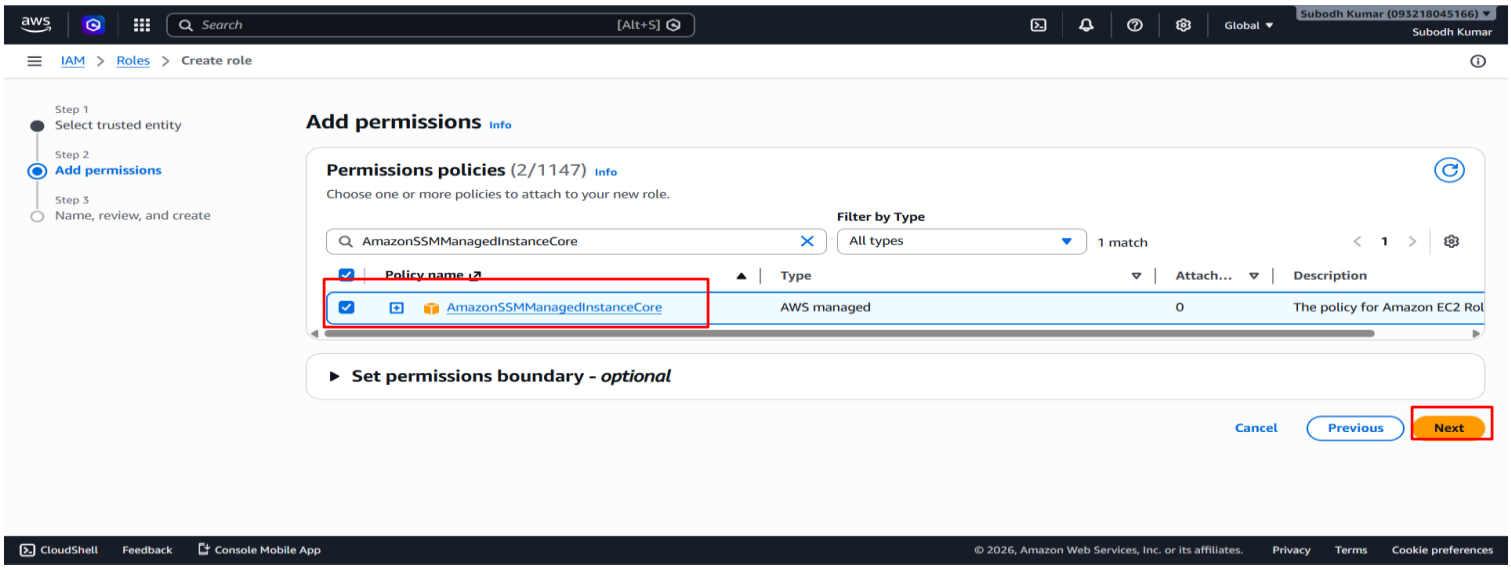
Search [Alt+S]
Global
Subodh Kumar (093218045166)
Subodh Kumar
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Add permissions Info
Permissions policies (2/1147) Info
Choose one or more policies to attach to your new role.
Filter by Type
AmazonSSMManagedInstanceCore
All types
1 match
Policy name
Type
Attach...
Description
AmazonSSMManagedInstanceCore
AWS managed
0
The policy for Amazon EC2 Rol
Set permissions boundary - optional
Cancel
Previous
Next
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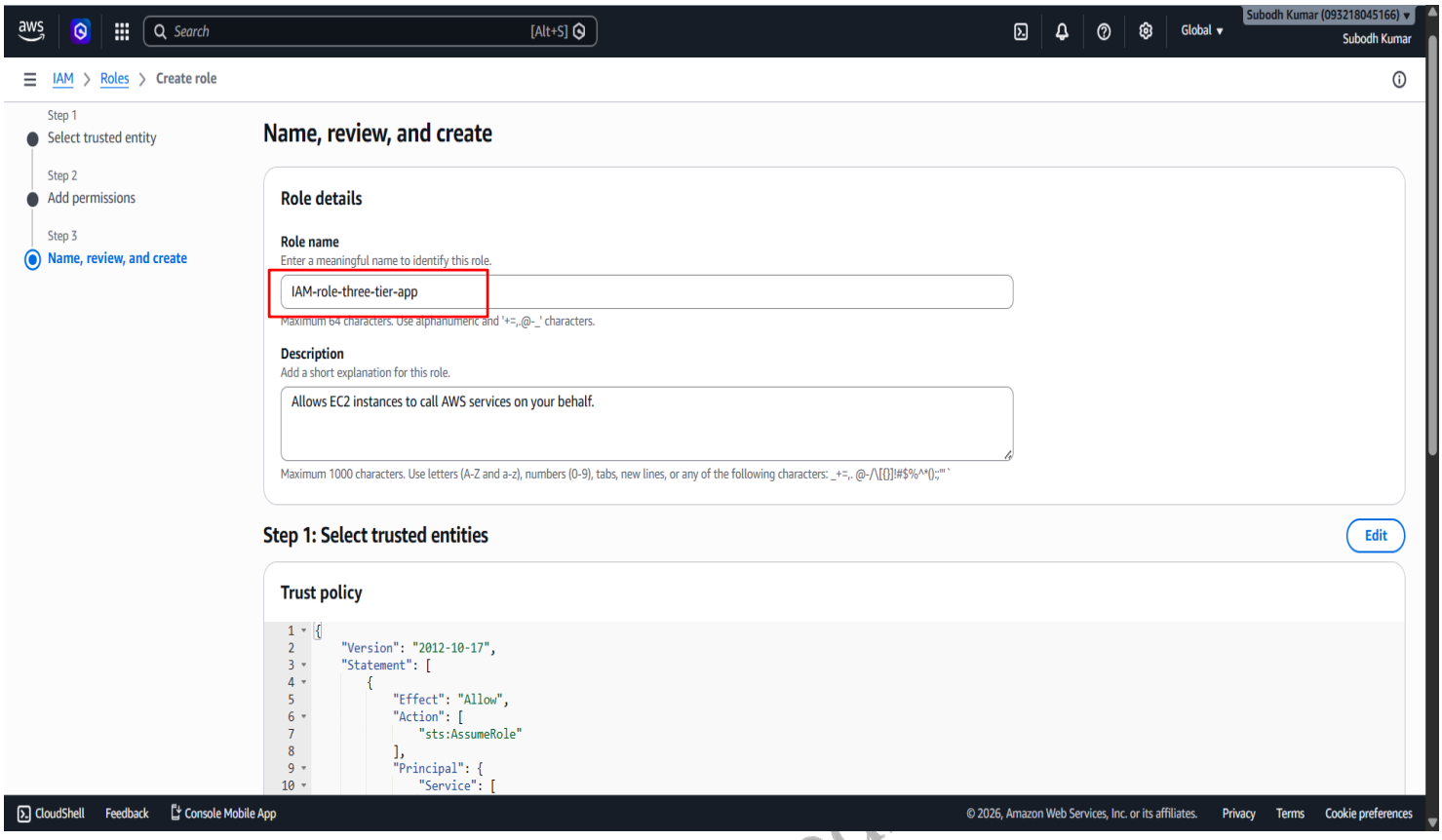
Search [Alt+S]
Global
Subodh Kumar (093218045166)
Subodh Kumar
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Name, review, and create
Role details
Role name
Enter a meaningful name to identify this role.
IAM-role-three-tier-app
Maximum 64 characters. Use alphanumeric and '+=,.@-_' characters.
Description
Add a short explanation for this role.
Allows EC2 instances to call AWS services on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=,. @-/\[{}]!#$%^*();"'`
Step 1: Select trusted entities
Edit
Trust policy
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": [
"sts:AssumeRole"
],
"Principal": {
"Service": [
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Feedback
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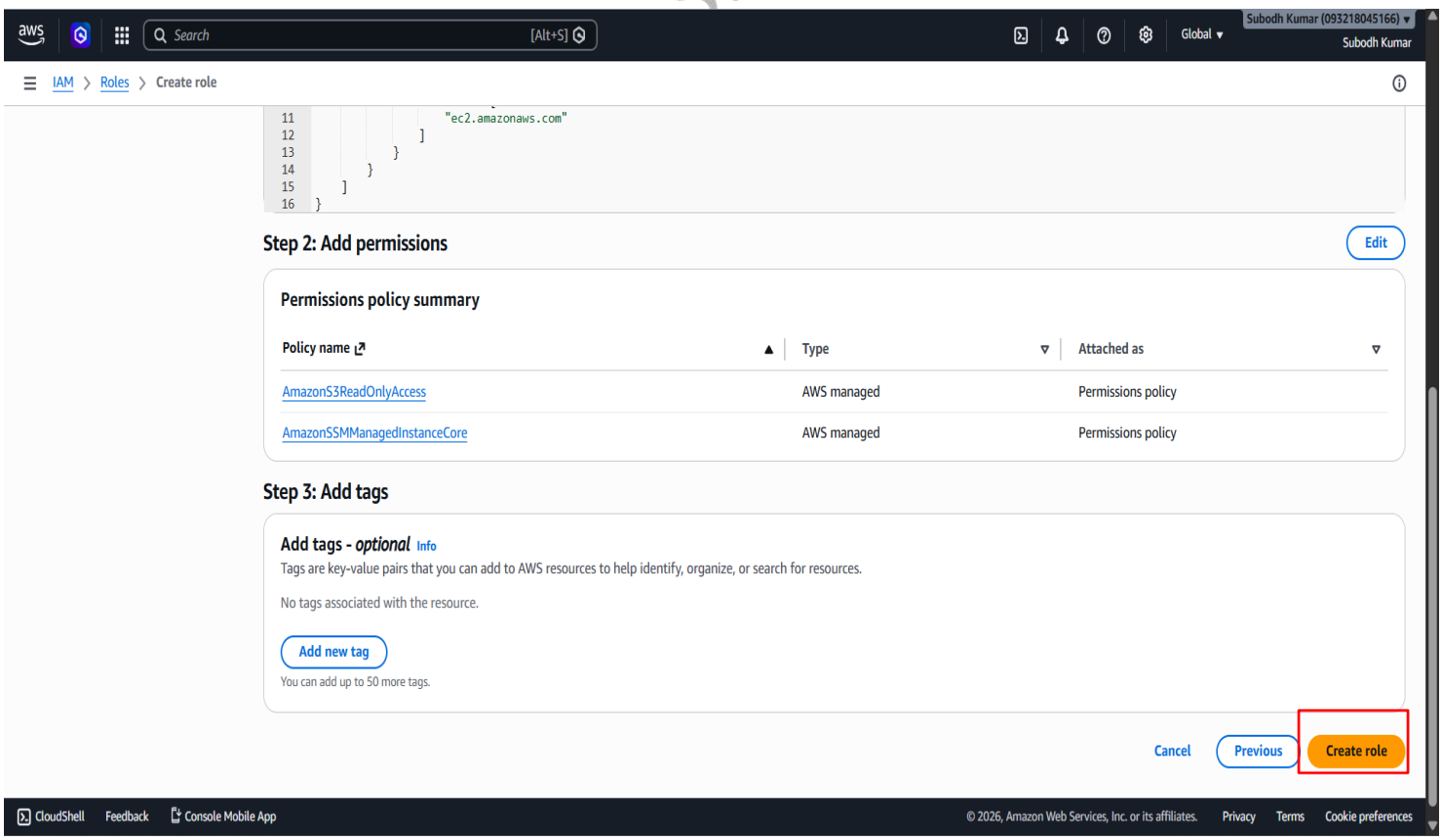
Search [Alt+S]
Global
Subodh Kumar (093218045166)
Subodh Kumar
IAM > Roles > Create role
"ec2.amazonaws.com"
]
}
}
]
}
Step 2: Add permissions
Edit
Permissions policy summary
Policy name
Type
Attached as
AmazonS3ReadOnlyAccess
AWS managed
Permissions policy
AmazonSSMManagedInstanceCore
AWS managed
Permissions policy
Step 3: Add tags
Add tags - optional Info
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Previous
Create role
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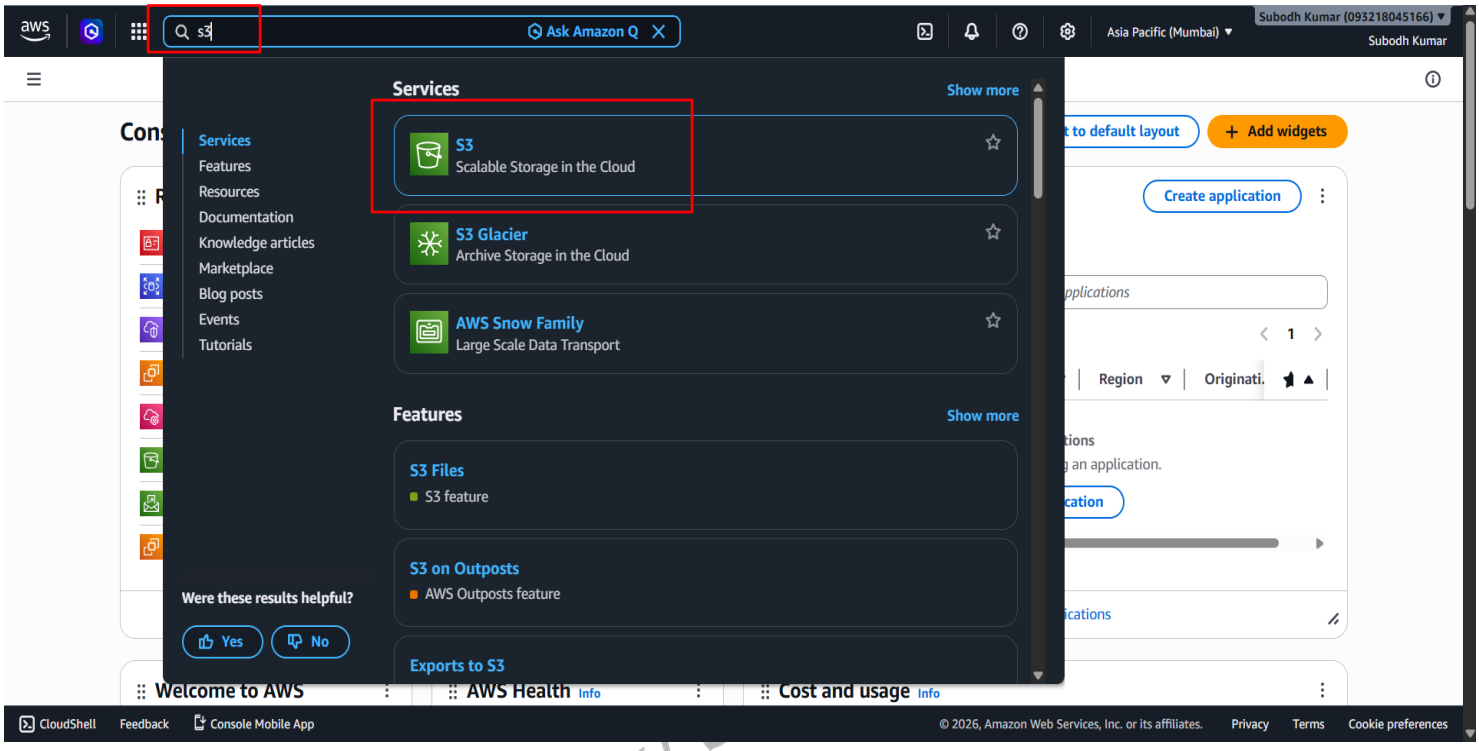
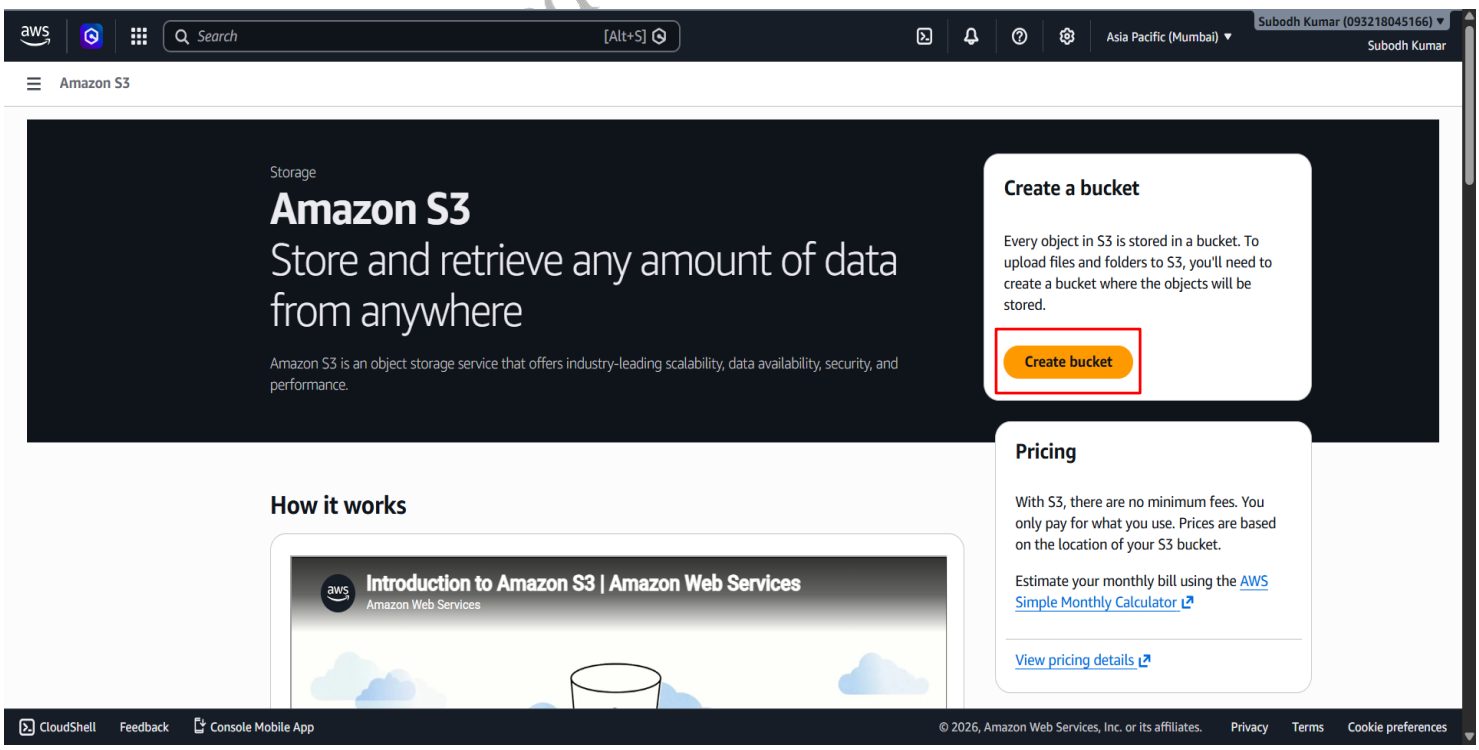
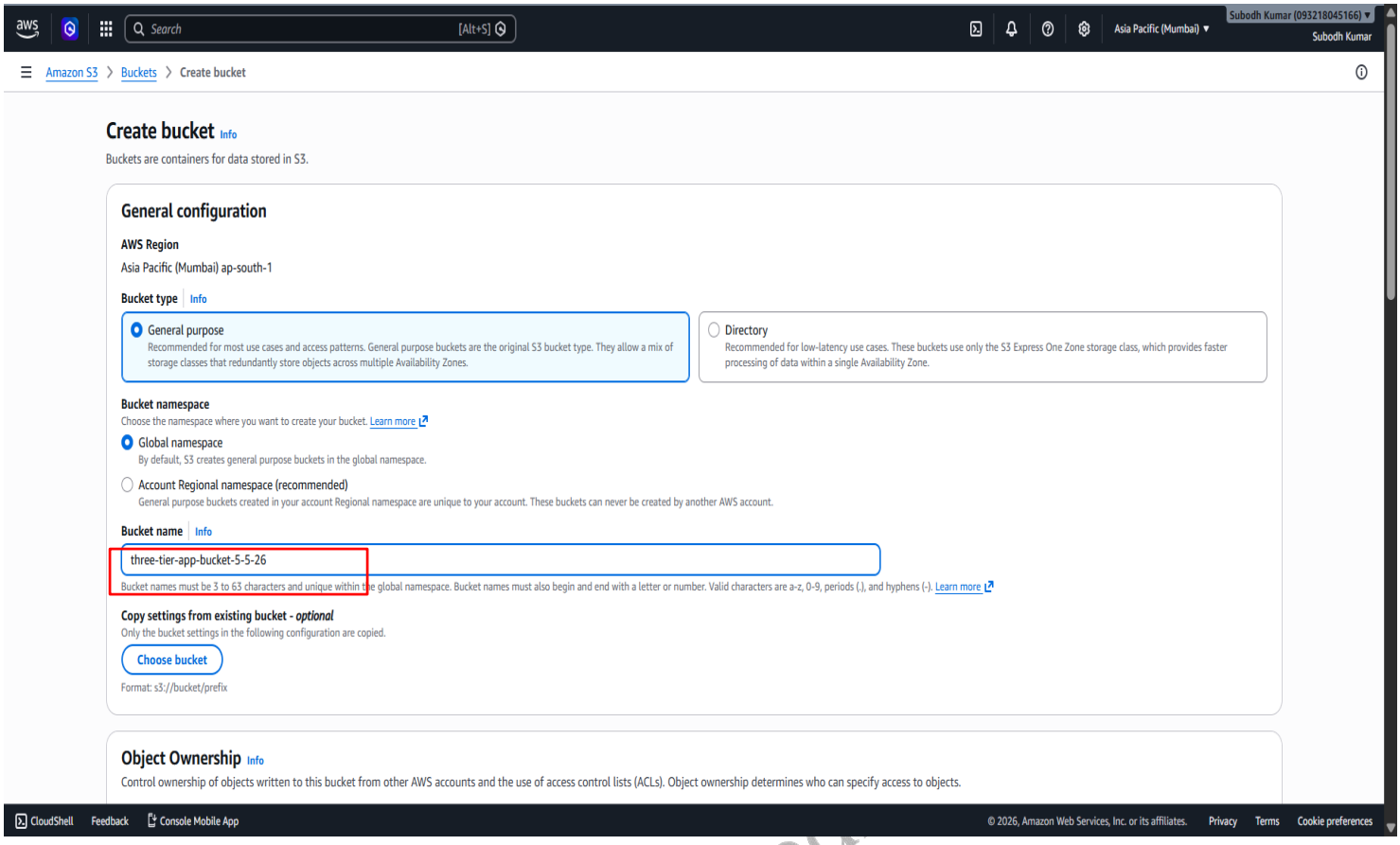
6. Create a S3 Bucket and upload the Source code

When we setup application on servers it is easy to if the source codes are in a s3 bucket

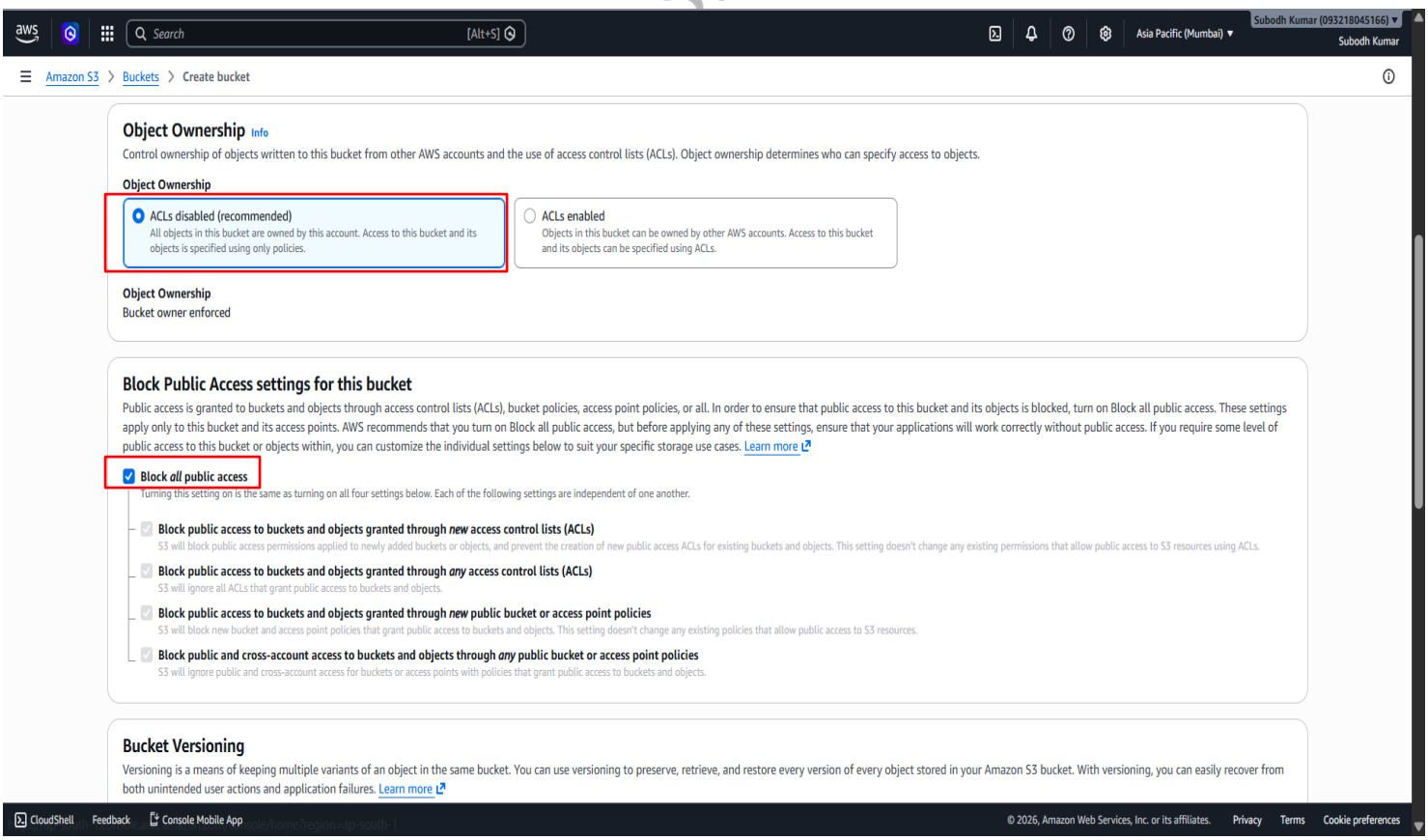
Go to **search bar** and search for **S3**. Then click **create bucket.** Bucket name should be unique one.

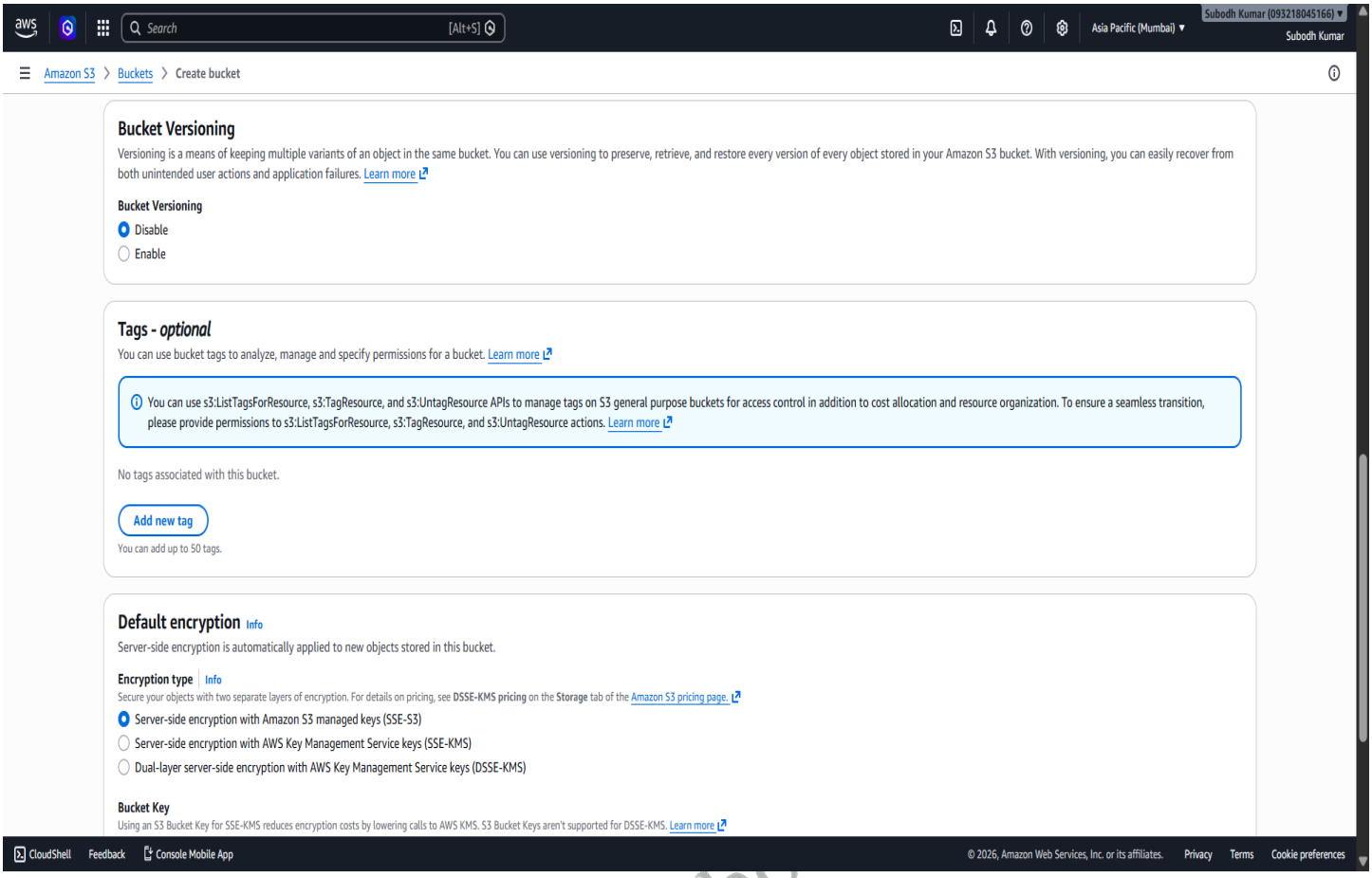






Let all other options default and click create bucket.




Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Amazon S3 > Buckets > Create bucket
Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. Learn more
Bucket Versioning
Disable
Enable
Tags - optional
You can use bucket tags to analyze, manage and specify permissions for a bucket. Learn more
You can use s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets for access control in addition to cost allocation and resource organization. To ensure a seamless transition, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. Learn more
No tags associated with this bucket.
Add new tag
You can add up to 50 tags.
Default encryption Info
Server-side encryption is automatically applied to new objects stored in this bucket.
Encryption type Info
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.
Server-side encryption with Amazon S3 managed keys (SSE-S3)
Server-side encryption with AWS Key Management Service keys (SSE-KMS)
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more
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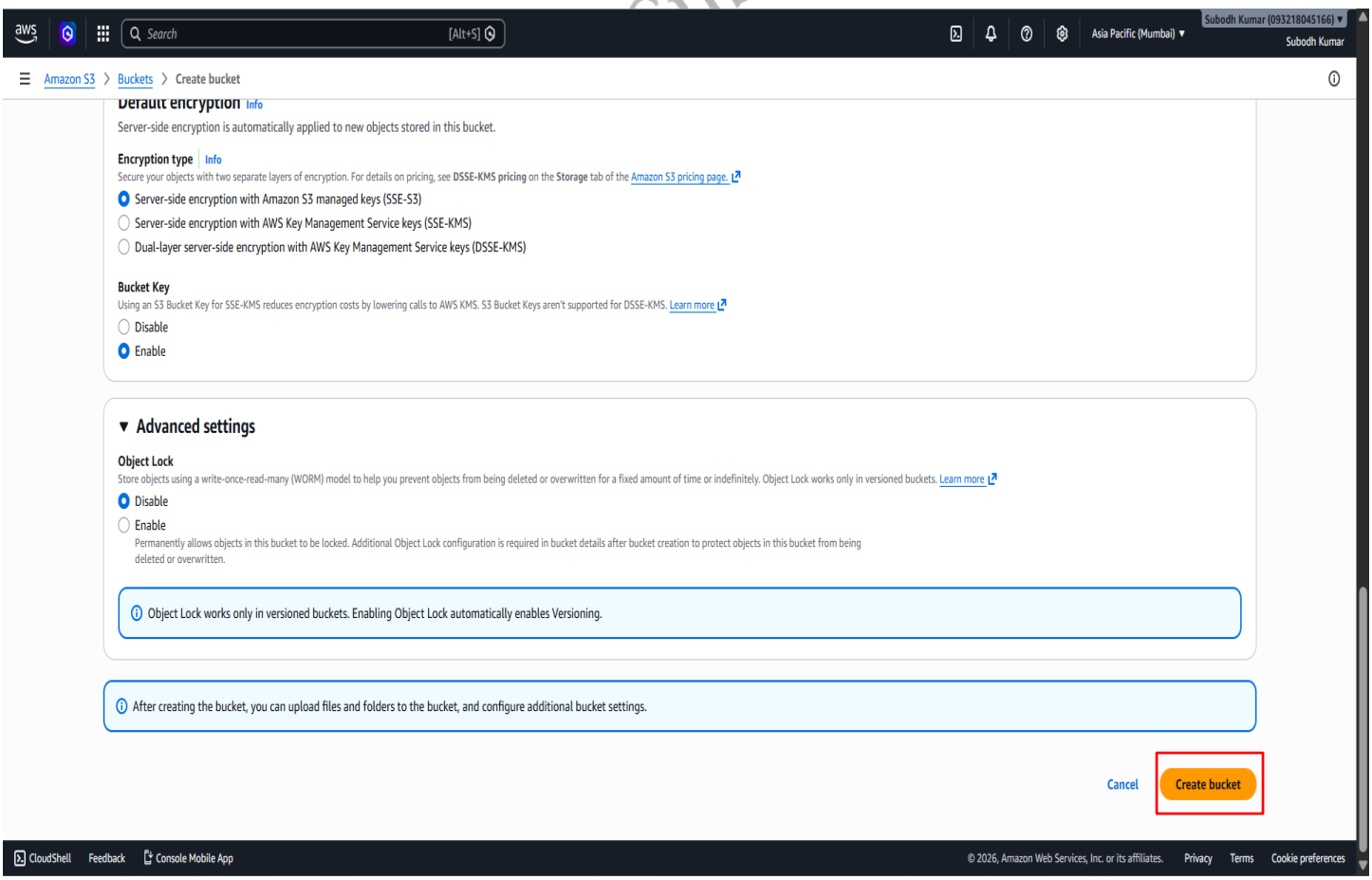
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Amazon S3 > Buckets > Create bucket
Default encryption Info
Server-side encryption is automatically applied to new objects stored in this bucket.
Encryption type Info
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.
Server-side encryption with Amazon S3 managed keys (SSE-S3)
Server-side encryption with AWS Key Management Service keys (SSE-KMS)
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more
Disable
Enable
▼ Advanced settings
Object Lock
Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. Learn more
Disable
Enable
Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.
Object Lock works only in versioned buckets. Enabling Object Lock automatically enables Versioning.
After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.
Cancel
Create bucket
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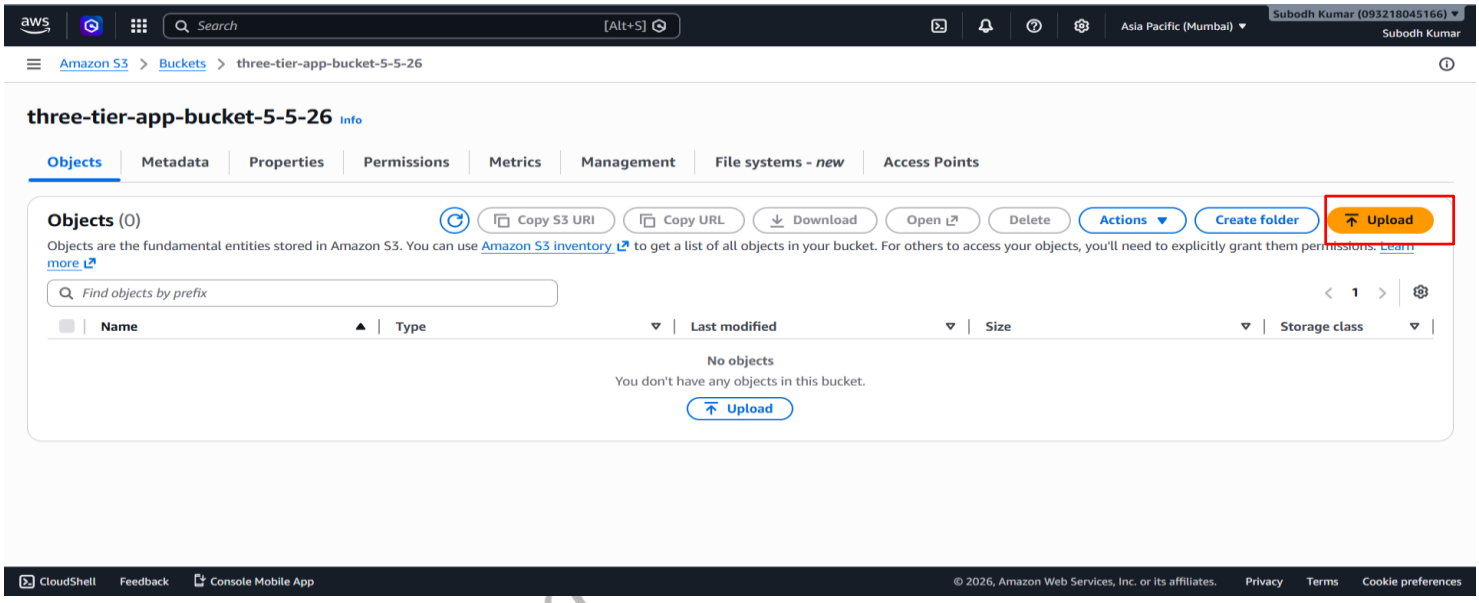
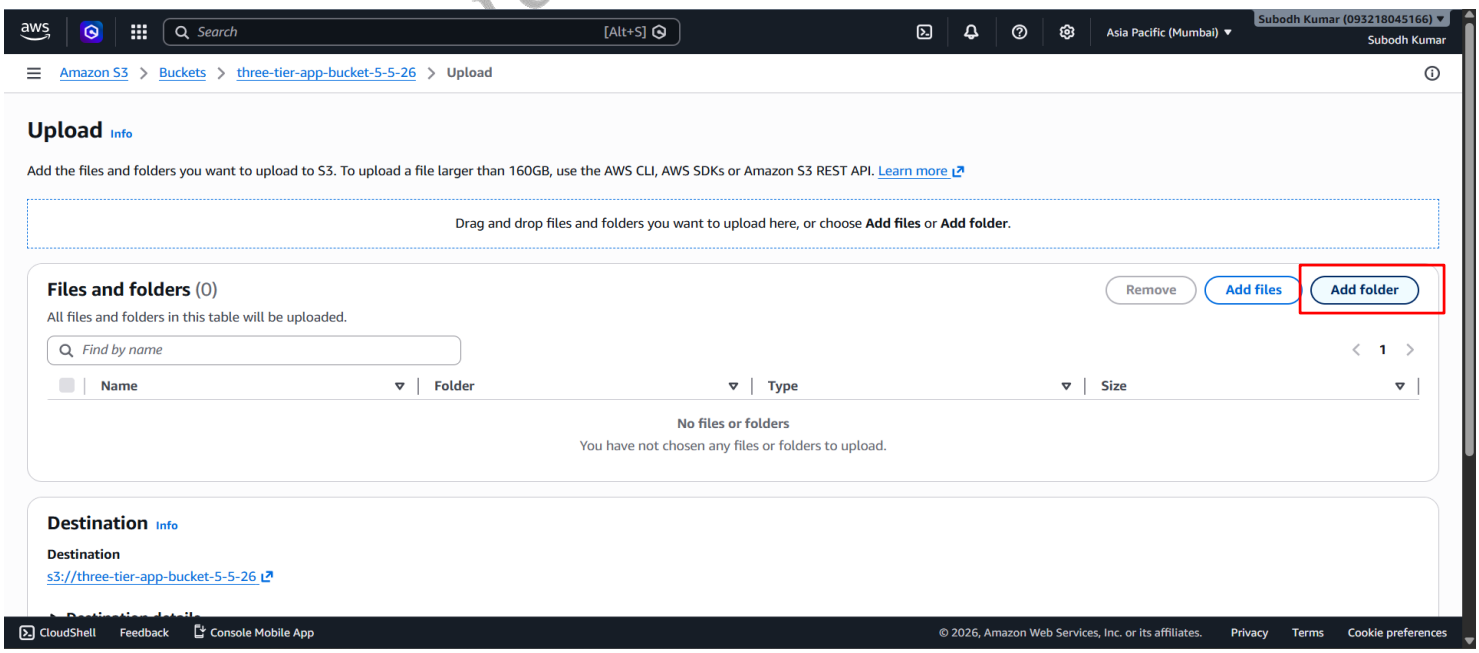
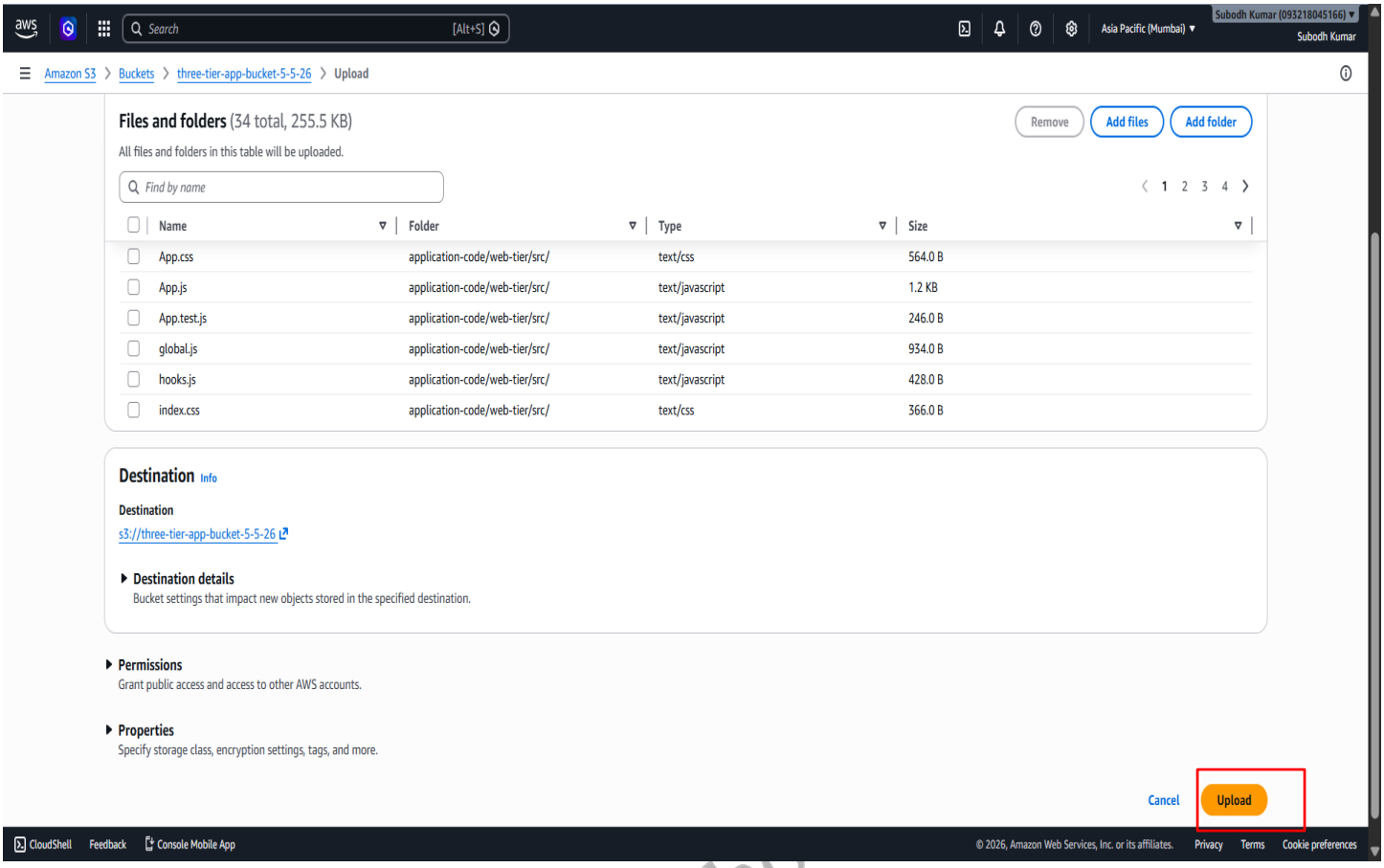
Now go to the following repository and clone it in your pc. It contains all source code.

https://github.com/SubodhK143/three-tier-app-on-aws.git

Now go to the bucket and upload the application-code folder. There are some changes to make in the files. We will do it in the upcoming steps.






Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Amazon S3 > Buckets > three-tier-app-bucket-5-5-26 > Upload
Files and folders (34 total, 255.5 KB)
All files and folders in this table will be uploaded.
Remove
Add files
Add folder
Find by name
1 2 3 4
Name | Folder | Type | Size
App.css | application-code/web-tier/src/ | text/css | 564.0 B
App.js | application-code/web-tier/src/ | text/javascript | 1.2 KB
App.test.js | application-code/web-tier/src/ | text/javascript | 246.0 B
global.js | application-code/web-tier/src/ | text/javascript | 934.0 B
hooks.js | application-code/web-tier/src/ | text/javascript | 428.0 B
index.css | application-code/web-tier/src/ | text/css | 366.0 B
Destination Info
Destination
s3://three-tier-app-bucket-5-5-26
Destination details
Bucket settings that impact new objects stored in the specified destination.
Permissions
Grant public access and access to other AWS accounts.
Properties
Specify storage class, encryption settings, tags, and more.
Cancel
Upload
CloudShell
Feedback
Console Mobile App
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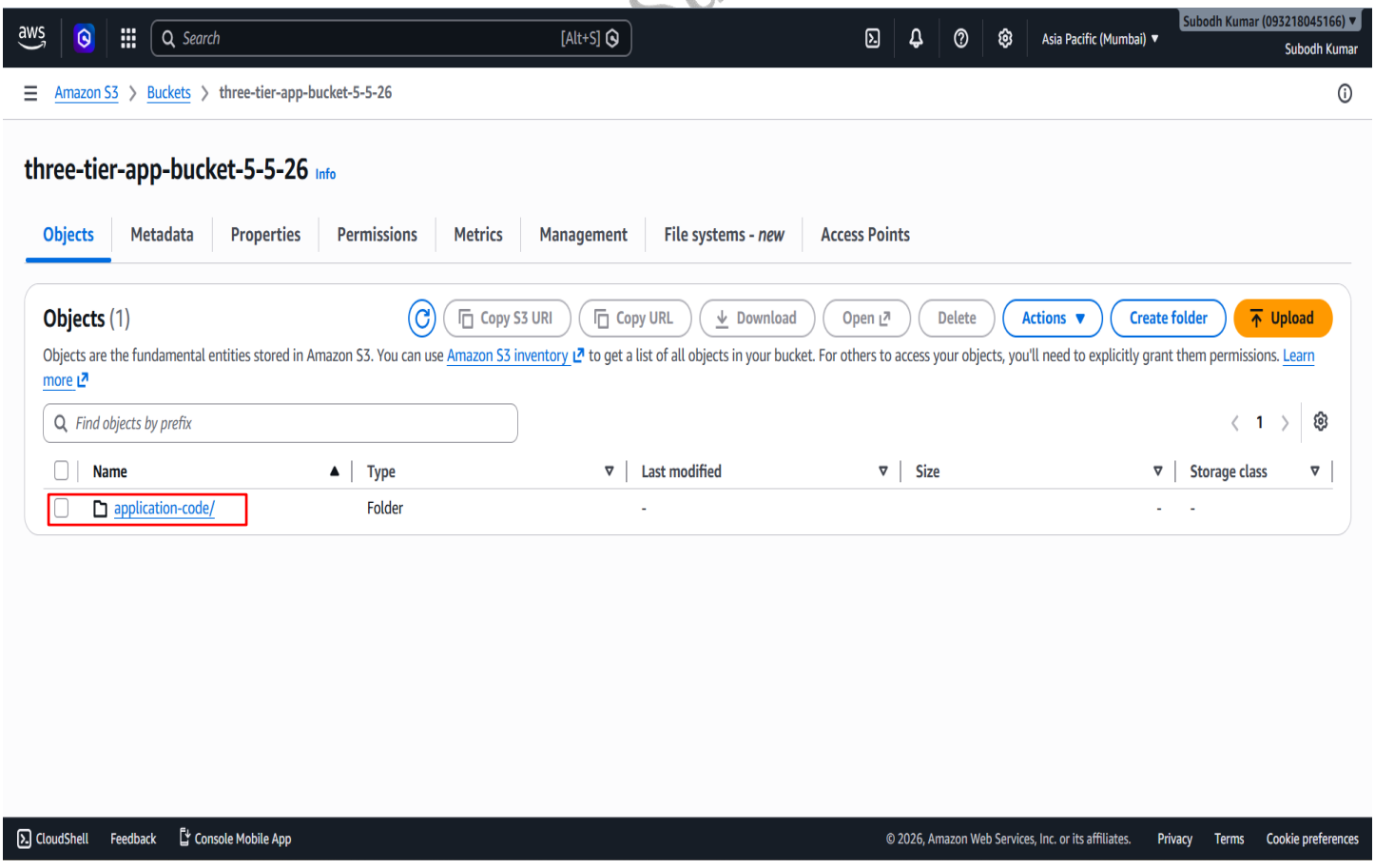
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
Amazon S3 > Buckets > three-tier-app-bucket-5-5-26
three-tier-app-bucket-5-5-26 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
1
Name | Type | Last modified | Size | Storage class
application-code/ | Folder | - | - | -
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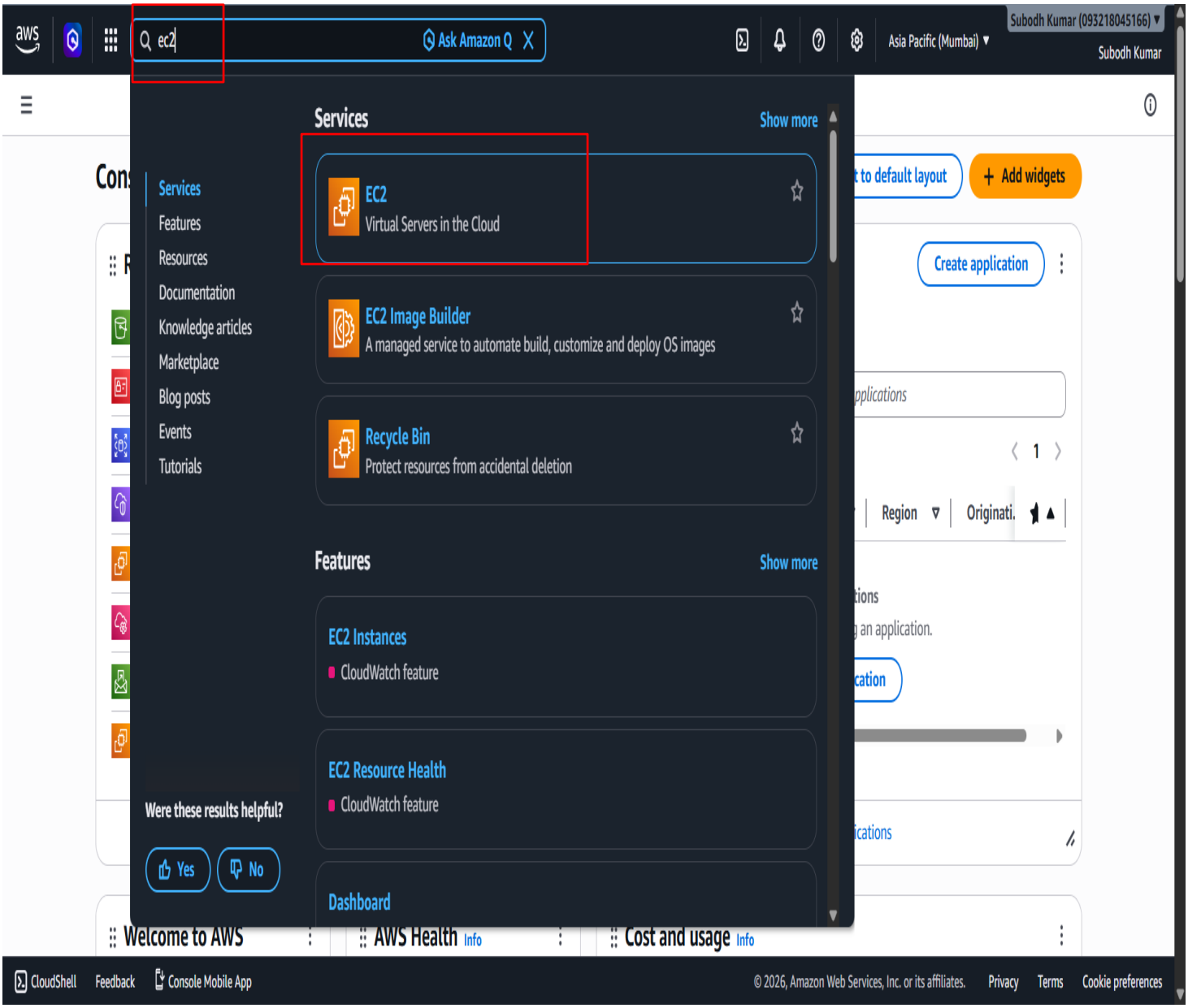
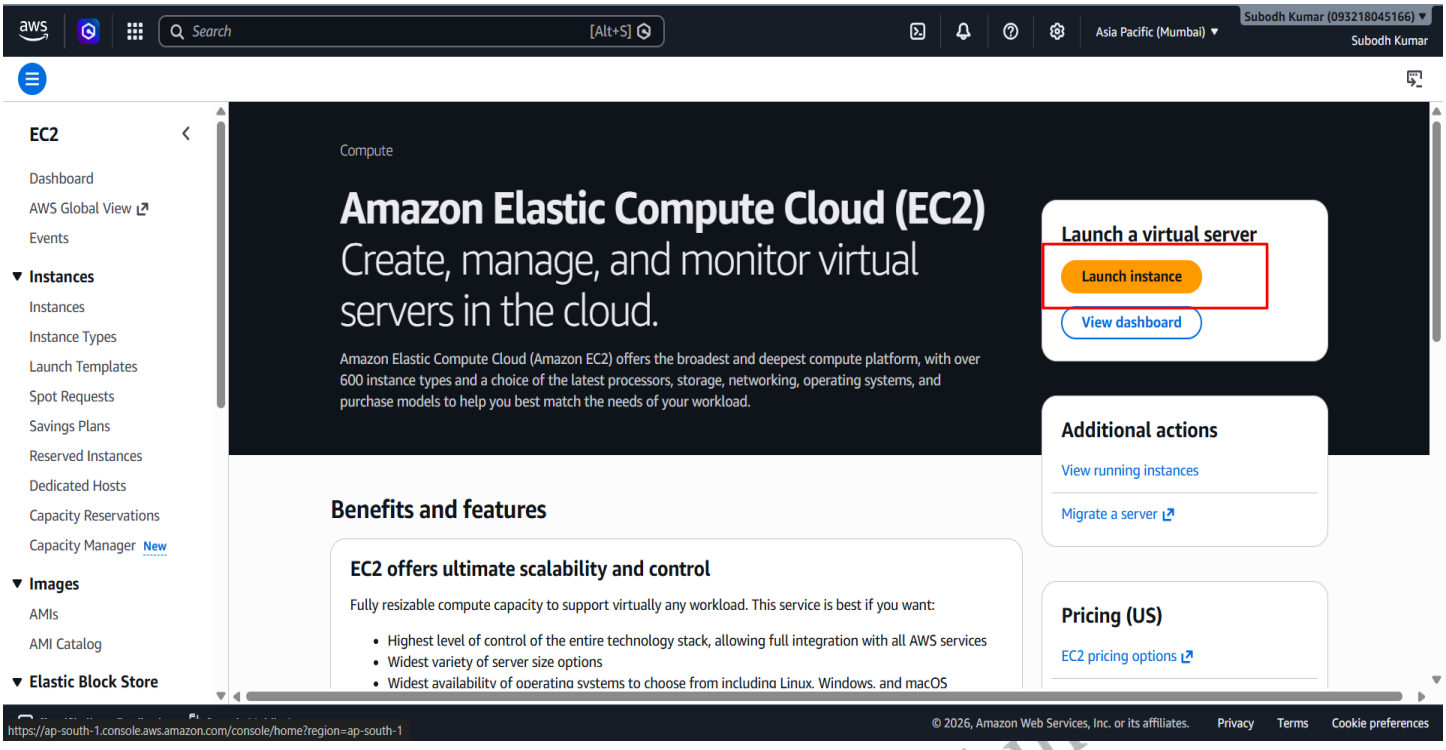
Our application should be highly scalable. For achieve it we have to use Auto Scaling Groups. For automatically creating servers and it configurations we use Launch templates so we need Amazon Machine Images (AMI) of our servers. For that we are going to create a test server using EC2 instance and configure application in that and then create an AMI using it. In this way we implement our Web Servers and App Servers.

Let's do

7. Create Test App Server and setup it

Search for EC2 in the search bar. Go to **Instances > Launch instances.** Setup it as follows.




Search [Alt+S]
Asia Pacific (Mumbai)
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EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Compute
Amazon Elastic Compute Cloud (EC2)
Create, manage, and monitor virtual servers in the cloud.
Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 600 instance types and a choice of the latest processors, storage, networking, operating systems, and purchase models to help you best match the needs of your workload.
Launch a virtual server
Launch instance
View dashboard
Additional actions
View running instances
Migrate a server
Pricing (US)
EC2 pricing options
Benefits and features
EC2 offers ultimate scalability and control
Fully resizable compute capacity to support virtually any workload. This service is best if you want:
Highest level of control of the entire technology stack, allowing full integration with all AWS services
Widest variety of server size options
Widest availability of operating systems to choose from including Linux, Windows, and macOS
https://ap-south-1.console.aws.amazon.com/console/home?region=ap-south-1
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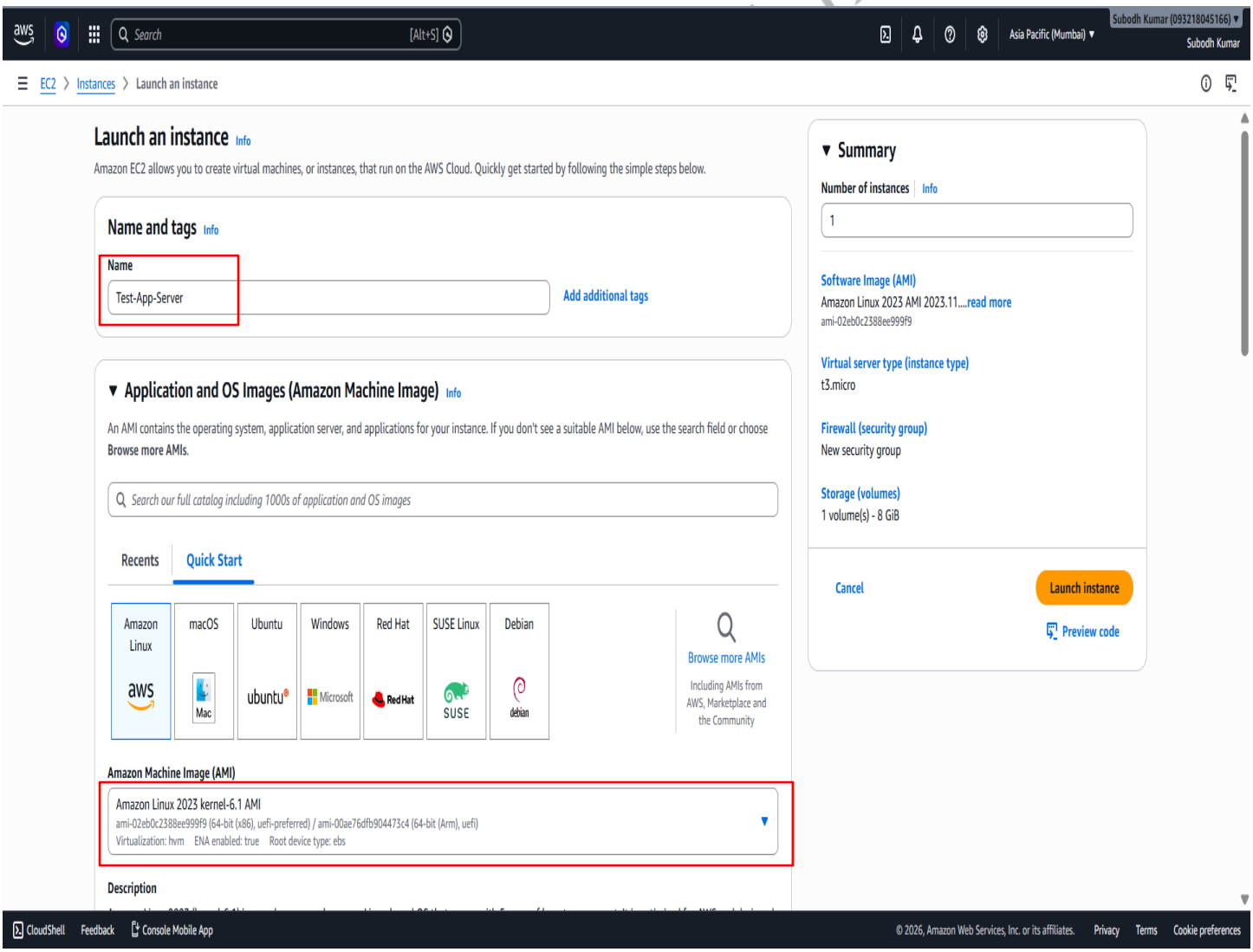
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Instances > Launch an instance
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
Test-App-Server
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.1 AMI
ami-02eb0c2388ee999f9 (64-bit (x86), uefi-preferred) / ami-00ae76dfb904473c4 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs
Description
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....read more
ami-02eb0c2388ee999f9
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
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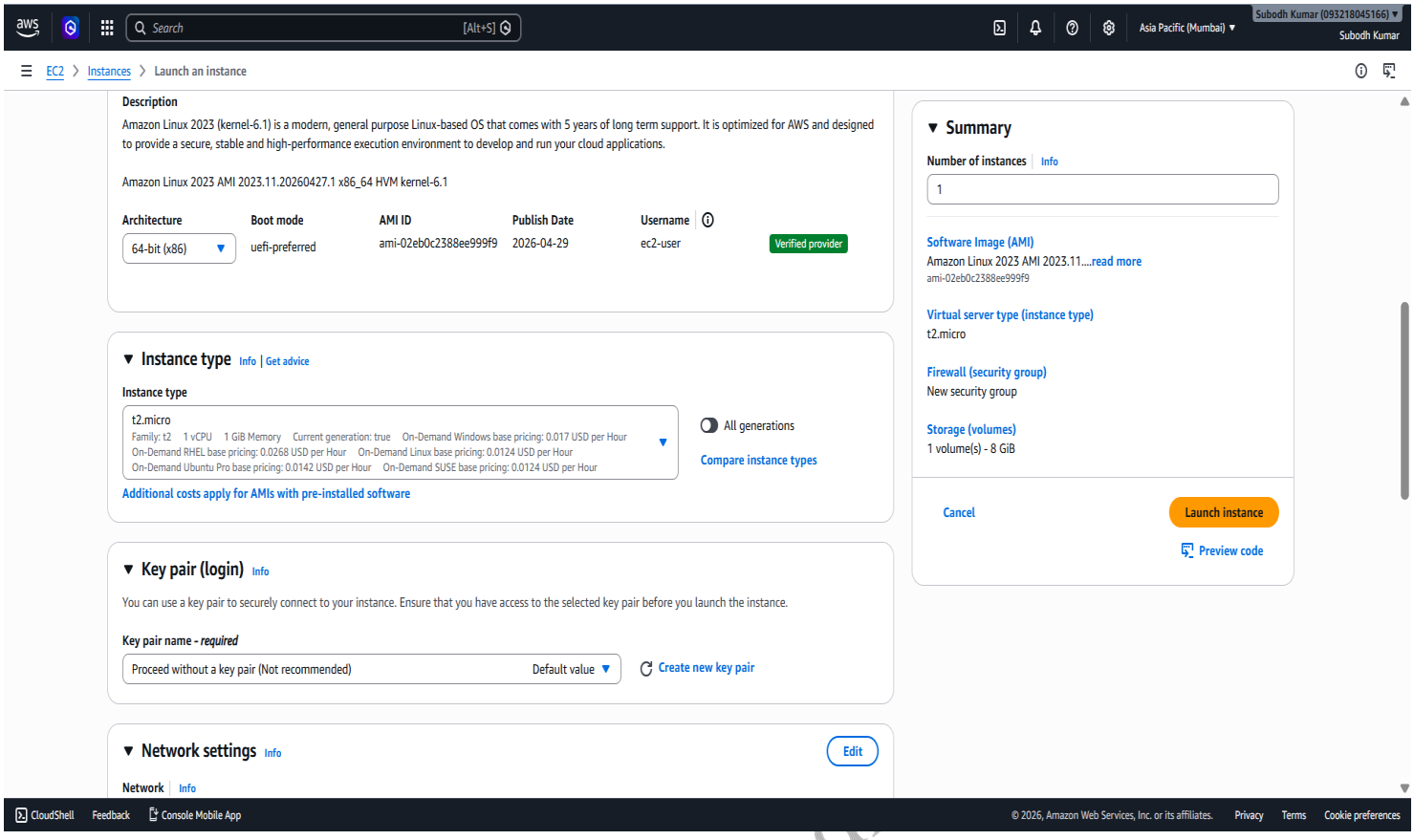
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Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Instances > Launch an instance
Description
Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.11.20260427.1 x86_64 HVM kernel-6.1
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-02eb0c2388ee999f9
Publish Date
2026-04-29
Username
ec2-user
Verified provider
Instance type Info | Get advice
Instance type
t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Proceed without a key pair (Not recommended)
Default value
Create new key pair
Network settings Info
Edit
Network Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....read more
ami-02eb0c2388ee999f9
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
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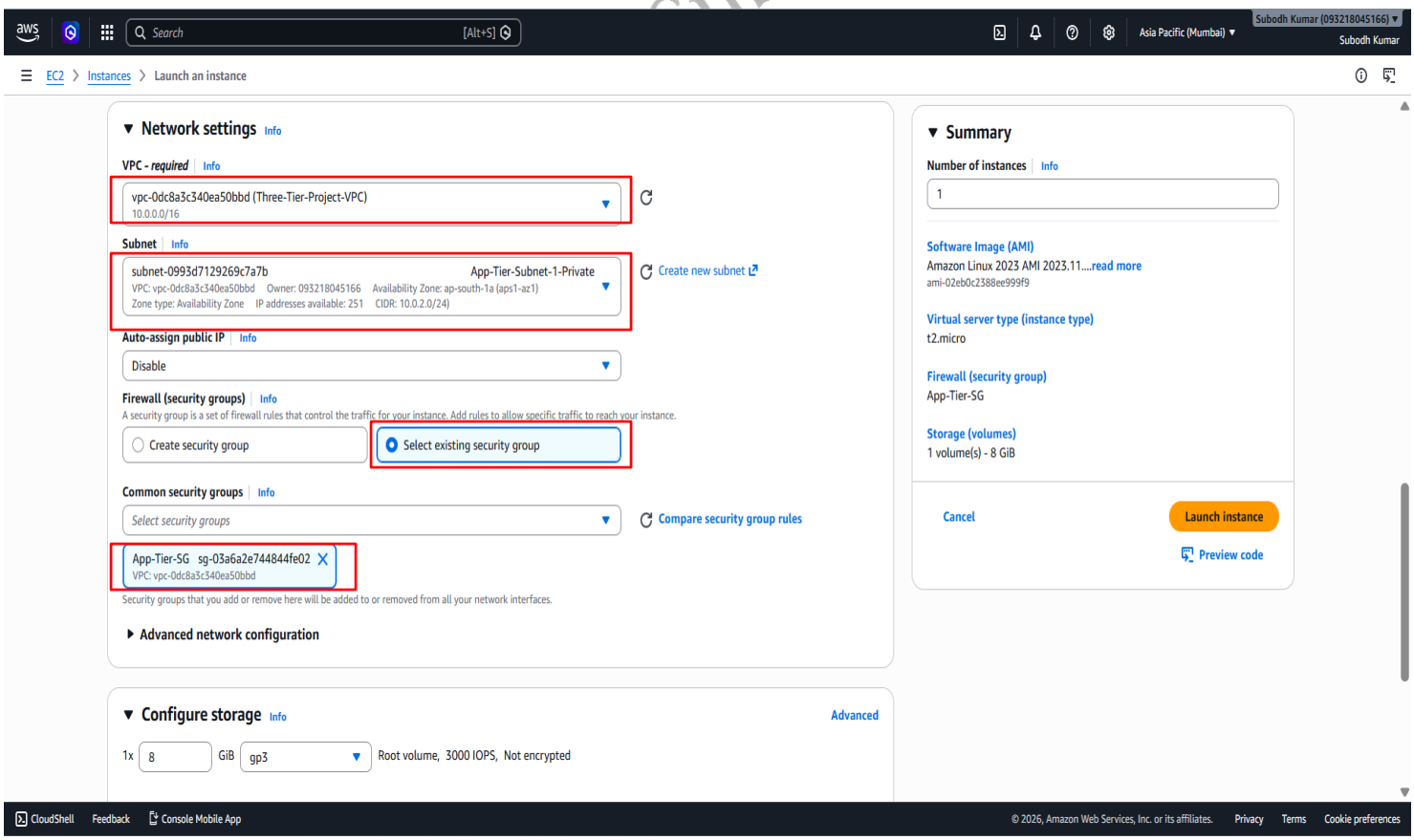
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EC2 > Instances > Launch an instance
Network settings Info
VPC - required Info
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Subnet Info
subnet-0993d7129269c7a7b App-Tier-Subnet-1-Private
VPC: vpc-0dc8a3c340ea50bbd Owner: 093218045166 Availability Zone: ap-south-1a (aps1-az1)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24)
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
App-Tier-SG sg-03a6a2e744844fe02
VPC: vpc-0dc8a3c340ea50bbd
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....read more
ami-02eb0c2388ee999f9
Virtual server type (instance type)
t2.micro
Firewall (security group)
App-Tier-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
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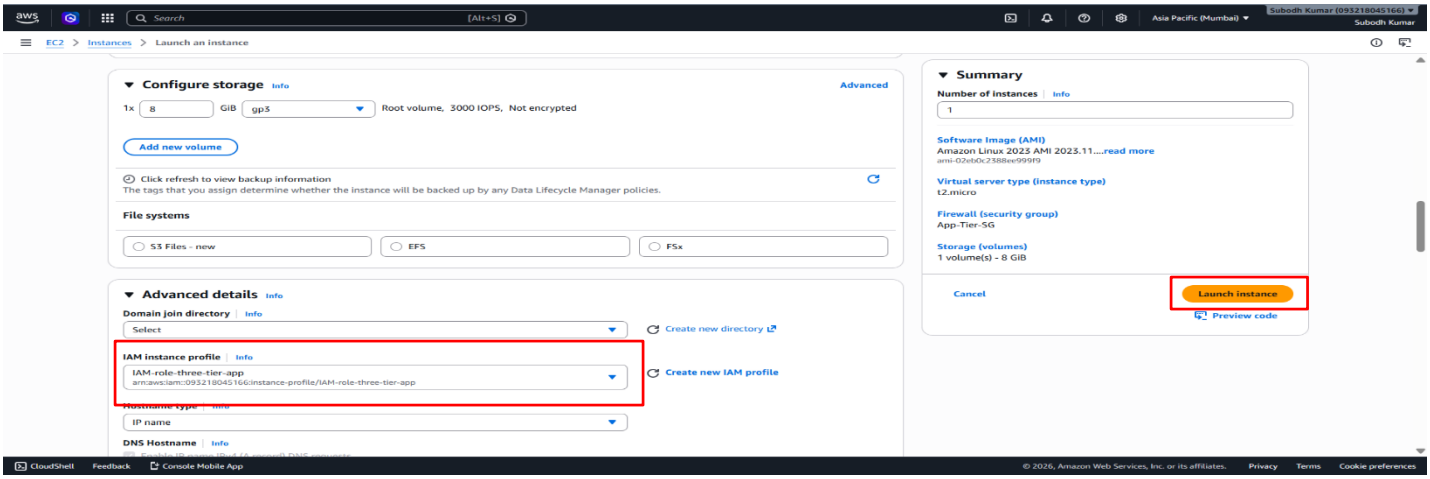
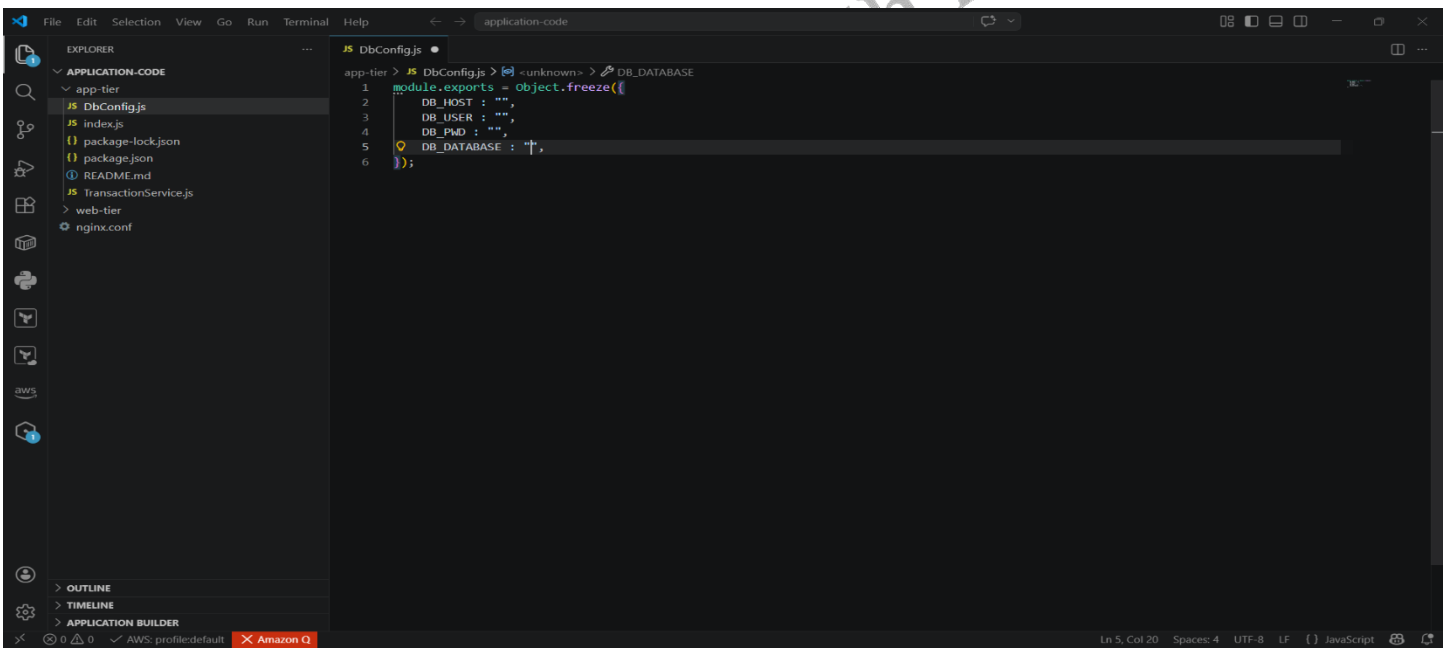


Keep other settings as default and click Launch Instance.

Now we have to connect to the server and configure the application tier.

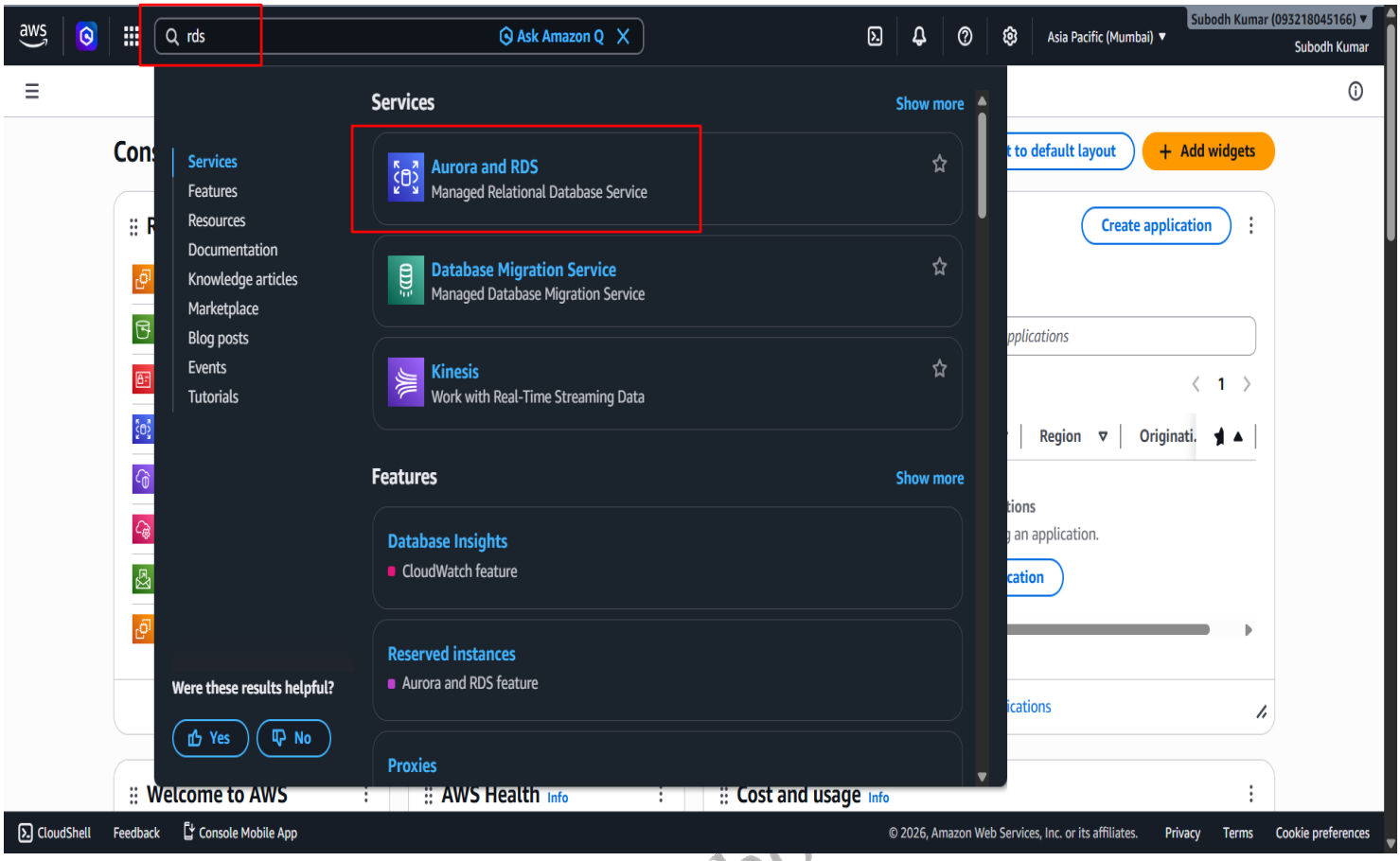
Before that let's do some changes in the source code. Go to the source code folder on your pc. Go to **application-code > app-tier.** Open **DbConfig.js** with your favorite editor. I use vs code.



We have to change these values.

Let's see how to get the **DB_HOST**

Go to search and go to **RDS > Databases**. Select Writer instance. Under connectivity and security, you can find a value called Endpoint. Copy it and paste it as the value for DB_HOST.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Database Migration Service
Managed Database Migration Service
Kinesis
Work with Real-Time Streaming Data
Features
Show more
Database Insights
CloudWatch feature
Reserved instances
Aurora and RDS feature
Proxies
Were these results helpful?
Yes
No
Add widgets
Create application
Welcome to AWS
AWS Health Info
Cost and usage Info
CloudShell
Feedback
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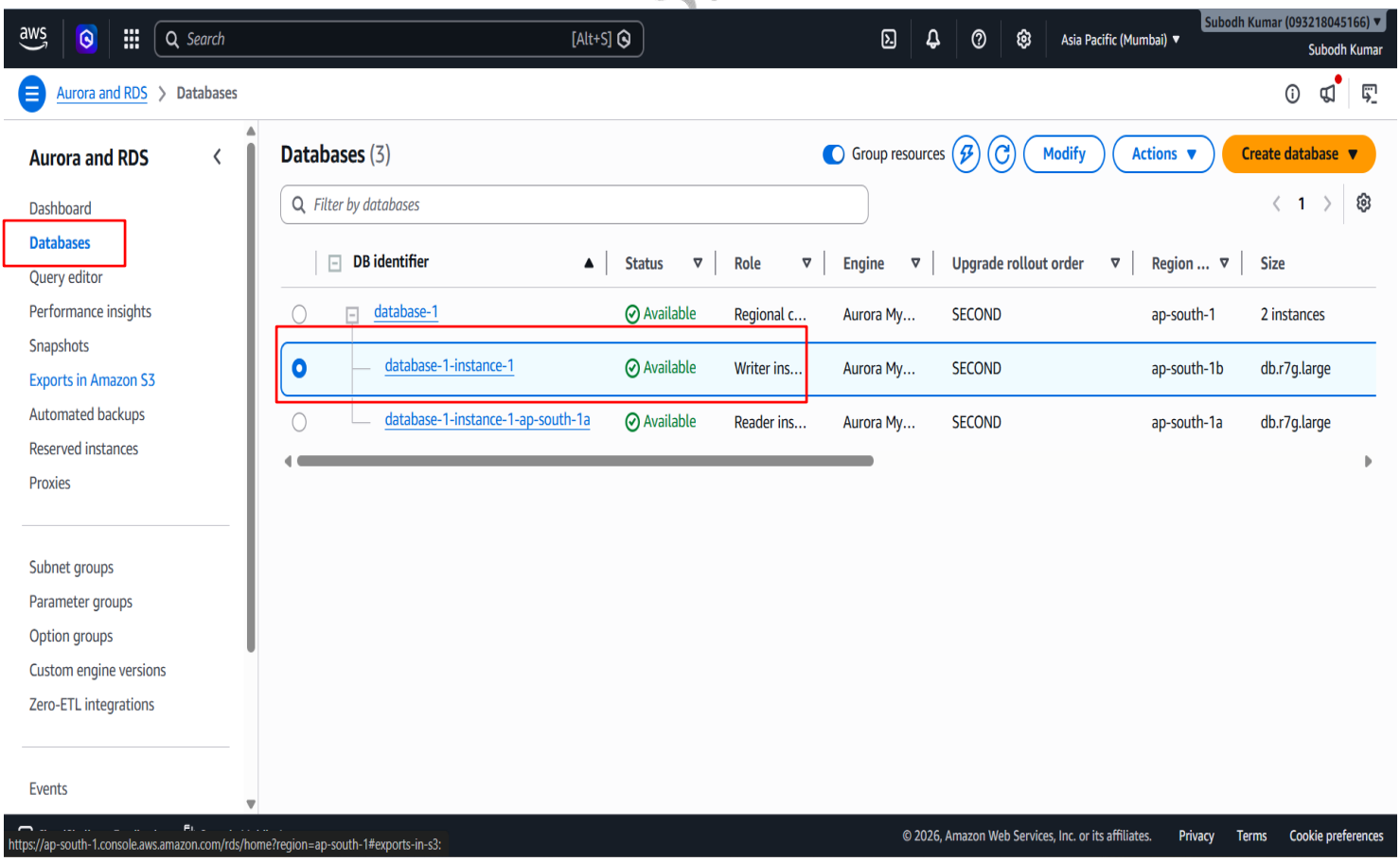
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Query editor
Performance insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies
Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations
Events
Databases (3)
Group resources
Modify
Actions
Create database
Filter by databases
DB identifier
Status
Role
Engine
Upgrade rollout order
Region ...
Size
database-1
Available
Regional c...
Aurora My...
SECOND
ap-south-1
2 instances
database-1-instance-1
Available
Writer ins...
Aurora My...
SECOND
ap-south-1b
db.r7g.large
database-1-instance-1-ap-south-1a
Available
Reader ins...
Aurora My...
SECOND
ap-south-1a
db.r7g.large
https://ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#exports-in-s3:
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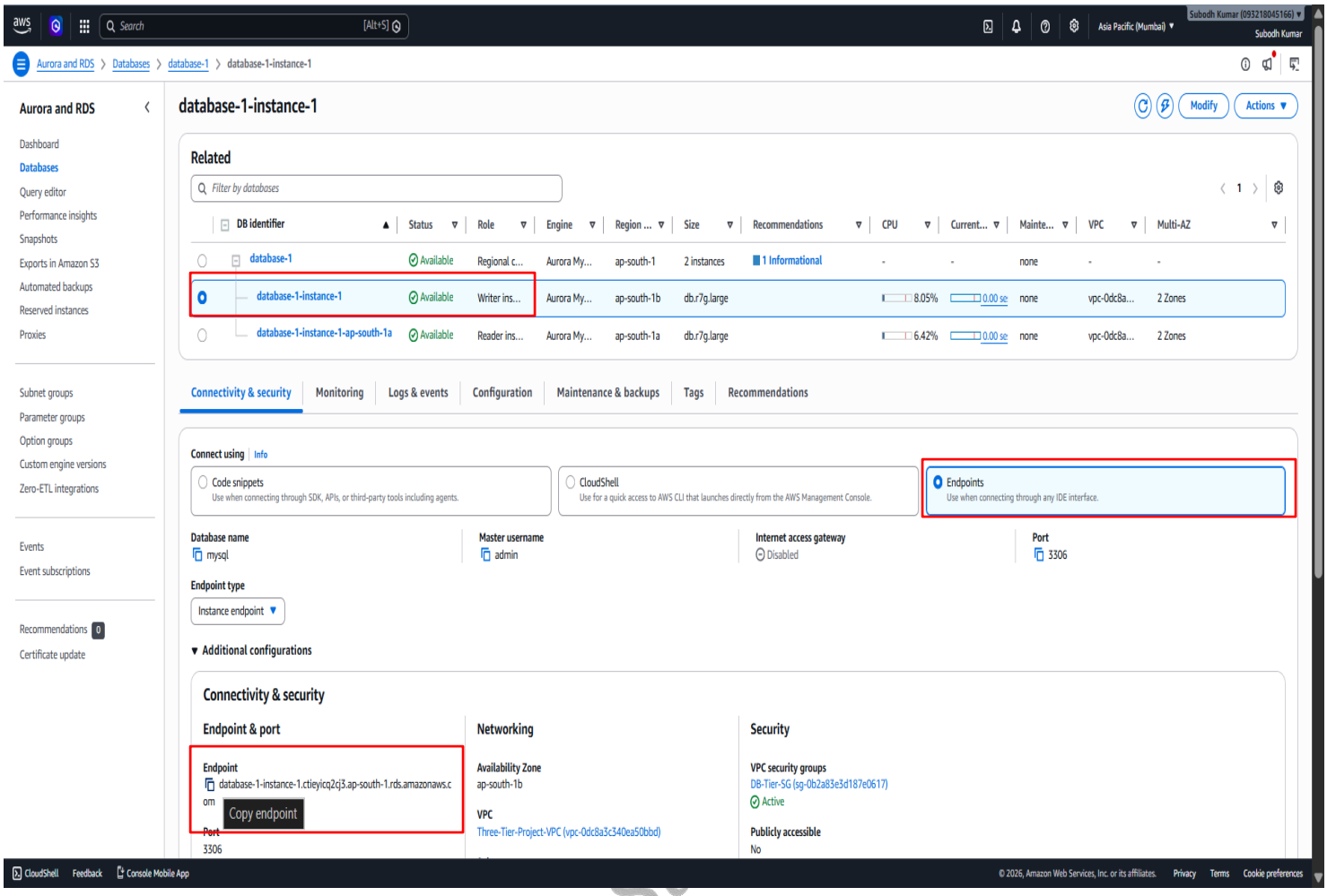
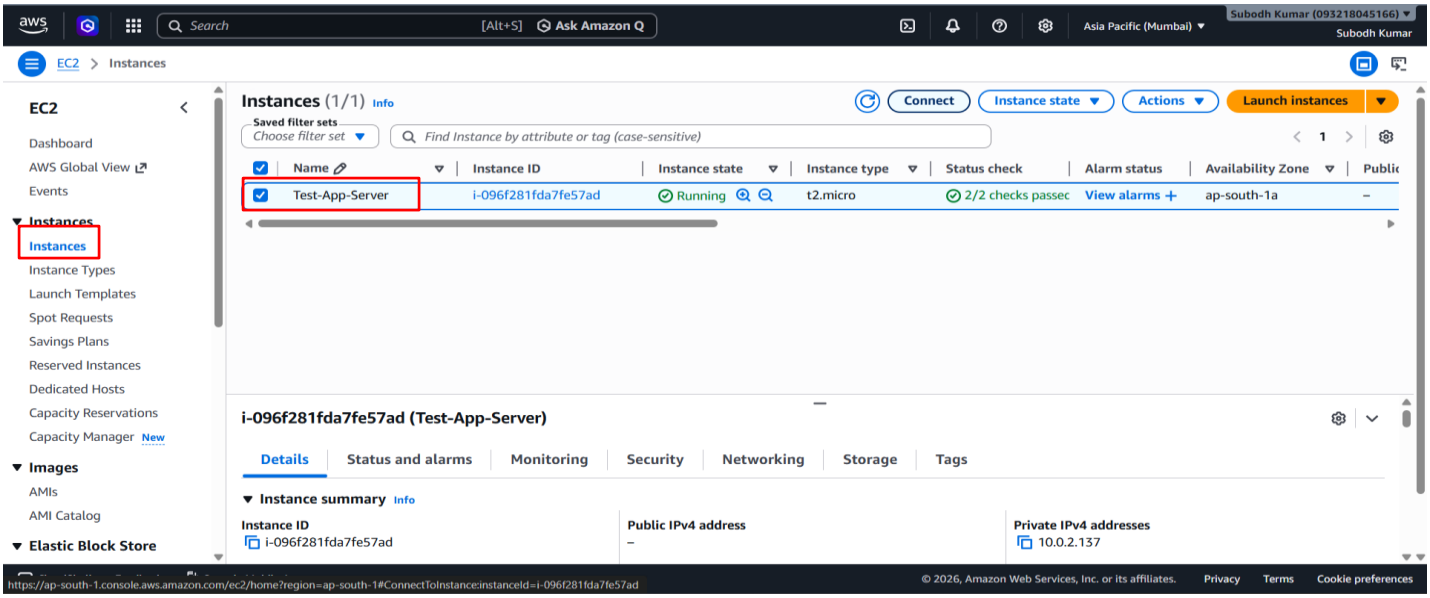


DB_USER is the username for DB server (**admin**). **DB_PWD** is the password we assigned (**subodhdatabase**). **DB_DATABASE** is **webappdb**.

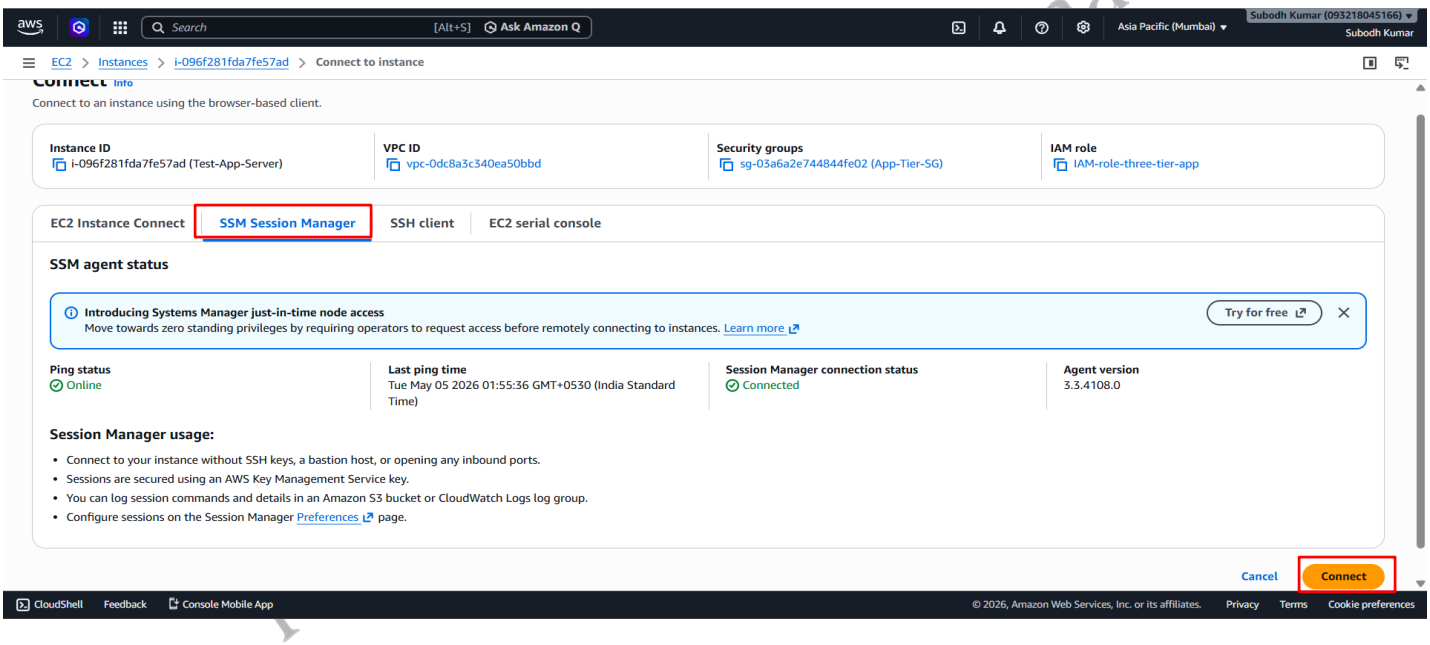
Now go to the S3 Bucket we created. Go to **application-code > app-tier** in the s3 bucket**.** Delete the DbConfig.js and then upload our newly updated DbConfig.js file.

Cool

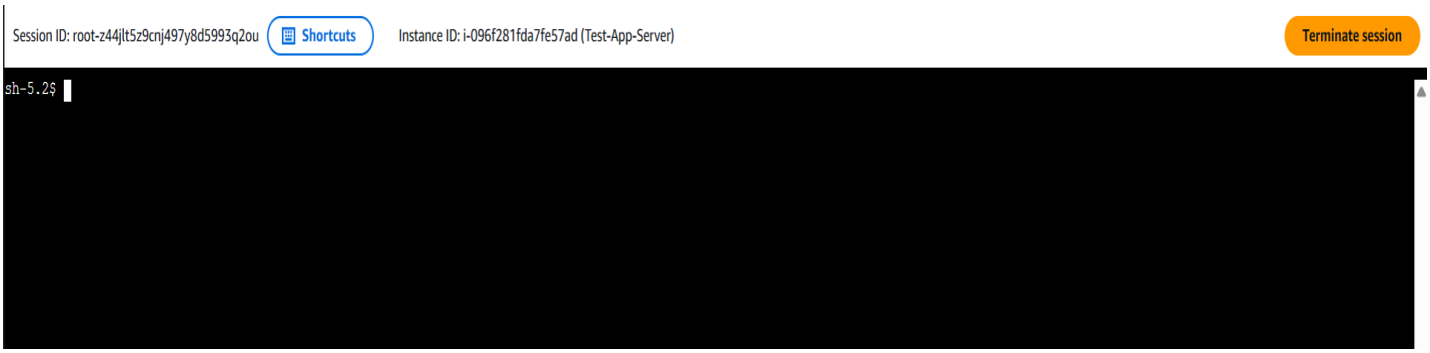
Now again go to EC2 instance and select Test-APP-Server we created click connect.



Go to Session Manager tab and click connect.



It will open a new tab like this. Which means you have connected to EC2 instance. IAM role we created let us connect to this EC2 instance even if it is in a private subnet.



Now let’s configure the app-tier

First Switch user to the ec2-user by running `sudo -su ec2-user`

Now install the MySQL using following commands.

```
sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm
sudo dnf install mysql80-community-release-el9-1.noarch.rpm -y
sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023
sudo dnf install mysql-community-client -y
```

Run `mysql --version` to confirm it installed correctly

```
[ec2-user@ip-10-0-2-137 bin]$ mysql --version
mysql  Ver 8.0.46 for Linux on x86_64 (MySQL Community Server - GPL)
[ec2-user@ip-10-0-2-137 bin]$
```

Now let's connect to Database server by running following command

```
mysql -h <RDS-Endpoint> -u <username> -p
```

RDS-Endpoint: Endpoint value of Writer instance copied from the RDS>Databases.

Username: admin

```
mysql -h database-1-instance-1.ctieyicq2cj3.ap-south-1.rds.amazonaws.com -u admin -p
```

Hit enter and input the password (**subodhdatabase**). Enter. You will see the following output.

```
[ec2-user@ip-10-0-2-137 bin]$ mysql -h database-1-instance-1.ctieyicq2cj3.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 294
Server version: 8.0.42 6252a59a

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Prepared by Subodh

Let's Create the database. Run following commands.

```
create DATABASE webappdb;

show databases;

use webappdb;
```

```
mysql> create DATABASE webappdb;
Query OK, 1 row affected (0.00 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| webappdb           |
+--------------------+
5 rows in set (0.00 sec)

mysql> use webappdb;
Database changed
mysql>
```

Prepared by Subodh Kumar

Now Create a table by running following command

```
CREATE TABLE IF NOT EXISTS transactions(id INT NOT NULL
AUTO_INCREMENT, amount DECIMAL(10,2), description
VARCHAR(100), PRIMARY KEY(id));
```

```
mysql> CREATE TABLE IF NOT EXISTS transactions(id INT NOT NULL
    -> AUTO_INCREMENT, amount DECIMAL(10,2), description
    -> VARCHAR(100), PRIMARY KEY(id));
Query OK, 0 rows affected (0.02 sec)

mysql> show tables;
+--------------------+
| Tables_in_webappdb |
+--------------------+
| transactions       |
+--------------------+
1 row in set (0.00 sec)

mysql>
```

Let’s create insert a sample value.

```
INSERT INTO transactions (amount, description) VALUES ('400', 'groceries');

SELECT * FROM transactions;
```

```
mysql> INSERT INTO transactions (amount,description) VALUES ('400','groceries');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM transactions;
+----+--------+-------------+
| id | amount | description |
+----+--------+-------------+
|  1 | 400.00 | groceries   |
+----+--------+-------------+
1 row in set (0.00 sec)

mysql>
```

Type `exit` and hit Enter

Next we copy application tier codes from s3 bucket. Replace <S3_BUCKET_NAME> with your s3 bucket name.

```
cd /home/ec2-user
sudo aws s3 cp s3://<YOUR-S3-BUCKET-NAME>/application-code/app-tier app-tier
--recursive

cd app-tier
sudo chown -R ec2-user:ec2-user /home/ec2-user/app-tier
sudo chmod -R 755 /home/ec2-user/app-tier
```

by Subodh Kumar

```
[ec2-user@ip-10-0-2-137 bin]$ cd /home/ec2-user
[ec2-user@ip-10-0-2-137 ~]$ sudo aws s3 cp s3://three-tier-app-bucket-5-5-26/application-code/app-tier app-tier --recursive
download: s3://three-tier-app-bucket-5-5-26/application-code/app-tier/README.md to app-tier/README.md
download: s3://three-tier-app-bucket-5-5-26/application-code/app-tier/DbConfig.js to app-tier/DbConfig.js
download: s3://three-tier-app-bucket-5-5-26/application-code/app-tier/package-lock.json to app-tier/package-lock.json
download: s3://three-tier-app-bucket-5-5-26/application-code/app-tier/package.json to app-tier/package.json
download: s3://three-tier-app-bucket-5-5-26/application-code/app-tier/index.js to app-tier/index.js
download: s3://three-tier-app-bucket-5-5-26/application-code/app-tier/TransactionService.js to app-tier/TransactionService.js
[ec2-user@ip-10-0-2-137 ~]$ cd app-tier
[ec2-user@ip-10-0-2-137 app-tier]$ sudo chown -R ec2-user:ec2-user /home/ec2-user/app-tier
[ec2-user@ip-10-0-2-137 app-tier]$ sudo chmod -R 755 /home/ec2-user/app-tier
[ec2-user@ip-10-0-2-137 app-tier]$
```

Now let’s install Node.js

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh | bash
```

```
source ~/.bashrc
```

```
nvm install 16
```

```
nvm use 16
```

```
npm install -g pm2
```

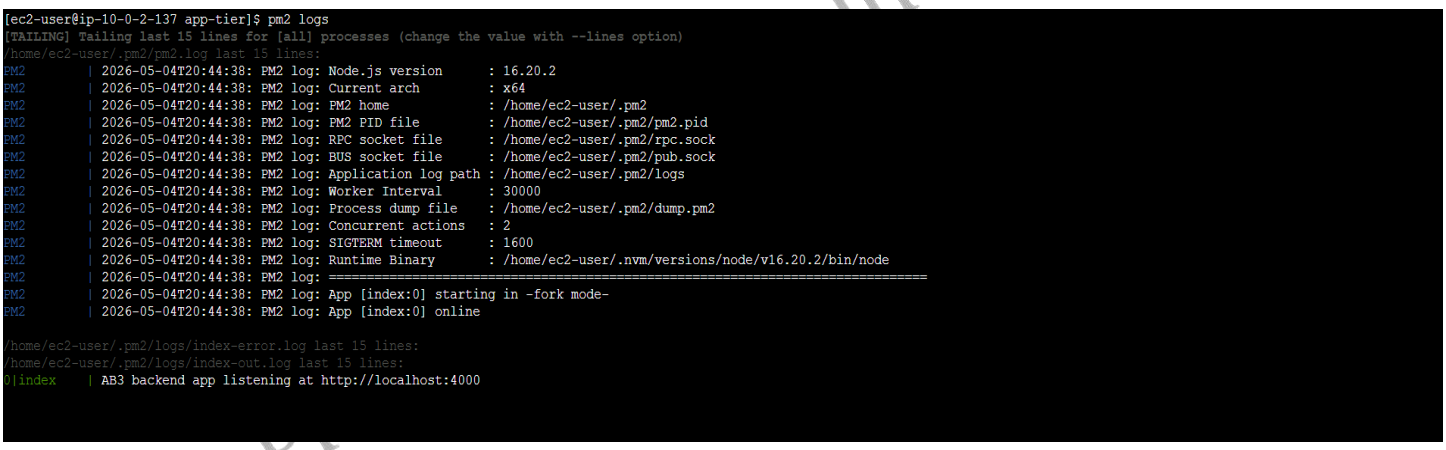
```
npm install
```

```
npm audit fix
```

After that lest run the Index.js file and then save the configurations.

```
pm2 start index.js
```

```
pm2 logs
```



If you got output like this. The backend is connected to port 4000. Press Ctrl+c to exit the loop. Then run the following commands.

```
pm2 startup
```

```
sudo env PATH=$PATH:/home/ec2-user/.nvm/versions/node/v16.20.2/bin /home/ec2-user/.nvm/versions/node/v16.20.2/lib/node_modules/pm2/bin/pm2 startup systemd -u ec2-user --hp /home/ec2-user
```

```
pm2 save
```

```
curl http://localhost:4000/health
```

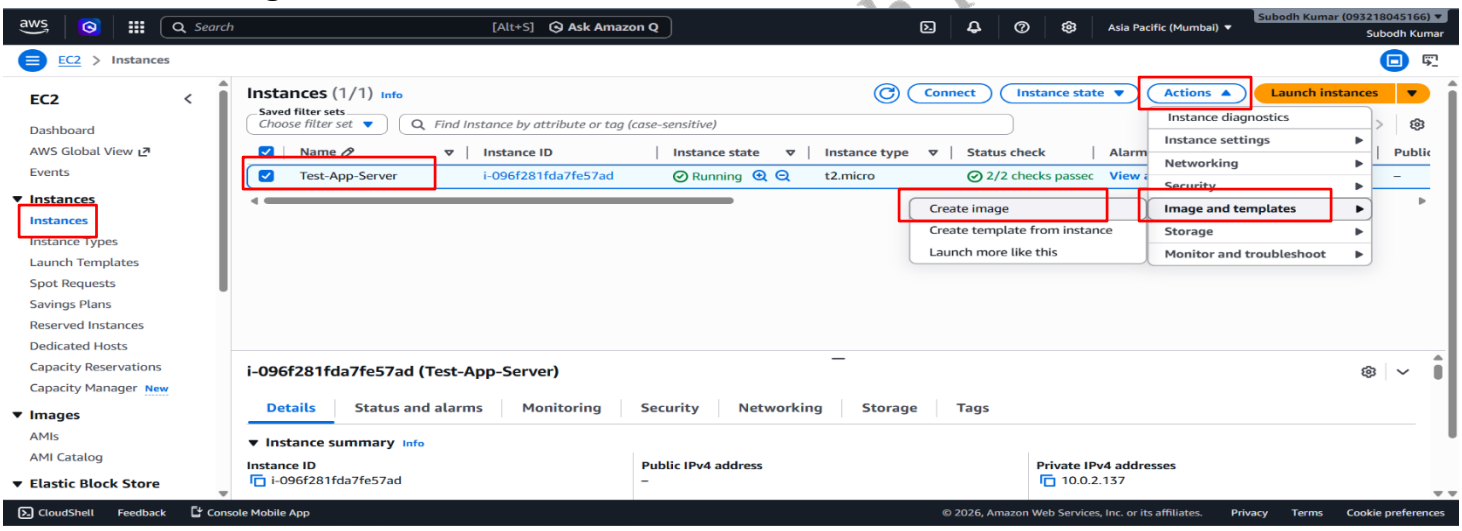
```
[PM2] Remove init script via:
$ pm2 unstartup systemd
[ec2-user@ip-10-0-2-137 app-tier]$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ec2-user/.pm2/dump.pm2
[ec2-user@ip-10-0-2-137 app-tier]$ curl http://localhost:4000/health
"This is the health check"[ec2-user@ip-10-0-2-137 app-tier]$
```

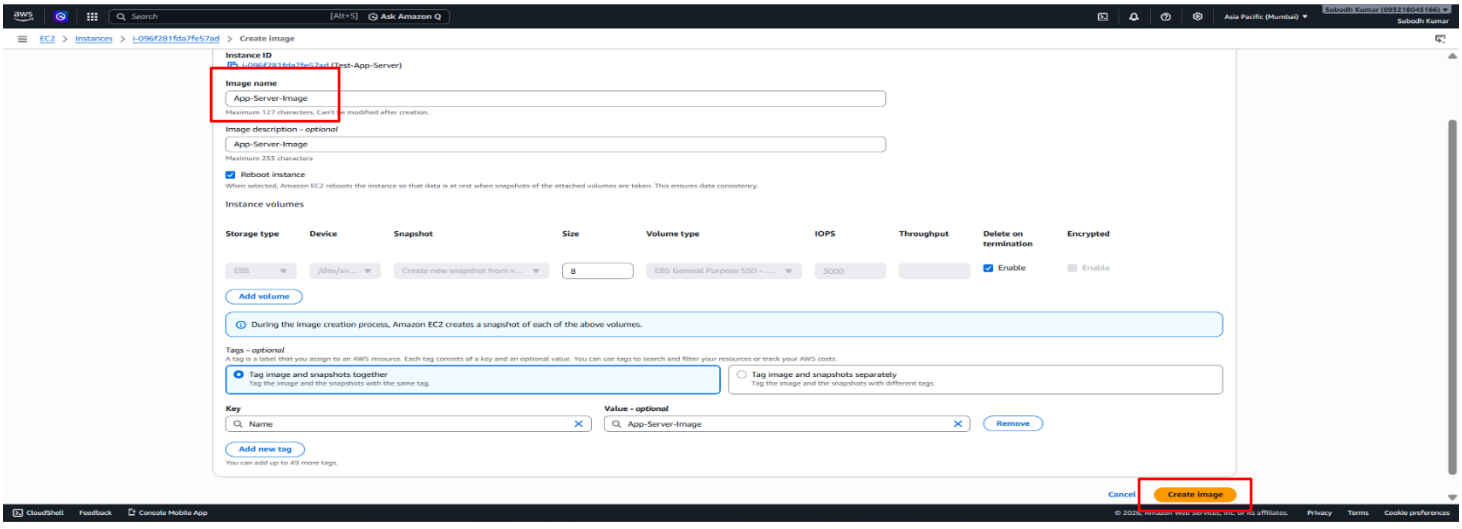
Cool You successfully setup the app server

8. Create AMI of App-Server

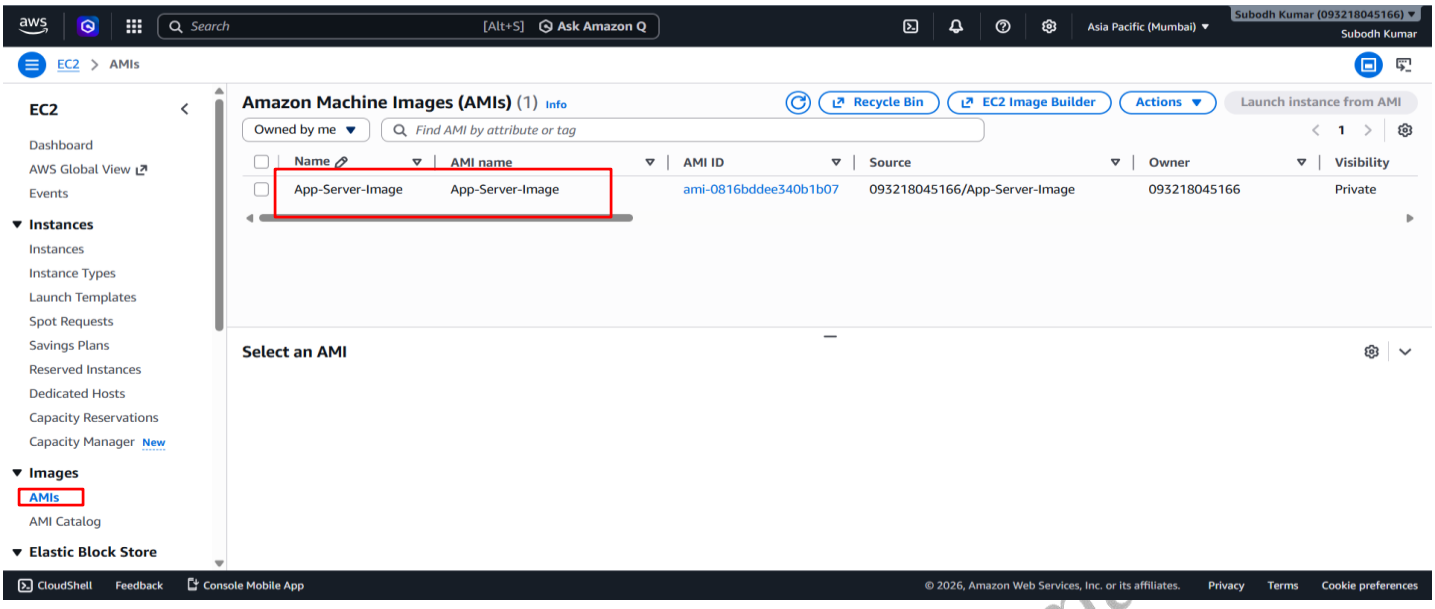
Next Let's create an image from this Test-App-Server instance.

Go to instances choose **Test-App-Server** go to **Actions > Image and templates > Create image.**





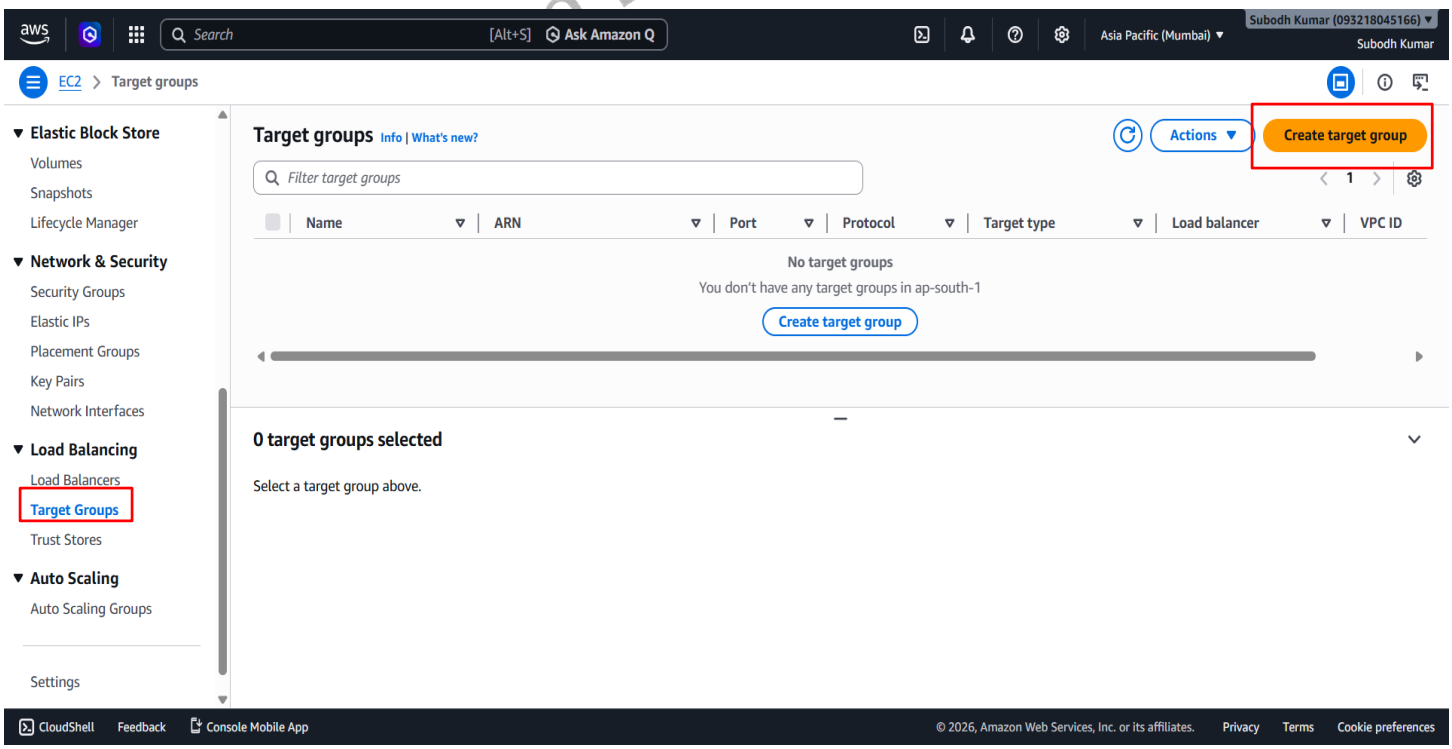
If you go to **AMIs section**, you can see that App-Server AMI is created



9. Create Target Group, Launch Template, Internal Load Balancer for App Tier

Create Target Group

Go to EC2 > Target Groups > Create Target Group change only following settings. Click Next. Click Create target group.



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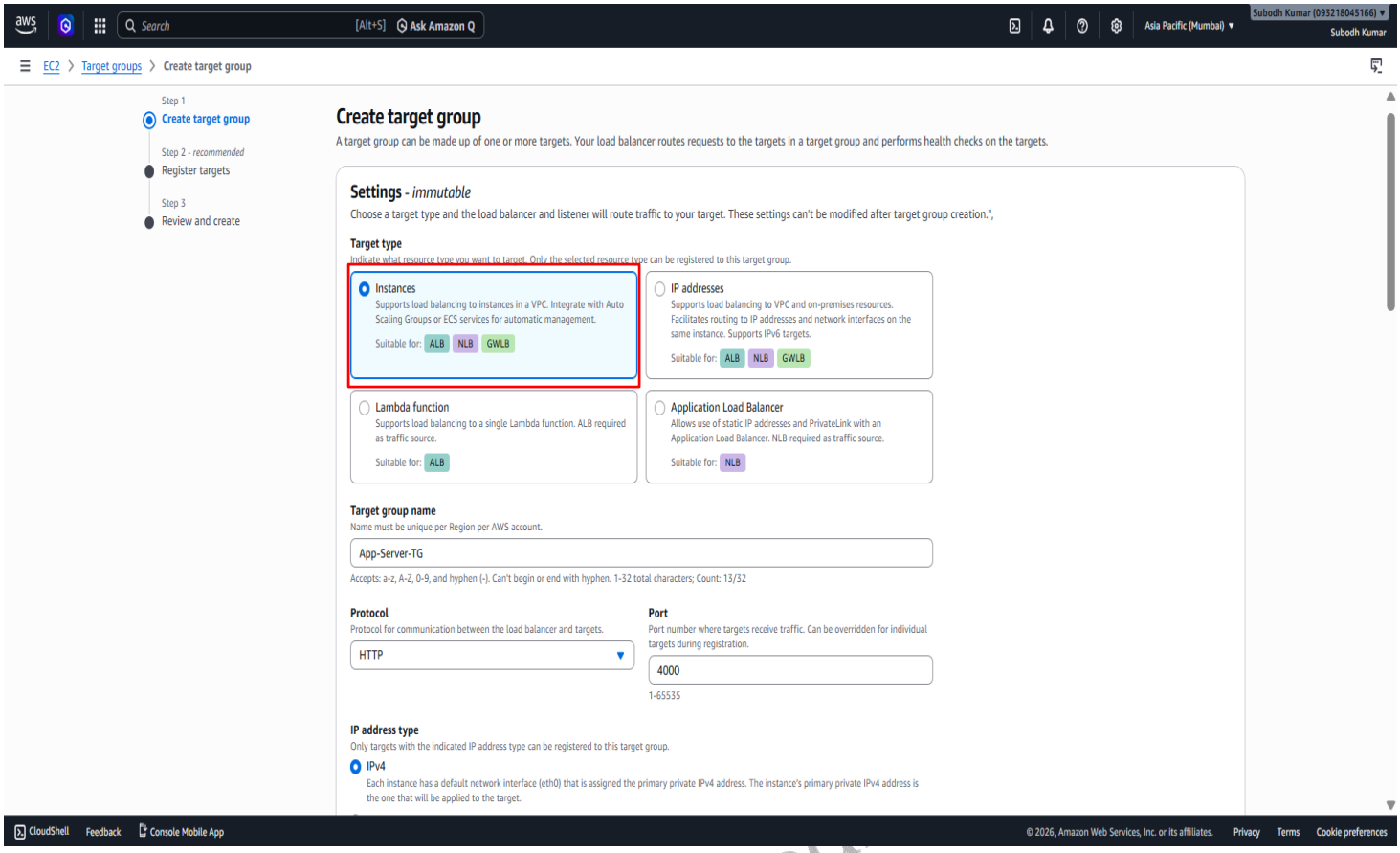
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EC2 > Target groups > Create target group
Step 1
Create target group
Step 2 - recommended
Register targets
Step 3
Review and create
Create target group
A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.
Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.",
Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.
Instances
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB
IP addresses
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB
Lambda function
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB
Application Load Balancer
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB
Target group name
Name must be unique per Region per AWS account.
App-Server-TG
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32
Protocol
Protocol for communication between the load balancer and targets.
HTTP
Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
4000
1-65535
IP address type
Only targets with the indicated IP address type can be registered to this target group.
IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
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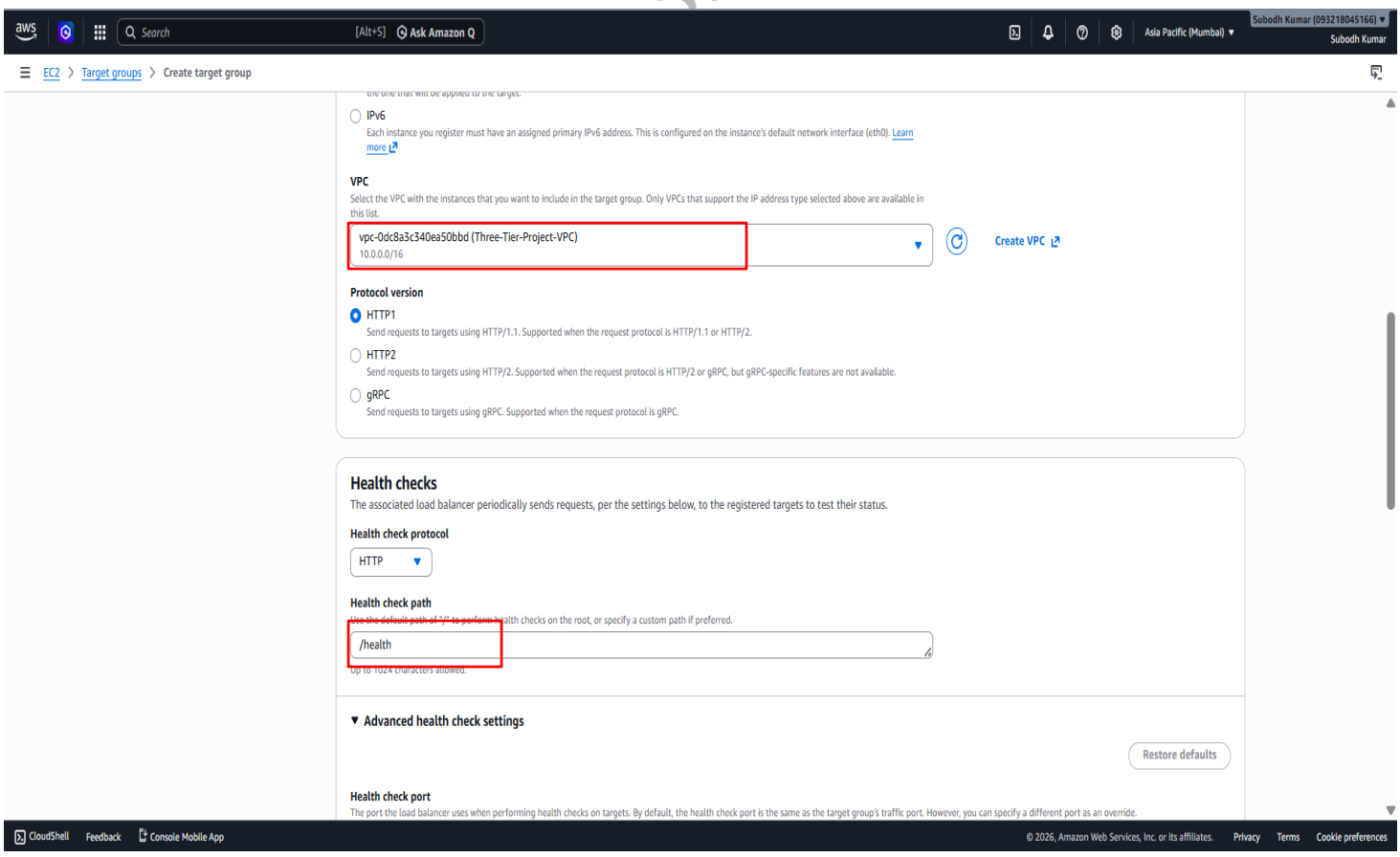
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EC2 > Target groups > Create target group
IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). Learn more
VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create VPC
Protocol version
HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.
gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.
Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.
Health check protocol
HTTP
Health check path
/health
Advanced health check settings
Restore defaults
Health check port
The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.
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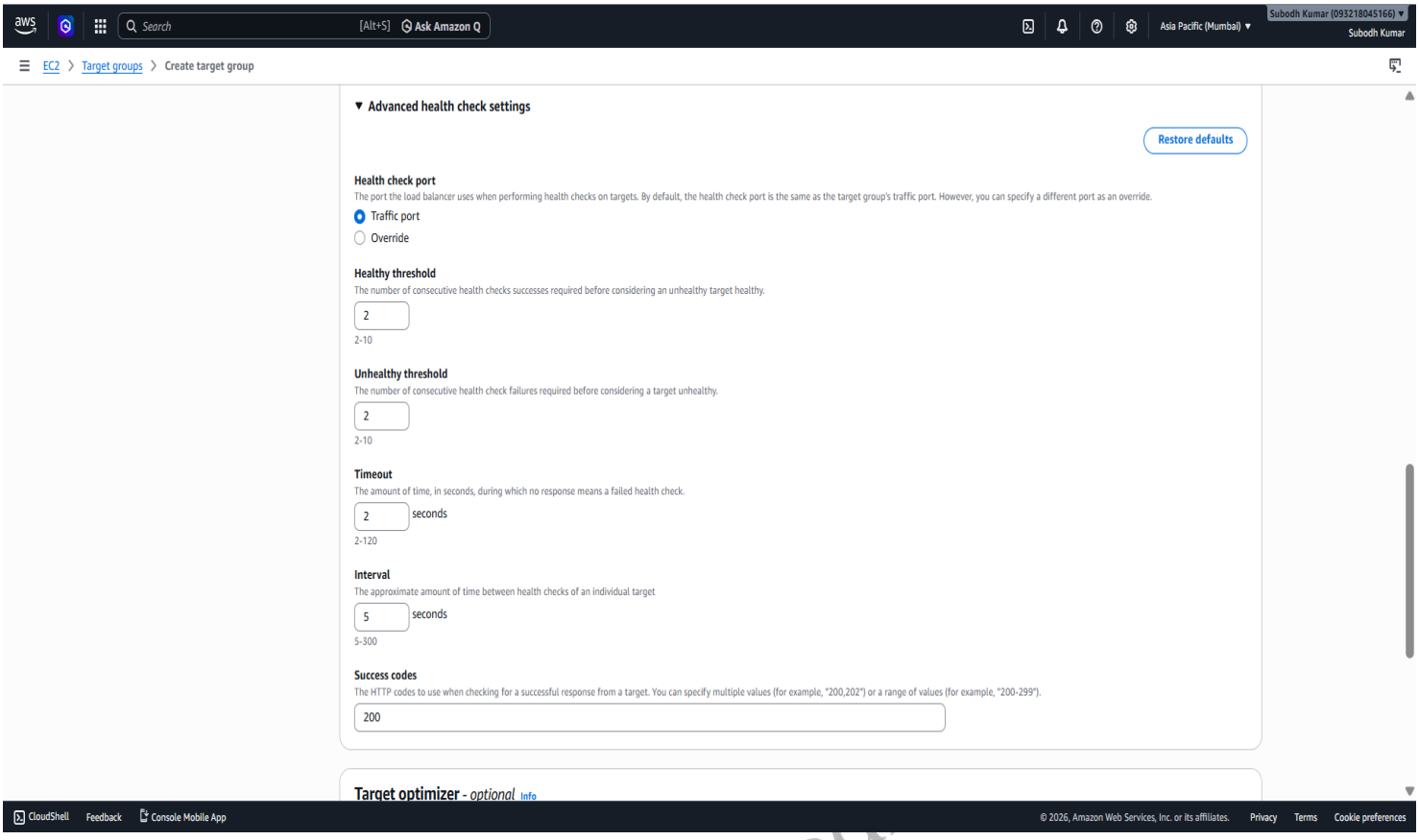
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Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Target groups > Create target group
Advanced health check settings
Restore defaults
Health check port
The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.
Traffic port
Override
Healthy threshold
The number of consecutive health checks successes required before considering an unhealthy target healthy.
2
2-10
Unhealthy threshold
The number of consecutive health check failures required before considering a target unhealthy.
2
2-10
Timeout
The amount of time, in seconds, during which no response means a failed health check.
2
seconds
2-120
Interval
The approximate amount of time between health checks of an individual target
5
seconds
5-300
Success codes
The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").
200
Target optimizer - optional Info
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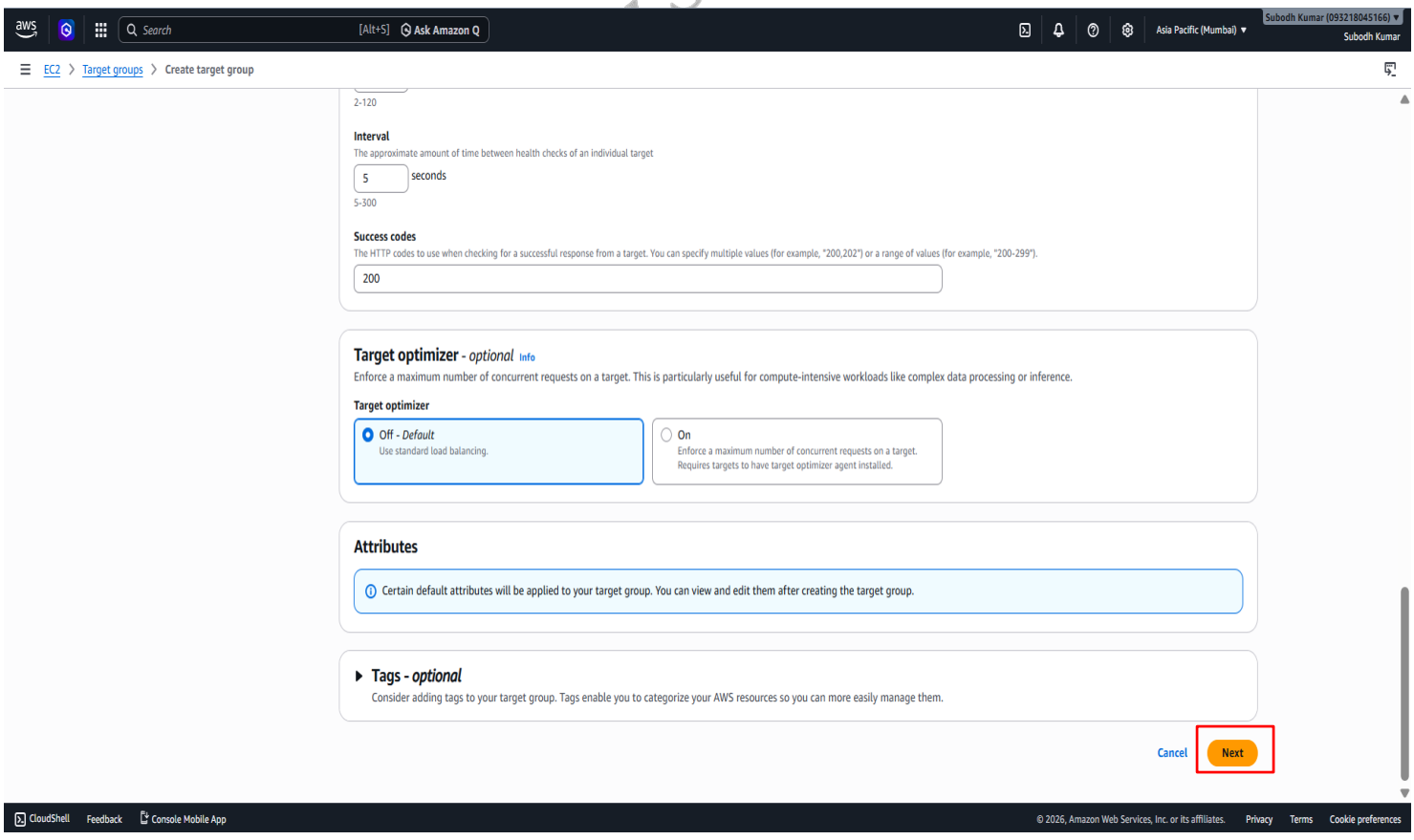
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Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Target groups > Create target group
2-120
Interval
The approximate amount of time between health checks of an individual target
5
seconds
5-300
Success codes
The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").
200
Target optimizer - optional Info
Enforce a maximum number of concurrent requests on a target. This is particularly useful for compute-intensive workloads like complex data processing or inference.
Target optimizer
Off - Default
Use standard load balancing.
On
Enforce a maximum number of concurrent requests on a target.
Requires targets to have target optimizer agent installed.
Attributes
Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.
Tags - optional
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.
Cancel
Next
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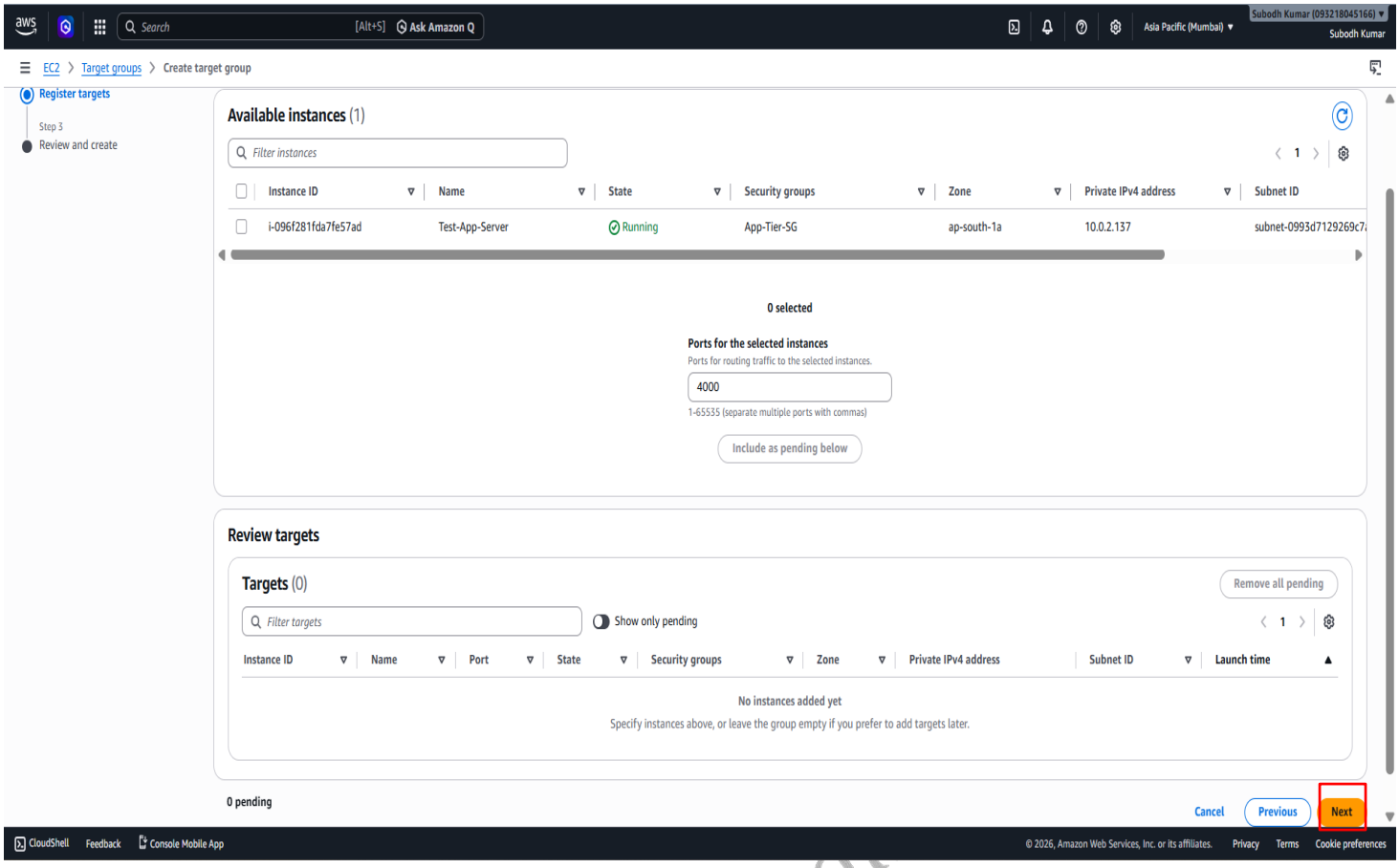
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Subodh Kumar
EC2 > Target groups > Create target group
Register targets
Step 3
Review and create
Available instances (1)
Filter instances
Instance ID
Name
State
Security groups
Zone
Private IPv4 address
Subnet ID
i-096f281fda7fe57ad
Test-App-Server
Running
App-Tier-SG
ap-south-1a
10.0.2.137
0 selected
Ports for the selected instances
Ports for routing traffic to the selected instances.
4000
1-65535 (separate multiple ports with commas)
Include as pending below
Review targets
Targets (0)
Remove all pending
Filter targets
Show only pending
Instance ID
Name
Port
State
Security groups
Zone
Private IPv4 address
Subnet ID
Launch time
No instances added yet
Specify instances above, or leave the group empty if you prefer to add targets later.
0 pending
Cancel
Previous
Next
CloudShell
Feedback
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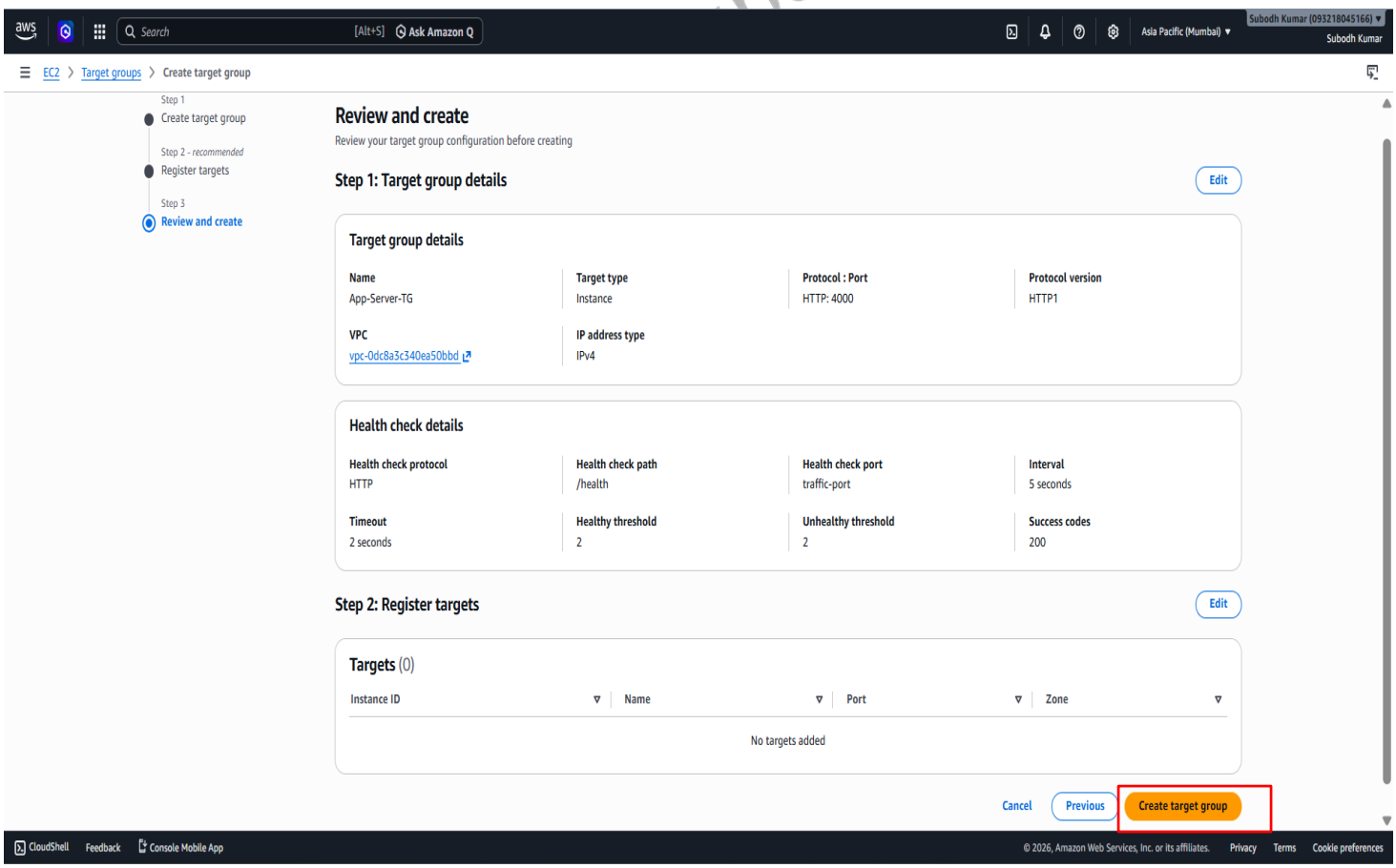
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EC2 > Target groups > Create target group
Step 1
Create target group
Step 2 - recommended
Register targets
Step 3
Review and create
Review and create
Review your target group configuration before creating
Step 1: Target group details
Edit
Target group details
Name
App-Server-TG
Target type
Instance
Protocol : Port
HTTP: 4000
Protocol version
HTTP1
VPC
vpc-0dc8a3c340ea50bbd
IP address type
IPv4
Health check details
Health check protocol
HTTP
Health check path
/health
Health check port
traffic-port
Interval
5 seconds
Timeout
2 seconds
Healthy threshold
2
Unhealthy threshold
2
Success codes
200
Step 2: Register targets
Edit
Targets (0)
Instance ID
Name
Port
Zone
No targets added
Cancel
Previous
Create target group
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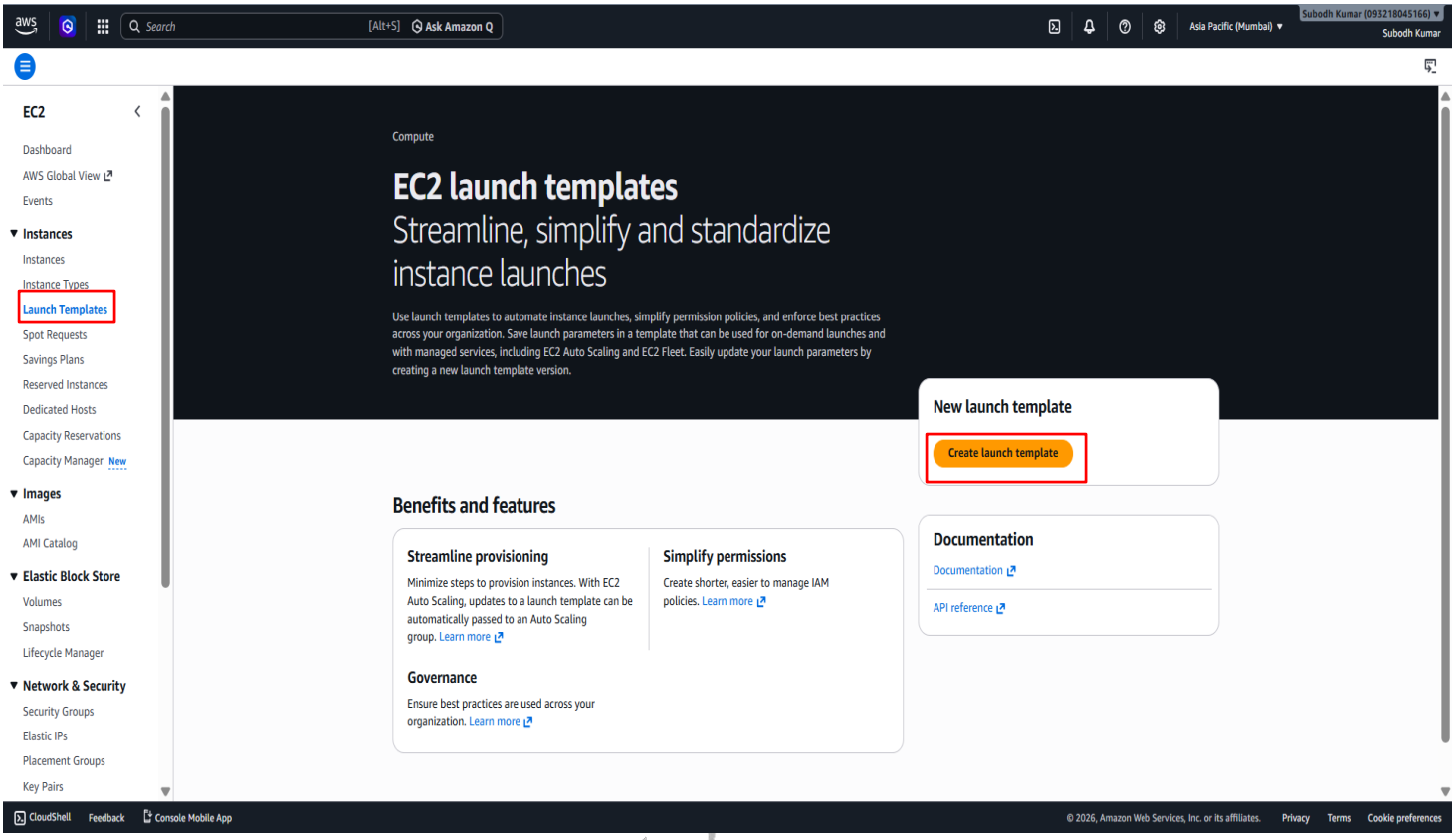
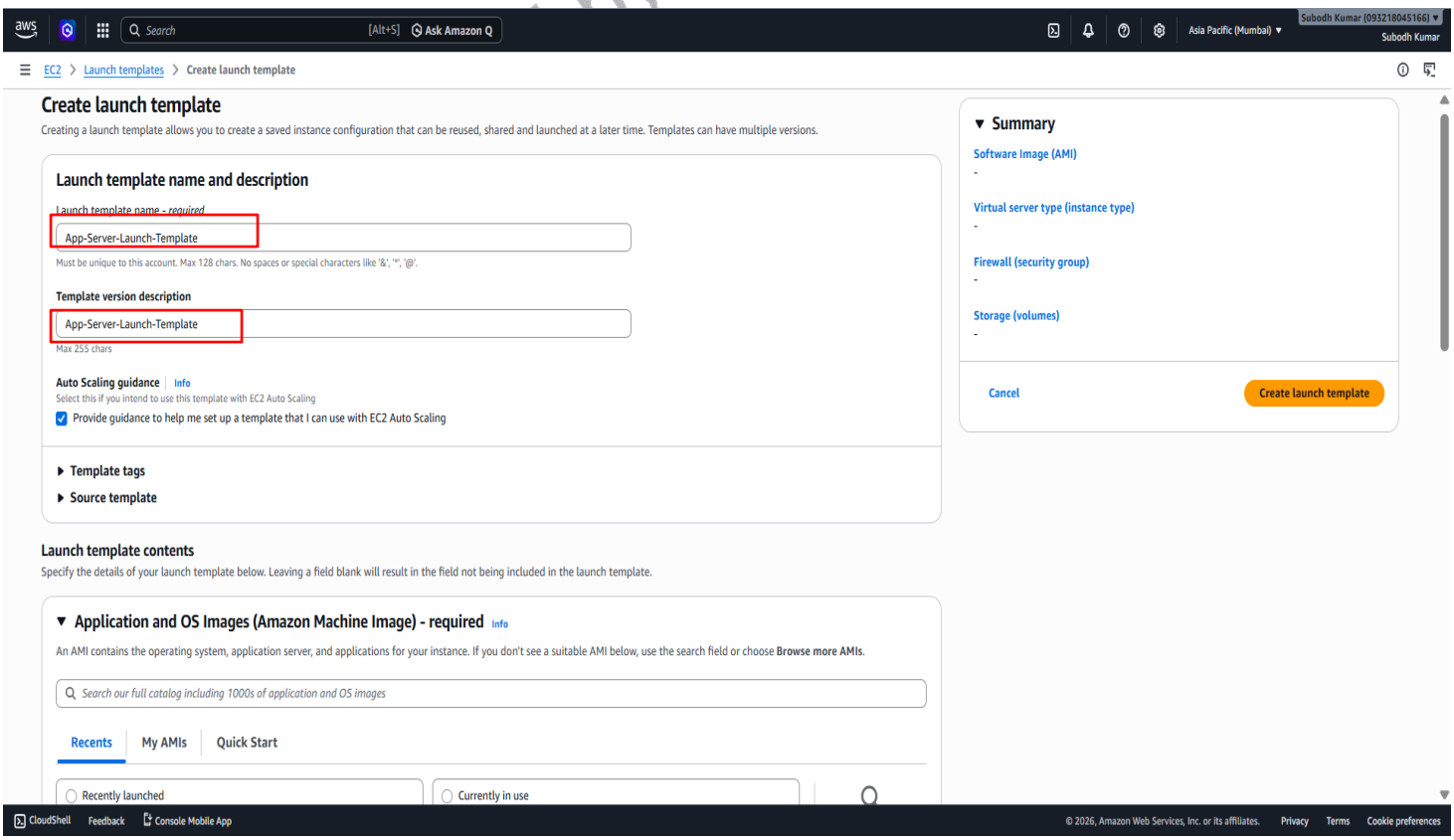
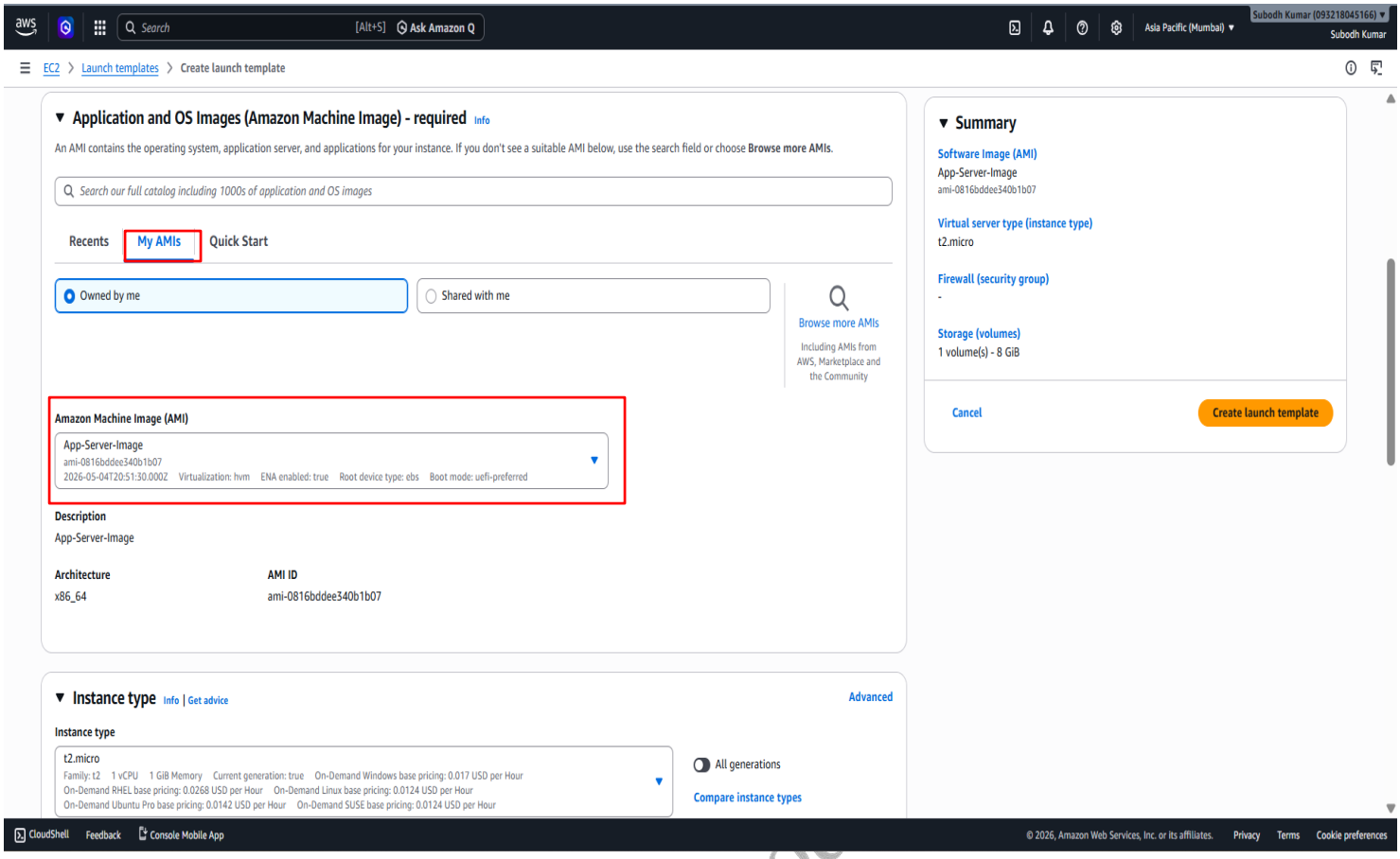
Create Launch Template

Go to EC2 > Launch Templates > Create launch template.



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EC2 > Launch templates > Create launch template
Application and OS Images (Amazon Machine Image) - required Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
My AMIs
Quick Start
Owned by me
Shared with me
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
App-Server-Image
ami-0816bddee340b1b07
2026-05-04T20:51:30.000Z Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred
Description
App-Server-Image
Architecture
x86_64
AMI ID
ami-0816bddee340b1b07
Instance type Info | Get advice
Advanced
Instance type
t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour
All generations
Compare instance types
Summary
Software Image (AMI)
App-Server-Image
ami-0816bddee340b1b07
Virtual server type (instance type)
t2.micro
Firewall (security group)
-
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
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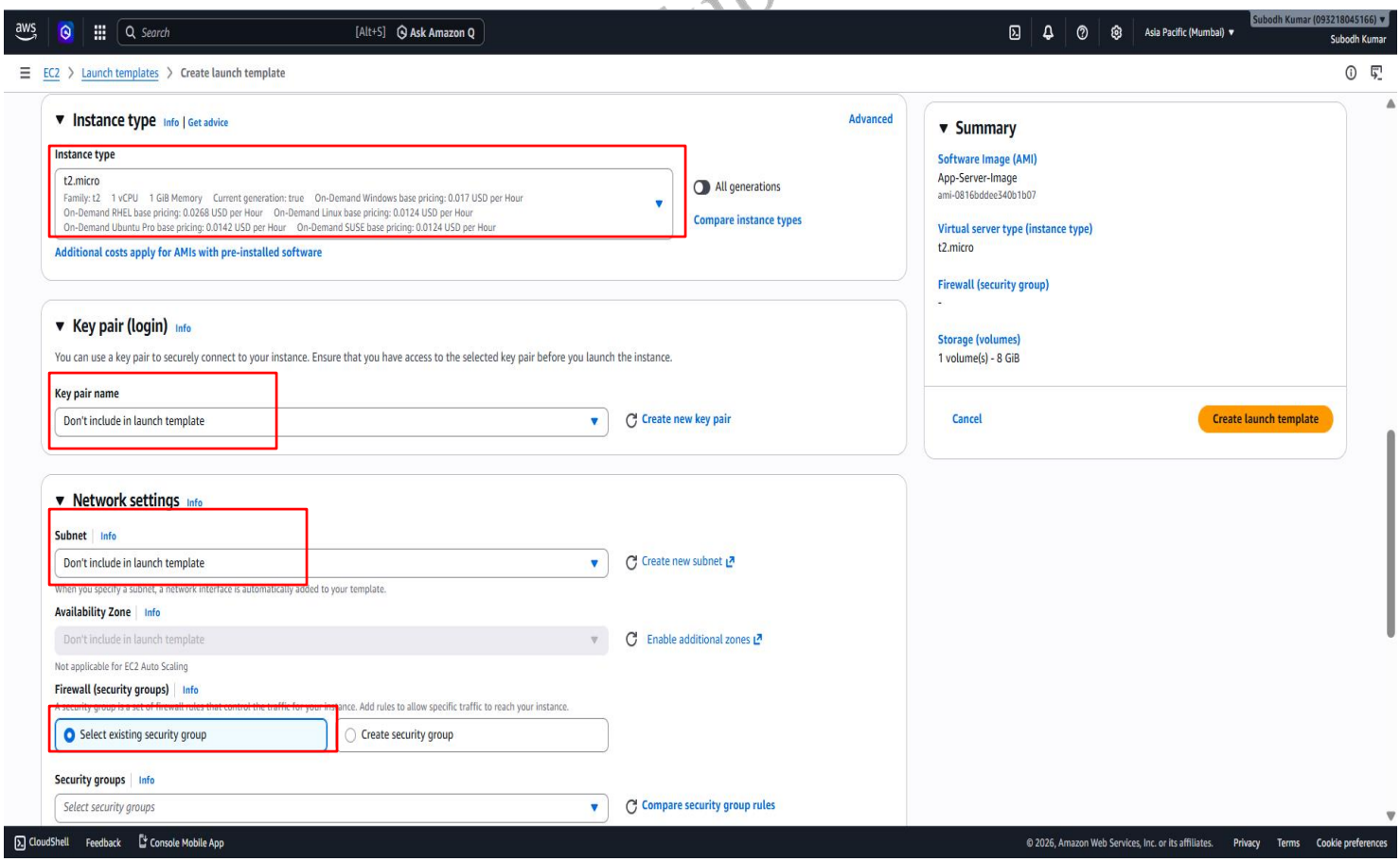
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EC2 > Launch templates > Create launch template
Instance type Info | Get advice
Advanced
Instance type
t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name
Don't include in launch template
Create new key pair
Network settings Info
Subnet Info
Don't include in launch template
Create new subnet
When you specify a subnet, a network interface is automatically added to your template.
Availability Zone Info
Don't include in launch template
Enable additional zones
Not applicable for EC2 Auto Scaling
Firewall (security groups) Info
Select existing security group
Create security group
Security groups Info
Select security groups
Compare security group rules
Summary
Software Image (AMI)
App-Server-Image
ami-0816bddee340b1b07
Virtual server type (instance type)
t2.micro
Firewall (security group)
-
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
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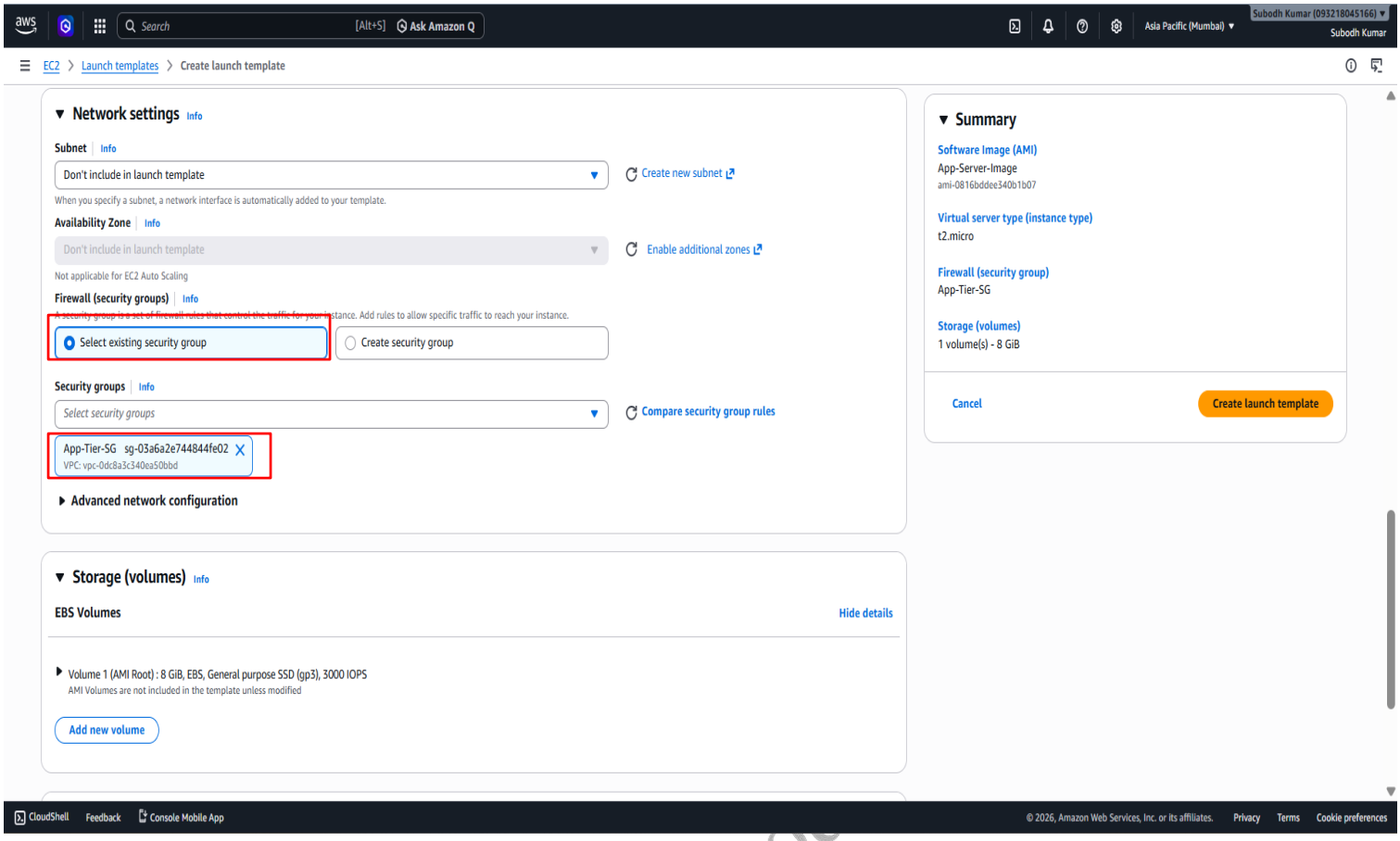
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EC2 > Launch templates > Create launch template
Network settings Info
Subnet Info
Don't include in launch template
Create new subnet
When you specify a subnet, a network interface is automatically added to your template.
Availability Zone Info
Don't include in launch template
Enable additional zones
Not applicable for EC2 Auto Scaling
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Select existing security group
Create security group
Security groups Info
Select security groups
Compare security group rules
App-Tier-SG sg-03a6a2e744844fe02
VPC: vpc-0dc8a3c340ea50bbd
Advanced network configuration
Storage (volumes) Info
EBS Volumes
Hide details
Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3), 3000 IOPS
AMI Volumes are not included in the template unless modified
Add new volume
Summary
Software Image (AMI)
App-Server-Image
ami-0816bddee340b1b07
Virtual server type (instance type)
t2.micro
Firewall (security group)
App-Tier-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
CloudShell Feedback Console Mobile App
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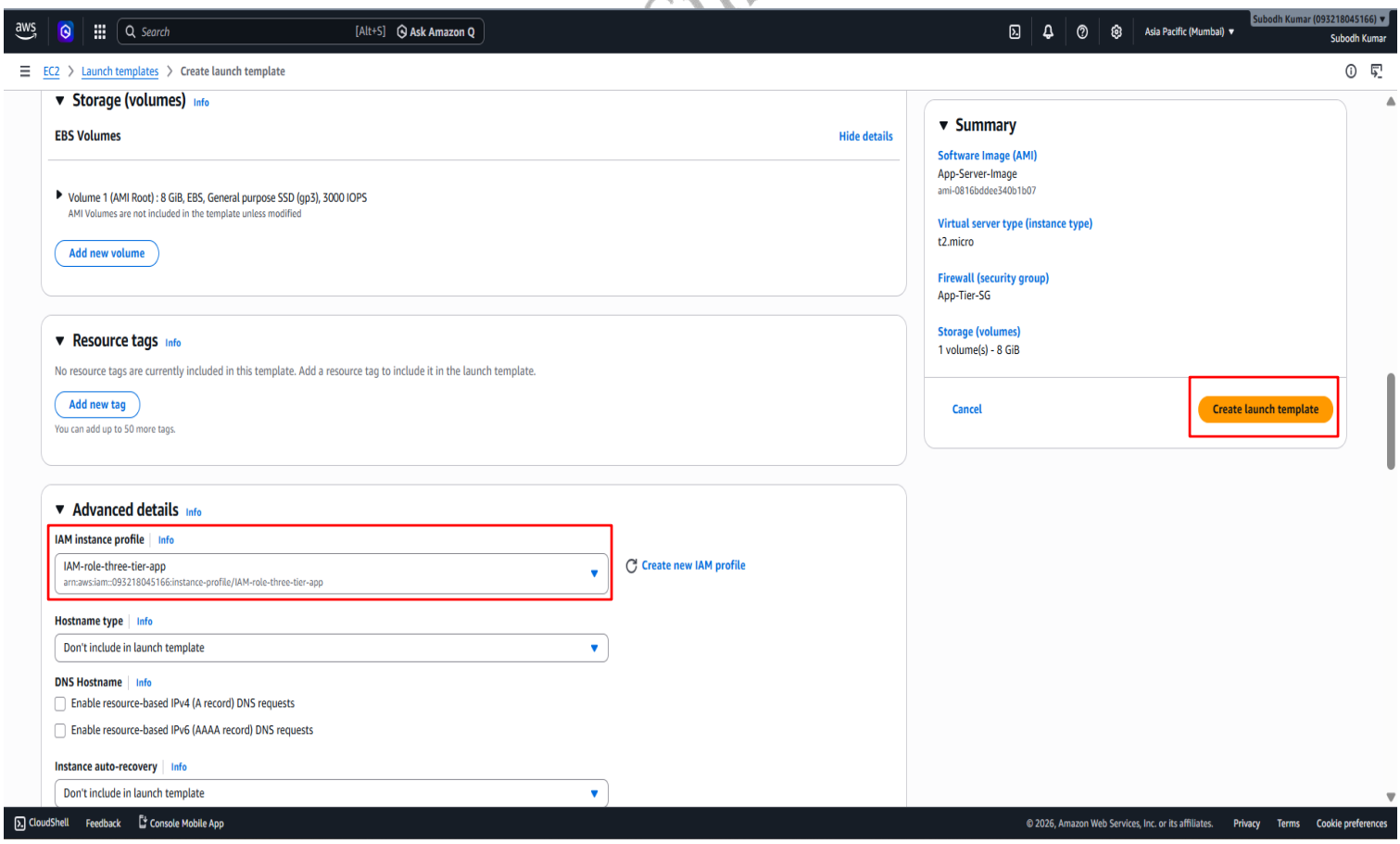
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EC2 > Launch templates > Create launch template
Storage (volumes) Info
EBS Volumes
Hide details
Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3), 3000 IOPS
AMI Volumes are not included in the template unless modified
Add new volume
Resource tags Info
No resource tags are currently included in this template. Add a resource tag to include it in the launch template.
Add new tag
You can add up to 50 more tags.
Advanced details Info
IAM instance profile Info
IAM-role-three-tier-app
arn:aws:iam::093218045166:instance-profile/IAM-role-three-tier-app
Create new IAM profile
Hostname type Info
Don't include in launch template
DNS Hostname Info
Enable resource-based IPv4 (A record) DNS requests
Enable resource-based IPv6 (AAAA record) DNS requests
Instance auto-recovery Info
Don't include in launch template
Summary
Software Image (AMI)
App-Server-Image
ami-0816bddee340b1b07
Virtual server type (instance type)
t2.micro
Firewall (security group)
App-Tier-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
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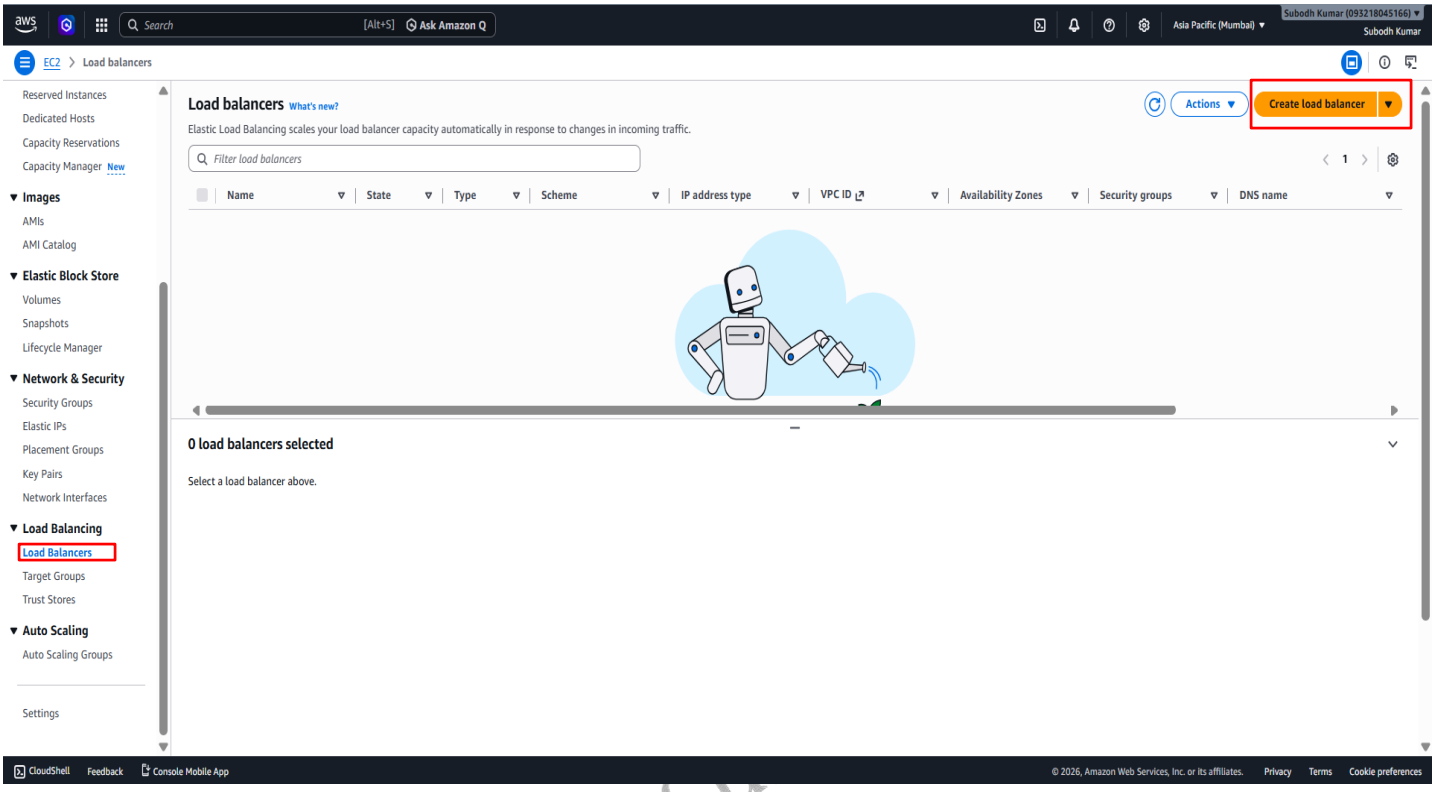
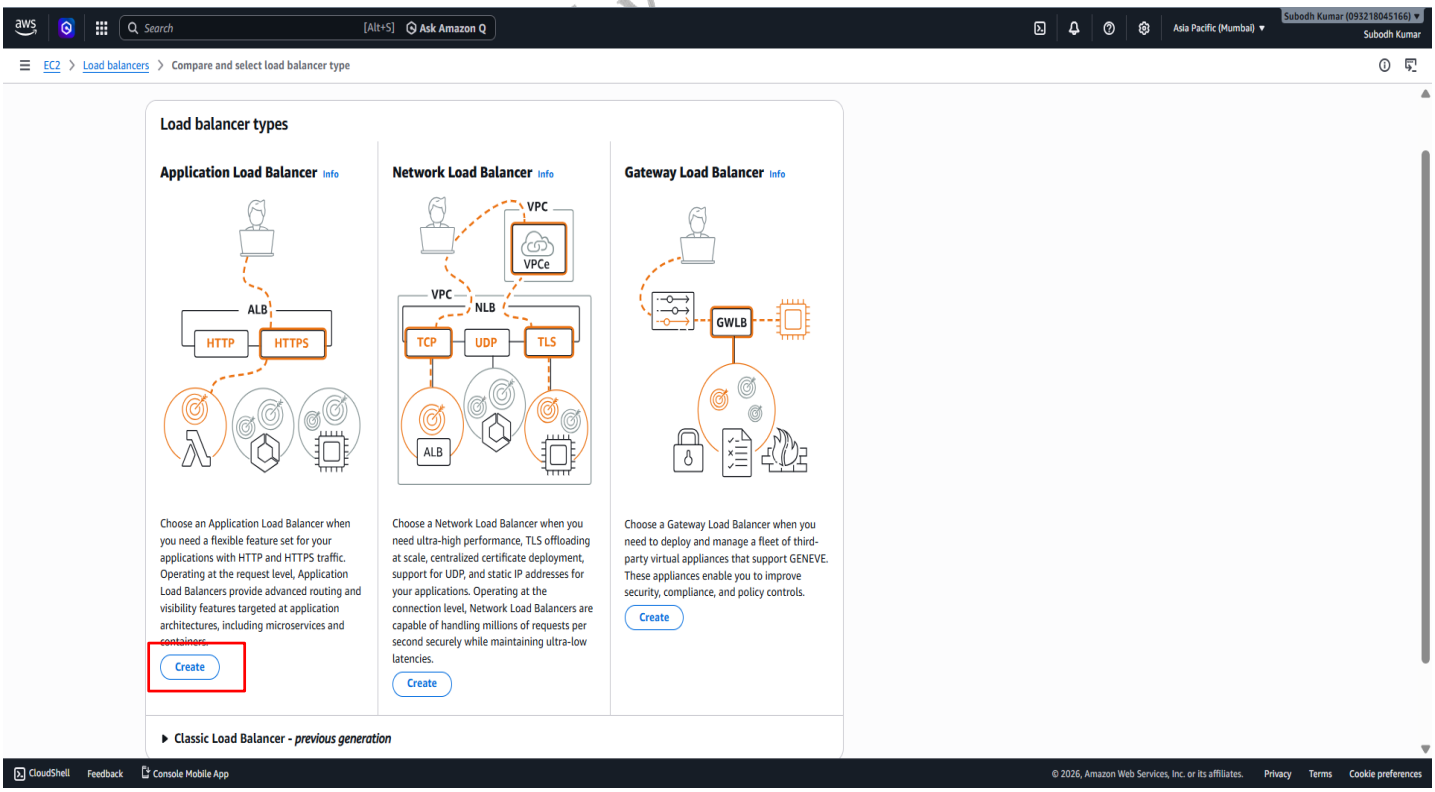
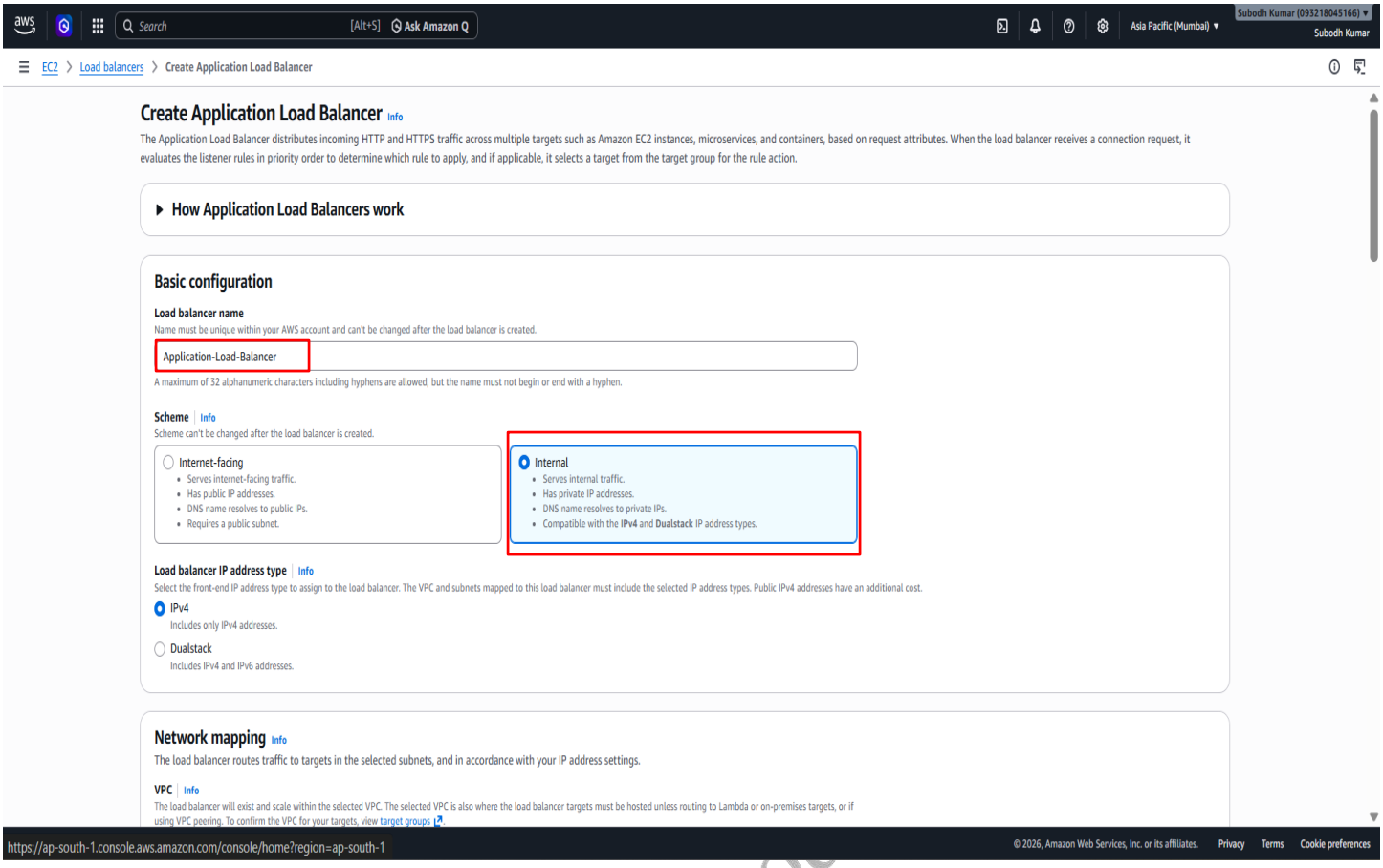
Create Application Load Balancer (Internal)

Go to EC2 > Load Balancer > Create Load Balancer create application load balancer.



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EC2 > Load balancers > Create Application Load Balancer
Create Application Load Balancer Info
The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.
How Application Load Balancers work
Basic configuration
Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
Application-Load-Balancer
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Scheme Info
Scheme can't be changed after the load balancer is created.
Internet-facing
Serves internet-facing traffic.
Has public IP addresses.
DNS name resolves to public IPs.
Requires a public subnet.
Internal
Serves internal traffic.
Has private IP addresses.
DNS name resolves to private IPs.
Compatible with the IPv4 and Dualstack IP address types.
Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.
IPv4
Includes only IPv4 addresses.
Dualstack
Includes IPv4 and IPv6 addresses.
Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.
VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups
https://ap-south-1.console.aws.amazon.com/console/home?region=ap-south-1
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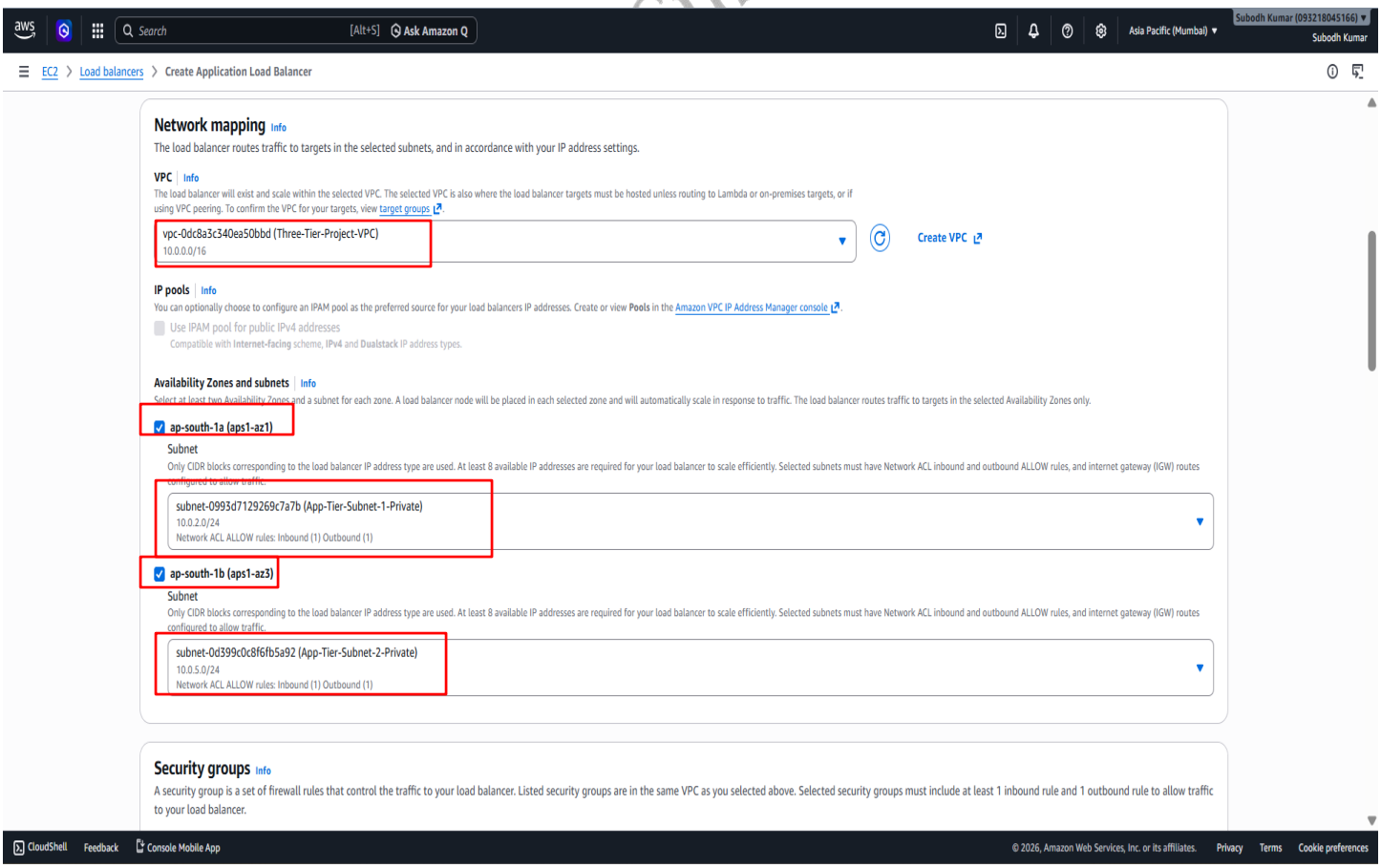
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EC2 > Load balancers > Create Application Load Balancer
Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.
VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create VPC
IP pools Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the Amazon VPC IP Address Manager console
Use IPAM pool for public IPv4 addresses
Compatible with Internet-facing scheme, IPv4 and Dualstack IP address types.
Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.
ap-south-1a (aps1-az1)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-0993d7129269c7a7b (App-Tier-Subnet-1-Private)
10.0.2.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1)
ap-south-1b (aps1-az3)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-0d399c0c8f6fb5a92 (App-Tier-Subnet-2-Private)
10.0.5.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1)
Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.
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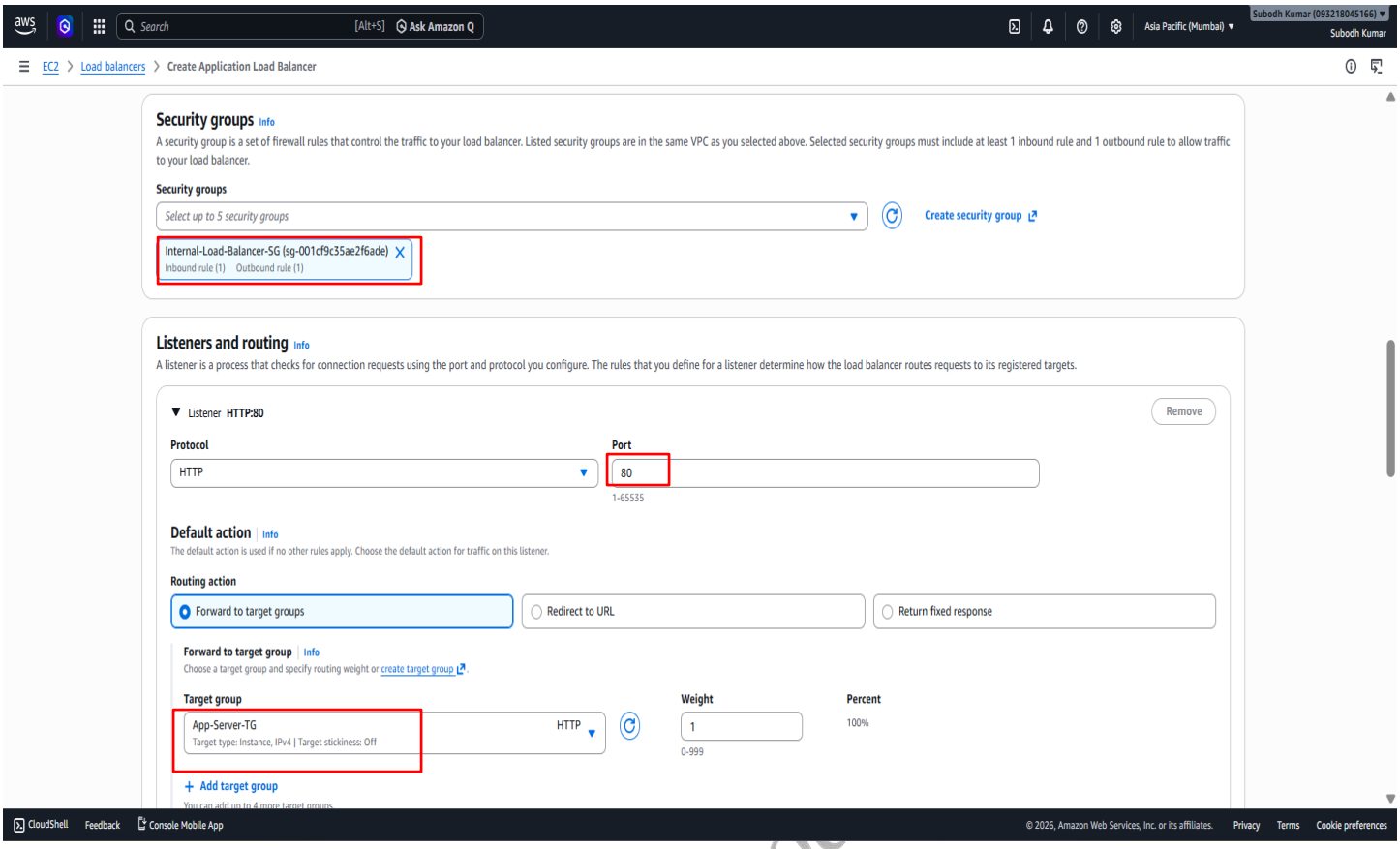
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Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Load balancers > Create Application Load Balancer
Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.
Security groups
Select up to 5 security groups
Create security group
Internal-Load-Balancer-SG (sg-001cf9c35ae2f6ade)
Inbound rule (1) Outbound rule (1)
Listeners and routing Info
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.
Listener HTTP:80
Remove
Protocol
HTTP
Port
80
1-65535
Default action Info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.
Routing action
Forward to target groups
Redirect to URL
Return fixed response
Forward to target group Info
Choose a target group and specify routing weight or create target group.
Target group
App-Server-TG
Target type: Instance, IPv4 | Target stickiness: Off
HTTP
Weight
1
0-999
Percent
100%
Add target group
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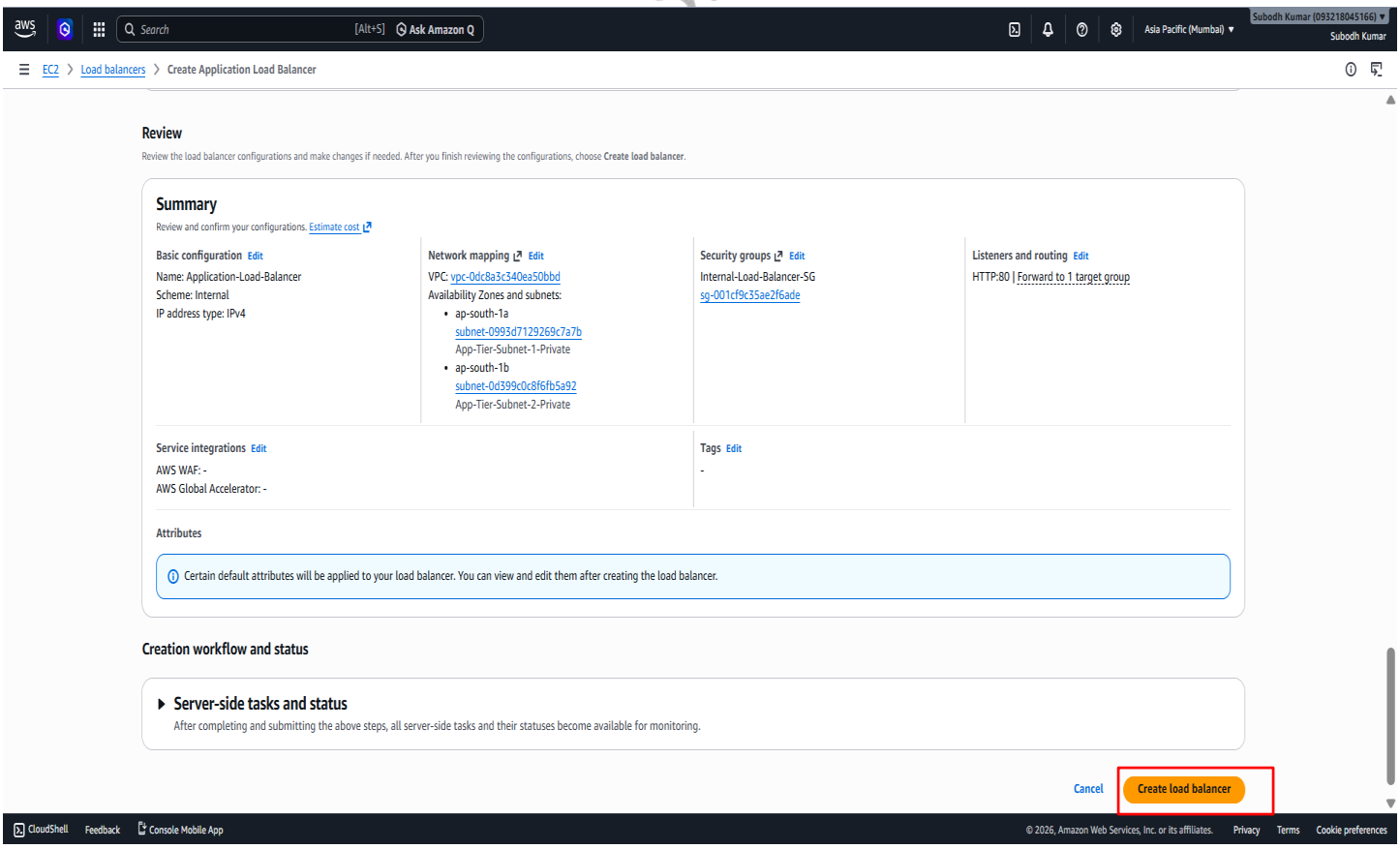
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Subodh Kumar
EC2 > Load balancers > Create Application Load Balancer
Review
Review the load balancer configurations and make changes if needed. After you finish reviewing the configurations, choose Create load balancer.
Summary
Review and confirm your configurations. Estimate cost
Basic configuration Edit
Name: Application-Load-Balancer
Scheme: Internal
IP address type: IPv4
Network mapping Edit
VPC: vpc-0dc8a3c340ea50bbd
Availability Zones and subnets:
ap-south-1a
subnet-0993d7129269c7a7b
App-Tier-Subnet-1-Private
ap-south-1b
subnet-0d399c0c8f6fb5a92
App-Tier-Subnet-2-Private
Security groups Edit
Internal-Load-Balancer-SG
sg-001cf9c35ae2f6ade
Listeners and routing Edit
HTTP:80 | Forward to 1 target group
Service integrations Edit
AWS WAF: -
AWS Global Accelerator: -
Tags Edit
-
Attributes
Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.
Creation workflow and status
Server-side tasks and status
After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.
Cancel
Create load balancer
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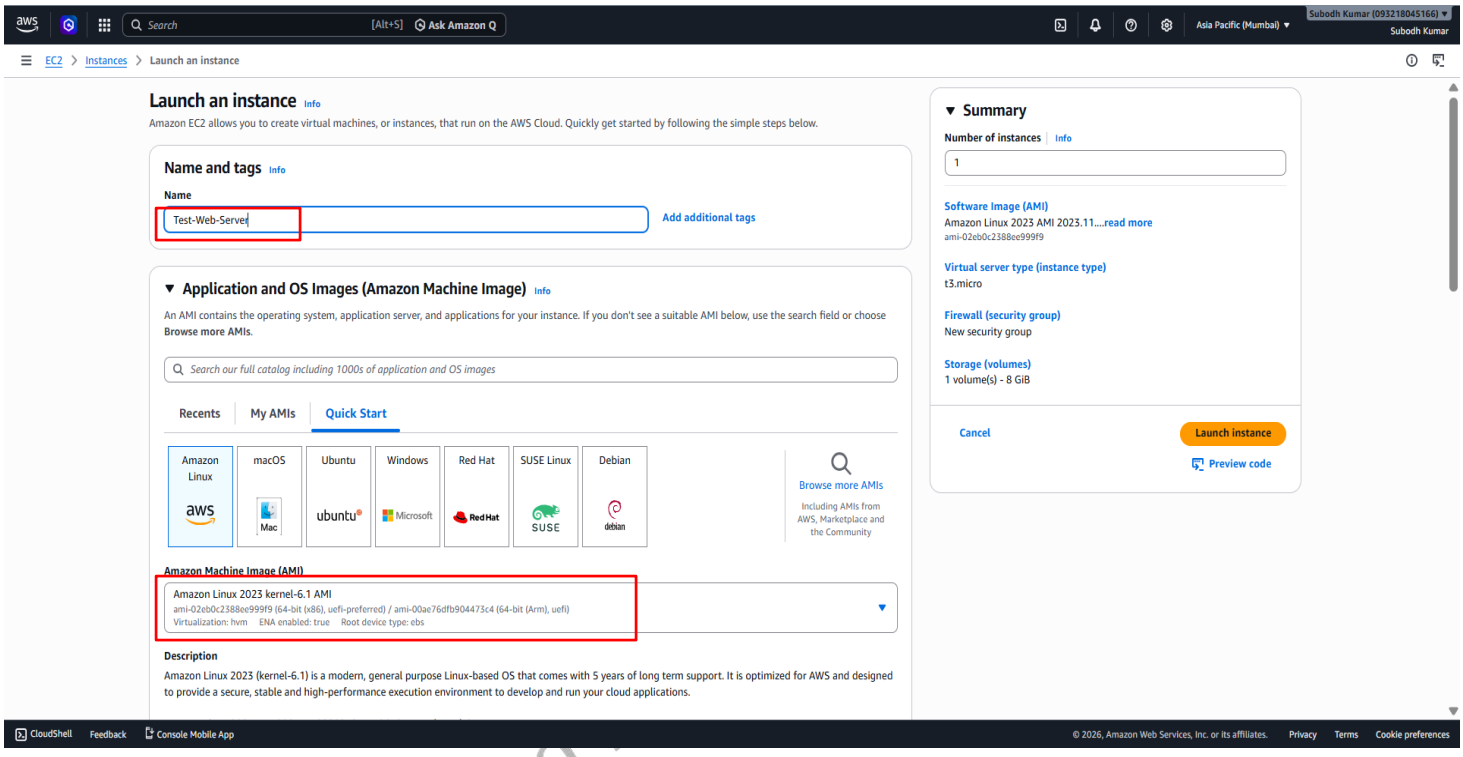
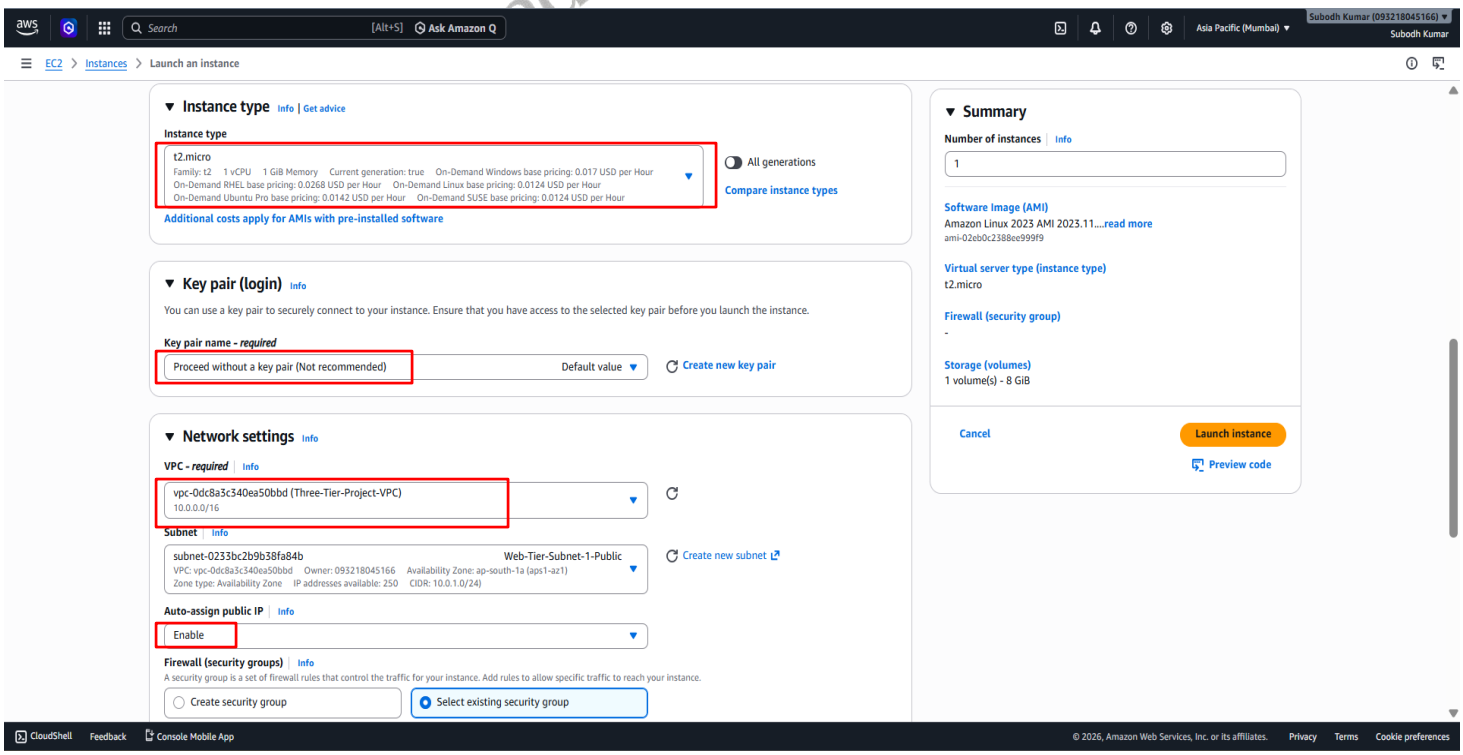
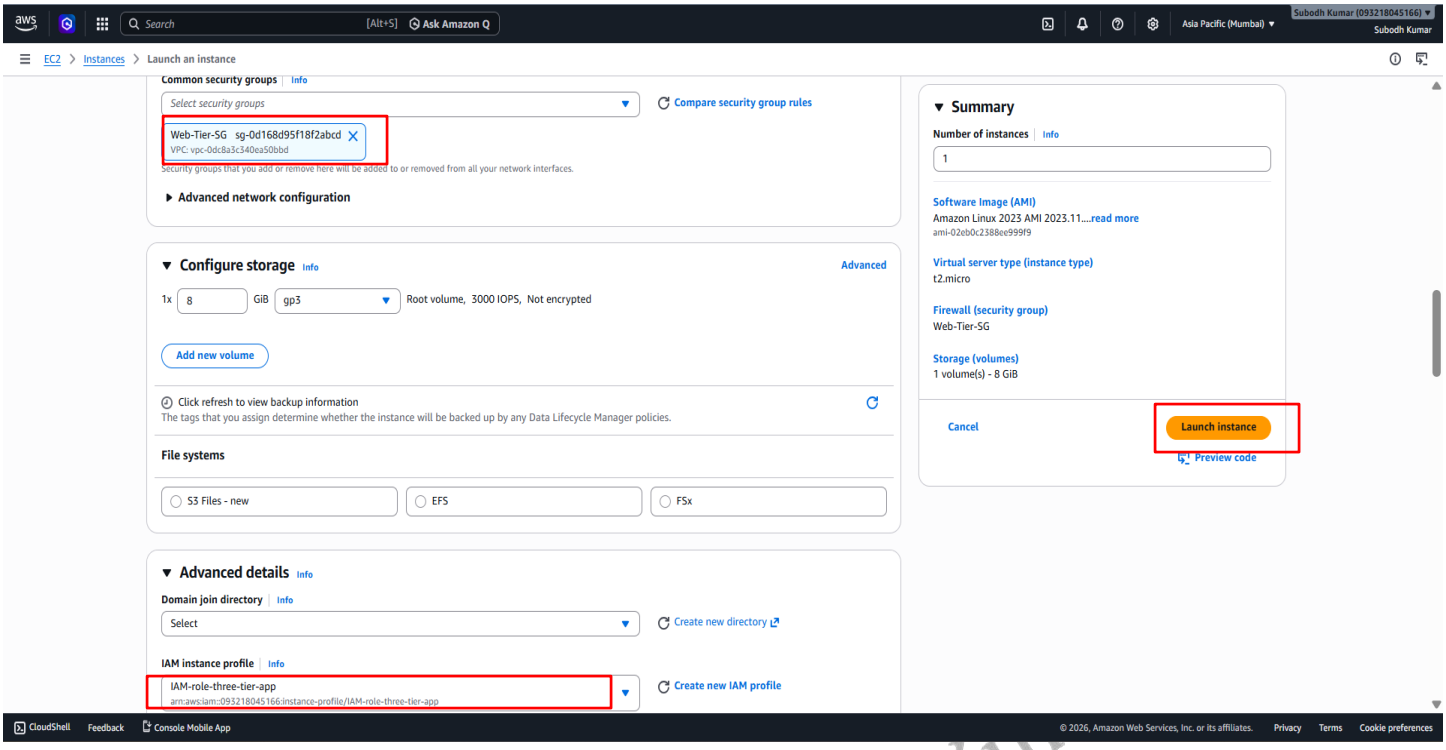
10. Create Test Web Server and setup it

We have to follow the same process as followed in App server. Only **EC2 configuration** part **is different**. Configurations will be given below.

Now got EC2 instance > Launch instance and create a instance for Test Web Server. Change name (Test-Web-Server), VPC, Subnet(Web-Tier-Subnet-1-Public), Security group (Web-Tier-SG) and also select IAM role under advanced details. Finally, Launch instance.

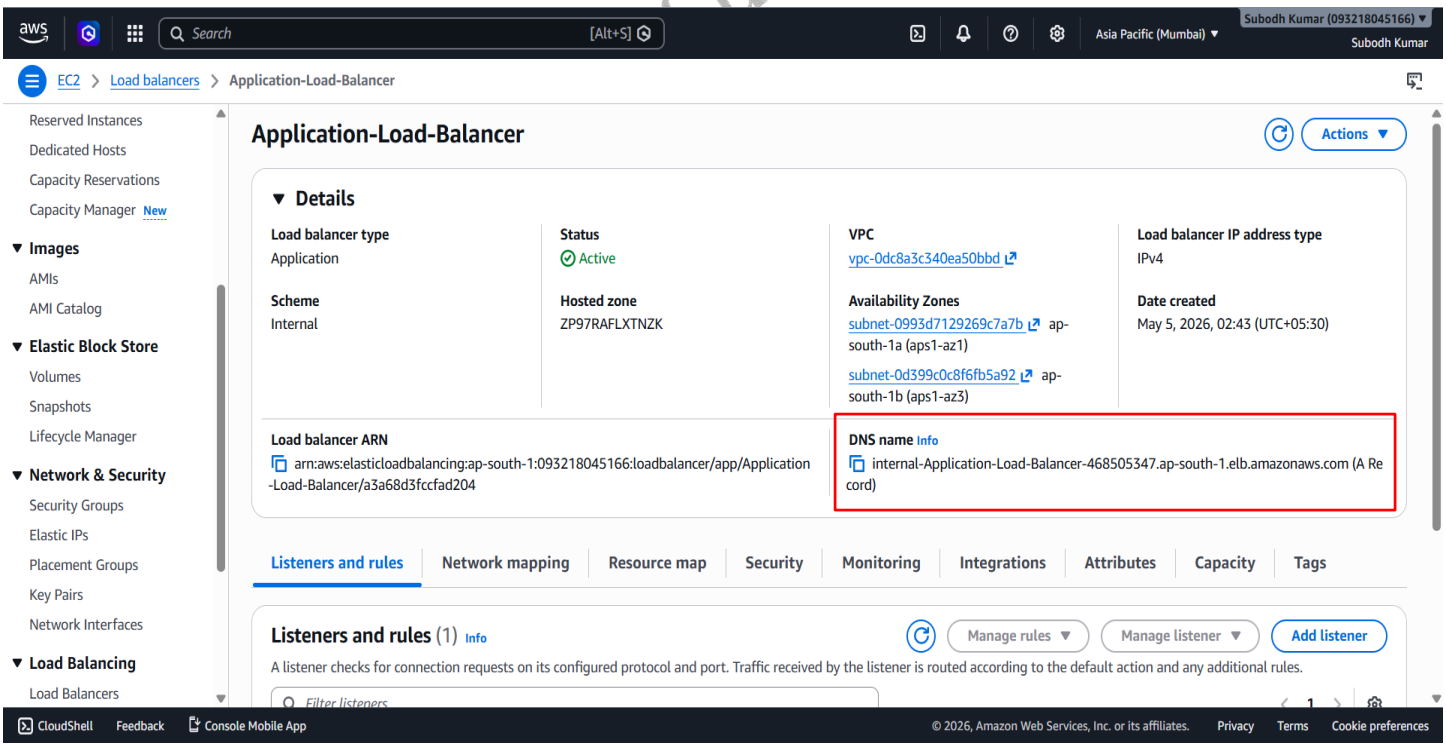




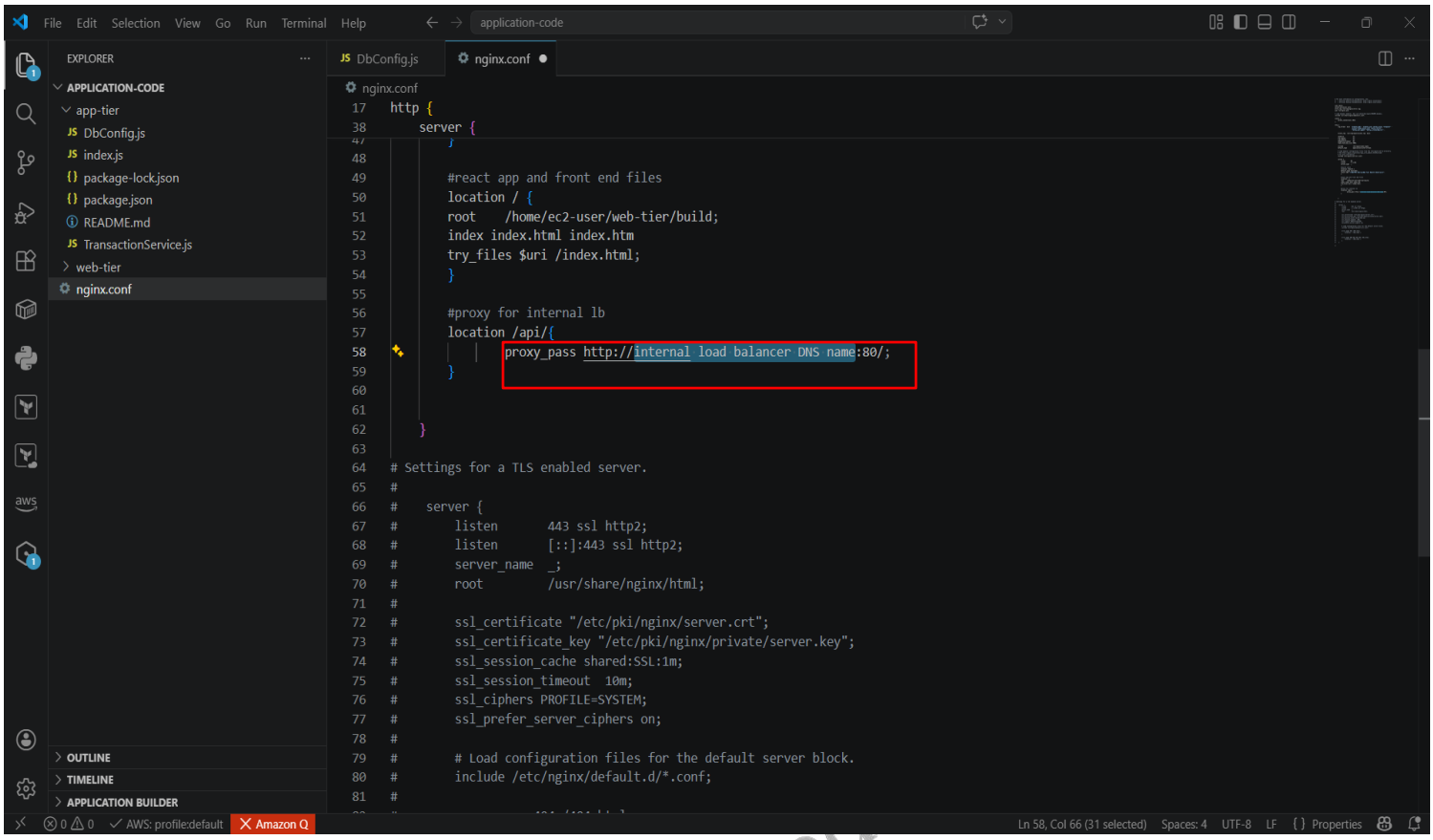


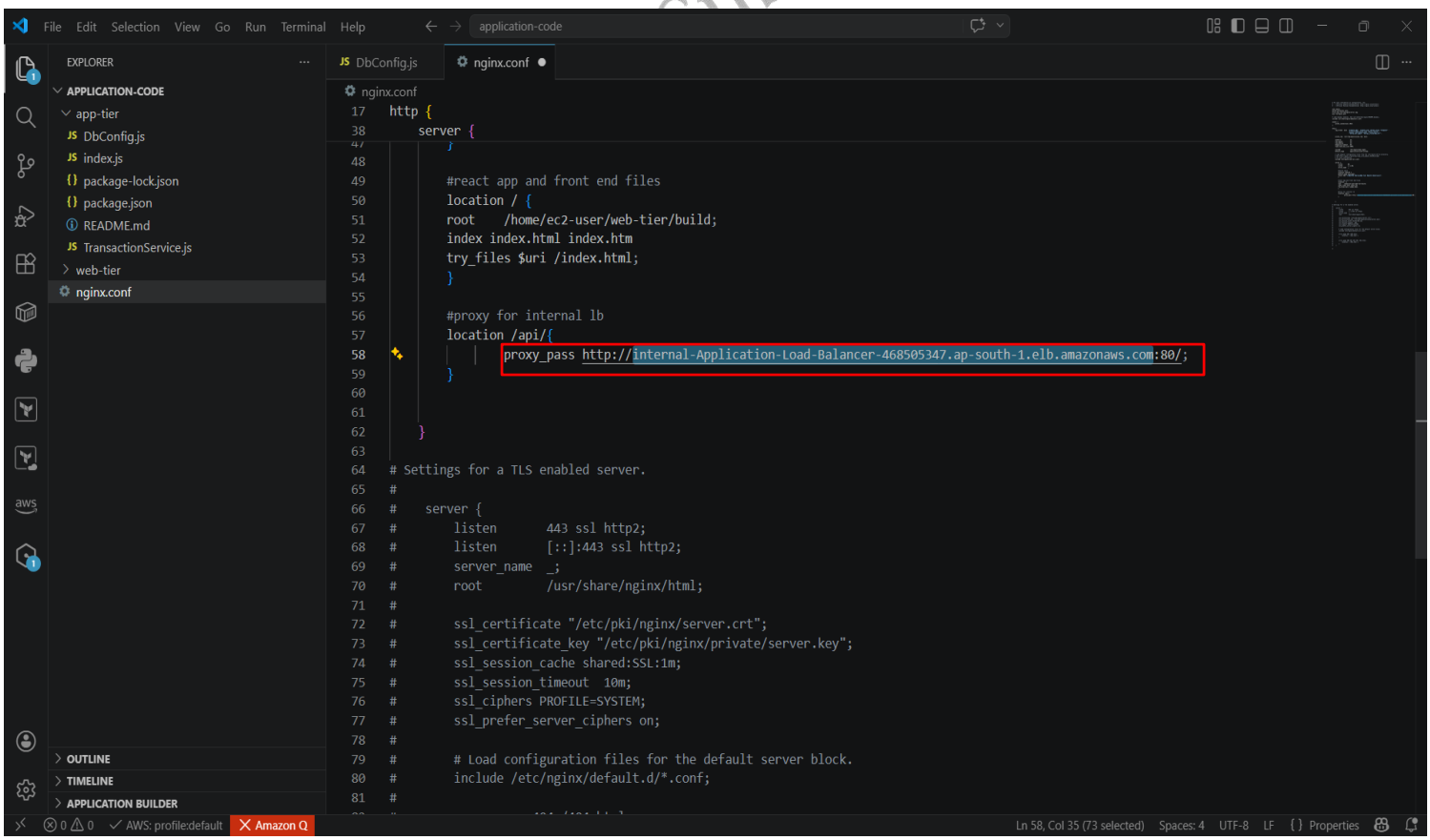
First go to the source code folder in your PC. Open **nginx.conf** file in an editor

Now go to the **internal load balancer** we created and copy the **DNS name** of it.



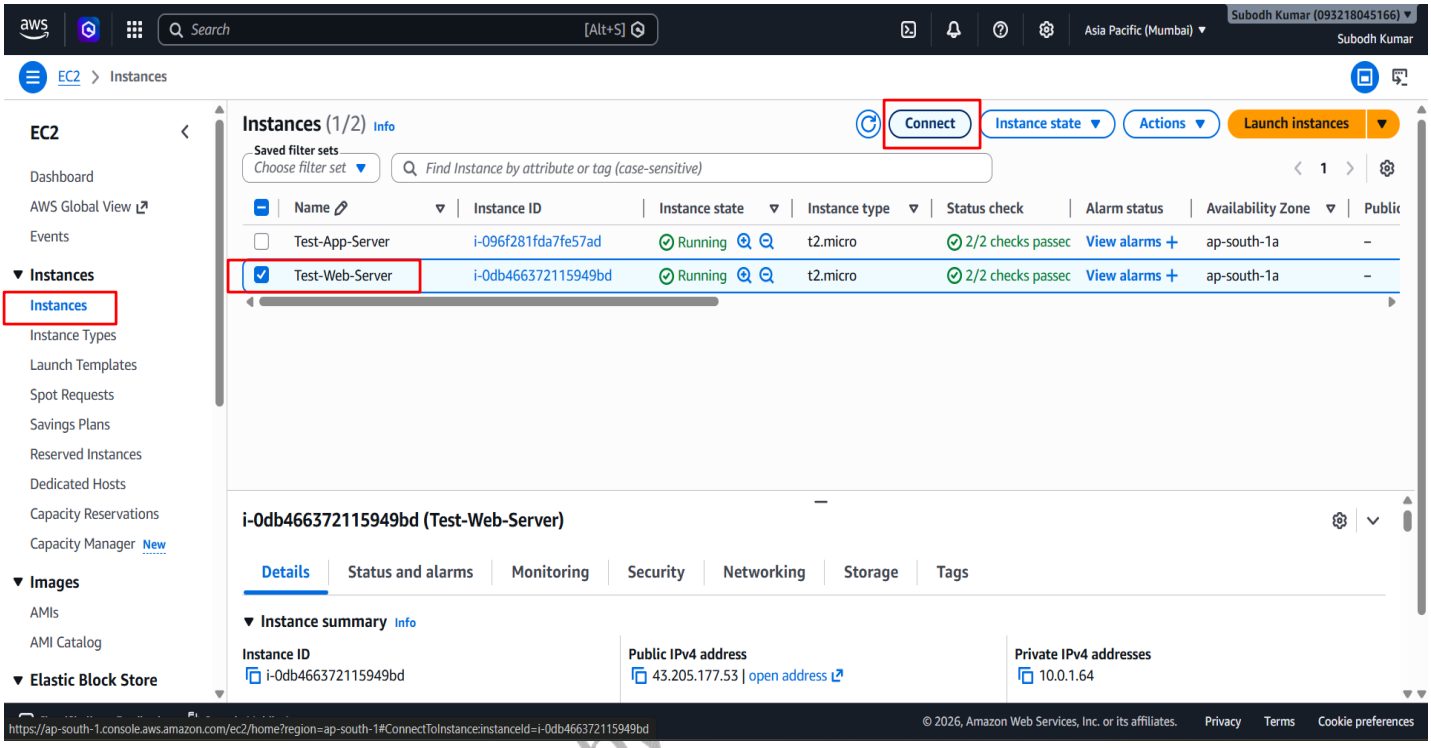
Go to the **nginx.conf** file and paste it in the following section.

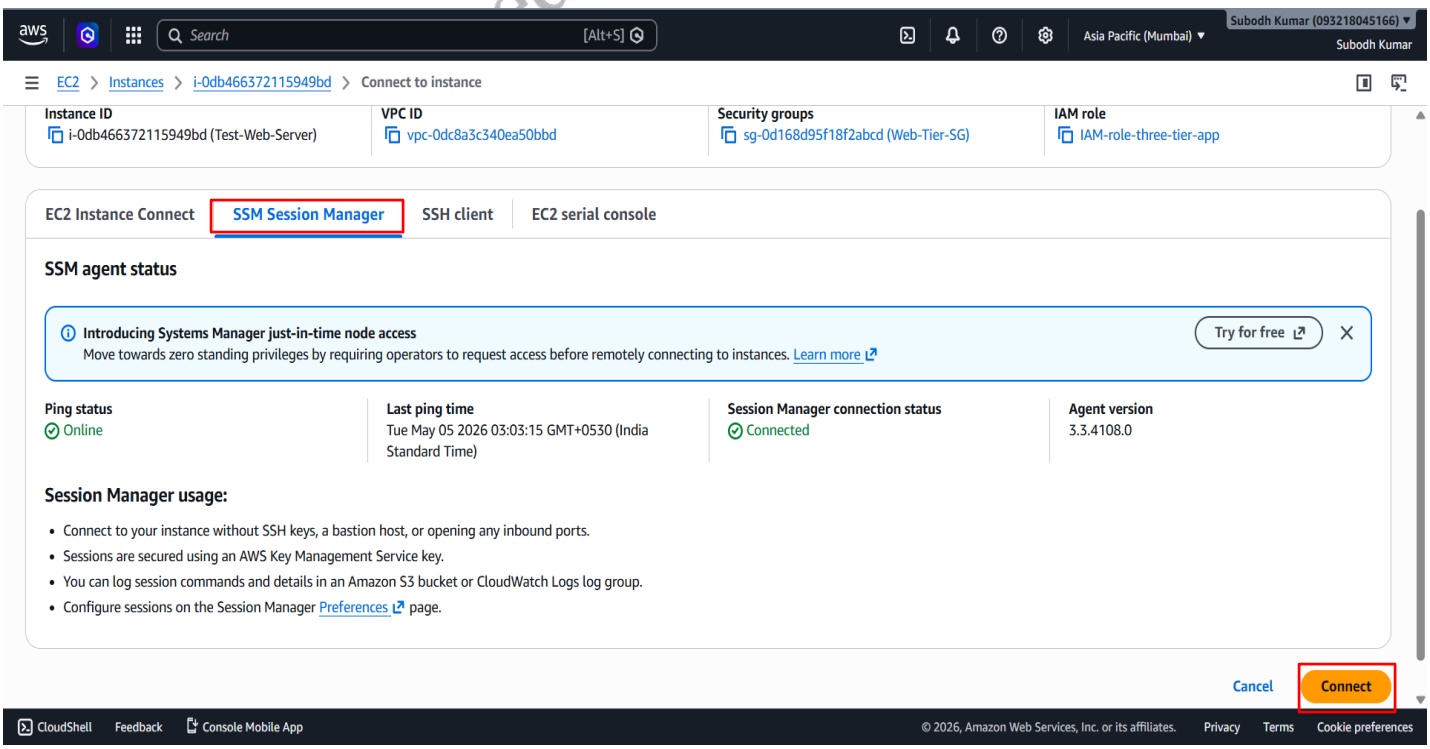

File Edit Selection View Go Run Terminal Help
application-code
EXPLORER
APPLICATION-CODE
app-tier
DbConfig.js
index.js
package-lock.json
package.json
README.md
TransactionService.js
web-tier
nginx.conf
DbConfig.js
nginx.conf
nginx.conf
http {
server {
#react app and front end files
location / {
root /home/ec2-user/web-tier/build;
index index.html index.htm
try_files $uri /index.html;
}
#proxy for internal lb
location /api/{
proxy_pass http://internal load balancer DNS name:80/;
}
}
# Settings for a TLS enabled server.
#
server {
listen 443 ssl http2;
listen [::]:443 ssl http2;
server_name _;
root /usr/share/nginx/html;
#
ssl_certificate "/etc/pki/nginx/server.crt";
ssl_certificate_key "/etc/pki/nginx/private/server.key";
ssl_session_cache shared:SSL:1m;
ssl_session_timeout 10m;
ssl_ciphers PROFILE=SYSTEM;
ssl_prefer_server_ciphers on;
#
Load configuration files for the default server block.
include /etc/nginx/default.d/*.conf;
#
OUTLINE
TIMELINE
APPLICATION BUILDER
AWS: profile:default
Amazon Q
Ln 58, Col 66 (31 selected)
Spaces: 4
UTF-8
LF
Properties


File Edit Selection View Go Run Terminal Help
application-code
EXPLORER
APPLICATION-CODE
app-tier
DbConfig.js
index.js
package-lock.json
package.json
README.md
TransactionService.js
web-tier
nginx.conf
DbConfig.js
nginx.conf
nginx.conf
http {
server {
#react app and front end files
location / {
root /home/ec2-user/web-tier/build;
index index.html index.htm
try_files $uri /index.html;
}
#proxy for internal lb
location /api/{
proxy_pass http://internal-Application-Load-Balancer-468505347.ap-south-1.elb.amazonaws.com:80/;
}
}
# Settings for a TLS enabled server.
#
server {
listen 443 ssl http2;
listen [::]:443 ssl http2;
server_name _;
root /usr/share/nginx/html;
#
ssl_certificate "/etc/pki/nginx/server.crt";
ssl_certificate_key "/etc/pki/nginx/private/server.key";
ssl_session_cache shared:SSL:1m;
ssl_session_timeout 10m;
ssl_ciphers PROFILE=SYSTEM;
ssl_prefer_server_ciphers on;
#
Load configuration files for the default server block.
include /etc/nginx/default.d/*.conf;
#
OUTLINE
TIMELINE
APPLICATION BUILDER
AWS: profile:default
Amazon Q
Ln 58, Col 35 (73 selected)
Spaces: 4
UTF-8
LF
Properties

Now go to S3 bucket and delete the **nginx.conf** file and upload the new one.

Now Connect to the instance (**Test-Web-Server**) and do configurations as follows.





Switch users

```
sudo -su ec2-user

cd /home/ec2-user
```

Copy content from the S3 bucket

```
sudo aws s3 cp s3://<S3_BUCKET_NAME>/application-code/web-tier web-tier –recursive

sudo chown -R ec2-user:ec2-user /home/ec2-user

sudo chmod -R 755 /home/ec2-user
```

Prepared by Subodh Kumar

Install Node js

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install.sh | bash

source ~/.bashrc

nvm install 16

nvm use 16

cd /home/ec2-user/web-tier

npm install
```

Build the app

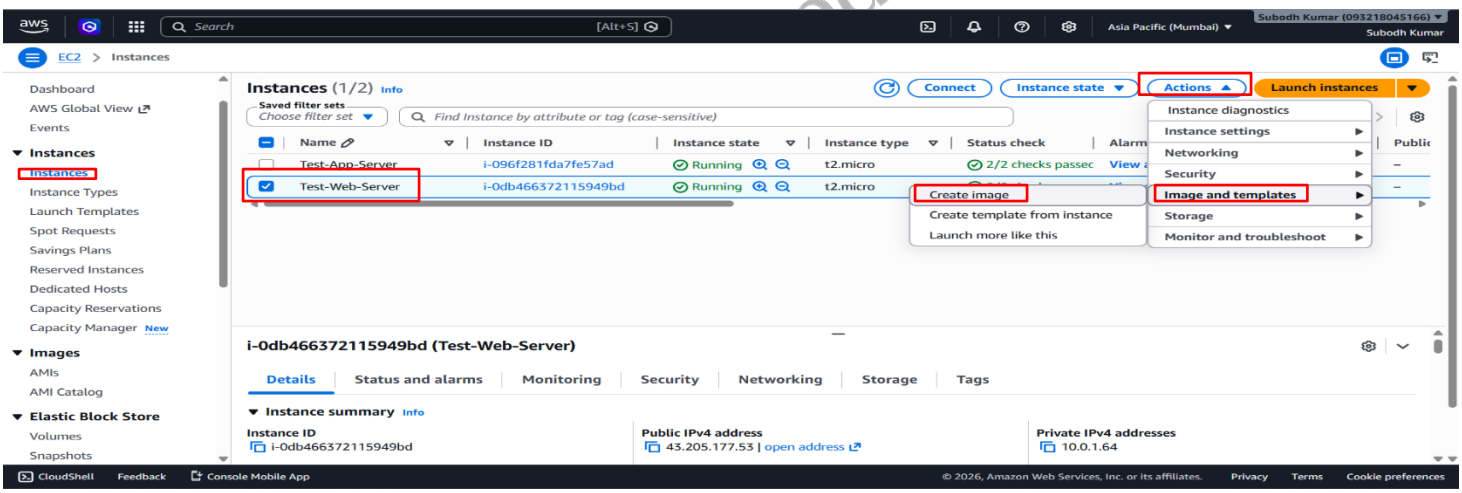
```
cd /home/ec2-user/web-tier

npm run build
```

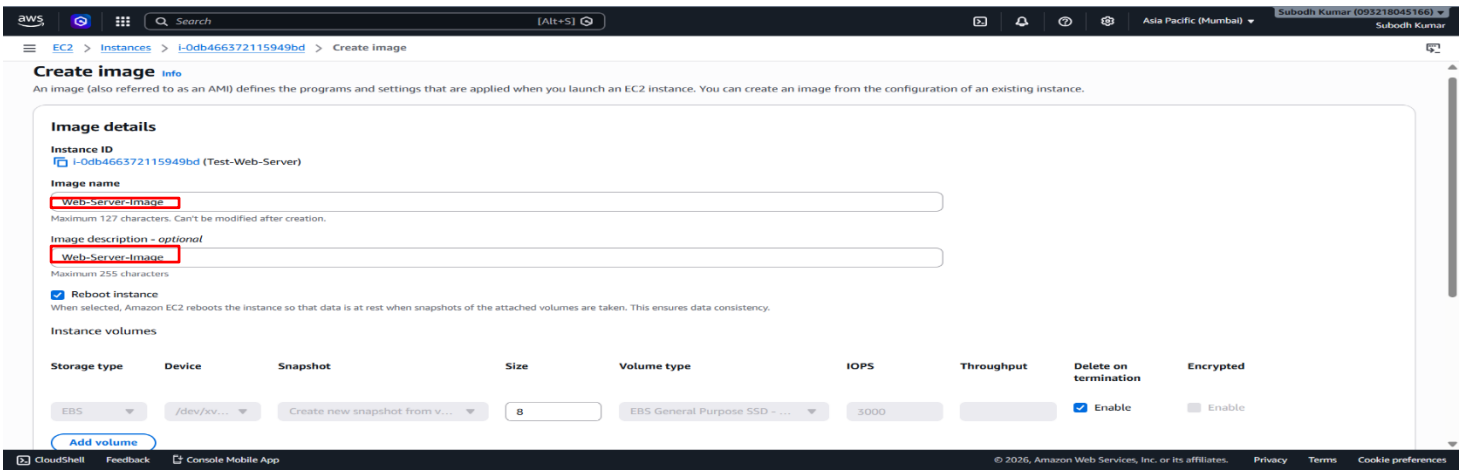
Install nginx (Webserver)

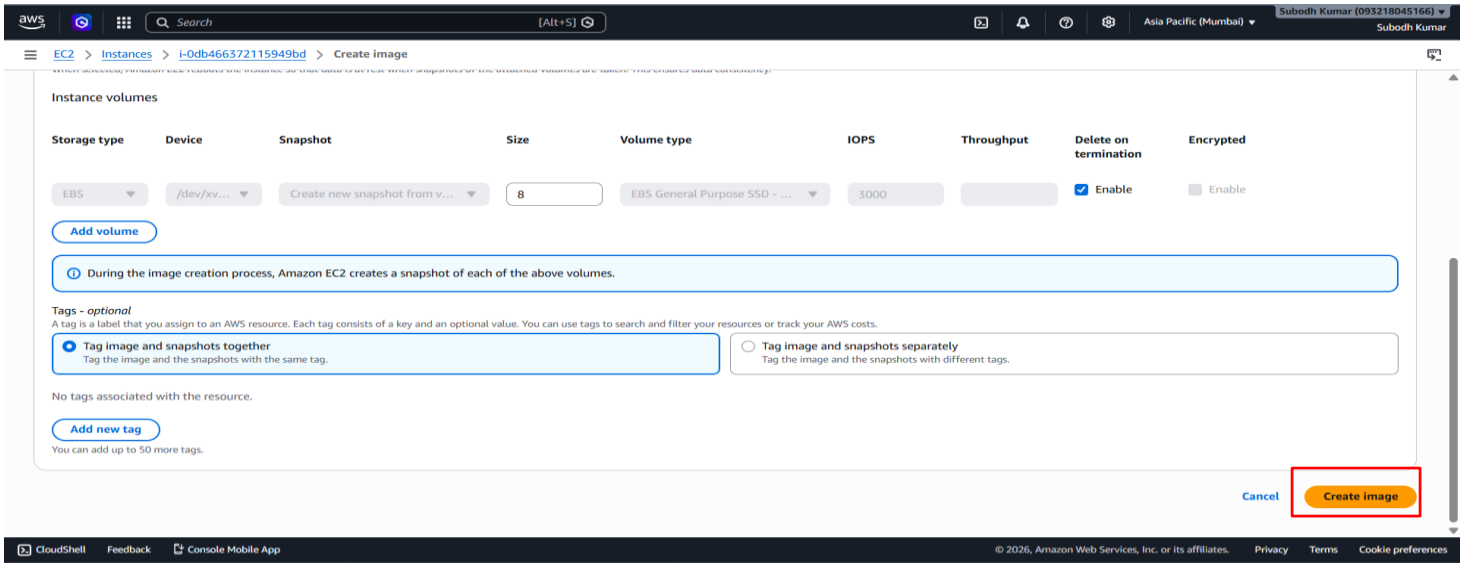
```
sudo yum install nginx -y

cd /etc/nginx

sudo mv nginx.conf nginx-backup.conf


sudo aws s3 cp s3://<S3_BUCKET_NAME>/application-code/nginx.conf .

sudo chmod -R 755 /home/ec2-user

sudo service nginx restart

sudo chkconfig nginx on
```

11. Create AMI of Web-Server

Go to instances and select Test-Web-Server > Actions > Image and templates > Create image.







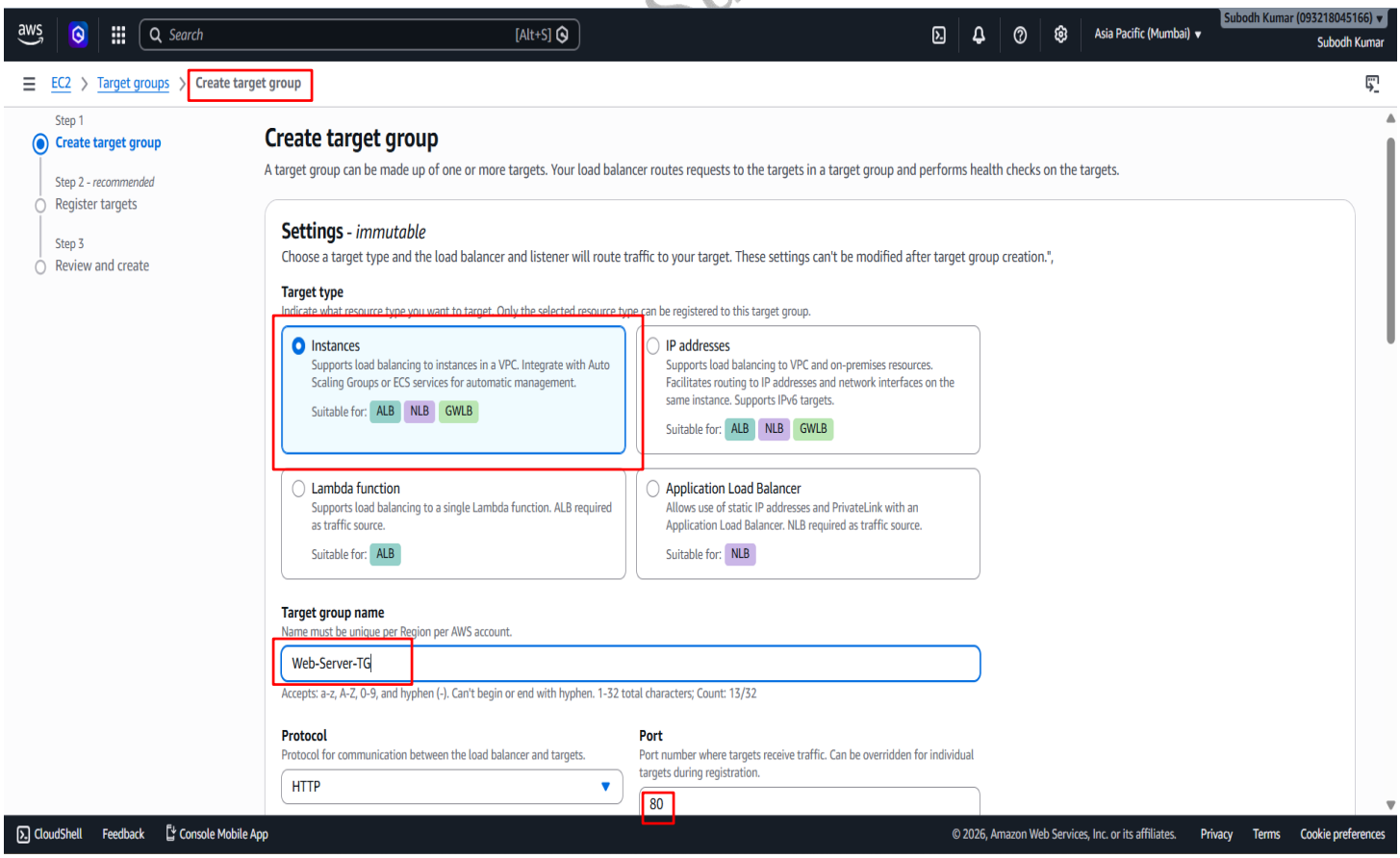
12. Create Target Group, Launch Template, External Load Balancer for Web Tier

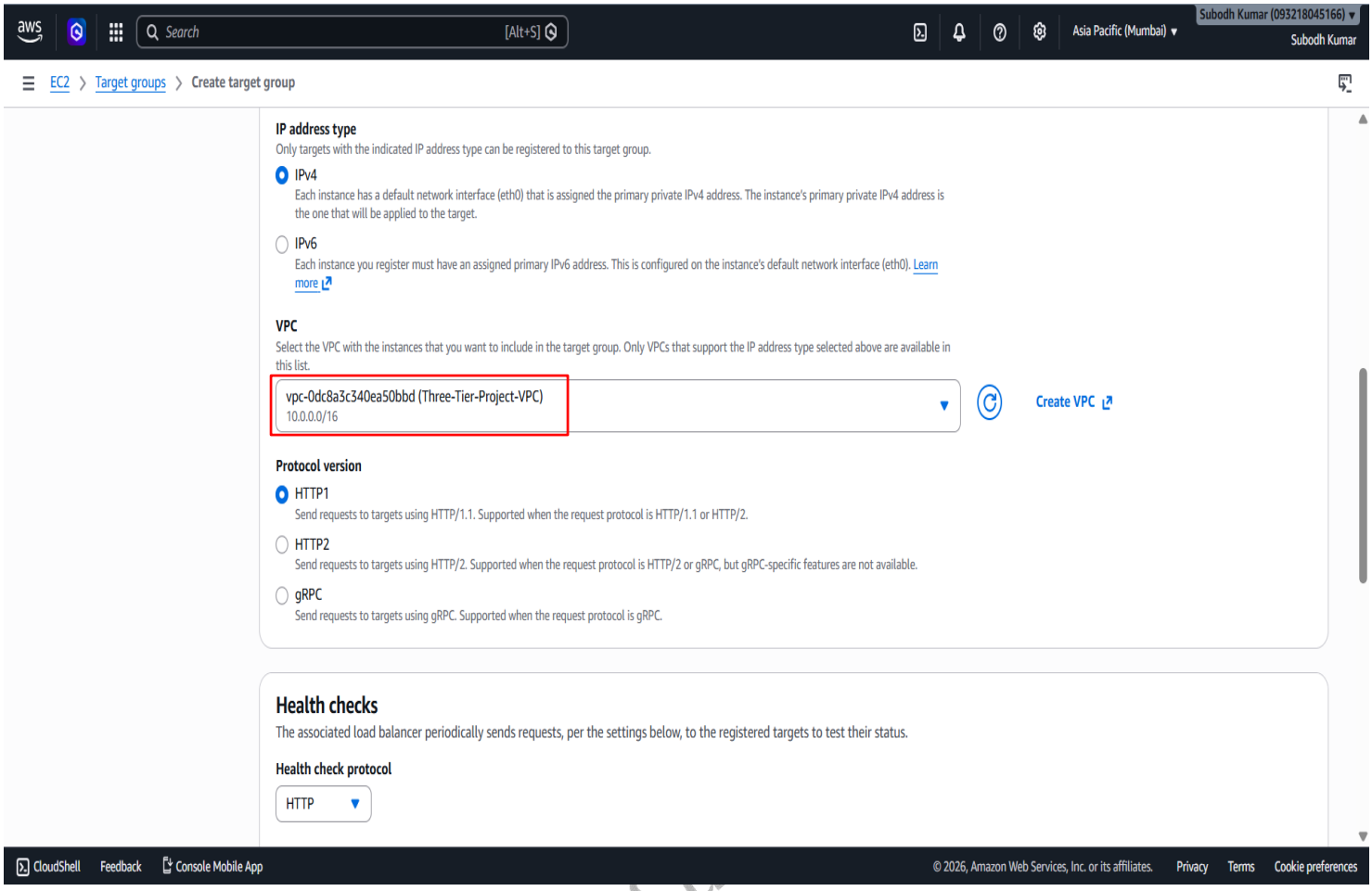
Same process

Subodh Kumar

Create Target Group

Change Protocol: Port to 80. Select VPC. Give name. Create.




Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Target groups > Create target group
IP address type
Only targets with the indicated IP address type can be registered to this target group.
IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). Learn more
VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create VPC
Protocol version
HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.
gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.
Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.
Health check protocol
HTTP
CloudShell
Feedback
Console Mobile App
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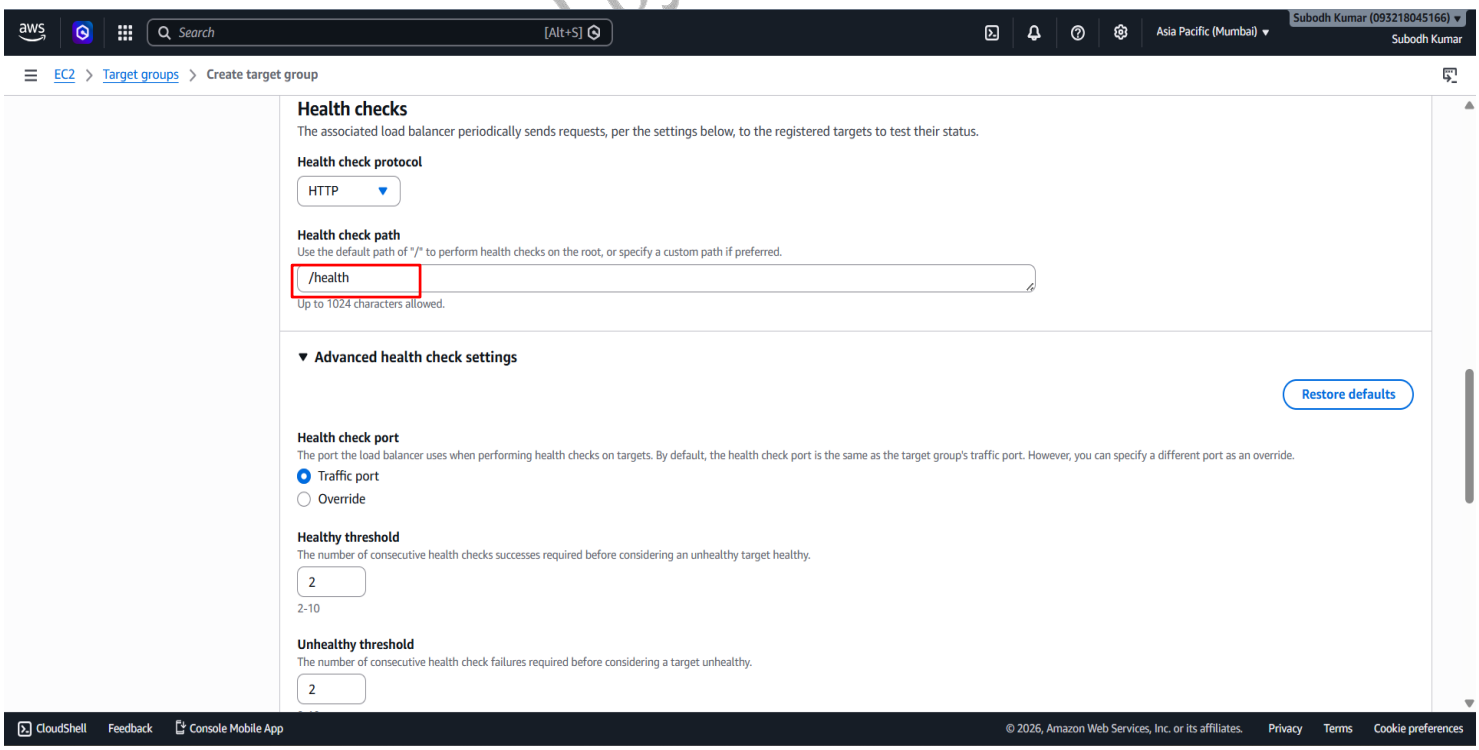
Search [Alt+S]
Asia Pacific (Mumbai)
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Subodh Kumar
EC2 > Target groups > Create target group
Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.
Health check protocol
HTTP
Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/health
Up to 1024 characters allowed.
Advanced health check settings
Restore defaults
Health check port
The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.
Traffic port
Override
Healthy threshold
The number of consecutive health checks successes required before considering an unhealthy target healthy.
2
2-10
Unhealthy threshold
The number of consecutive health check failures required before considering a target unhealthy.
2
CloudShell
Feedback
Console Mobile App
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Privacy
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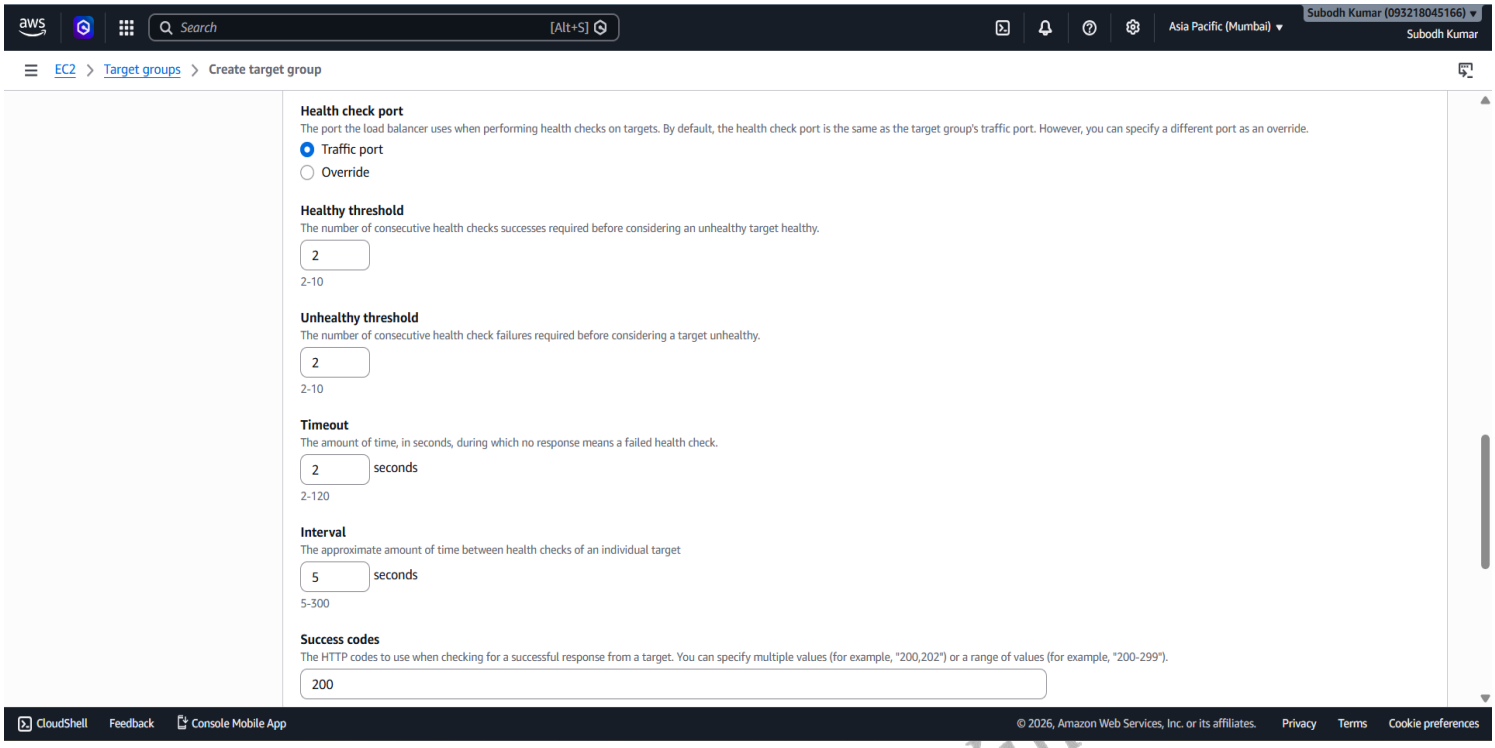
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Asia Pacific (Mumbai)
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Subodh Kumar
EC2 > Target groups > Create target group
Health check port
The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.
Traffic port
Override
Healthy threshold
The number of consecutive health checks successes required before considering an unhealthy target healthy.
2
2-10
Unhealthy threshold
The number of consecutive health check failures required before considering a target unhealthy.
2
2-10
Timeout
The amount of time, in seconds, during which no response means a failed health check.
2
seconds
2-120
Interval
The approximate amount of time between health checks of an individual target
5
seconds
5-300
Success codes
The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").
200
CloudShell
Feedback
Console Mobile App
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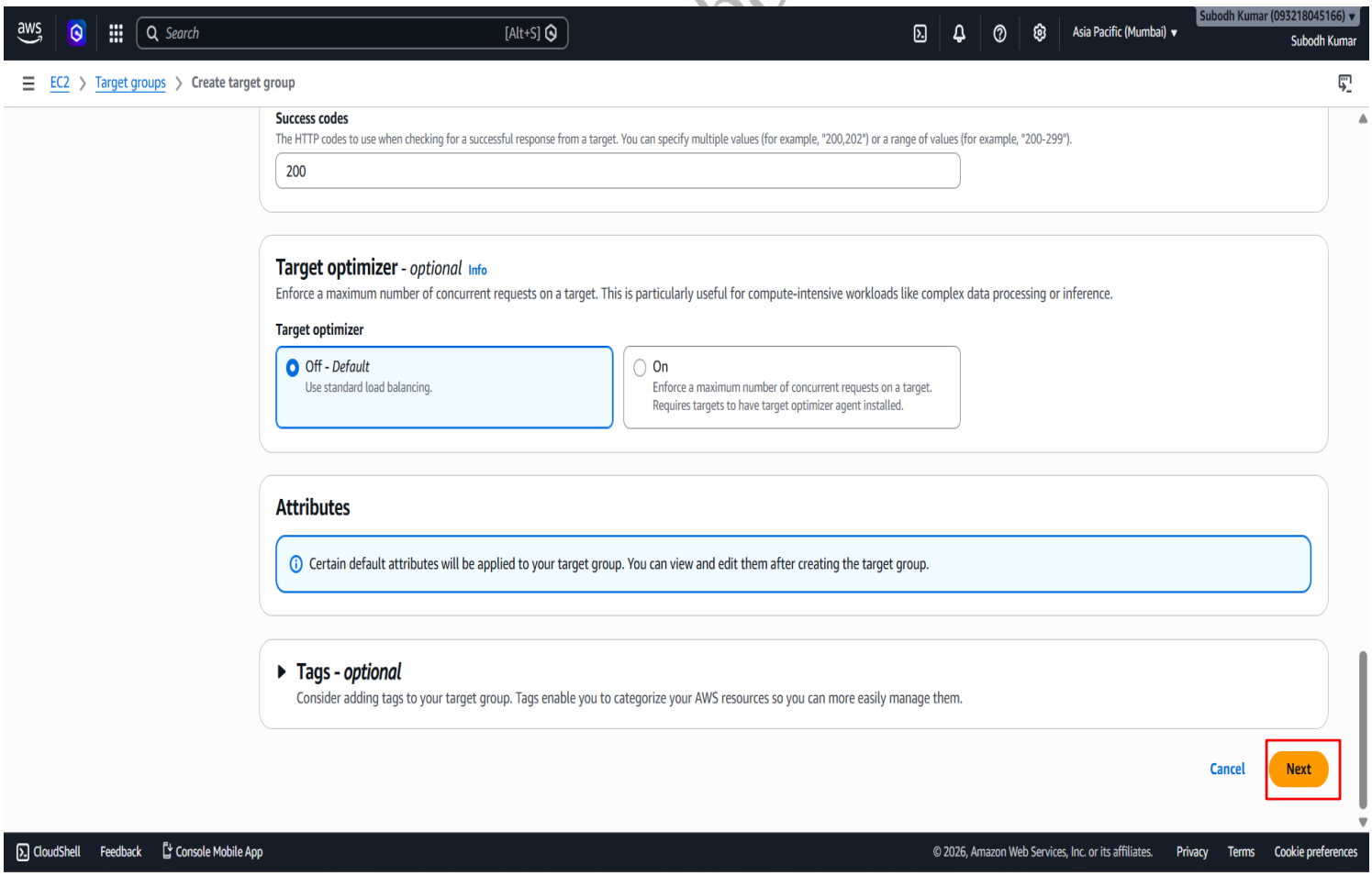
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Target groups > Create target group
Success codes
The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").
200
Target optimizer - optional Info
Enforce a maximum number of concurrent requests on a target. This is particularly useful for compute-intensive workloads like complex data processing or inference.
Target optimizer
Off - Default
Use standard load balancing.
On
Enforce a maximum number of concurrent requests on a target.
Requires targets to have target optimizer agent installed.
Attributes
Certain default attributes will be applied to your target group. You can view and edit them after creating the target group.
Tags - optional
Consider adding tags to your target group. Tags enable you to categorize your AWS resources so you can more easily manage them.
Cancel
Next
CloudShell
Feedback
Console Mobile App
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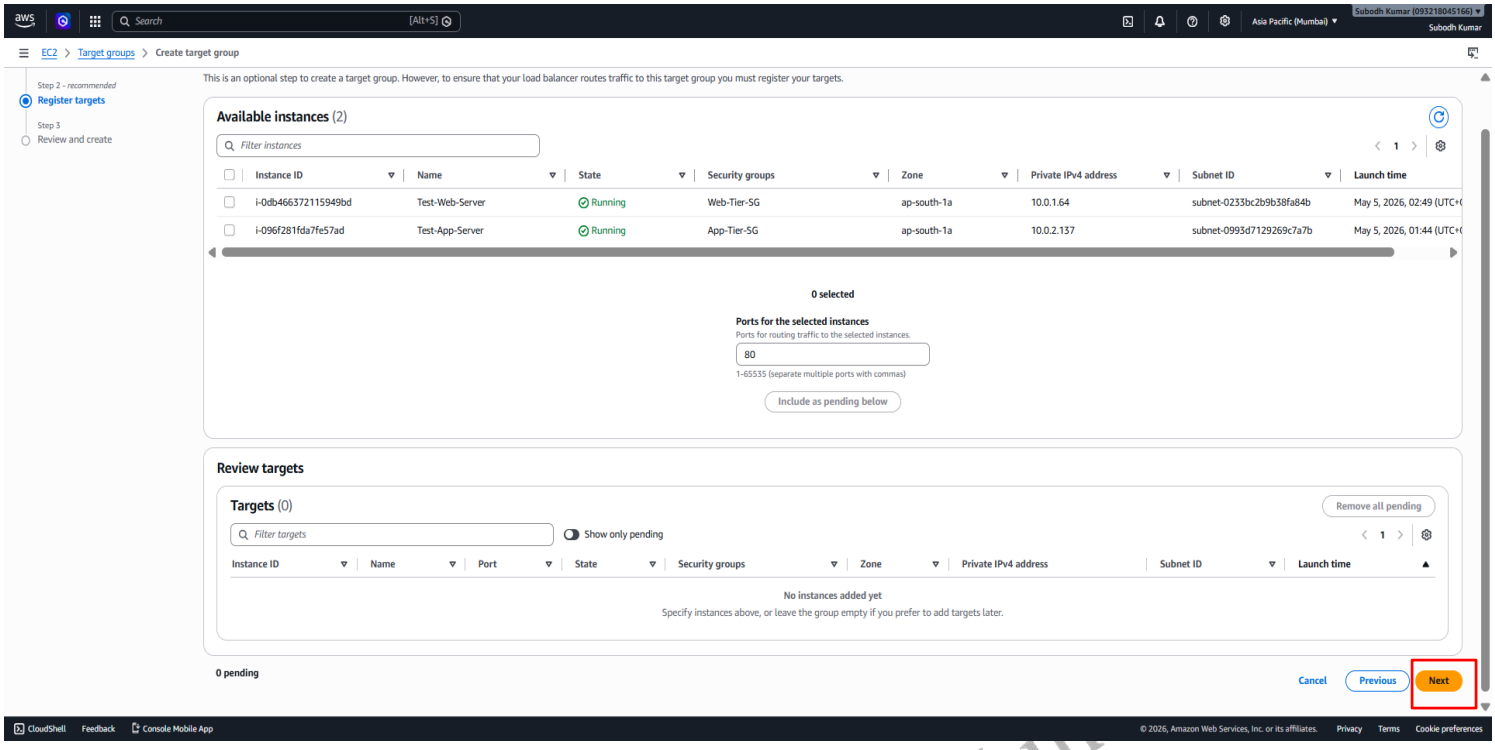
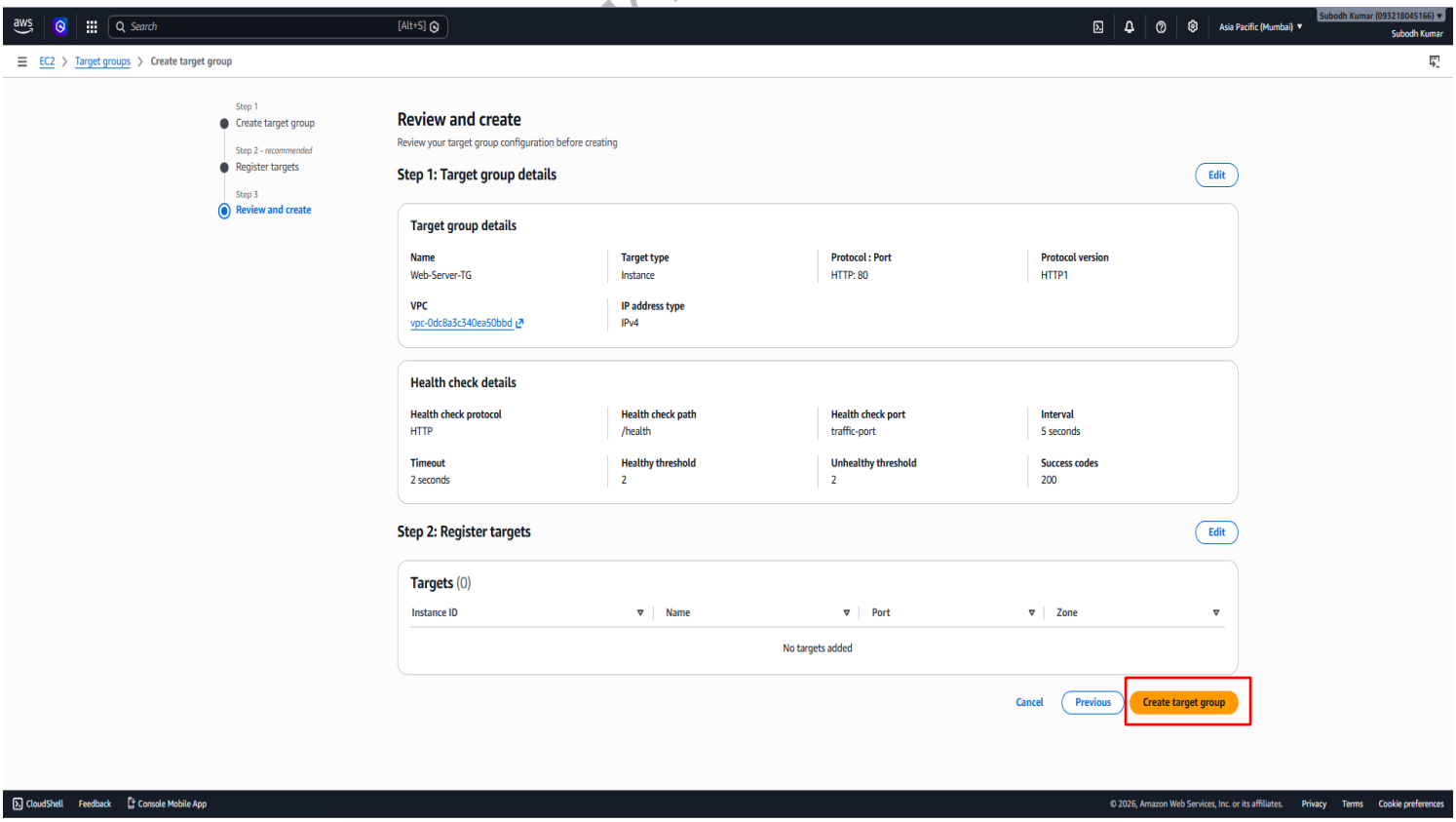
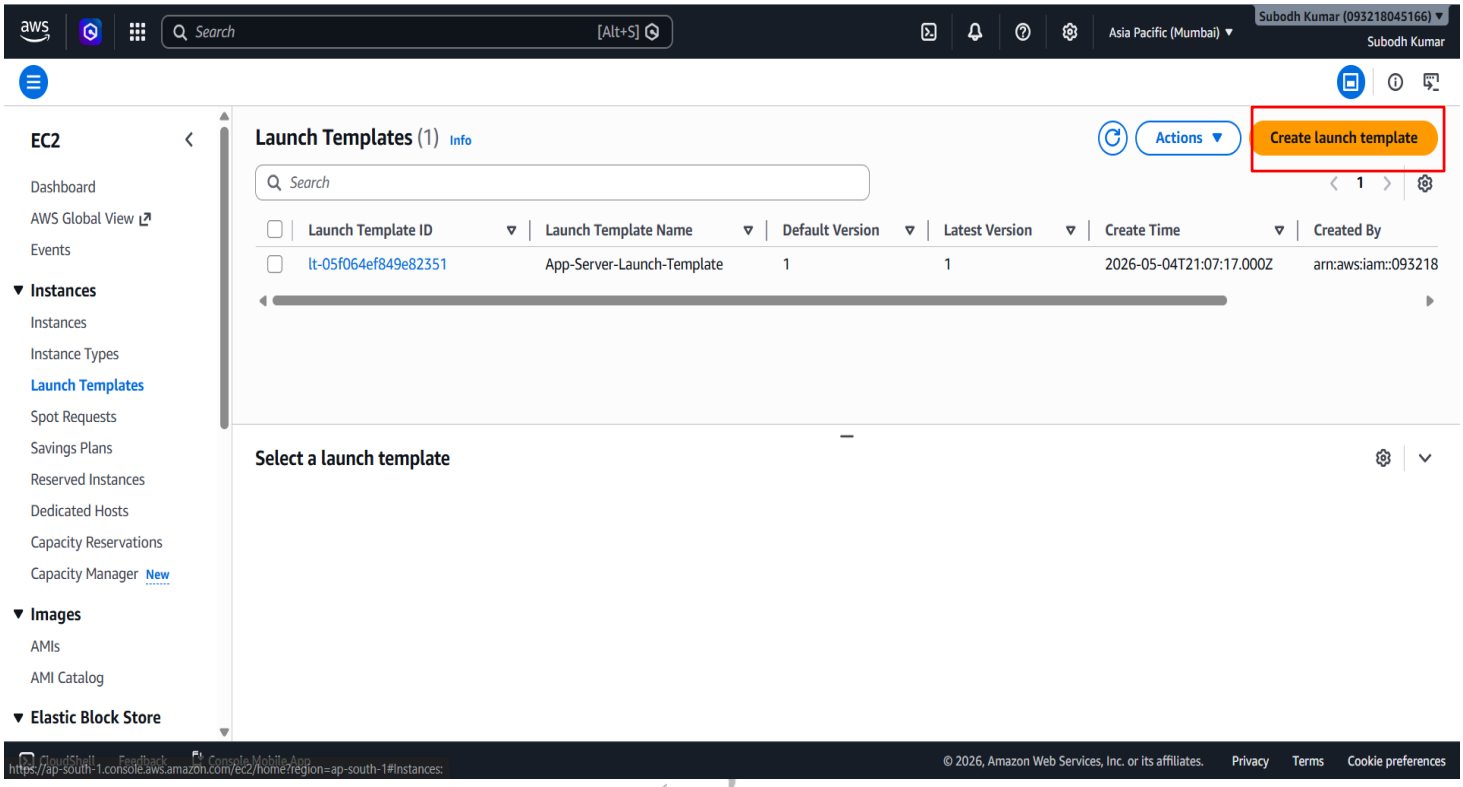


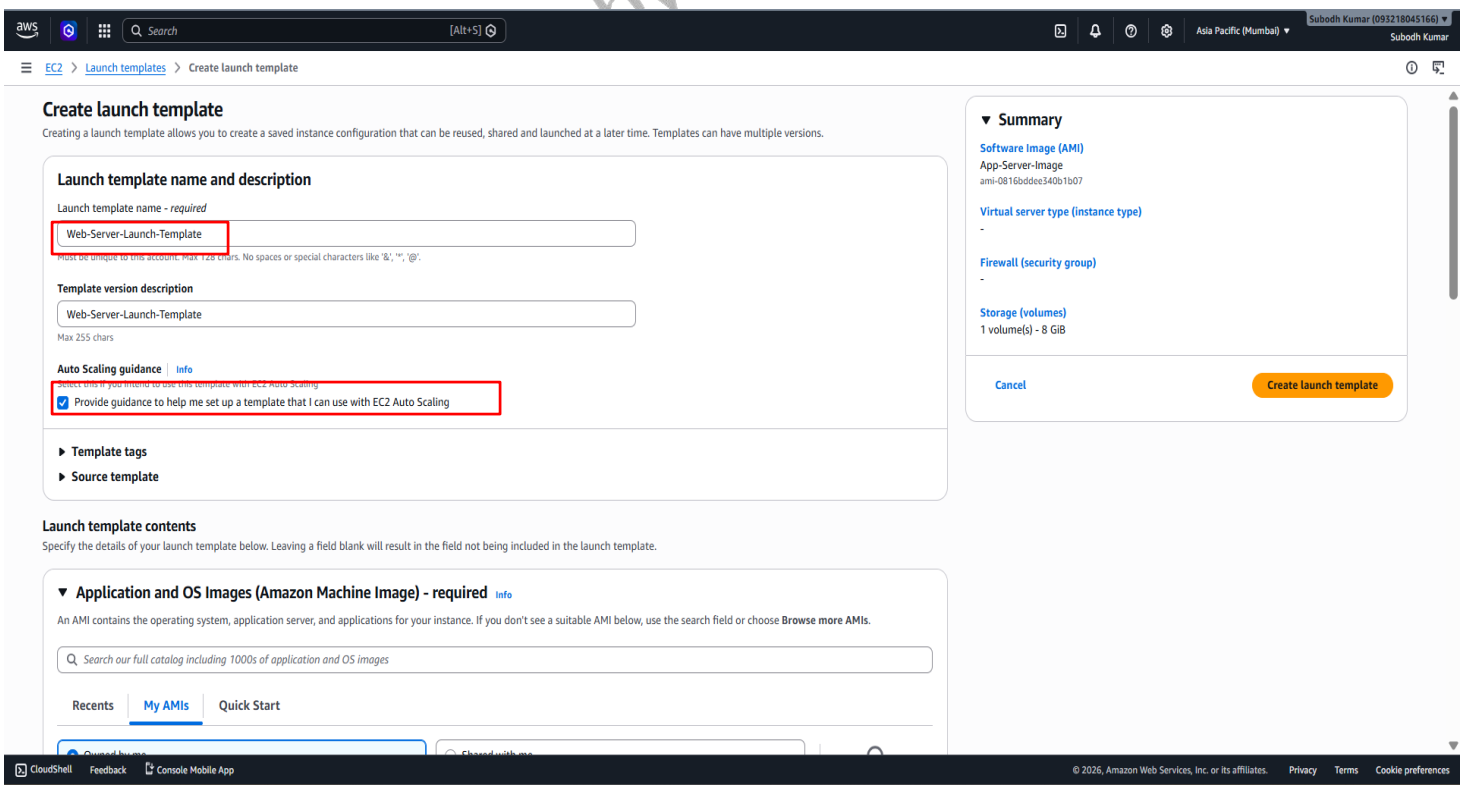


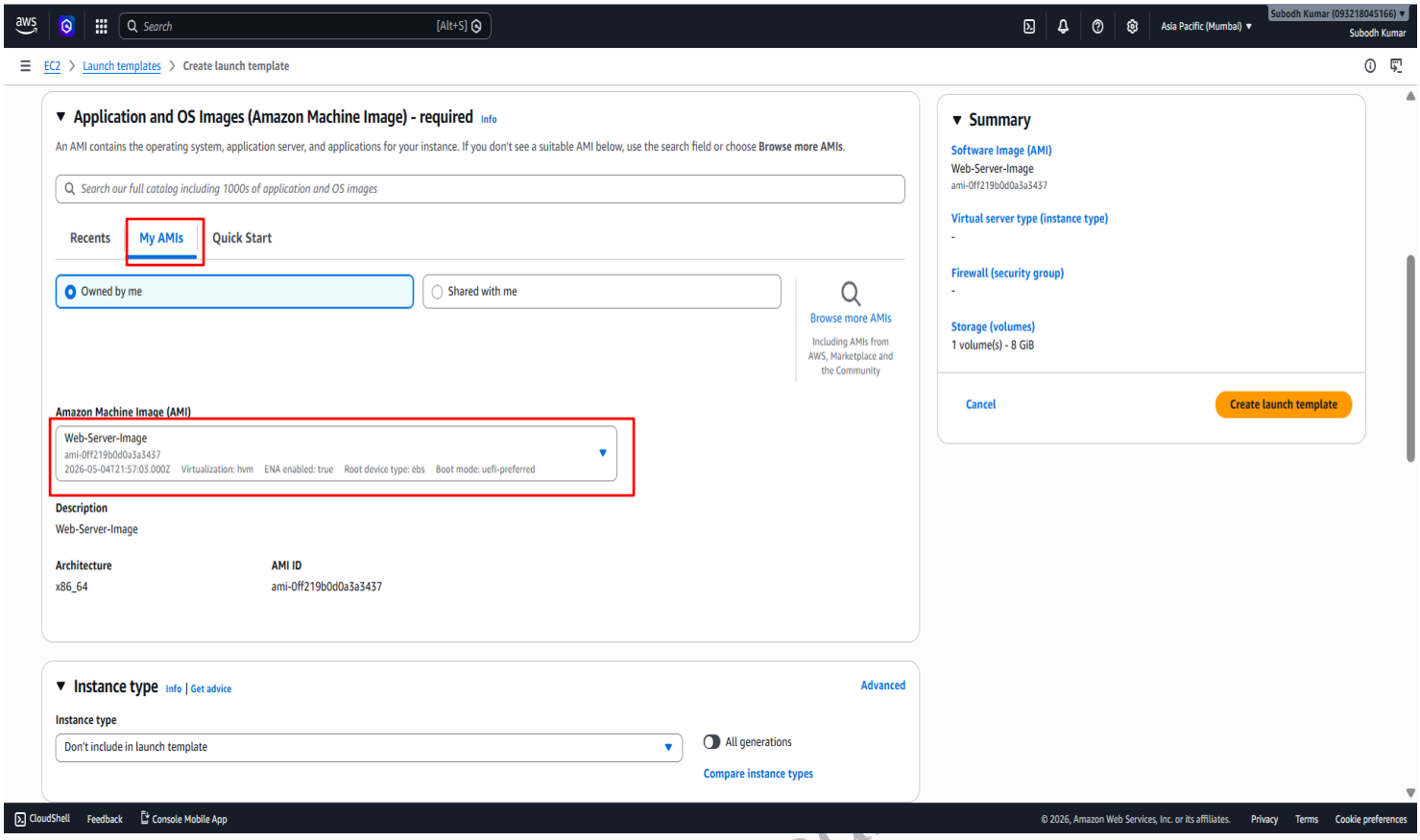
Subodh Kumar

Create Launch Template

Choose Web-Server-Image for AMI and choose Web-Tier-SG. Other settings are same as above app tier.






Search [Alt+S]
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Subodh Kumar
EC2 > Launch templates > Create launch template
Application and OS Images (Amazon Machine Image) - required Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
My AMIs
Quick Start
Owned by me
Shared with me
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Web-Server-Image
ami-0ff219b0d0a3a3437
2026-05-04T21:57:03.000Z Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred
Description
Web-Server-Image
Architecture
x86_64
AMI ID
ami-0ff219b0d0a3a3437
Instance type Info | Get advice
Advanced
Instance type
Don't include in launch template
All generations
Compare instance types
Summary
Software Image (AMI)
Web-Server-Image
ami-0ff219b0d0a3a3437
Virtual server type (instance type)
-
Firewall (security group)
-
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
CloudShell
Feedback
Console Mobile App
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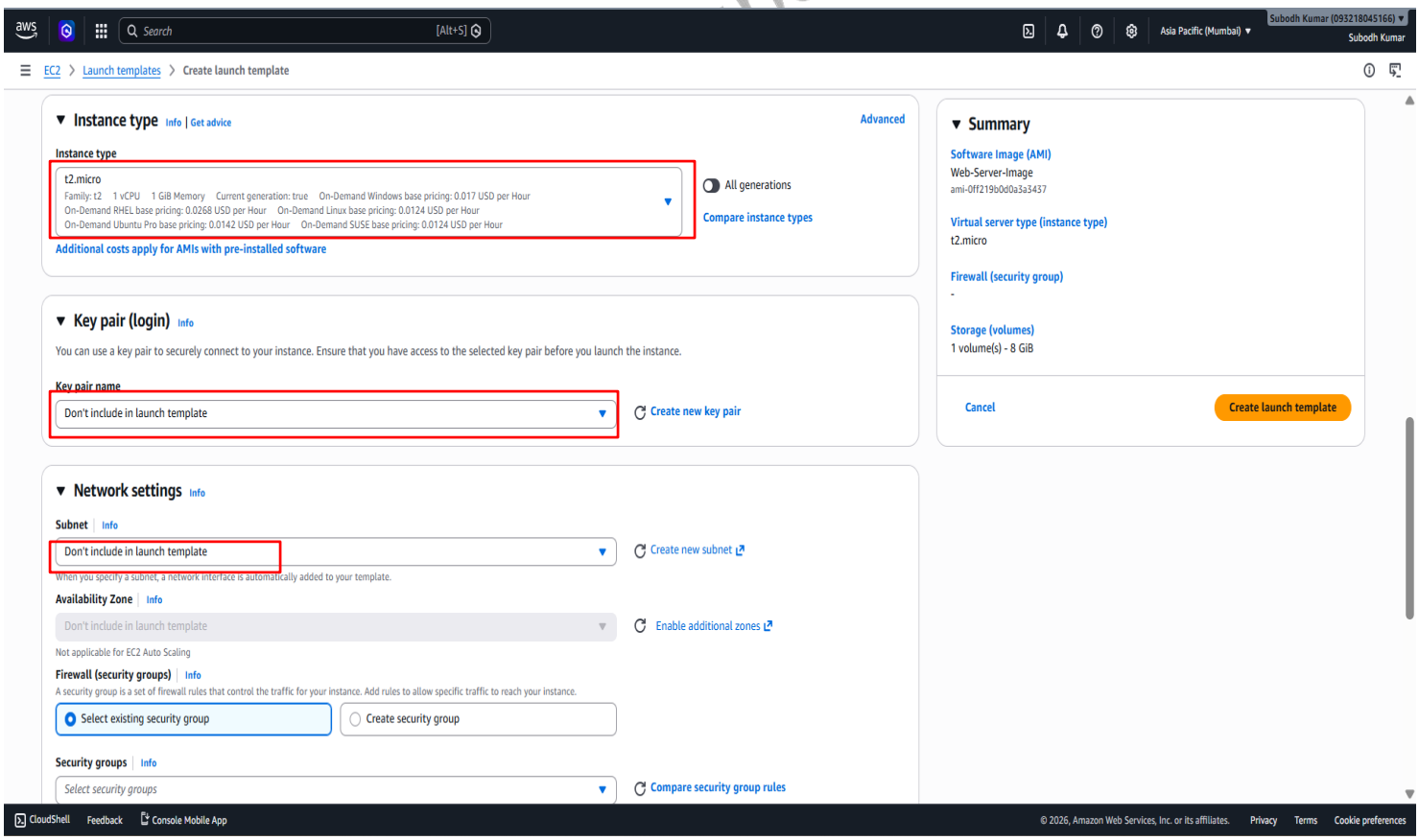
Search [Alt+S]
Asia Pacific (Mumbai)
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Subodh Kumar
EC2 > Launch templates > Create launch template
Instance type Info | Get advice
Advanced
Instance type
t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name
Don't include in launch template
Create new key pair
Network settings Info
Subnet Info
Don't include in launch template
Create new subnet
When you specify a subnet, a network interface is automatically added to your template.
Availability Zone Info
Don't include in launch template
Enable additional zones
Not applicable for EC2 Auto Scaling
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Select existing security group
Create security group
Security groups Info
Select security groups
Compare security group rules
Summary
Software Image (AMI)
Web-Server-Image
ami-0ff219b0d0a3a3437
Virtual server type (instance type)
t2.micro
Firewall (security group)
-
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
CloudShell
Feedback
Console Mobile App
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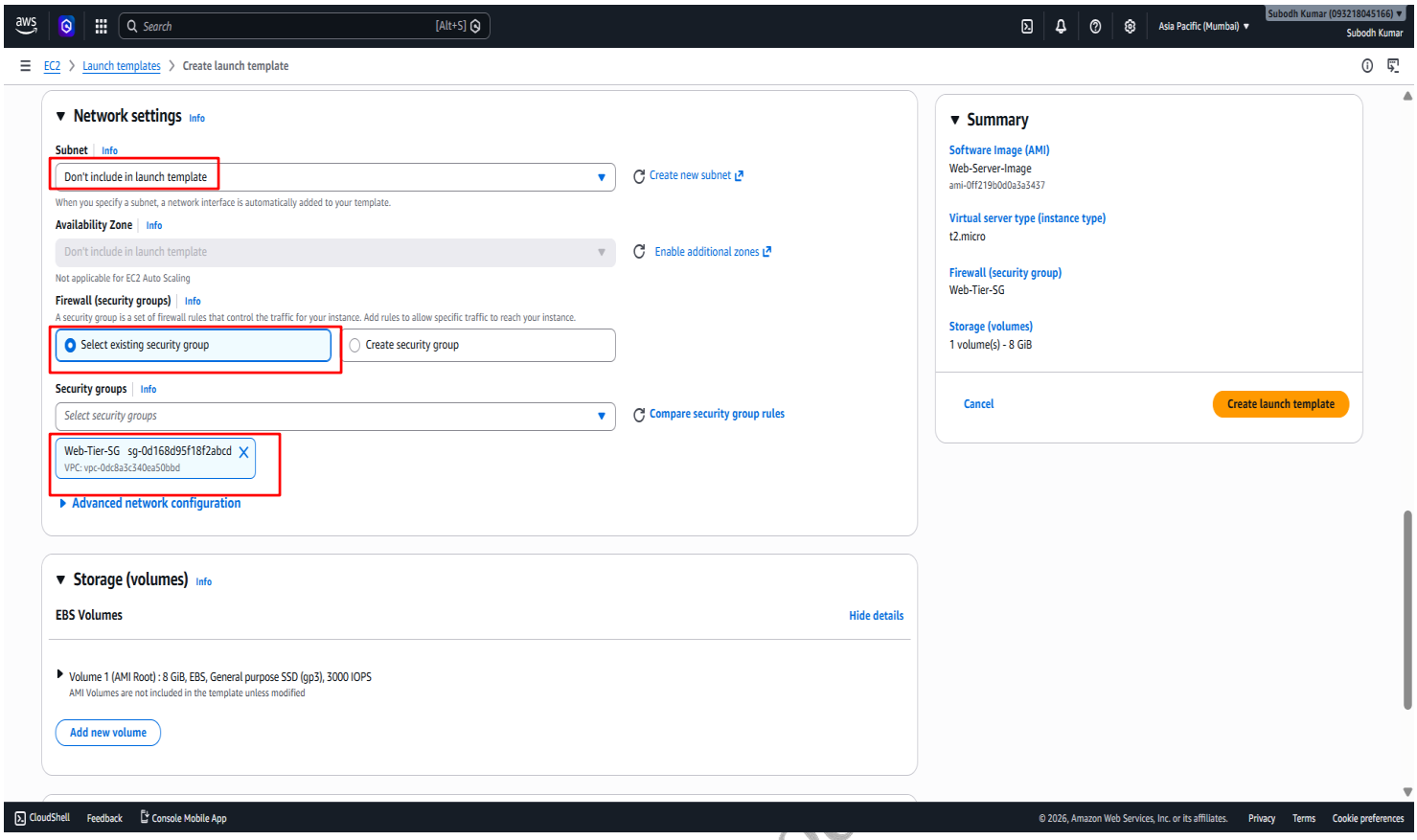
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Launch templates > Create launch template
Network settings Info
Subnet Info
Don't include in launch template
Create new subnet
When you specify a subnet, a network interface is automatically added to your template.
Availability Zone Info
Don't include in launch template
Enable additional zones
Not applicable for EC2 Auto Scaling
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Select existing security group
Create security group
Security groups Info
Select security groups
Compare security group rules
Web-Tier-SG sg-0d168d95f18f2abcd
VPC: vpc-0dc8a3c340ea50bbd
Advanced network configuration
Storage (volumes) Info
EBS Volumes
Hide details
Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3), 3000 IOPS
AMI Volumes are not included in the template unless modified
Add new volume
Summary
Software Image (AMI)
Web-Server-Image
ami-0ff219b0d0a3a3437
Virtual server type (instance type)
t2.micro
Firewall (security group)
Web-Tier-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
CloudShell
Feedback
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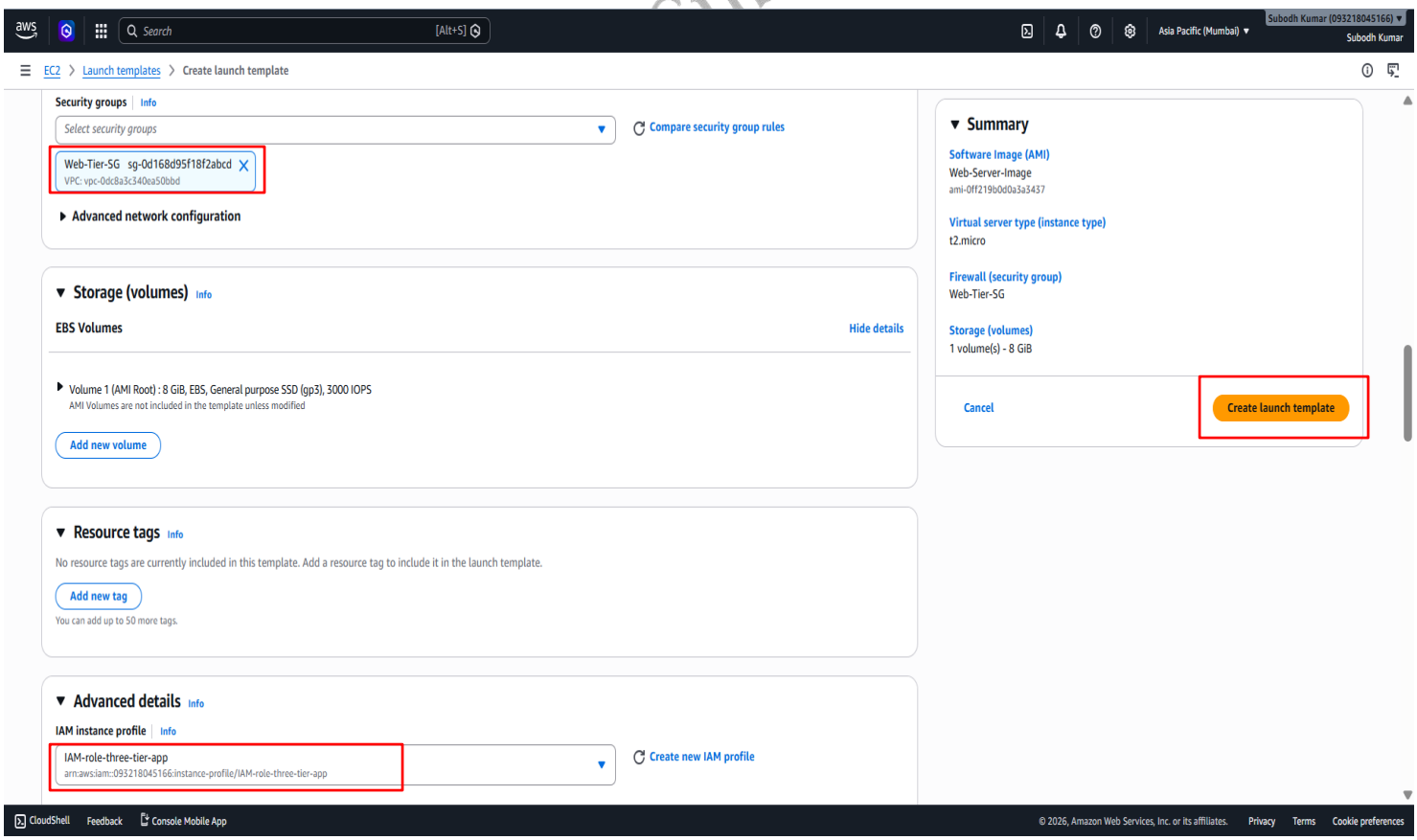
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Launch templates > Create launch template
Security groups Info
Select security groups
Compare security group rules
Web-Tier-SG sg-0d168d95f18f2abcd
VPC: vpc-0dc8a3c340ea50bbd
Advanced network configuration
Storage (volumes) Info
EBS Volumes
Hide details
Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3), 3000 IOPS
AMI Volumes are not included in the template unless modified
Add new volume
Resource tags Info
No resource tags are currently included in this template. Add a resource tag to include it in the launch template.
Add new tag
You can add up to 50 more tags.
Advanced details Info
IAM instance profile Info
IAM-role-three-tier-app
arn:aws:iam::093218045166:instance-profile/IAM-role-three-tier-app
Create new IAM profile
Summary
Software Image (AMI)
Web-Server-Image
ami-0ff219b0d0a3a3437
Virtual server type (instance type)
t2.micro
Firewall (security group)
Web-Tier-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Create launch template
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Feedback
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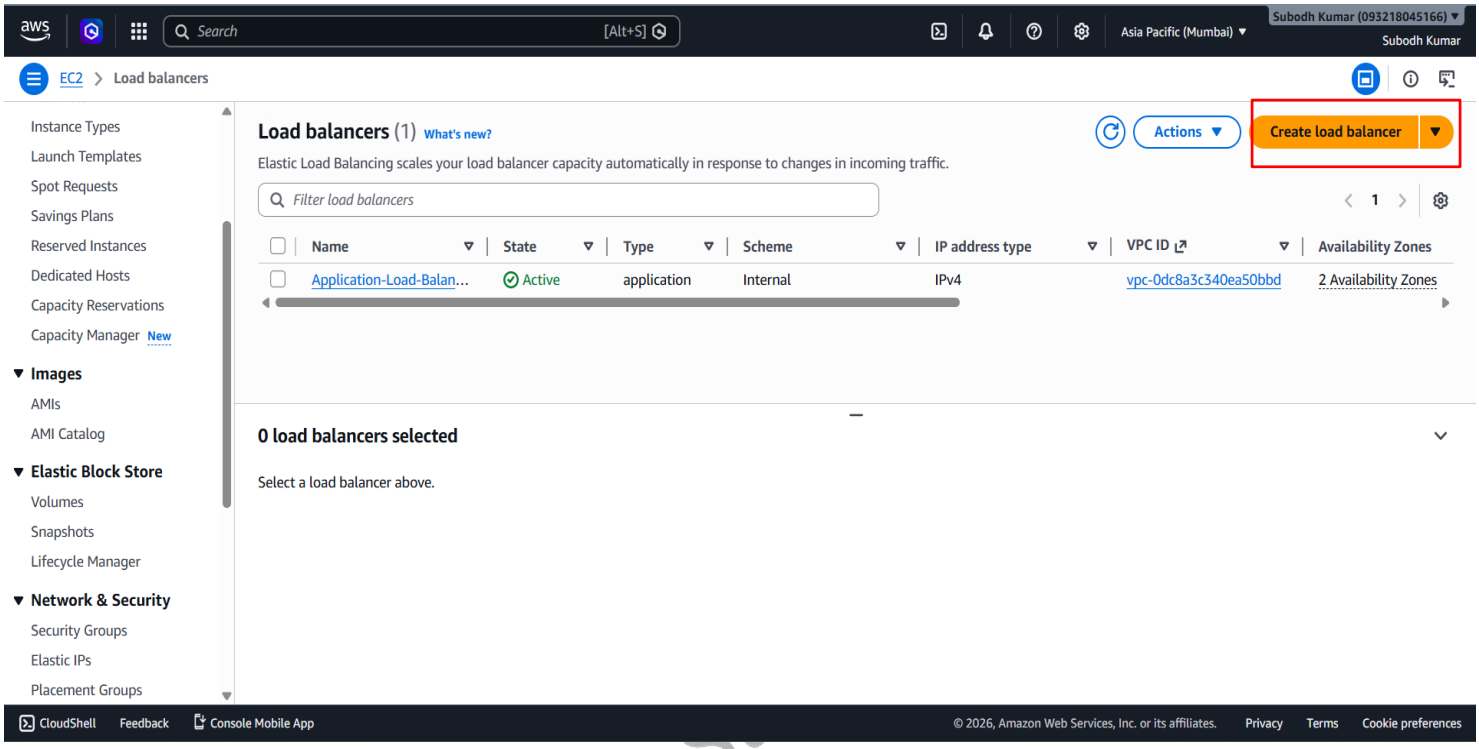
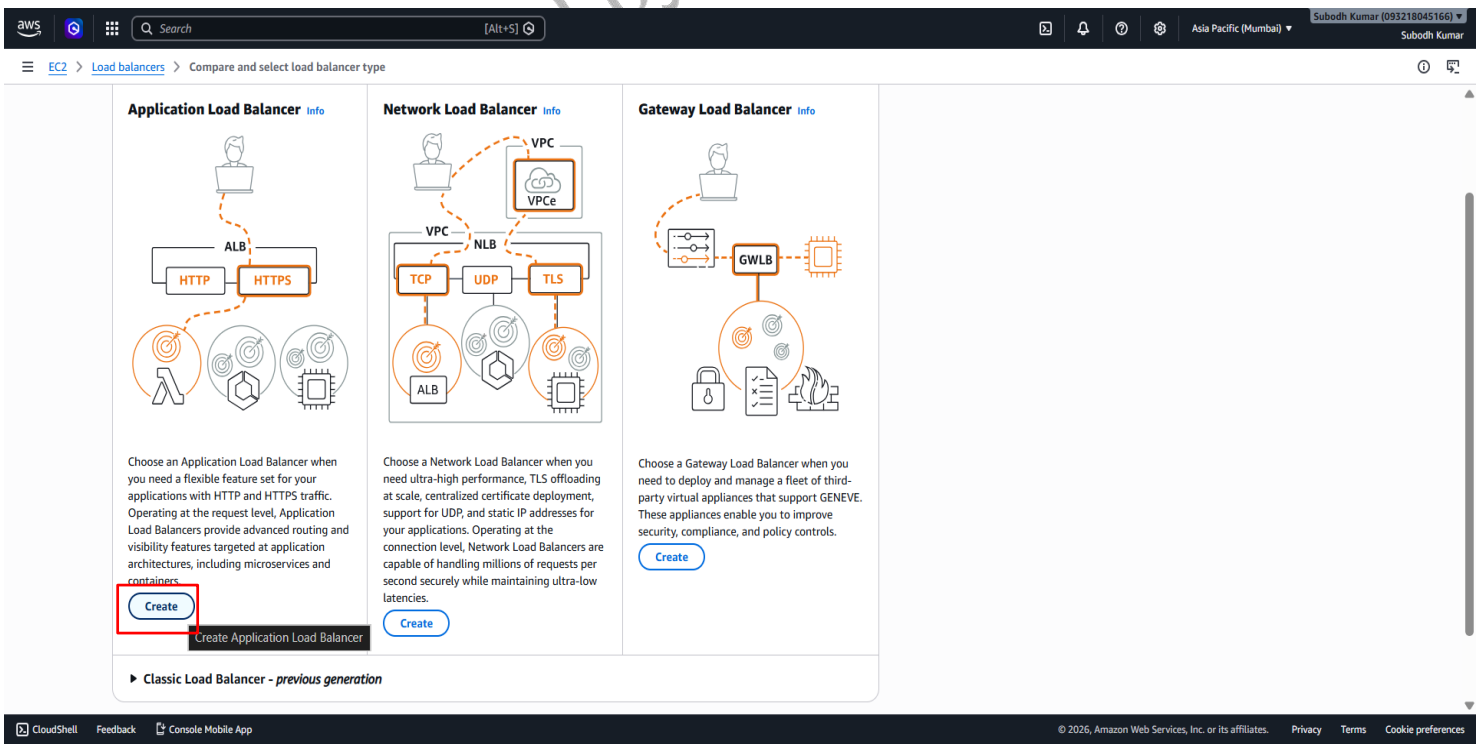
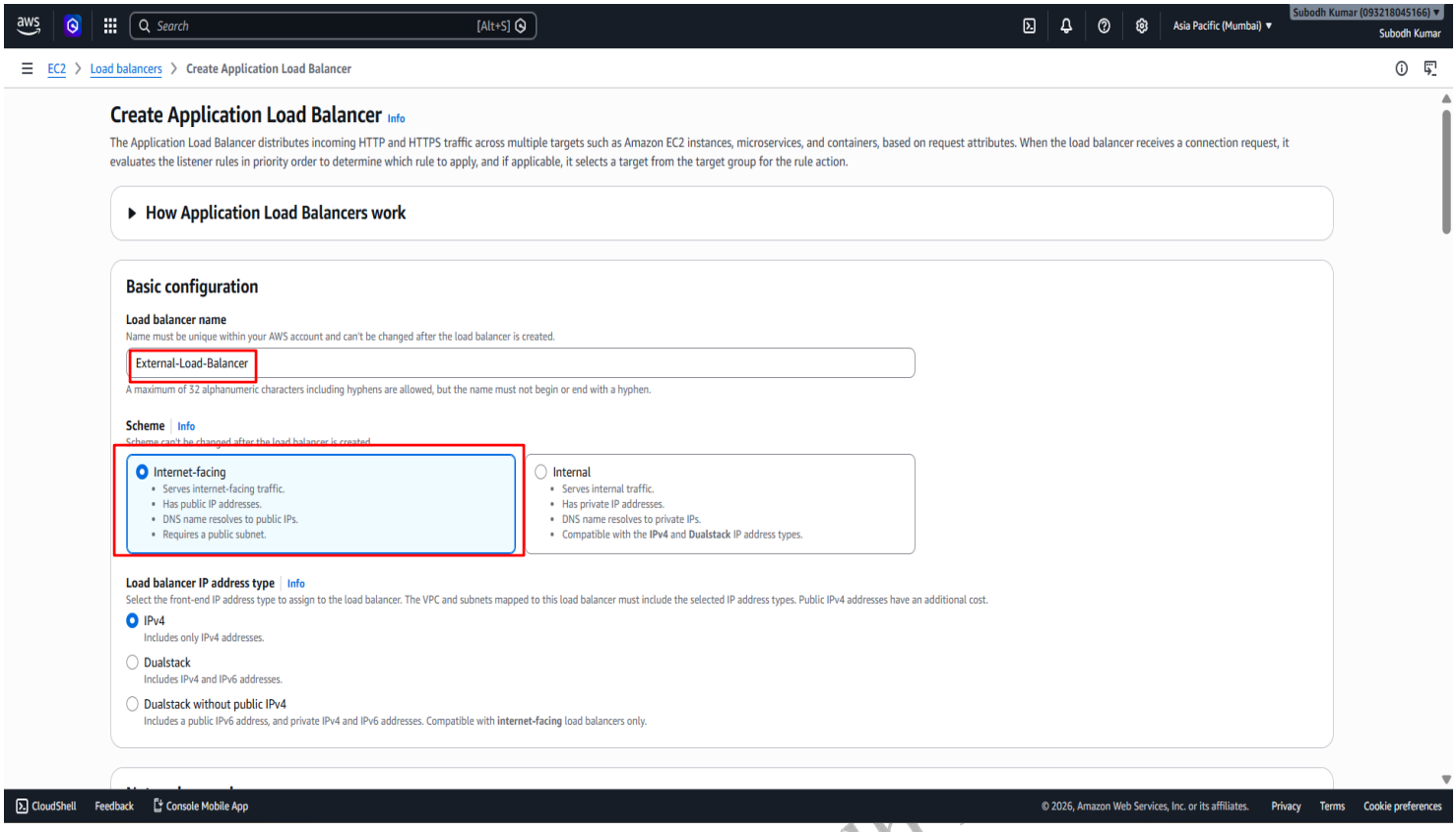
Create External Load Balancer

Choose **Scheme** as **Internet-facing**. Choose vpc. Choose **Availability Zones** of **web- tier**. Choose **External load balancer Security Group**. Choose target group in Listeners and routing. Create load balancer.






Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Load balancers > Create Application Load Balancer
Create Application Load Balancer Info
The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.
How Application Load Balancers work
Basic configuration
Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
External-Load-Balancer
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Scheme Info
Scheme can't be changed after the load balancer is created.
Internet-facing
Serves internet-facing traffic.
Has public IP addresses.
DNS name resolves to public IPs.
Requires a public subnet.
Internal
Serves internal traffic.
Has private IP addresses.
DNS name resolves to private IPs.
Compatible with the IPv4 and Dualstack IP address types.
Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.
IPv4
Includes only IPv4 addresses.
Dualstack
Includes IPv4 and IPv6 addresses.
Dualstack without public IPv4
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.
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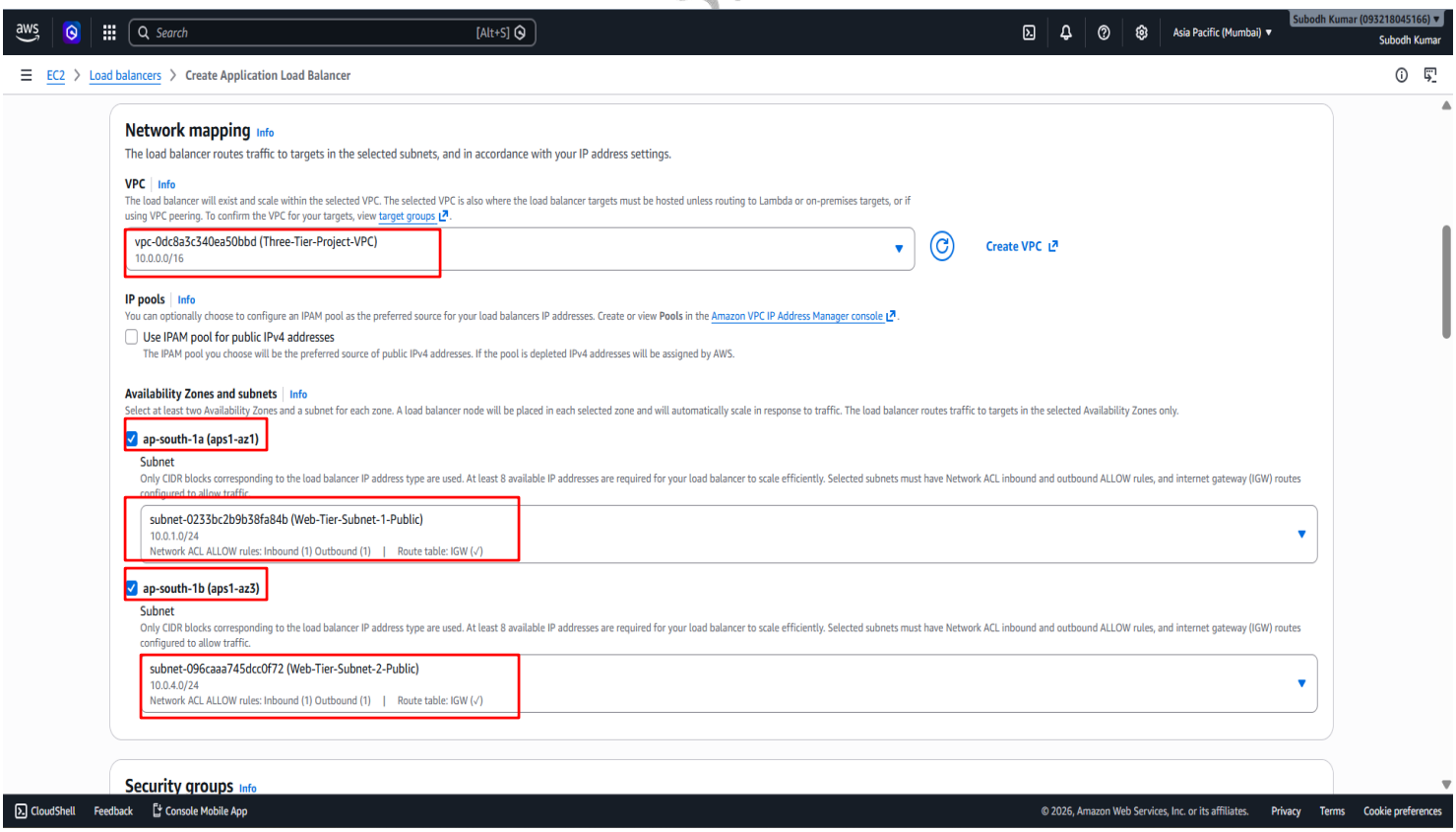
Search [Alt+S]
Asia Pacific (Mumbai)
Subodh Kumar (093218045166)
Subodh Kumar
EC2 > Load balancers > Create Application Load Balancer
Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.
VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create VPC
IP pools Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the Amazon VPC IP Address Manager console.
Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.
Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.
ap-south-1a (aps1-az1)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-0233bc2b9b38fa84b (Web-Tier-Subnet-1-Public)
10.0.1.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)
ap-south-1b (aps1-az3)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.
subnet-096caaa745dcc0f72 (Web-Tier-Subnet-2-Public)
10.0.4.0/24
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)
Security groups Info
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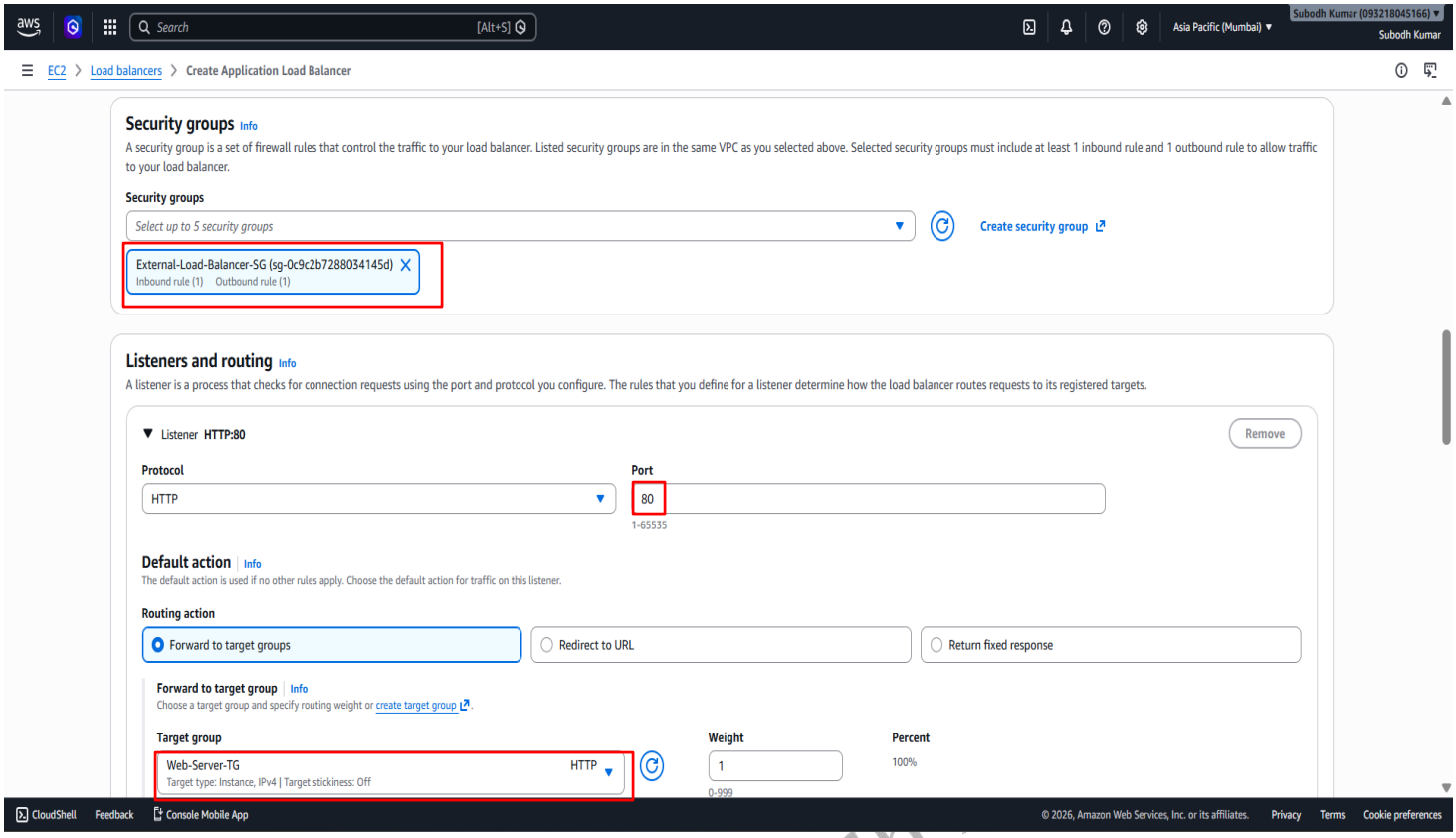
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Subodh Kumar
EC2 > Load balancers > Create Application Load Balancer
Security groups Info
A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups must include at least 1 inbound rule and 1 outbound rule to allow traffic to your load balancer.
Security groups
Select up to 5 security groups
Create security group
External-Load-Balancer-SG (sg-0c9c2b7288034145d)
Inbound rule (1) Outbound rule (1)
Listeners and routing Info
A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.
Listener HTTP:80
Remove
Protocol
HTTP
Port
80
1-65535
Default action Info
The default action is used if no other rules apply. Choose the default action for traffic on this listener.
Routing action
Forward to target groups
Redirect to URL
Return fixed response
Forward to target group Info
Choose a target group and specify routing weight or create target group.
Target group
Web-Server-TG
HTTP
Target type: Instance, IPv4 | Target stickiness: Off
Weight
1
0-999
Percent
100%
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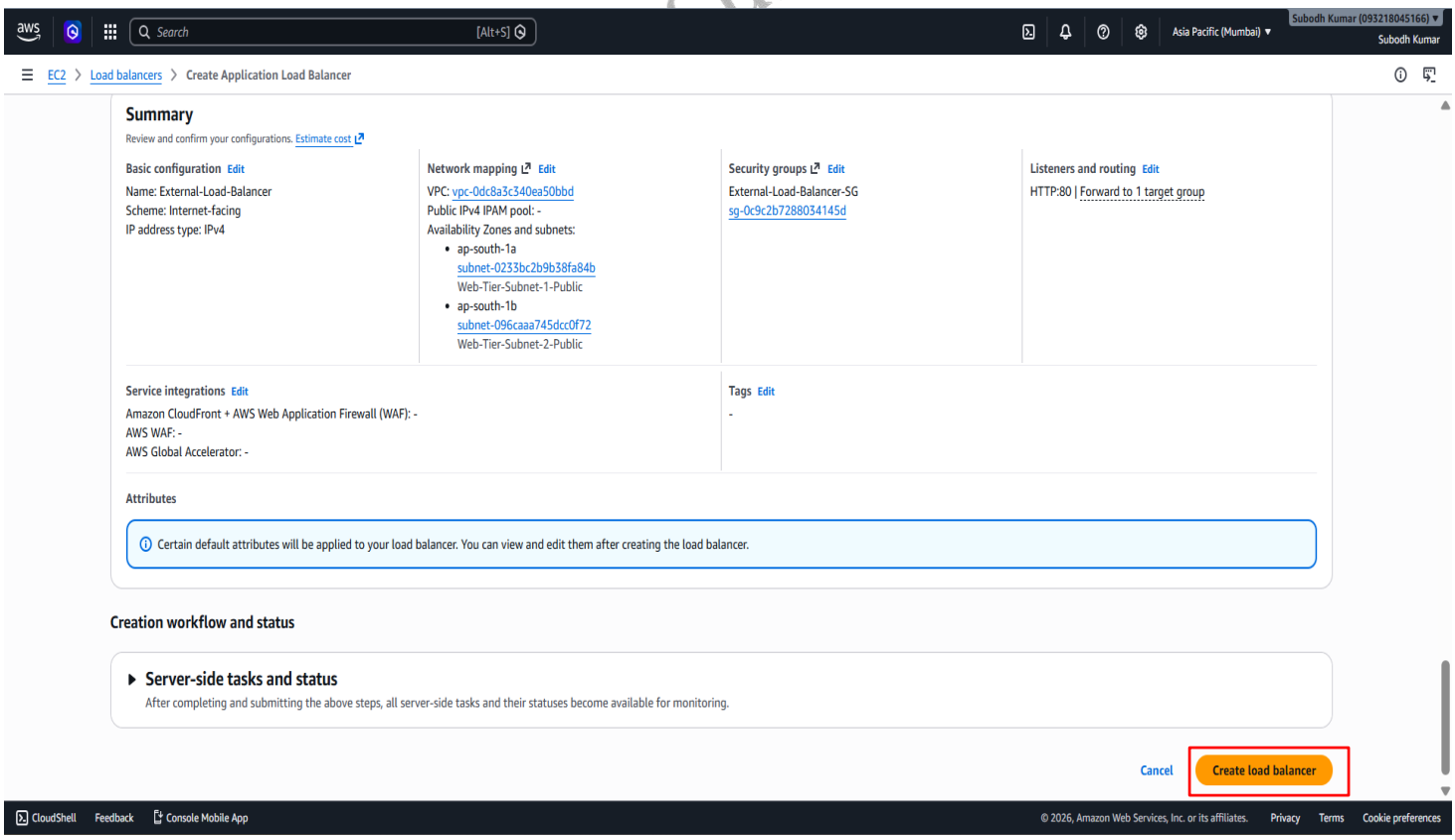
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EC2 > Load balancers > Create Application Load Balancer
Summary
Review and confirm your configurations. Estimate cost
Basic configuration Edit
Name: External-Load-Balancer
Scheme: Internet-facing
IP address type: IPv4
Network mapping Edit
VPC: vpc-0dc8a3c340ea50bbd
Public IPv4 IPAM pool: -
Availability Zones and subnets:
ap-south-1a
subnet-0233bc2b9b38fa84b
Web-Tier-Subnet-1-Public
ap-south-1b
subnet-096caaa745dcc0f72
Web-Tier-Subnet-2-Public
Security groups Edit
External-Load-Balancer-SG
sg-0c9c2b7288034145d
Listeners and routing Edit
HTTP:80 | Forward to 1 target group
Service integrations Edit
Amazon CloudFront + AWS Web Application Firewall (WAF): -
AWS WAF: -
AWS Global Accelerator: -
Tags Edit
-
Attributes
Certain default attributes will be applied to your load balancer. You can view and edit them after creating the load balancer.
Creation workflow and status
Server-side tasks and status
After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.
Cancel
Create load balancer
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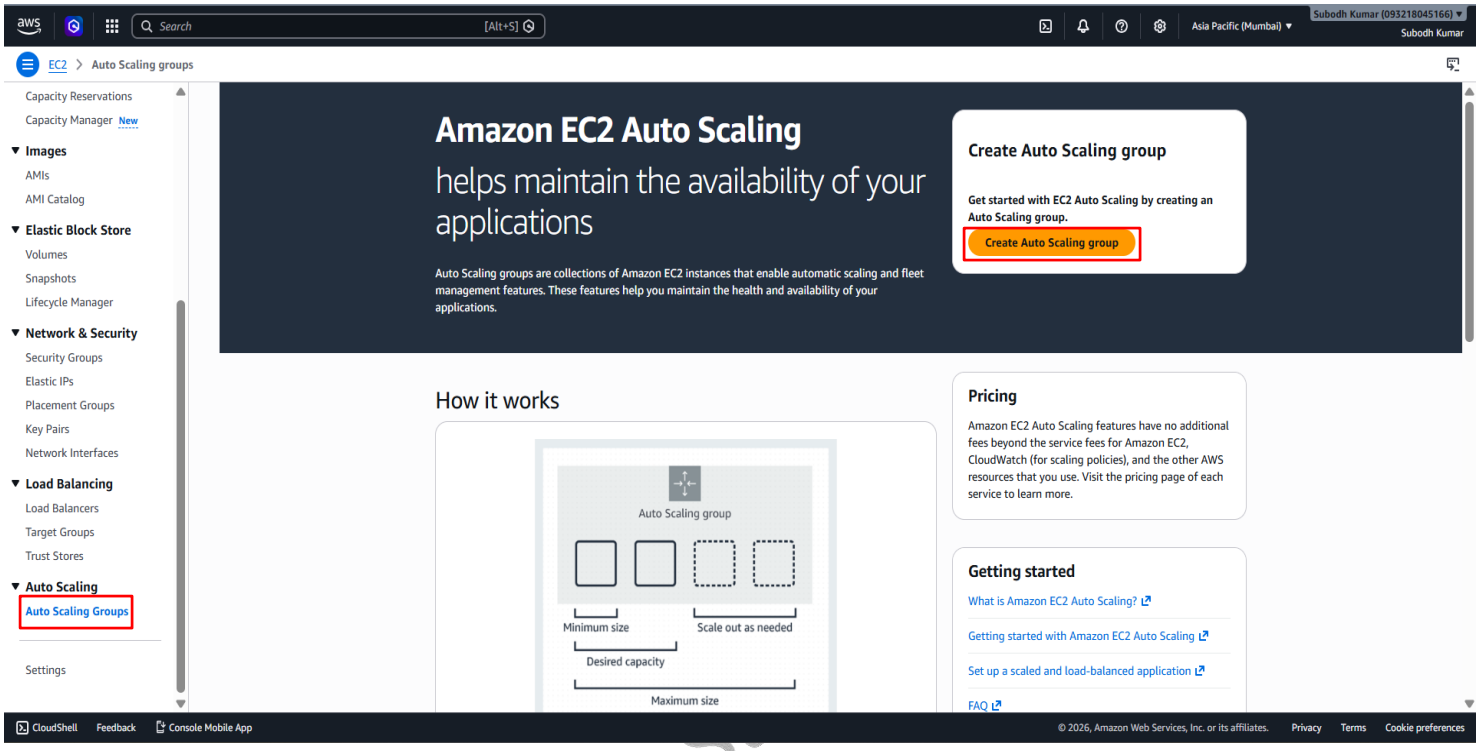
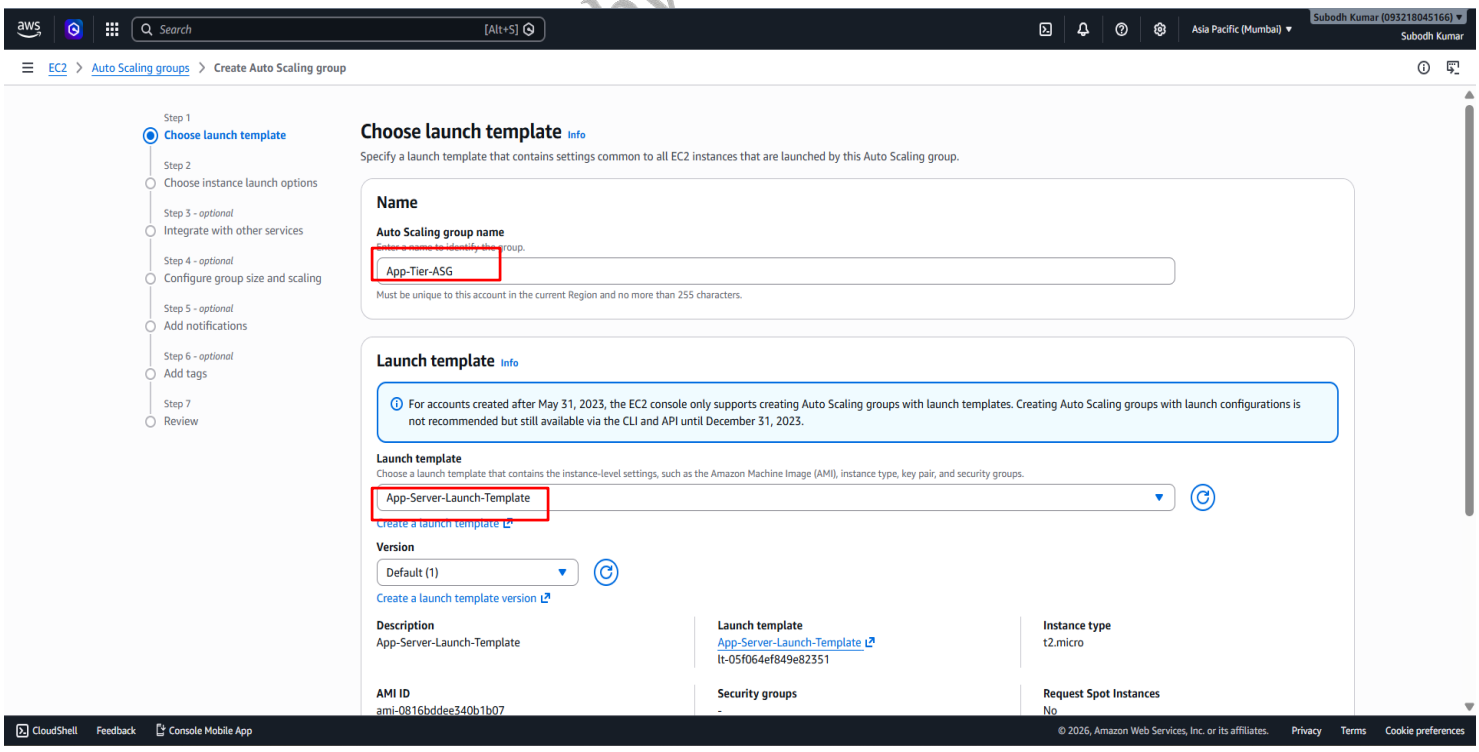
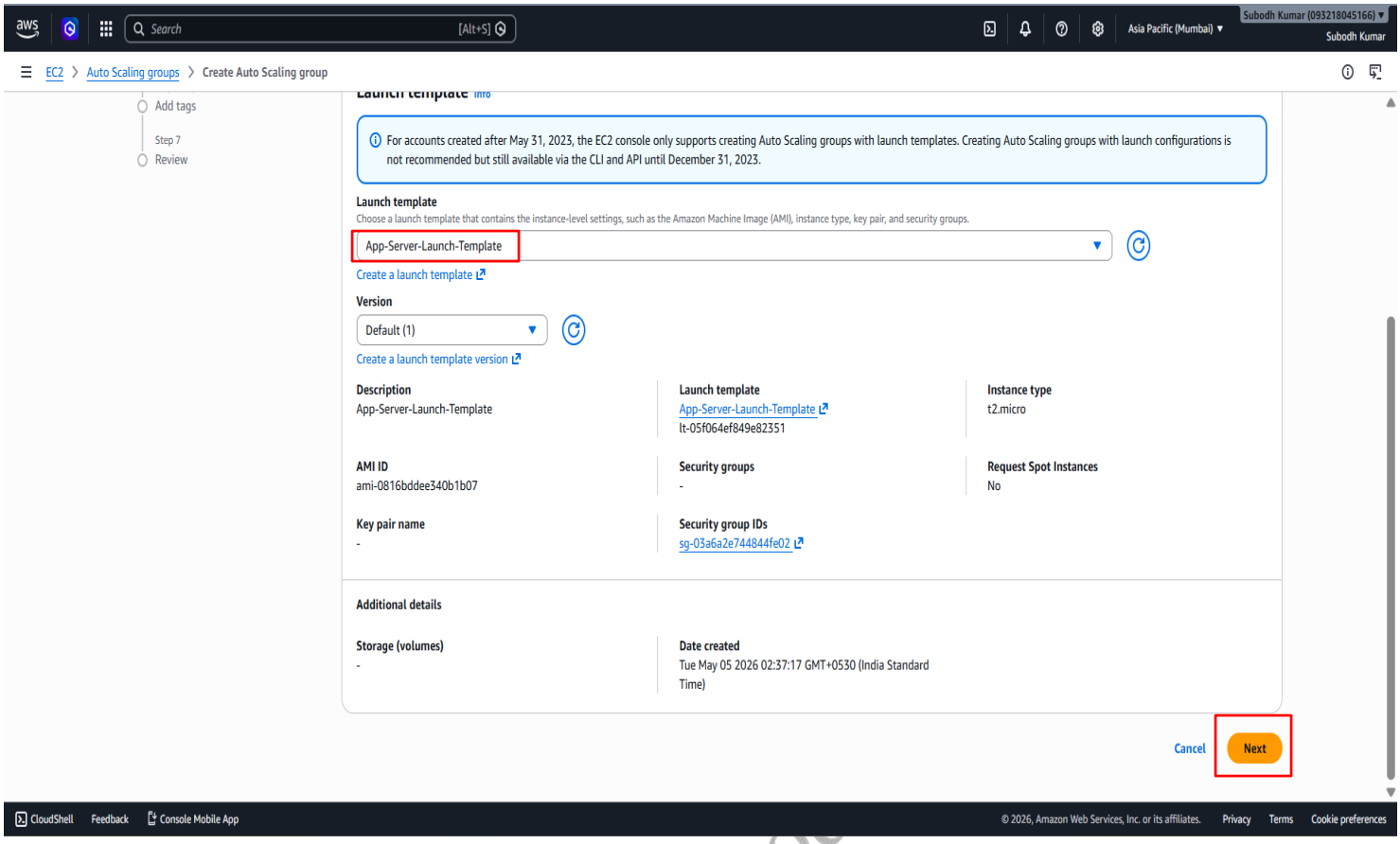
13.Create Auto Scaling groups

Auto Scaling group for App Tier

Go to EC2 > Create Auto Scaling group



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EC2 > Auto Scaling groups > Create Auto Scaling group
Add tags
Step 7
Review
For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.
Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
App-Server-Launch-Template
Create a launch template
Version
Default (1)
Create a launch template version
Description
App-Server-Launch-Template
Launch template
App-Server-Launch-Template
lt-05f064ef849e82351
Instance type
t2.micro
AMI ID
ami-0816bddee340b1b07
Security groups
-
Request Spot Instances
No
Key pair name
-
Security group IDs
sg-03a6a2e744844fe02
Additional details
Storage (volumes)
-
Date created
Tue May 05 2026 02:37:17 GMT+0530 (India Standard Time)
Cancel
Next
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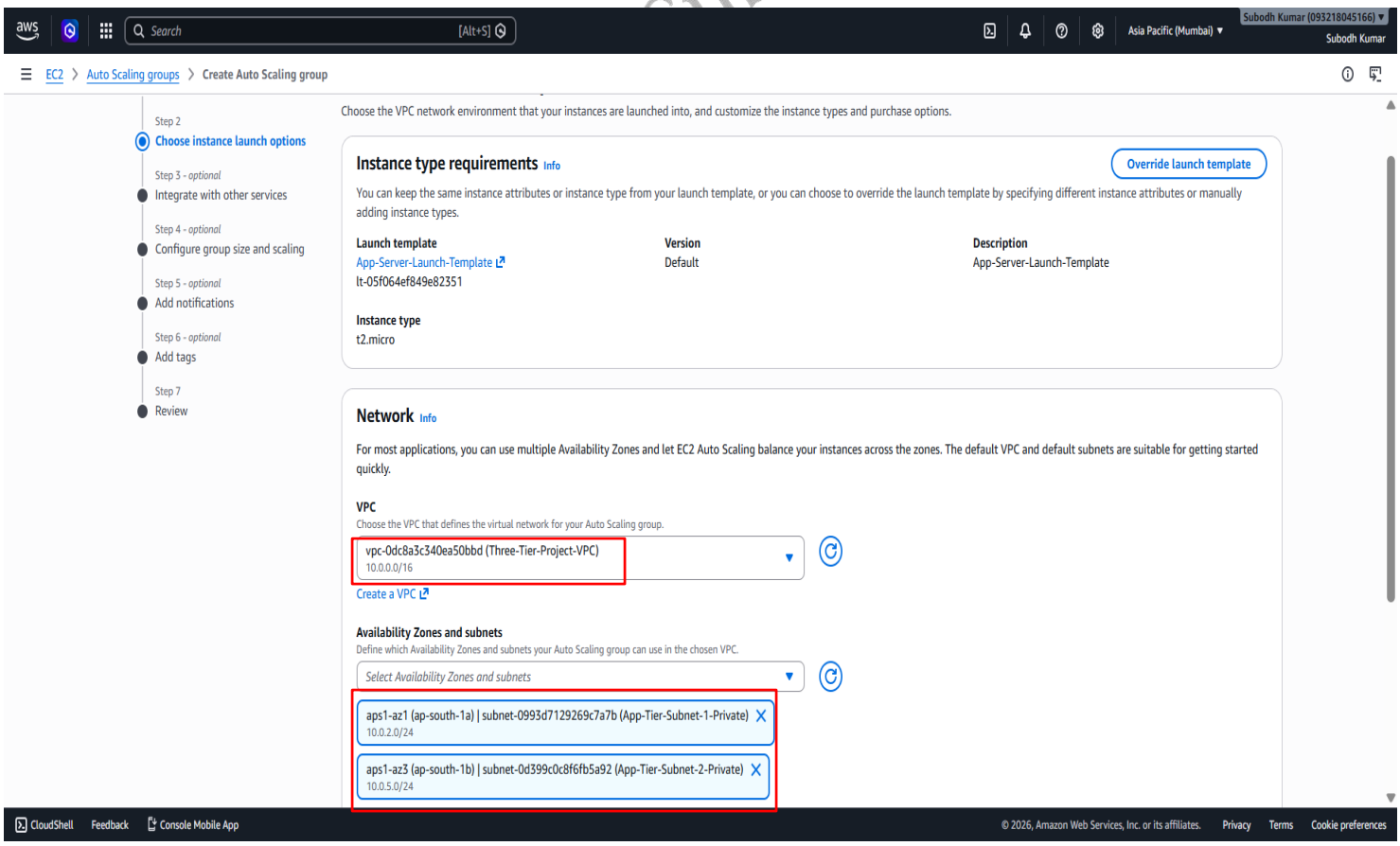
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.
Instance type requirements
Override launch template
You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.
Launch template
App-Server-Launch-Template
lt-05f064ef849e82351
Version
Default
Description
App-Server-Launch-Template
Instance type
t2.micro
Network
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.
VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create a VPC
Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
aps1-az1 (ap-south-1a) | subnet-0993d7129269c7a7b (App-Tier-Subnet-1-Private)
10.0.2.0/24
aps1-az3 (ap-south-1b) | subnet-0d399c0c8f6fb5a92 (App-Tier-Subnet-2-Private)
10.0.5.0/24
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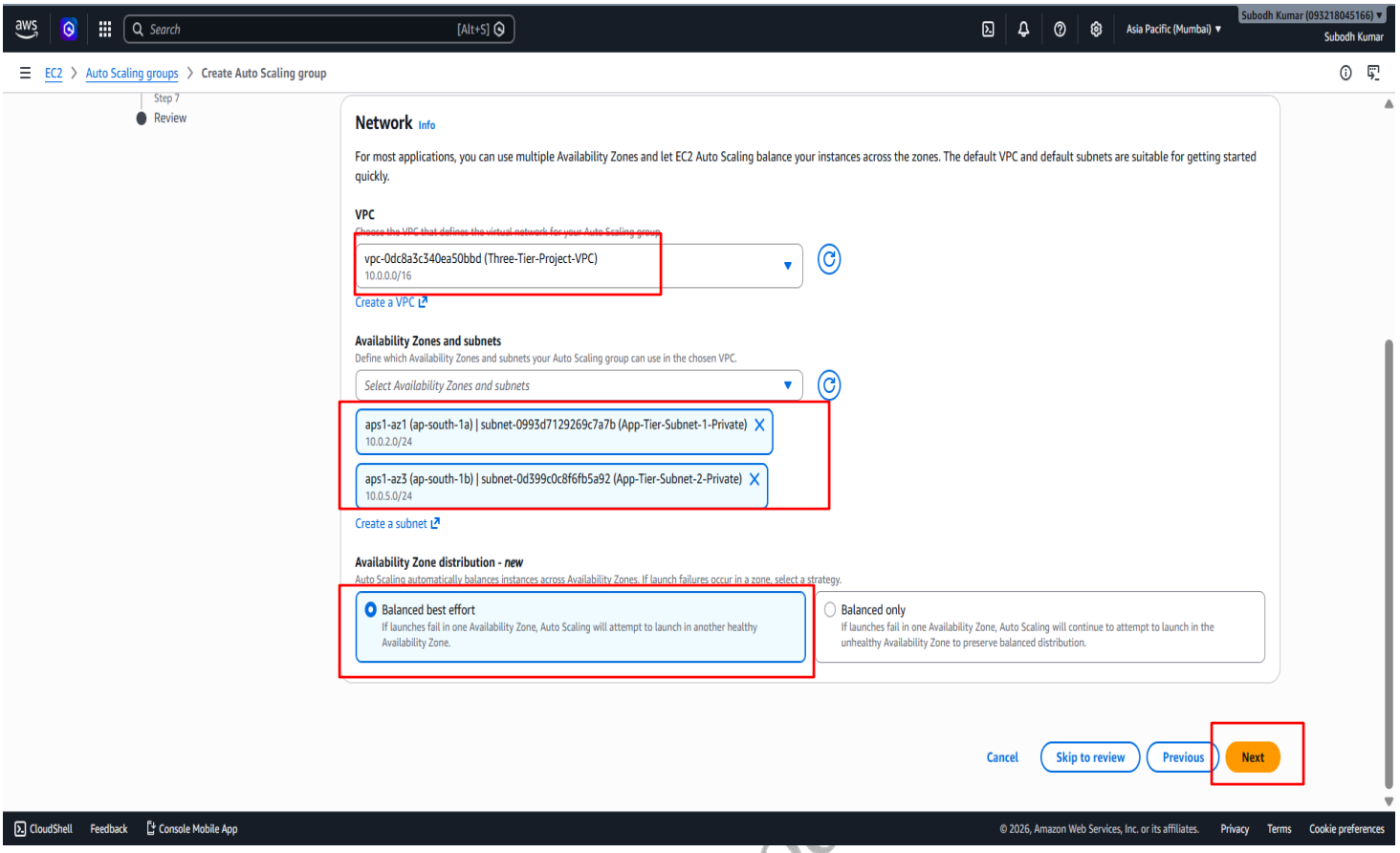
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 7
Review
Network Info
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.
VPC
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create a VPC
Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
aps1-az1 (ap-south-1a) | subnet-0993d7129269c7a7b (App-Tier-Subnet-1-Private)
10.0.2.0/24
aps1-az3 (ap-south-1b) | subnet-0d399c0c8f6fb5a92 (App-Tier-Subnet-2-Private)
10.0.5.0/24
Create a subnet
Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.
Balanced best effort
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.
Balanced only
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.
Cancel
Skip to review
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Next
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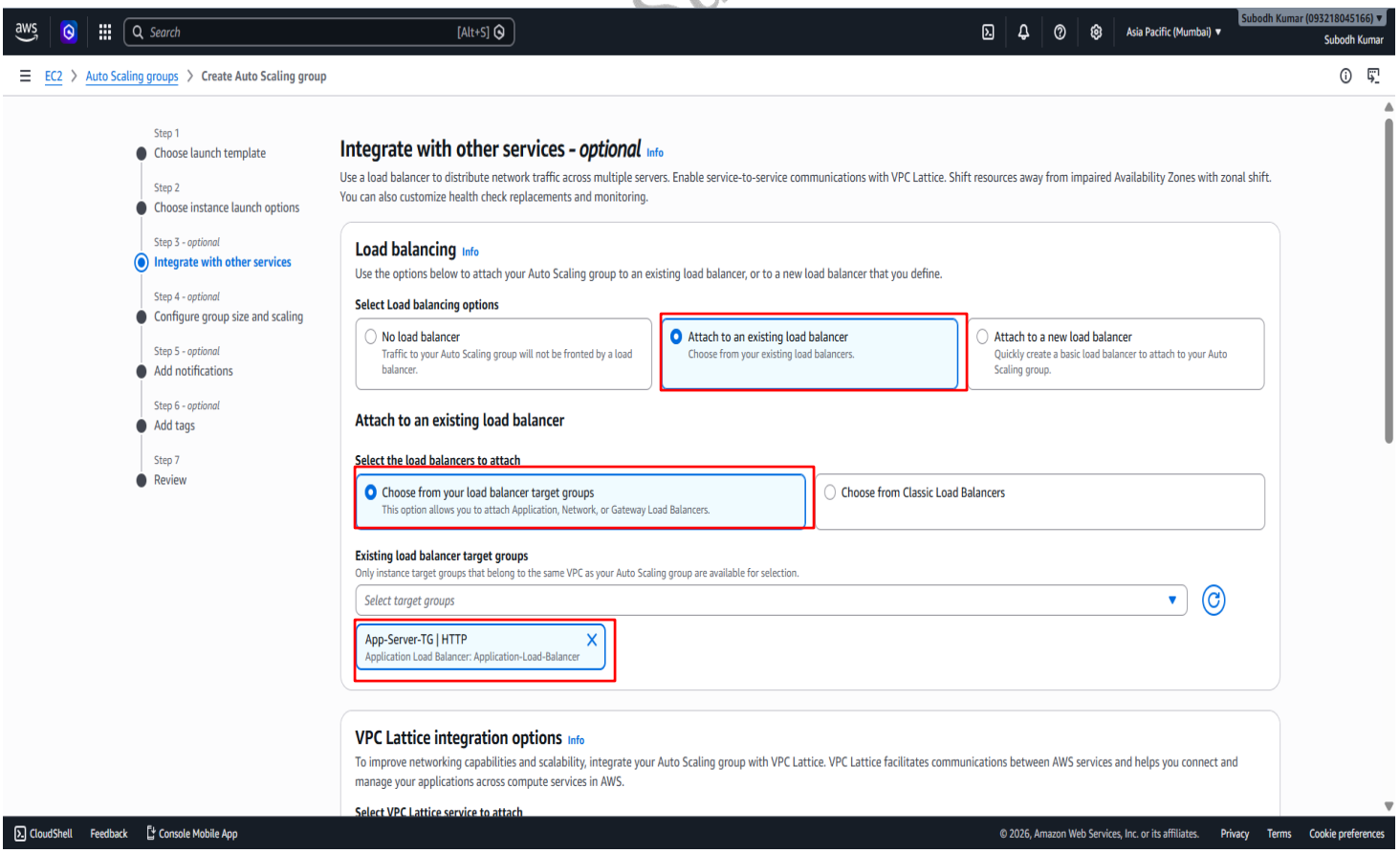
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Integrate with other services - optional Info
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.
Load balancing Info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.
Select Load balancing options
No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
Attach to an existing load balancer
Choose from your existing load balancers.
Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.
Attach to an existing load balancer
Select the load balancers to attach
Choose from your load balancer target groups
This option allows you to attach Application, Network, or Gateway Load Balancers.
Choose from Classic Load Balancers
Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.
Select target groups
App-Server-TG | HTTP
Application Load Balancer: Application-Load-Balancer
VPC Lattice integration options Info
To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.
Select VPC Lattice service to attach
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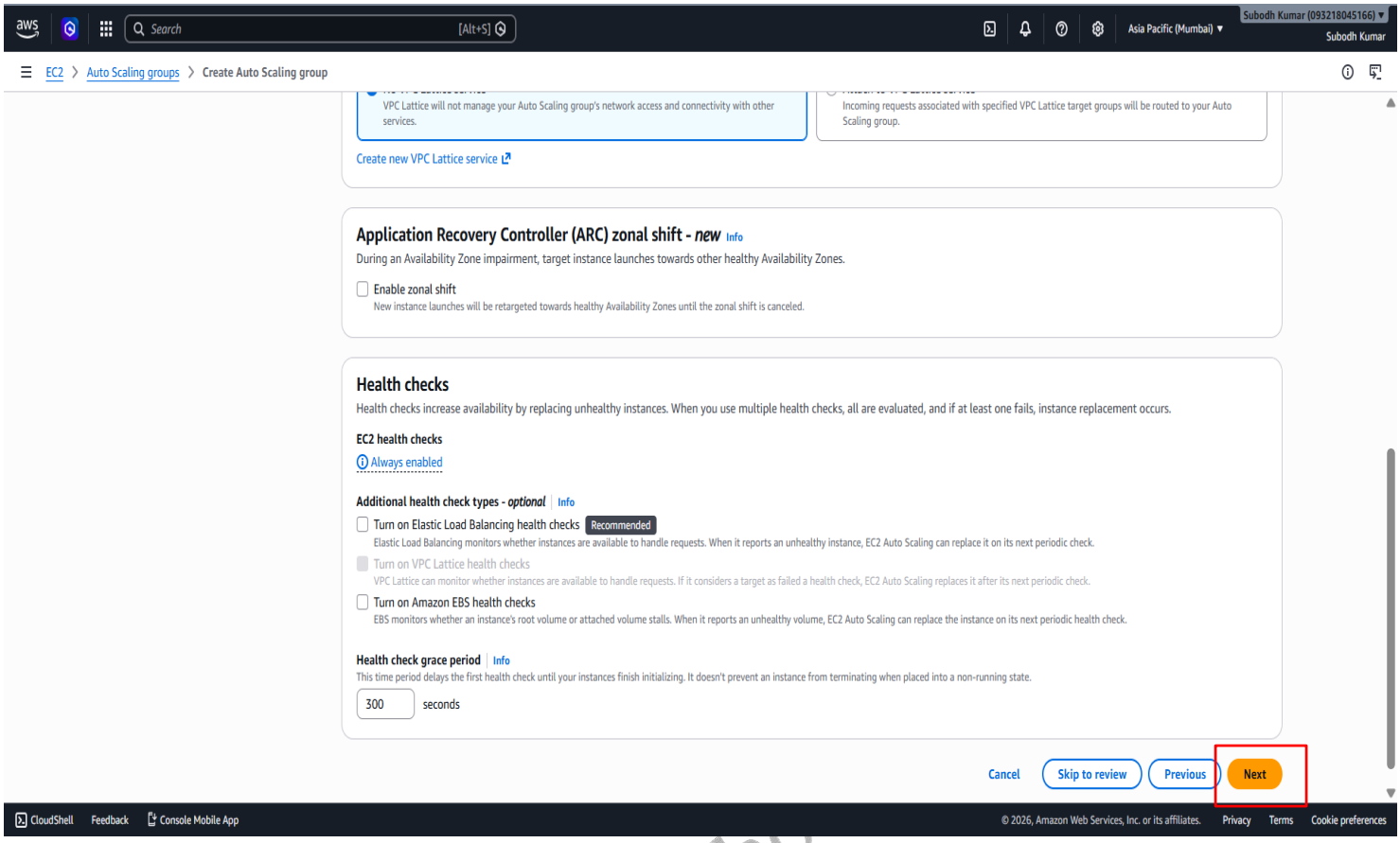
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EC2 > Auto Scaling groups > Create Auto Scaling group
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.
Create new VPC Lattice service
Application Recovery Controller (ARC) zonal shift - new Info
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.
Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.
Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.
EC2 health checks
Always enabled
Additional health check types - optional Info
Turn on Elastic Load Balancing health checks Recommended
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.
Turn on VPC Lattice health checks
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.
Turn on Amazon EBS health checks
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.
Health check grace period Info
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.
300 seconds
Cancel
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Previous
Next
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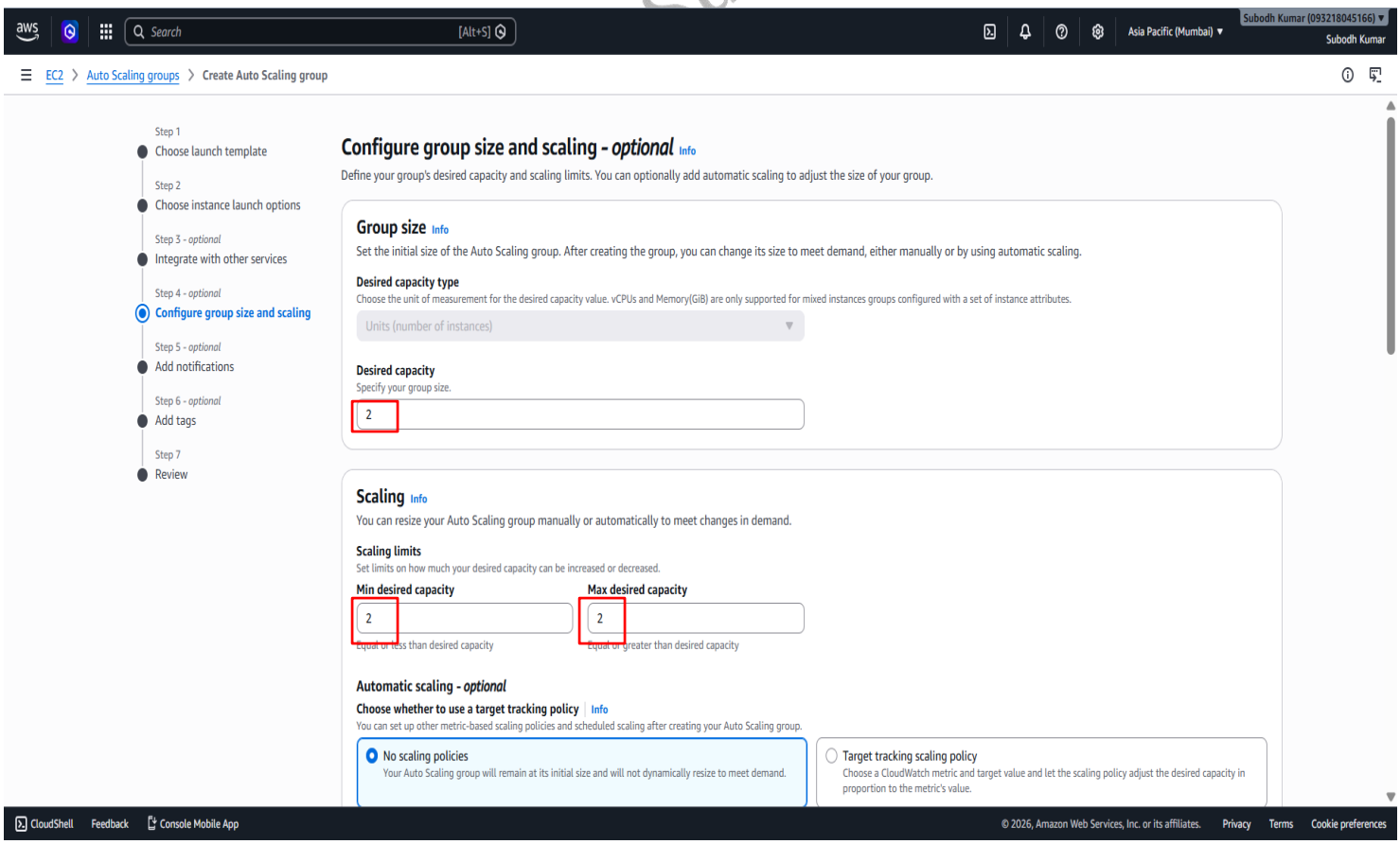
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Configure group size and scaling - optional Info
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.
Group size Info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.
Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)
Desired capacity
Specify your group size.
2
Scaling Info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.
Scaling limits
Set limits on how much your desired capacity can be increased or decreased.
Min desired capacity
2
Equal or less than desired capacity
Max desired capacity
2
Equal or greater than desired capacity
Automatic scaling - optional
Choose whether to use a target tracking policy Info
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.
No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.
Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.
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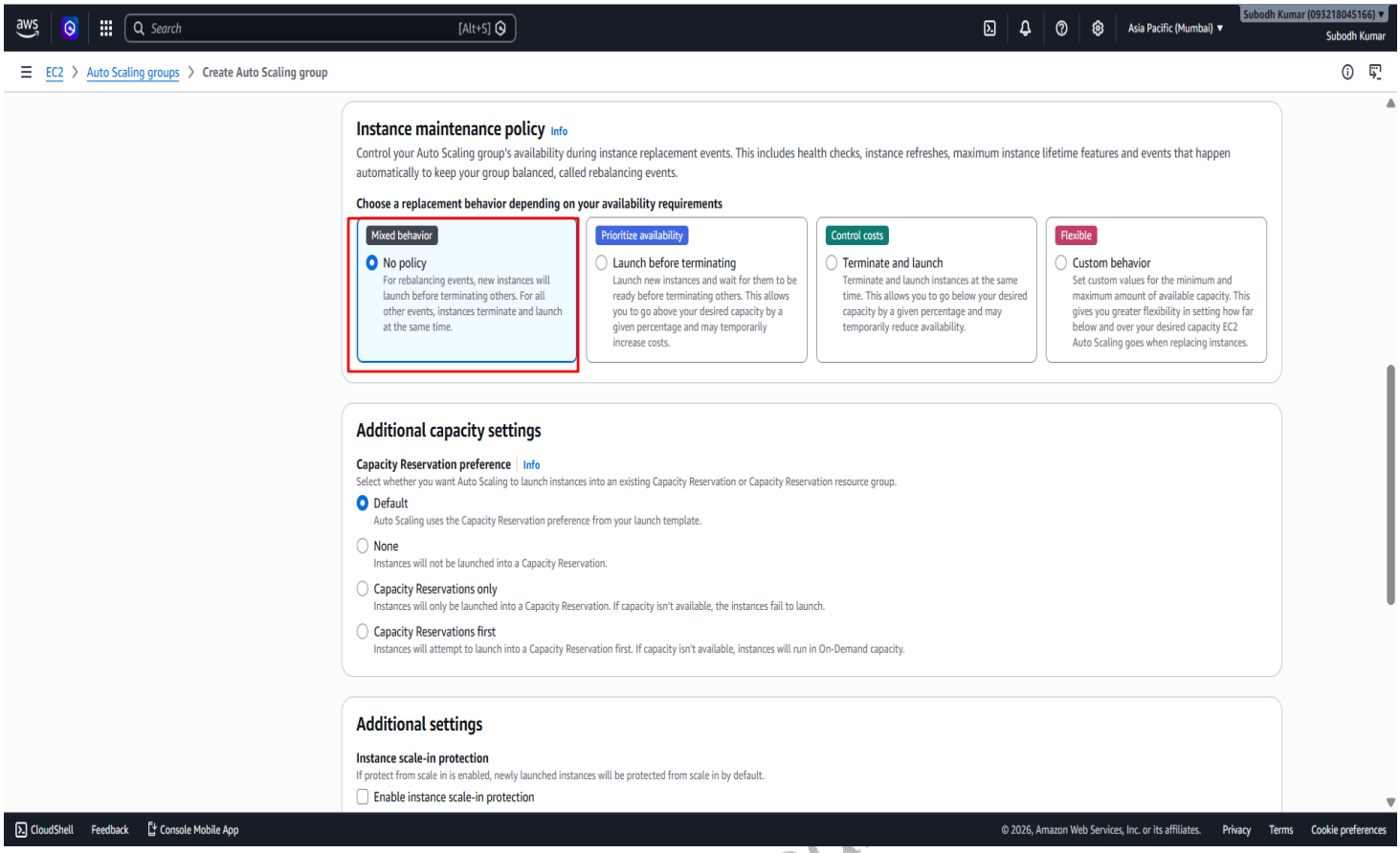
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EC2 > Auto Scaling groups > Create Auto Scaling group
Instance maintenance policy Info
Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.
Choose a replacement behavior depending on your availability requirements
Mixed behavior
No policy
For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.
Prioritize availability
Launch before terminating
Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.
Control costs
Terminate and launch
Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.
Flexible
Custom behavior
Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.
Additional capacity settings
Capacity Reservation preference Info
Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.
Default
Auto Scaling uses the Capacity Reservation preference from your launch template.
None
Instances will not be launched into a Capacity Reservation.
Capacity Reservations only
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.
Capacity Reservations first
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.
Additional settings
Instance scale-in protection
If protect from scale in is enabled, newly launched instances will be protected from scale in by default.
Enable instance scale-in protection
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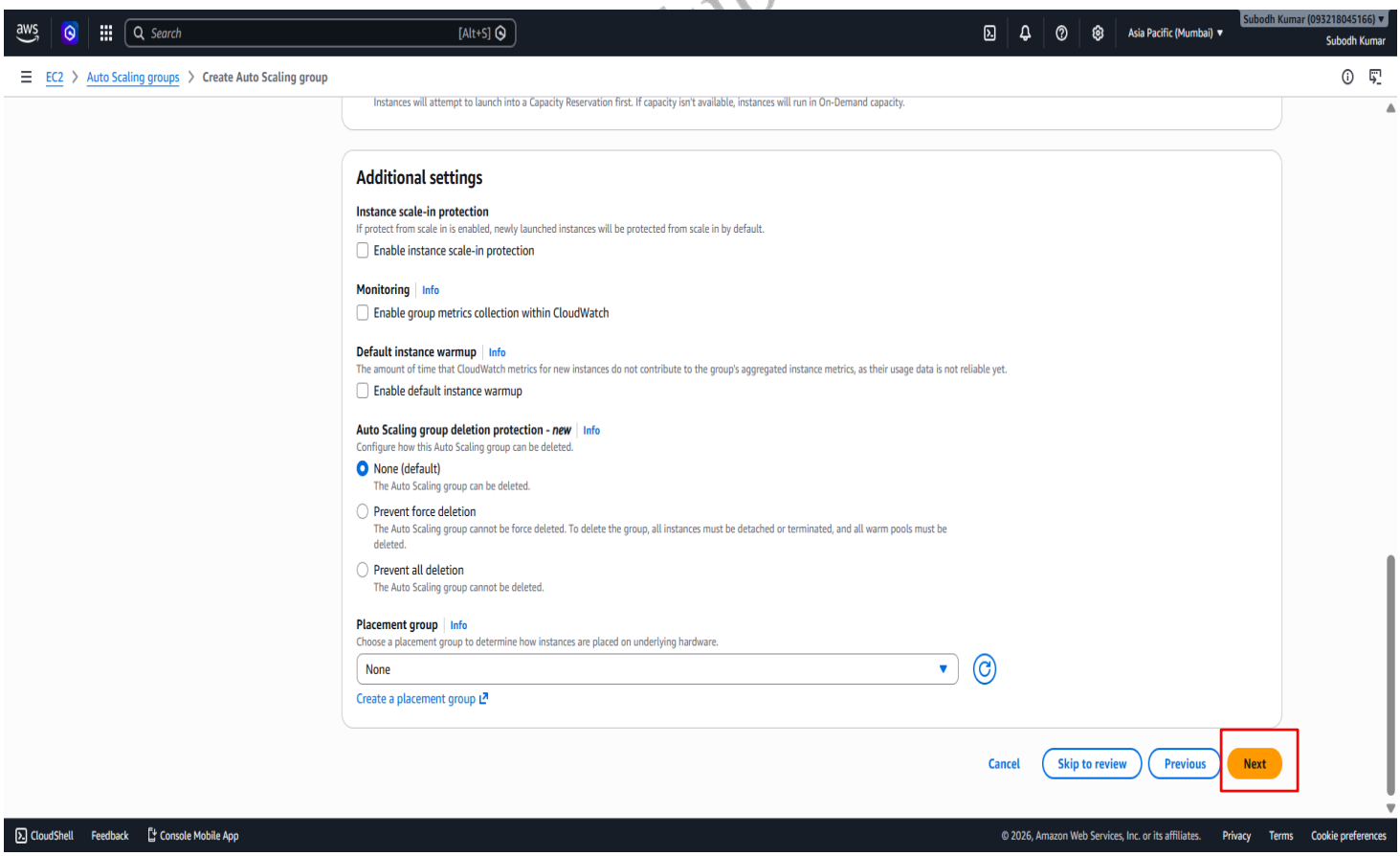
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EC2 > Auto Scaling groups > Create Auto Scaling group
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.
Additional settings
Instance scale-in protection
If protect from scale in is enabled, newly launched instances will be protected from scale in by default.
Enable instance scale-in protection
Monitoring Info
Enable group metrics collection within CloudWatch
Default instance warmup Info
The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.
Enable default instance warmup
Auto Scaling group deletion protection - new Info
Configure how this Auto Scaling group can be deleted.
None (default)
The Auto Scaling group can be deleted.
Prevent force deletion
The Auto Scaling group cannot be force deleted. To delete the group, all instances must be detached or terminated, and all warm pools must be deleted.
Prevent all deletion
The Auto Scaling group cannot be deleted.
Placement group Info
Choose a placement group to determine how instances are placed on underlying hardware.
None
Create a placement group
Cancel
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Previous
Next
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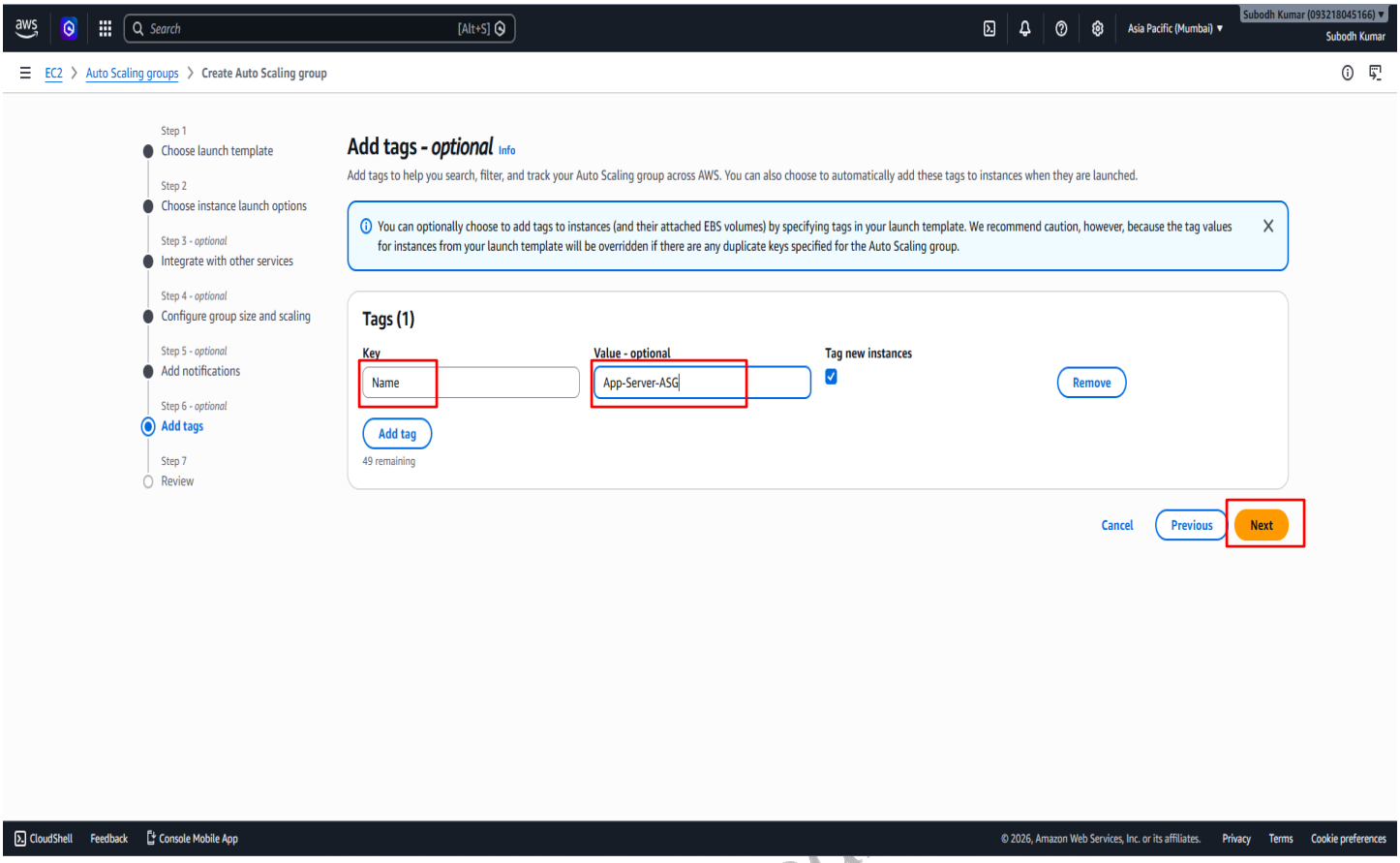
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1 Choose launch template
Step 2 Choose instance launch options
Step 3 - optional Integrate with other services
Step 4 - optional Configure group size and scaling
Step 5 - optional Add notifications
Step 6 - optional Add tags
Step 7 Review
Add tags - optional Info
Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.
You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.
Tags (1)
Key
Value - optional
Tag new instances
Name
App-Server-ASG
Remove
Add tag
49 remaining
Cancel
Previous
Next
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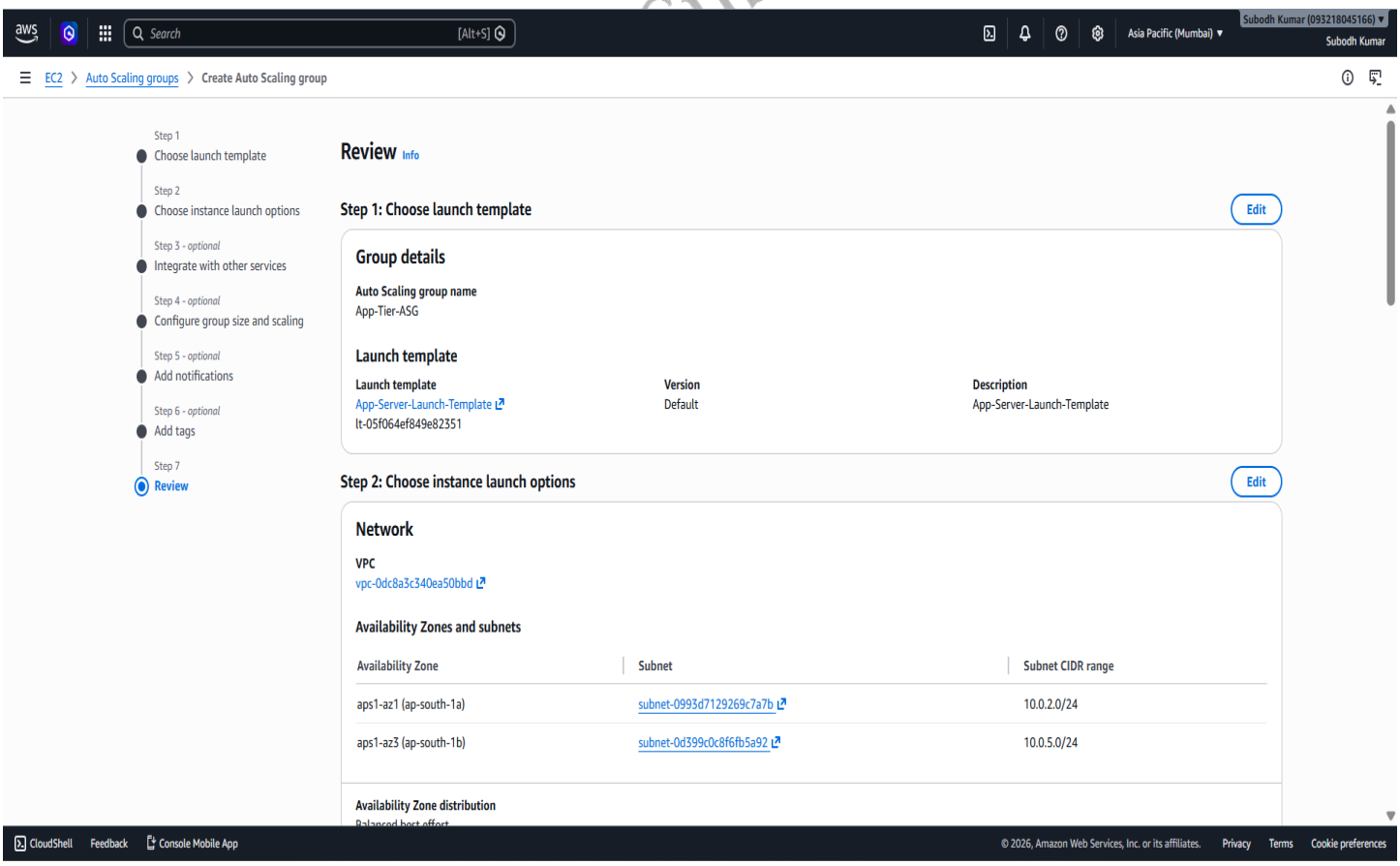
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1 Choose launch template
Step 2 Choose instance launch options
Step 3 - optional Integrate with other services
Step 4 - optional Configure group size and scaling
Step 5 - optional Add notifications
Step 6 - optional Add tags
Step 7 Review
Review Info
Step 1: Choose launch template
Edit
Group details
Auto Scaling group name
App-Tier-ASG
Launch template
Launch template
App-Server-Launch-Template
lt-05f064ef849e82351
Version
Default
Description
App-Server-Launch-Template
Step 2: Choose instance launch options
Edit
Network
VPC
vpc-0dc8a3c340ea50bbd
Availability Zones and subnets
Availability Zone
Subnet
Subnet CIDR range
aps1-az1 (ap-south-1a)
subnet-0993d7129269c7a7b
10.0.2.0/24
aps1-az3 (ap-south-1b)
subnet-0d399c0c8f6fb5a92
10.0.5.0/24
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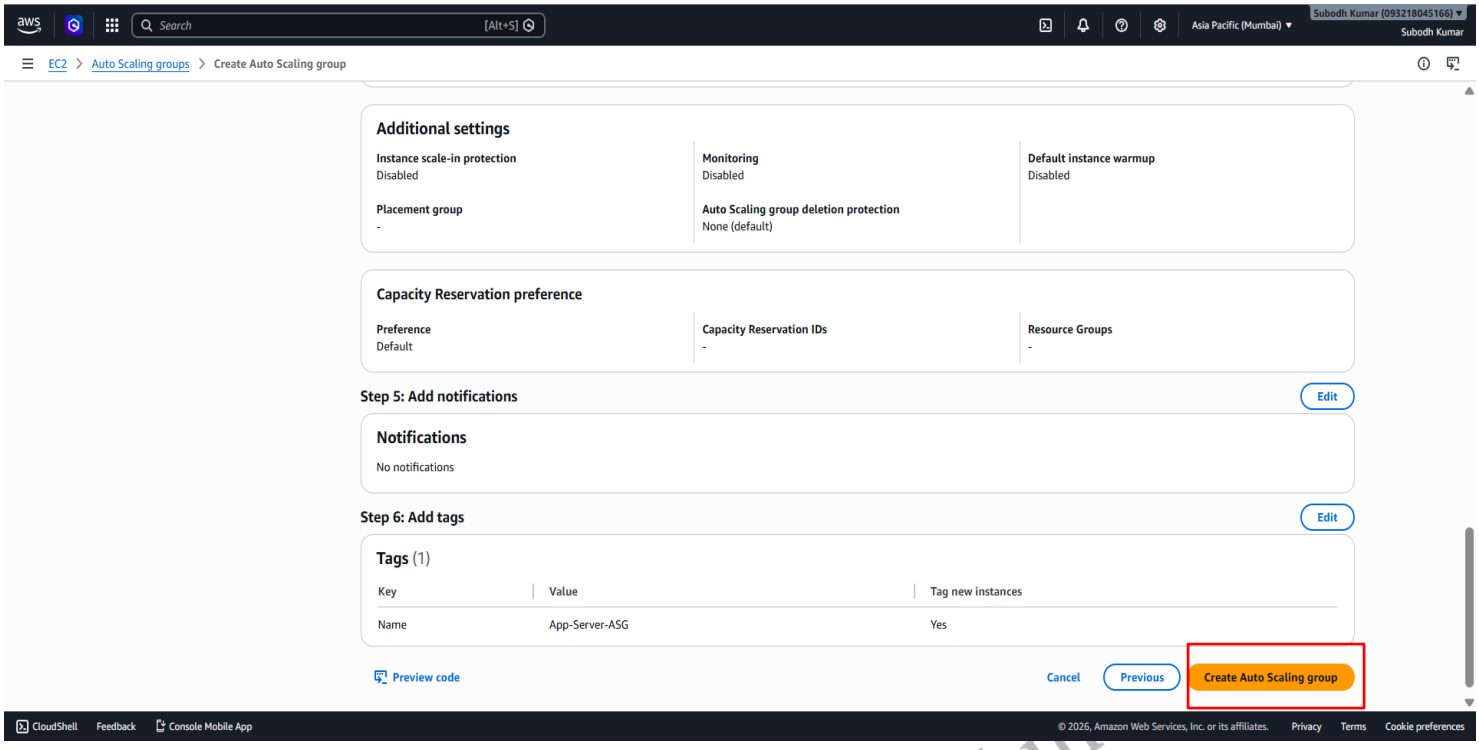
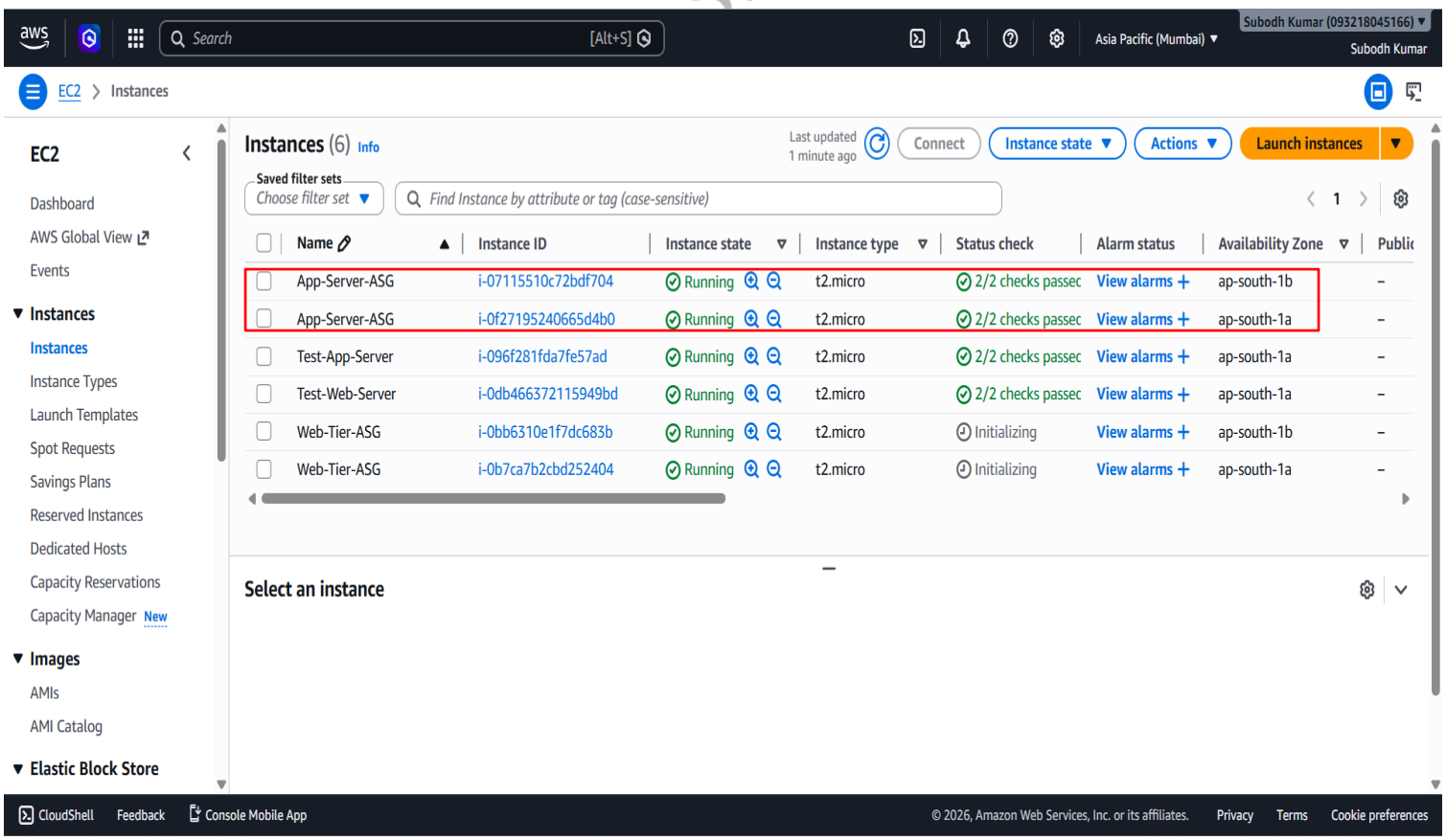


Keep other services as default and Create Auto scaling group.

Now if you **check** the **instance list** you Can see that **two App Servers** are **creating**



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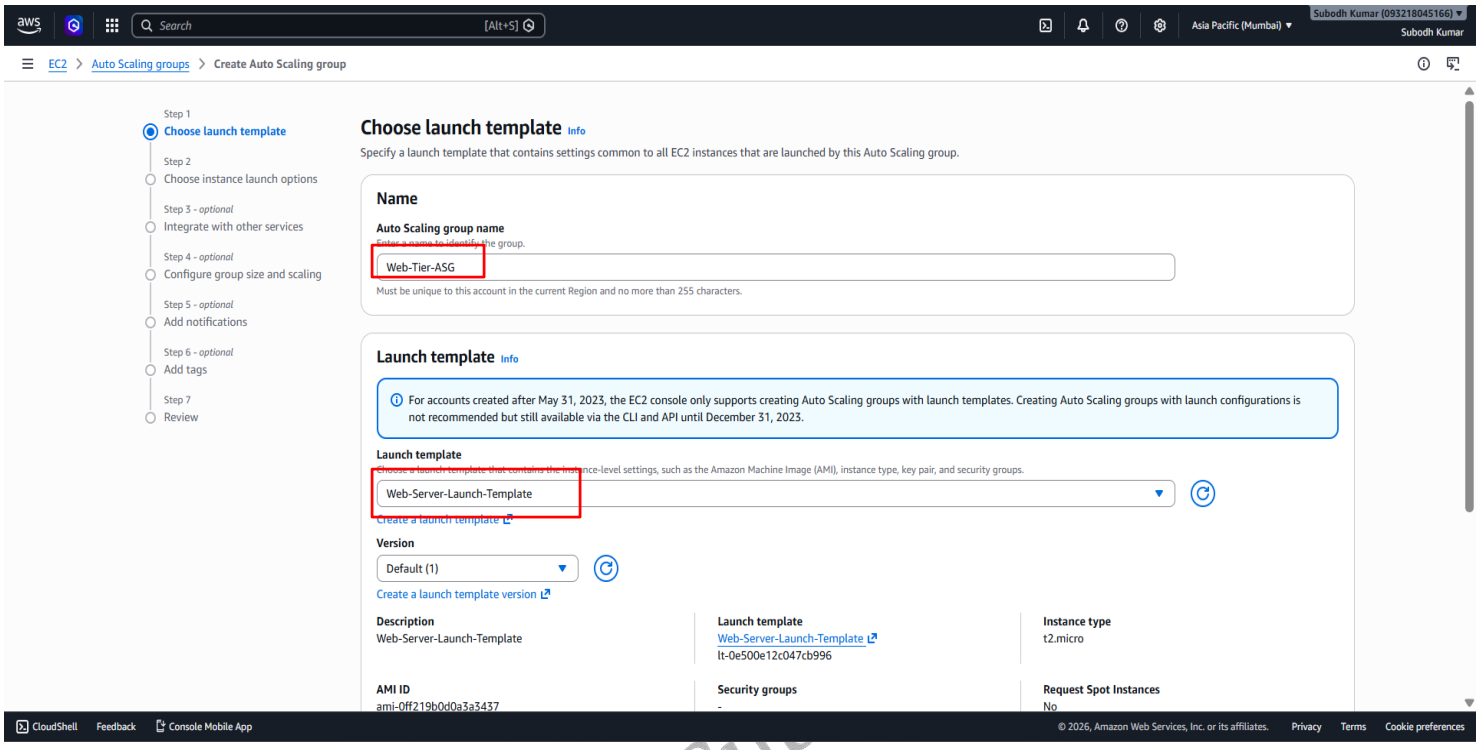
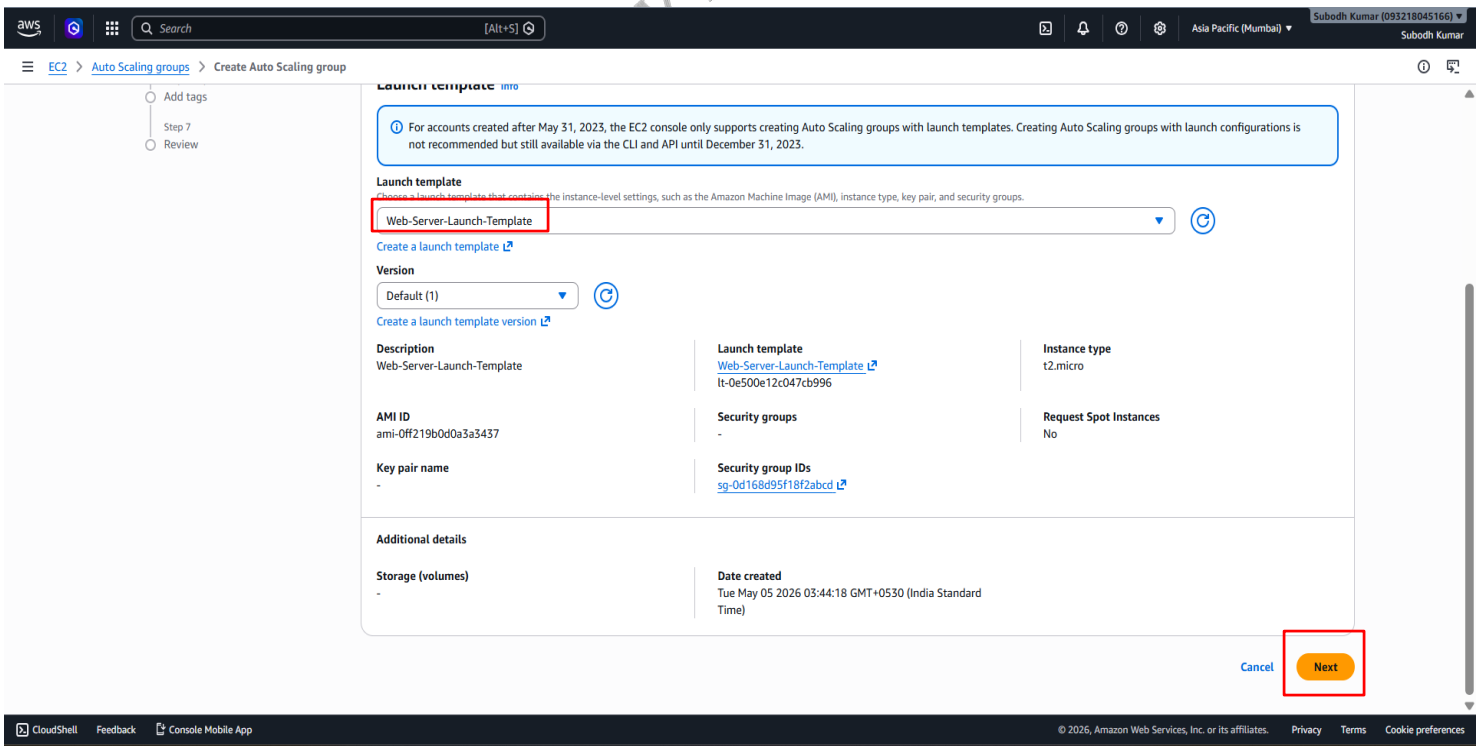
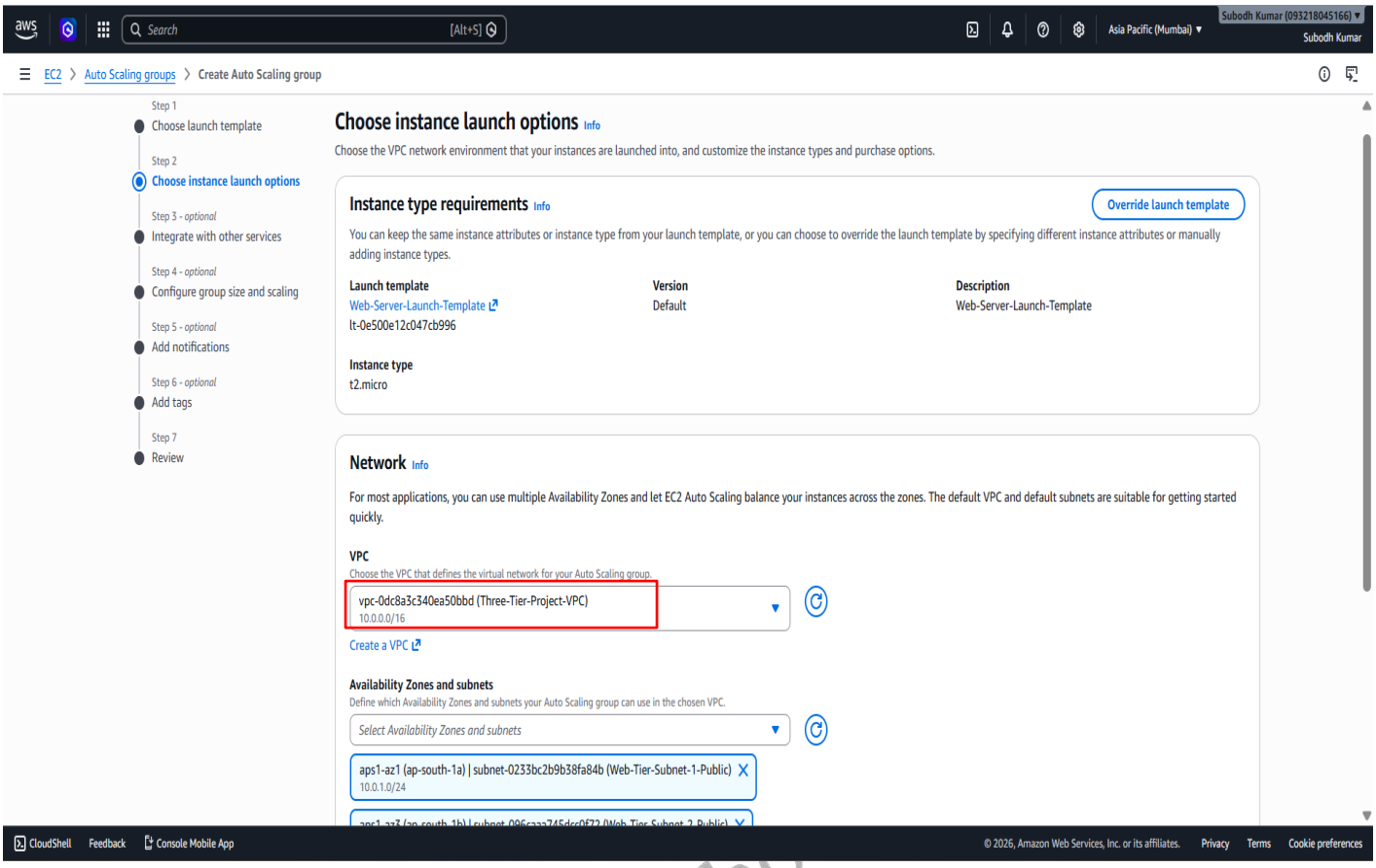
Auto Scaling group for Web Tier

The process is the same. Choose components related to Web Tier.



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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Choose instance launch options Info
Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.
Instance type requirements Info
Override launch template
You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.
Launch template
Web-Server-Launch-Template
lt-0e500e12c047cb996
Version
Default
Description
Web-Server-Launch-Template
Instance type
t2.micro
Network Info
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.
VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create a VPC
Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
aps1-az1 (ap-south-1a) | subnet-0233bc2b9b38fa84b (Web-Tier-Subnet-1-Public)
10.0.1.0/24
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Feedback
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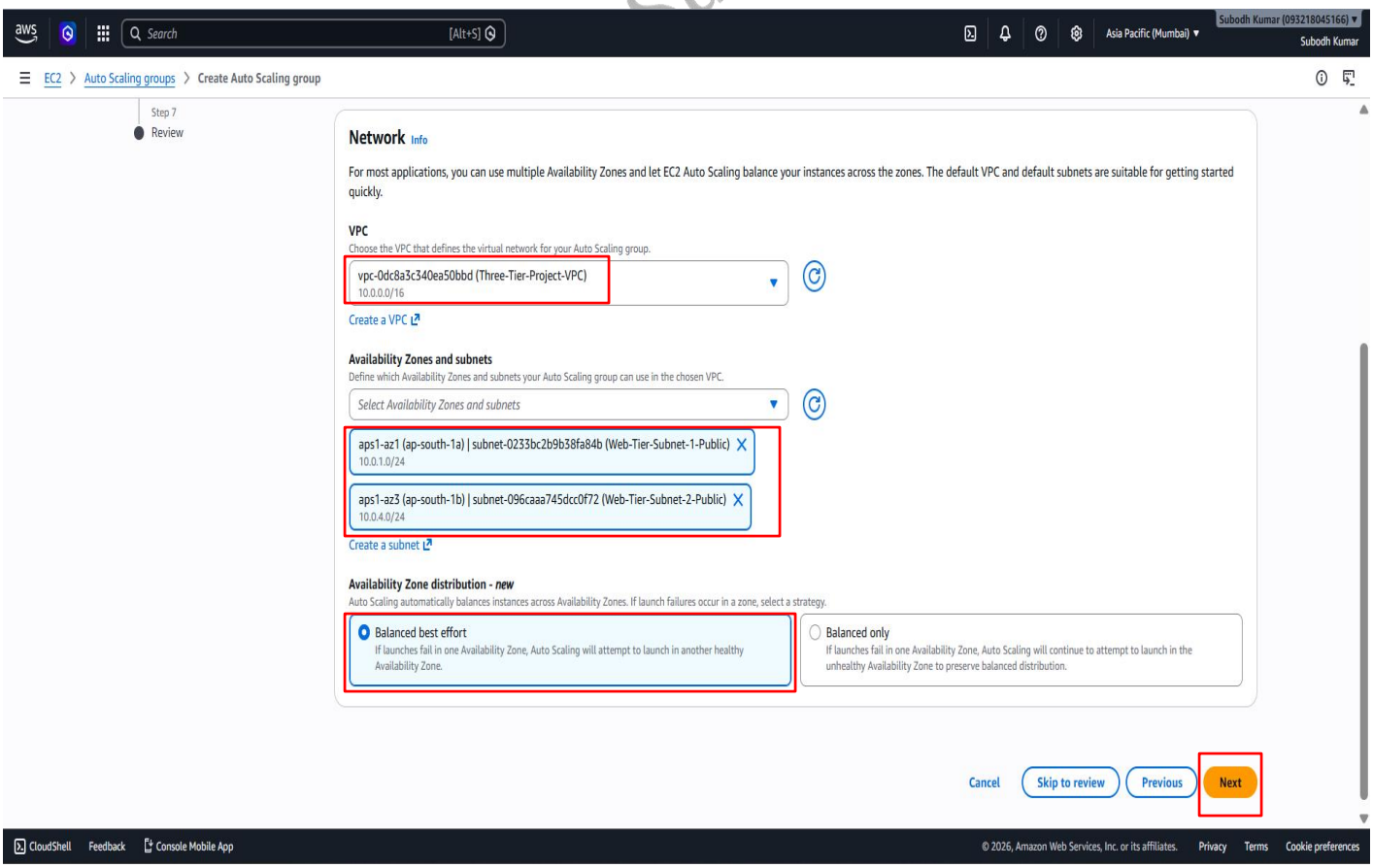
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Step 7
Review
Network Info
For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.
VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.
vpc-0dc8a3c340ea50bbd (Three-Tier-Project-VPC)
10.0.0.0/16
Create a VPC
Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.
Select Availability Zones and subnets
aps1-az1 (ap-south-1a) | subnet-0233bc2b9b38fa84b (Web-Tier-Subnet-1-Public)
10.0.1.0/24
aps1-az3 (ap-south-1b) | subnet-096caaa745dcc0f72 (Web-Tier-Subnet-2-Public)
10.0.4.0/24
Create a subnet
Availability Zone distribution - new
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.
Balanced best effort
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.
Balanced only
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.
Cancel
Skip to review
Previous
Next
CloudShell
Feedback
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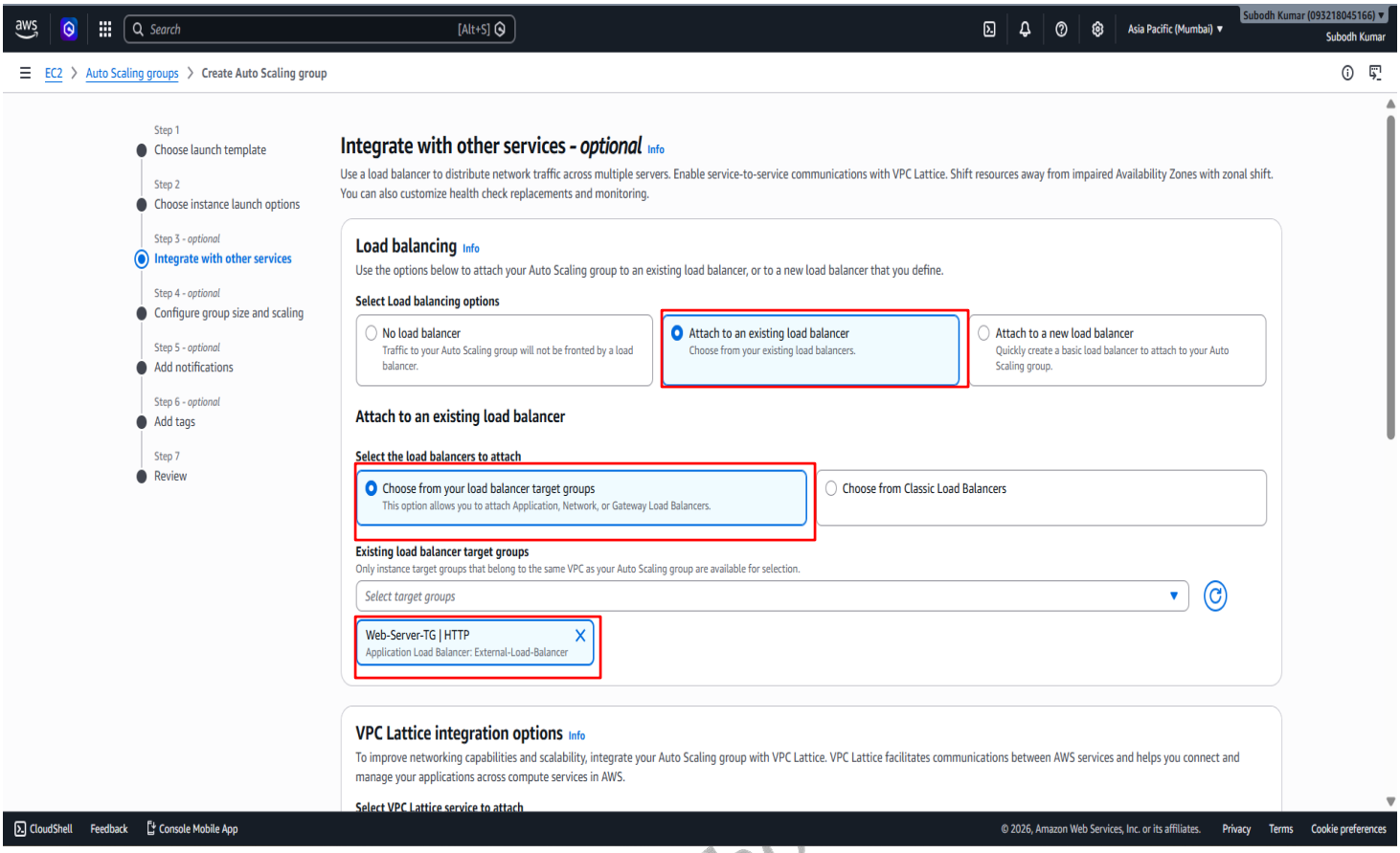
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Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
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Configure group size and scaling
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Add notifications
Step 6 - optional
Add tags
Step 7
Review
Integrate with other services - optional Info
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.
Load balancing Info
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.
Select Load balancing options
No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
Attach to an existing load balancer
Choose from your existing load balancers.
Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.
Attach to an existing load balancer
Select the load balancers to attach
Choose from your load balancer target groups
This option allows you to attach Application, Network, or Gateway Load Balancers.
Choose from Classic Load Balancers
Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.
Select target groups
Web-Server-TG | HTTP
Application Load Balancer: External-Load-Balancer
VPC Lattice integration options Info
To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.
Select VPC Lattice service to attach
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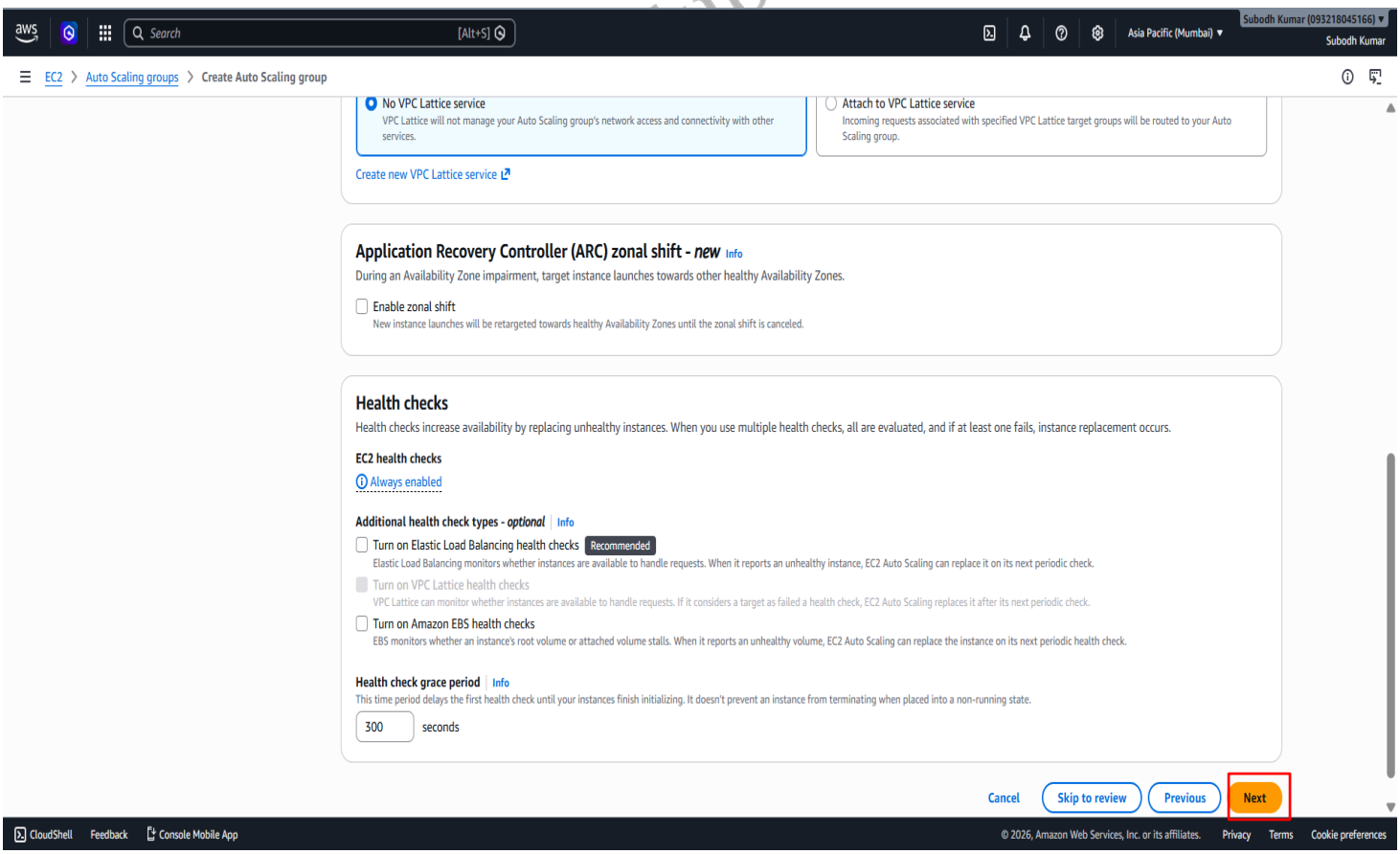
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EC2 > Auto Scaling groups > Create Auto Scaling group
No VPC Lattice service
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.
Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.
Create new VPC Lattice service
Application Recovery Controller (ARC) zonal shift - new Info
During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.
Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.
Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.
EC2 health checks
Always enabled
Additional health check types - optional Info
Turn on Elastic Load Balancing health checks
Recommended
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.
Turn on VPC Lattice health checks
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.
Turn on Amazon EBS health checks
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.
Health check grace period Info
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.
300
seconds
Cancel
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Previous
Next
CloudShell
Feedback
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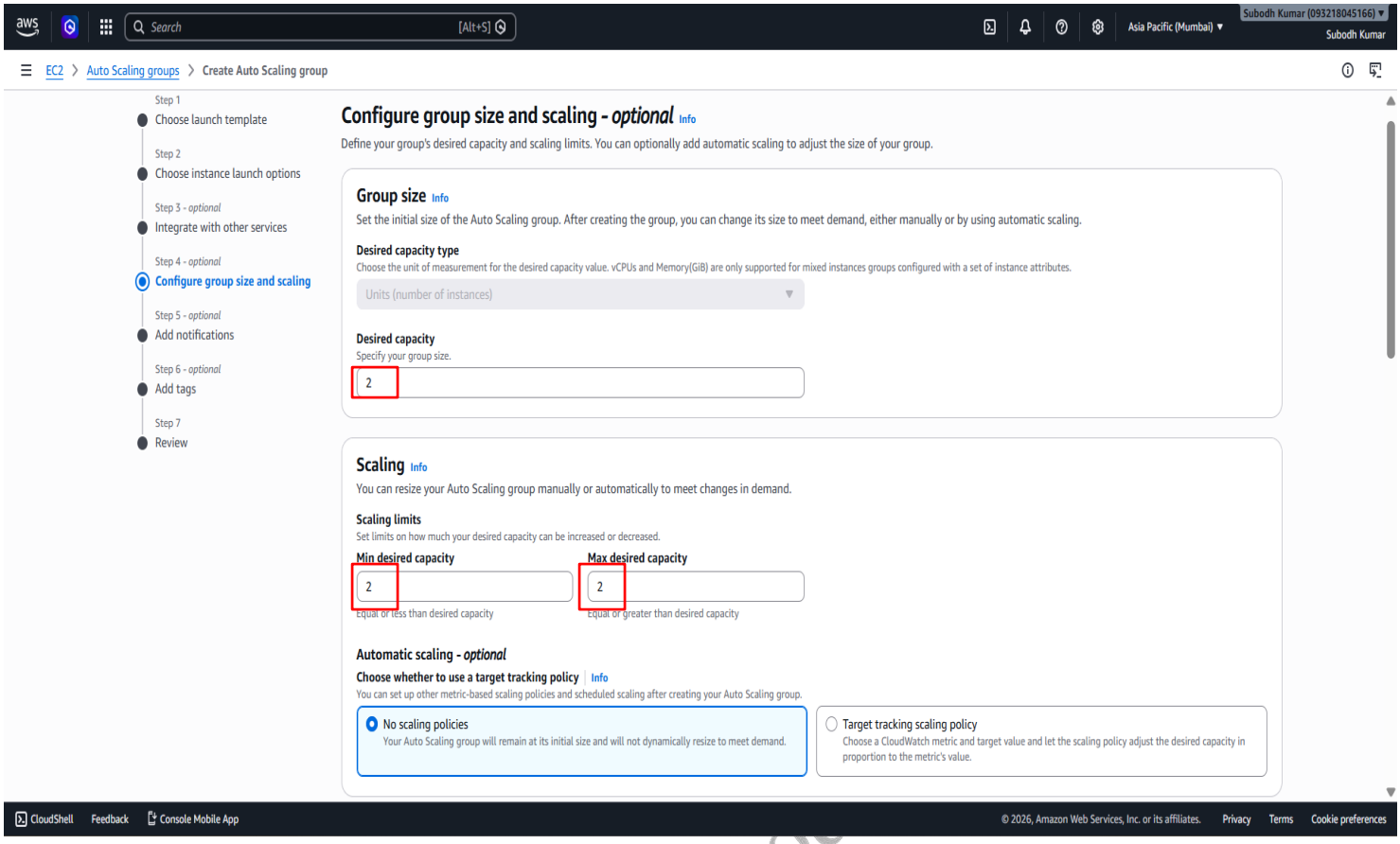
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Configure group size and scaling - optional Info
Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.
Group size Info
Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.
Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)
Desired capacity
Specify your group size.
2
Scaling Info
You can resize your Auto Scaling group manually or automatically to meet changes in demand.
Scaling limits
Set limits on how much your desired capacity can be increased or decreased.
Min desired capacity
2
Equal or less than desired capacity
Max desired capacity
2
Equal or greater than desired capacity
Automatic scaling - optional
Choose whether to use a target tracking policy Info
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.
No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.
Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.
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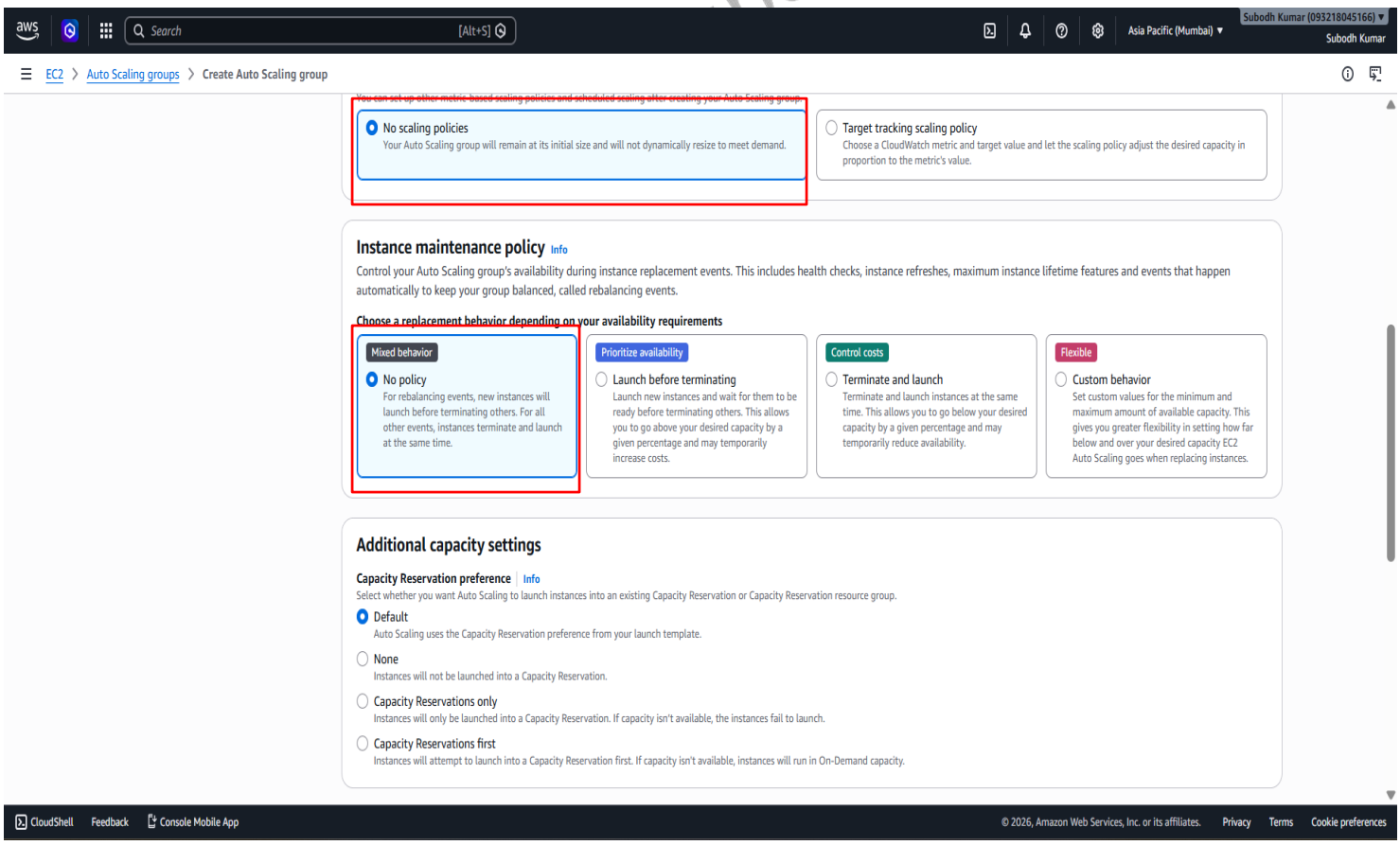
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EC2 > Auto Scaling groups > Create Auto Scaling group
No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.
Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.
Instance maintenance policy Info
Control your Auto Scaling group's availability during instance replacement events. This includes health checks, instance refreshes, maximum instance lifetime features and events that happen automatically to keep your group balanced, called rebalancing events.
Choose a replacement behavior depending on your availability requirements
Mixed behavior
No policy
For rebalancing events, new instances will launch before terminating others. For all other events, instances terminate and launch at the same time.
Prioritize availability
Launch before terminating
Launch new instances and wait for them to be ready before terminating others. This allows you to go above your desired capacity by a given percentage and may temporarily increase costs.
Control costs
Terminate and launch
Terminate and launch instances at the same time. This allows you to go below your desired capacity by a given percentage and may temporarily reduce availability.
Flexible
Custom behavior
Set custom values for the minimum and maximum amount of available capacity. This gives you greater flexibility in setting how far below and over your desired capacity EC2 Auto Scaling goes when replacing instances.
Additional capacity settings
Capacity Reservation preference Info
Select whether you want Auto Scaling to launch instances into an existing Capacity Reservation or Capacity Reservation resource group.
Default
Auto Scaling uses the Capacity Reservation preference from your launch template.
None
Instances will not be launched into a Capacity Reservation.
Capacity Reservations only
Instances will only be launched into a Capacity Reservation. If capacity isn't available, the instances fail to launch.
Capacity Reservations first
Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.
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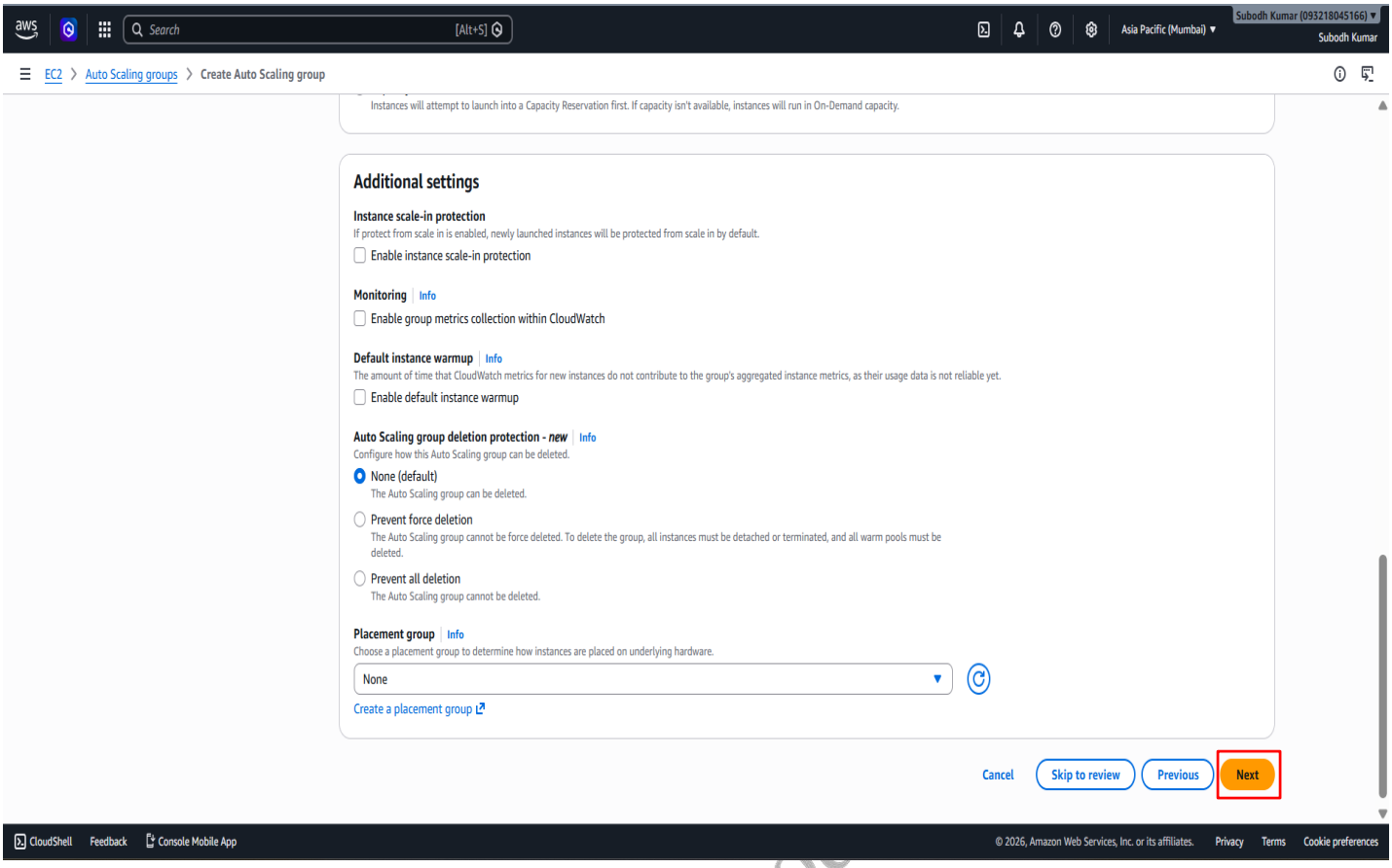
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Instances will attempt to launch into a Capacity Reservation first. If capacity isn't available, instances will run in On-Demand capacity.
Additional settings
Instance scale-in protection
If protect from scale in is enabled, newly launched instances will be protected from scale in by default.
Enable instance scale-in protection
Monitoring | Info
Enable group metrics collection within CloudWatch
Default instance warmup | Info
The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.
Enable default instance warmup
Auto Scaling group deletion protection - new | Info
Configure how this Auto Scaling group can be deleted.
None (default)
The Auto Scaling group can be deleted.
Prevent force deletion
The Auto Scaling group cannot be force deleted. To delete the group, all instances must be detached or terminated, and all warm pools must be deleted.
Prevent all deletion
The Auto Scaling group cannot be deleted.
Placement group | Info
Choose a placement group to determine how instances are placed on underlying hardware.
None
Create a placement group
Cancel
Skip to review
Previous
Next
CloudShell
Feedback
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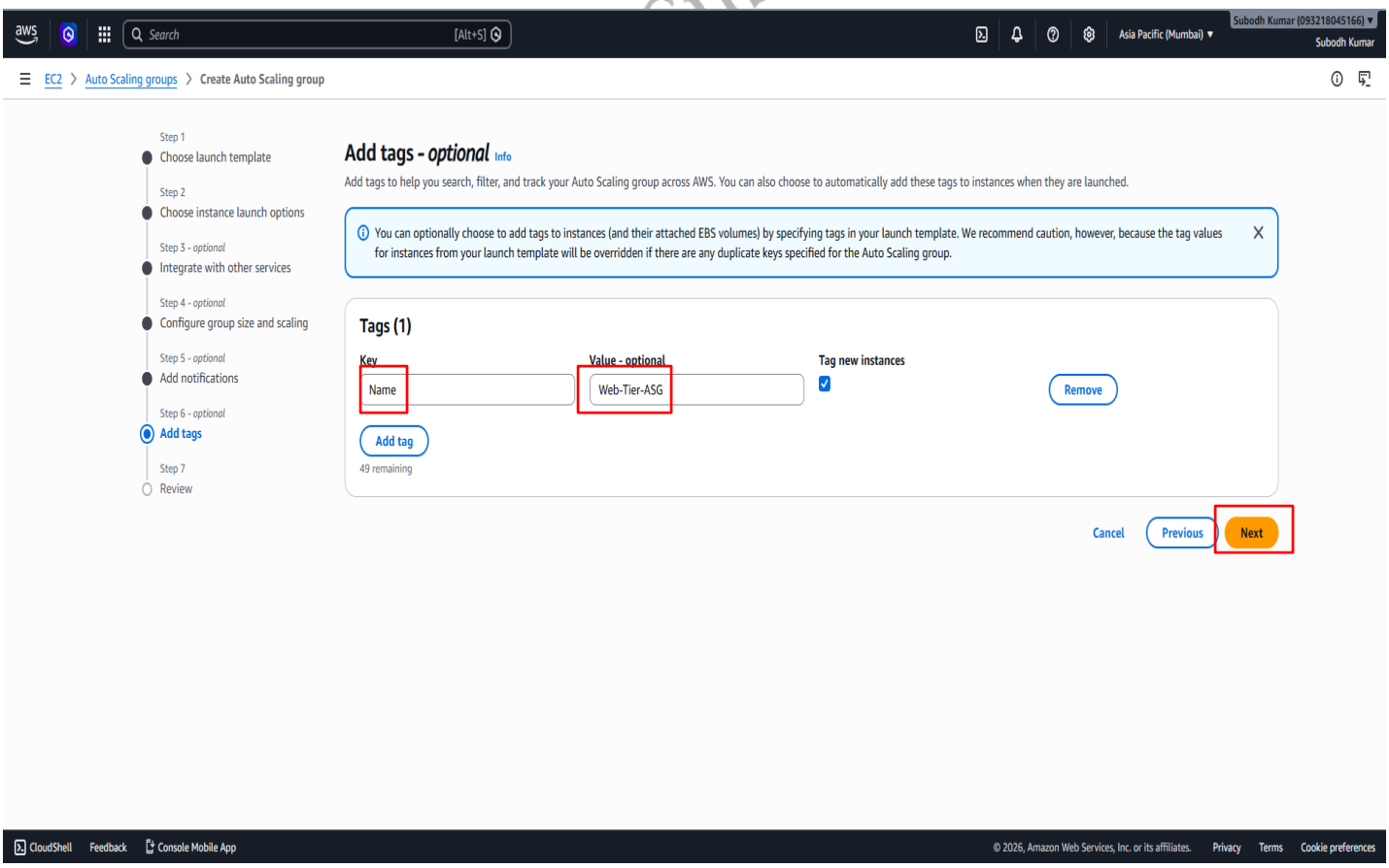
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1
Choose launch template
Step 2
Choose instance launch options
Step 3 - optional
Integrate with other services
Step 4 - optional
Configure group size and scaling
Step 5 - optional
Add notifications
Step 6 - optional
Add tags
Step 7
Review
Add tags - optional Info
Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.
You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.
Tags (1)
Key
Name
Value - optional
Web-Tier-ASG
Tag new instances
Remove
Add tag
49 remaining
Cancel
Previous
Next
CloudShell
Feedback
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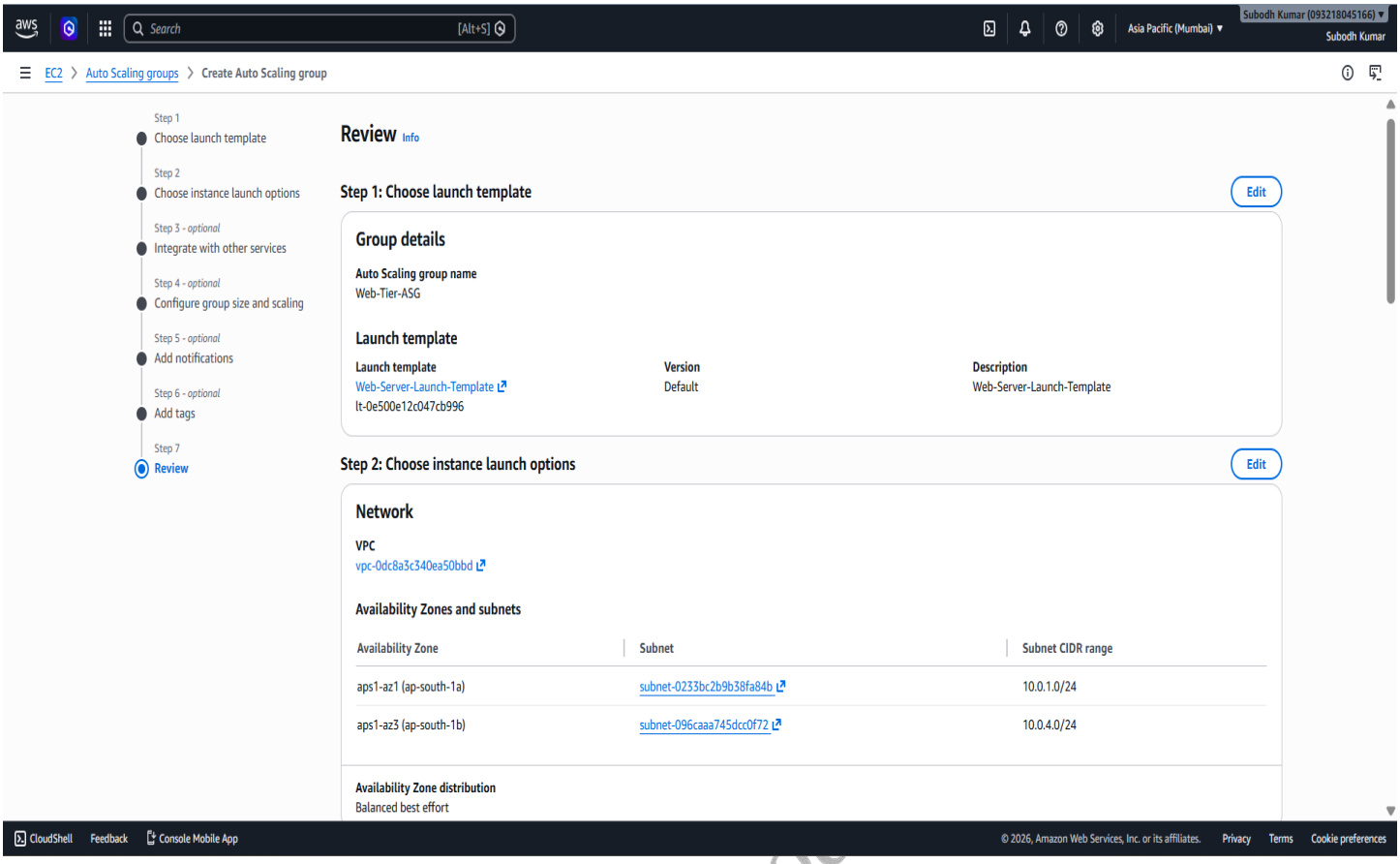
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EC2 > Auto Scaling groups > Create Auto Scaling group
Step 1 Choose launch template
Step 2 Choose instance launch options
Step 3 - optional Integrate with other services
Step 4 - optional Configure group size and scaling
Step 5 - optional Add notifications
Step 6 - optional Add tags
Step 7 Review
Review Info
Step 1: Choose launch template
Edit
Group details
Auto Scaling group name
Web-Tier-ASG
Launch template
Launch template
Web-Server-Launch-Template
lt-0e500e12c047cb996
Version
Default
Description
Web-Server-Launch-Template
Step 2: Choose instance launch options
Edit
Network
VPC
vpc-0dc8a3c340ea50bbd
Availability Zones and subnets
Availability Zone
Subnet
Subnet CIDR range
aps1-az1 (ap-south-1a)
subnet-0233bc2b9b38fa84b
10.0.1.0/24
aps1-az3 (ap-south-1b)
subnet-096caaa745dcc0f72
10.0.4.0/24
Availability Zone distribution
Balanced best effort
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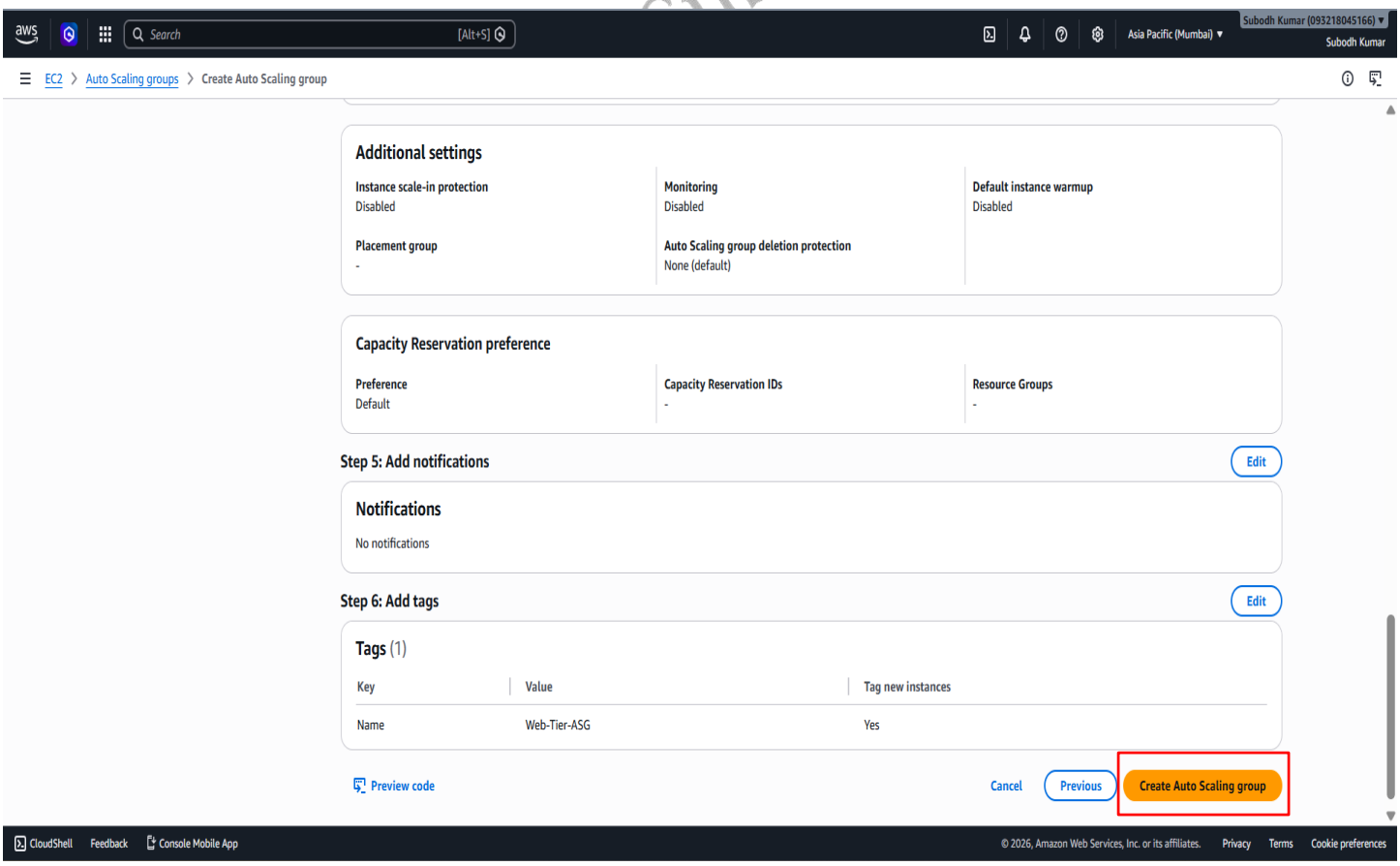
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EC2 > Auto Scaling groups > Create Auto Scaling group
Additional settings
Instance scale-in protection
Disabled
Monitoring
Disabled
Default instance warmup
Disabled
Placement group
-
Auto Scaling group deletion protection
None (default)
Capacity Reservation preference
Preference
Default
Capacity Reservation IDs
-
Resource Groups
-
Step 5: Add notifications
Edit
Notifications
No notifications
Step 6: Add tags
Edit
Tags (1)
Key
Value
Tag new instances
Name
Web-Tier-ASG
Yes
Preview code
Cancel
Previous
Create Auto Scaling group
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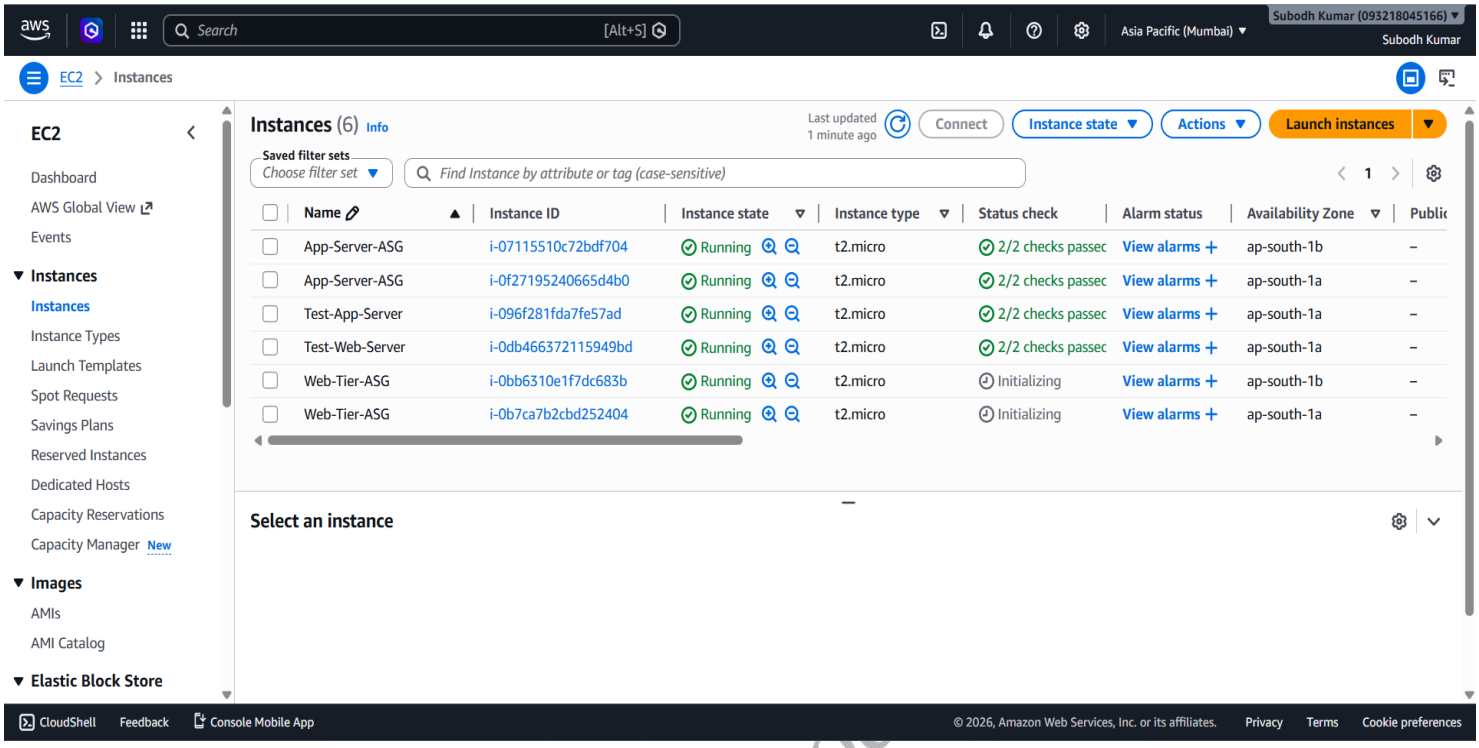
After you are creating it you can see 4 instance and test instance on the list. Now you can delete the test instances .

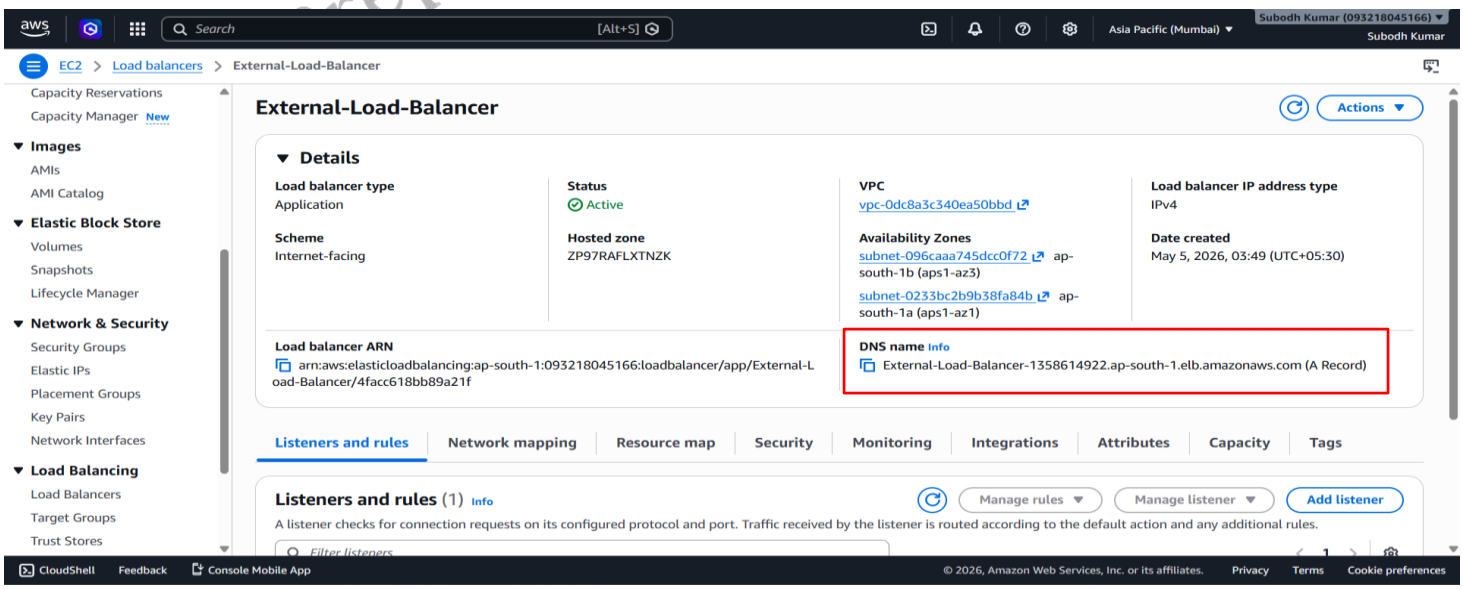


Cool You did a great job

Now We Can access our Web application.

14. Access the Application

Go to **EC2 > Load Balancers**. Select **External Load Balancer**. Copy the **DNS name** of it.



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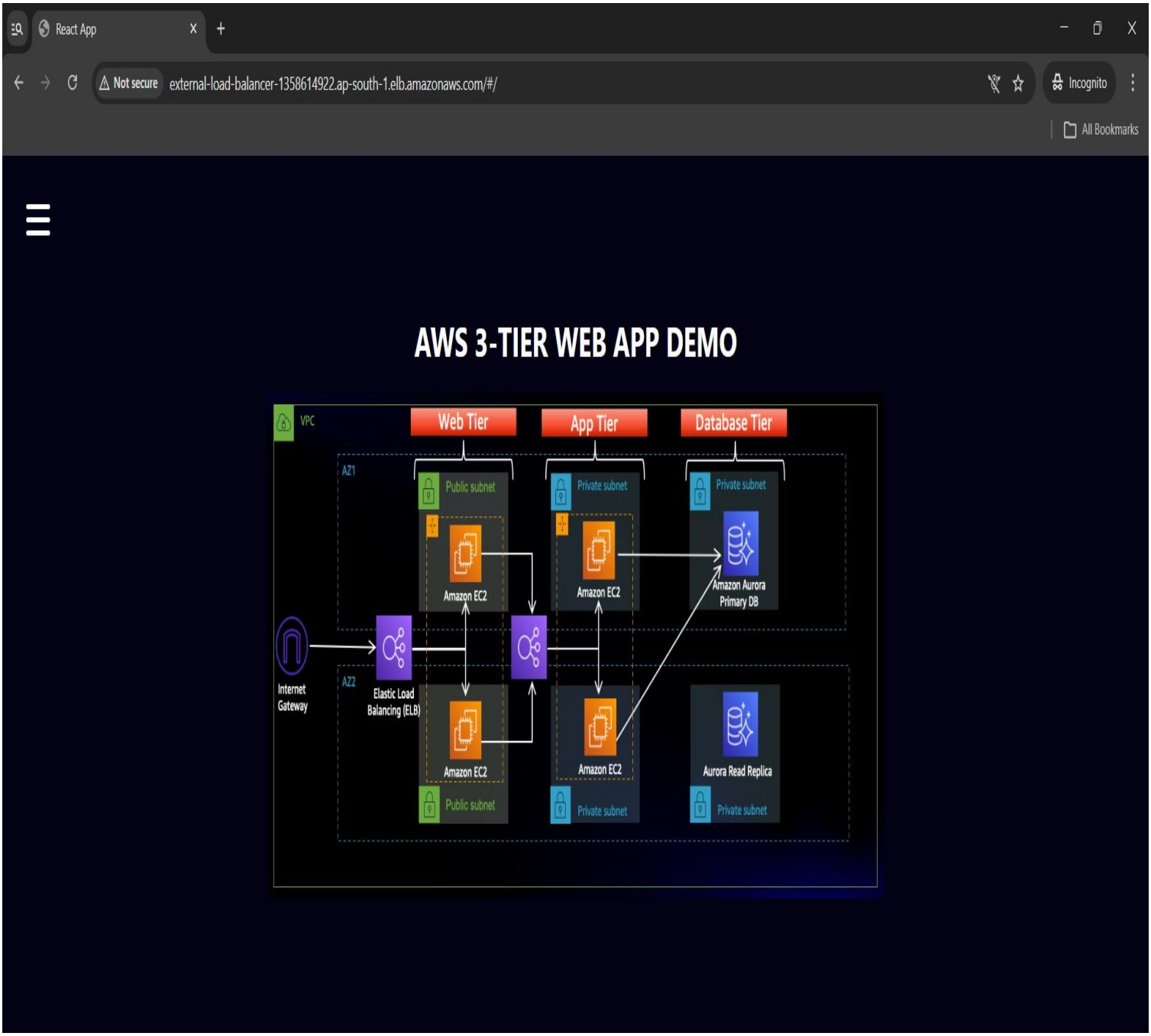
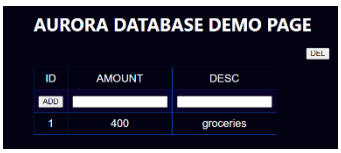
Paste It into your **Browser**. You will **see** the **Application** is **running**

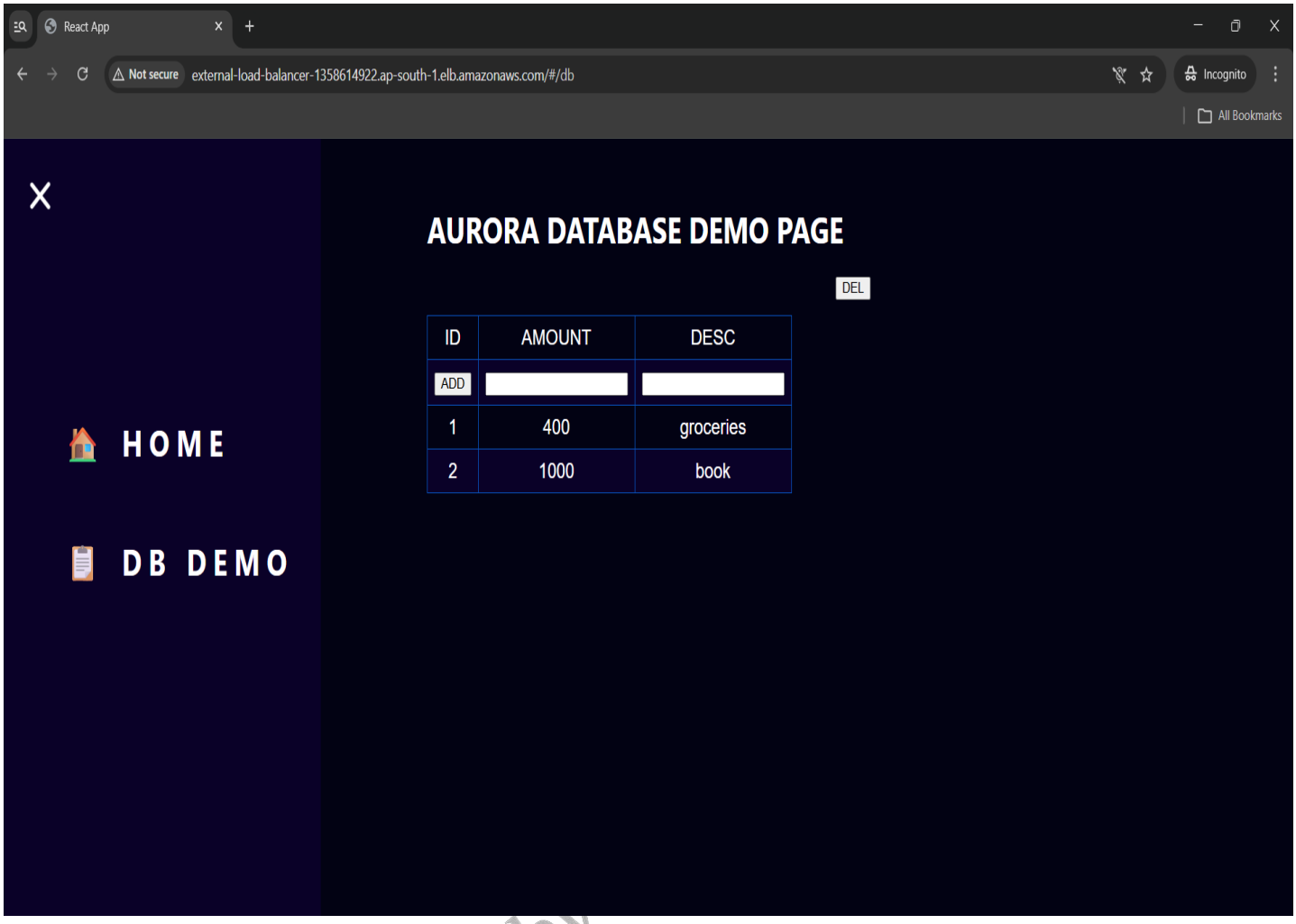
Note:- If not open please open in Incognito mode.



Go to DB DEMO section and check whether DB is working.



Here you can see the Example data we entered when setup the DB which means it receiving data from DB.



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You can add and delete items

Congratulations!!! You have successfully implemented Three Tier Web Application on AWS

Note: In hear we use External Load Balancer DNS name as the URL. But it is not user friendly. So, We can purchase a domain name from domain name service and setup it using Route 53 to access the web site using that domain name.

15. Cleaning the Infrastructure

Since some services we used charge cost we have to delete our demo project.

Let's **delete** them according to the following order.

1. Auto Scaling Groups
2. In Application Load Balancers delete listeners first and then Load balancers
3. Target Groups
4. Launch Templates
5. AMIs
6. Snapshots
7. EC2 Instances
8. Aurora Database (First go to modify and disable delete protection. Then when deleting do not create snapshot. Tick I acknowledge)
9. NAT gateways
10. Elastic Ips
11. Route Tables (Remove all subnet associations first)
12. Internet Gateways
13. VPC
14. S3 Bucket
15. IAM Role

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THANK YOU!

for going through this project documentation

★ ★ ★

Throughout this project, I interacted with the following components on AWS.

1. Amazon Elastic Compute Cloud (EC2)
2. Amazon Simple Storage Service (S3)
3. Virtual Private Cloud (VPC)
4. Auto Scaling Group (ASG)
5. Application Load Balancer (ALB)
6. Identity and Access Management (IAM)
7. Amazon Relational Database Service (RDS)
8. Amazon Machine Image (AMI)
9. Launch Templates
10. Target Groups
11. Internet Gateway (IGW)
12. NAT Gateway
13. Route Table
14. Security Groups

A big thank you to everyone who supported and guided me throughout this journey.
This project has been a valuable learning experience in building **HIGHLY AVAILABLE, SECURE, SCALABLE** and **FAULT TOLERANT** three-tier web application on AWS.

Prepared by

Subodh Kumar

AWS & DevOps Enthusiast

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